Datasheet

MCU Embedded and Configurable 3-phase Motor Controller FU6832

Fortior Technology (Shenzhen) Co., Ltd

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Explanation of Symbols

- The [ ] after a register indicates the bit in the register. E.g., ABCD[XY] indicates the XY bit in ABCD register
- The x in a register name indicates the similar registers. E.g., TIMx_CR0 indicate TIM3_CR0 and TIM4_CR0
- [m:n] indicates a range of bits. E.g., [3:0] means the bits from bit3 to bit0
- Pm.n indicates the nth port of the Portm. E.g., P0.0 indicates the 0th port of port0
- Registers read and write symbols:
  - R: Read only
  - W: Write only
  - R/W: Read and write
  - W0: Only 0 can be written
  - W1: Only 1 can be written
- The symbol of - indicates an uncertainty value
- The RMW instruction cannot be used for registers which have different functions when read and written to
- Q-value is a format for using fixed-point value to store floating-point value. MSB is the sign bit, followed by bits of integer and bits of fraction, where the bits value of fraction is Q and the bits value of integer is the rest. E.g., Q12, bit15 is the sign bit, bit14 ~ bit12 are bits of integer and bit11 ~ bit0 are bits of fraction. The Q12 number has a decimal range —8 ~ 7.9998(corresponding to 0x8000 ~ 0x7FFF).

Abbreviations

ADC: Analog Digital Convertor

BEMF: Back Electromotive Force

BLDC: Brushless Direct Current

CRC: Cyclic Redundancy Check

DAC: Digital Analog Convertor

DMA: Direct Memory Access

FG: Frequency Generator

FICE: Fortior Interactive Connectivity Establishment

FOC: Field Oriented Control

FOSC: Fast Oscillator

GPIO: General Purpose Input Output

I2C: Inter Integrated Circuit

IC: Integrated Circuit

IRAM: Internal RAM

IDE: Integrated Development Environment

LDO: Low Dropout Regulator

LIN: Local Interconnect Network

LPF: Low Pass Filter

LSB: Least Significant Bit

LVD: Low Voltage Detection

MDU: Multiplication Division Unit

ME: Motor Engine

MSB: Most Significant Bit

MOSFET: Metal Oxide Semiconductor Field Effect Transistor

NC: Not Connected

PGA: Programmable Gain Amplifier

PI/PID: Proportional Integral/Proportional Integral Derivative

PLL: Phase Locked Loop

PWM: Pulse width modulation

QEP: Quadrature Encoder Pulse

RAM: Random Access Memory

RMW: Read Modified Write

ROM: Read Only Memory

RSD: Rotating State Detection

RTC: Real Time Clock

SCL: Serial Clock Line

SDA: Serial Data Line

SFR: Special Function Register

SMO: Sliding Mode Observer

SOSC: Slow Oscillator

SPI: Serial Peripheral Interface

SVPWM: Space Vector PWM

TSD: Temperature Sensor Detect

UART: Universal Asynchronous Receiver/Transmitter

WDT: Watch Dog Timer

XRAM: External RAM

XSFR: External SFR

ZCP: Zero Cross Point

1 System Introduction

1.1 Features

- Power supply:
  - Single power supply high voltage mode: VCC_MODE = 0, external power supply inputs 5V ~ 28V voltage from VCC, and VDD5 is generated by internal LDO
  - Single power supply low voltage mode: VCC_MODE = 1, external power supply inputs 3V ~ 5.5V voltage from VDD5, and VDD5 is shorted to VCC
  - Dual power supply mode: VCC_MODE = 1, external power supply 1 inputs 5V ~ 36V voltage from VCC, and external power supply 2 input 5V voltage from VDD5
- Dual core: 8051 core and ME core
- Instruction cycles are mostly 1 or 2 system clock
- 16kB Flash ROM with CRC, self program and code protection
- 256 bytes IRAM, 768 bytes XRAM
- ME: Core with PID, FOC module, MDU and LPF
- 16 interrupt sources, with 4 priority levels
- GPIO:
  - FU6832L: 35 GPIO
  - FU6832N: 22 GPIO
  - FU6832S: 13 GPIO
  - FU6832F: 13 GPIO
- Timer:
  - Timer1: Timer, dedicated to square wave motor drive, supports square wave drive timing control, automatic phase commutation, cycle-by-cycle current limiting and Hall/BEMF based position detection
  - Timer2: Timer supports PWM generation, detection of duty cycle and period of input PWM wave, detection of the time of set PWM wave numbers, QEP decoding, RSD and rotation direction and speed detection of step motor.
  - Timer3/Timer4: Timers support PWM generation, detection of duty cycle and period of input PWM wave. Timer4 supports FG output and Timer3 supports 48Mhz mode
  - Systick Timer
  - RTC Timer
- Communication interface:
  - 1 SPI
  - 1 I2C
  - 2 UART, supports single wire mode

- 1 LIN
- 2-channels DMA supports I2C, SPI, UART and LIN

- Analogue peripherals:
  - 12-bit ADC, with 1μs conversion time and selectable internal VREF or external VREF as reference voltage
  - ADC channel numbers:
    - FU6832L: 14
    - FU6832N: 11
    - FU6832S: 7
    - FU6832F: 7
  - Internal VREF reference, with configurable 3V, 4V, 4.5V or VDD5 (FU6832N/S/F must configure VDD5 as internal reference)
  - Internal VHALF (VREF/2) reference output (except FU6832S/F)
  - 3 independent operational amplifiers (FU6832S/F only include AMP0), where the gain of AMP0 can be configured
  - 3 comparator groups
  - DAC: 1 channel 9-bit, 1 channel 6-bit
- Built-in 3P3N pre-driver
- FOC module supports single/dual/triple resistance current sampling (FU6832S/F only support single resistance current sampling)
- Built-in oscillator:
  - 24MHz ± 2% high speed oscillator
  - 32.8kHz low speed oscillator
- WDT
- LVD
- TSD
- Two-wire FICE protocol based real time emulation

1.2 Applications

The chip can be used for sensorless or sensor based drive of BLDC motors, single-phase/3-phase induction motors and servo motor.

Applications: Ceiling fans, floor fans, vacuum cleaner, hairdryers, industrial fans, pumps, power tools and drones.

1.3 Overview

FU6832 series are high-performance motor drive dedicated chips with integrated ME core and 8051 core. ME core integrates modules, FOC, MDU, LPF, PID, SVPWM, etc., which can automatically complete the calculation of FOC or square wave control by hardware used for sensor/sensorless based BLDC motors/PMSM. 8051 core is used for parameter configuration and routine operation processing. Most of 8051 core instruction cycle is 1 or 2 system clock. The dual cores work in parallel to achieve various high-performance motor control. The chip is integrated with high speed operational amplifiers, comparators, pre-driver, high speed ADC, CRC, SPI, I2C, UART, LIN, Timer and built-in high voltage LDO, which are essential for FOC or square wave control based BLDC motor drive.

FU6832 package types: FU6832L (LQFP48), FU6832N (QFN32), FU6832S (SSOP24) and FU6832F (QFN24).

1.4 Functional Block Diagram of System

1.4.1 Functional Block Diagram of FU6832L

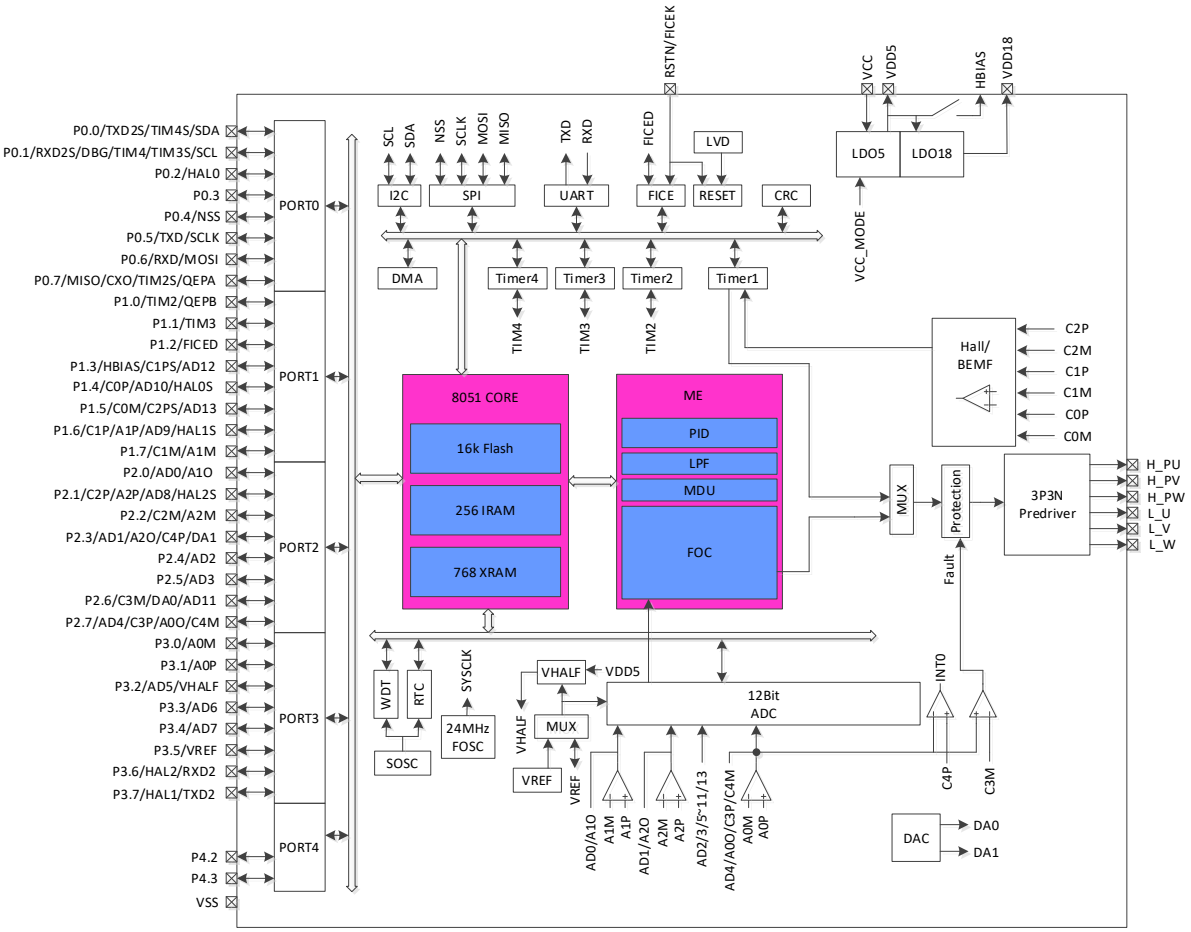


Figure 1-1 Functional Block Diagram of FU6832L

1.4.2 Functional Block Diagram of FU6832N

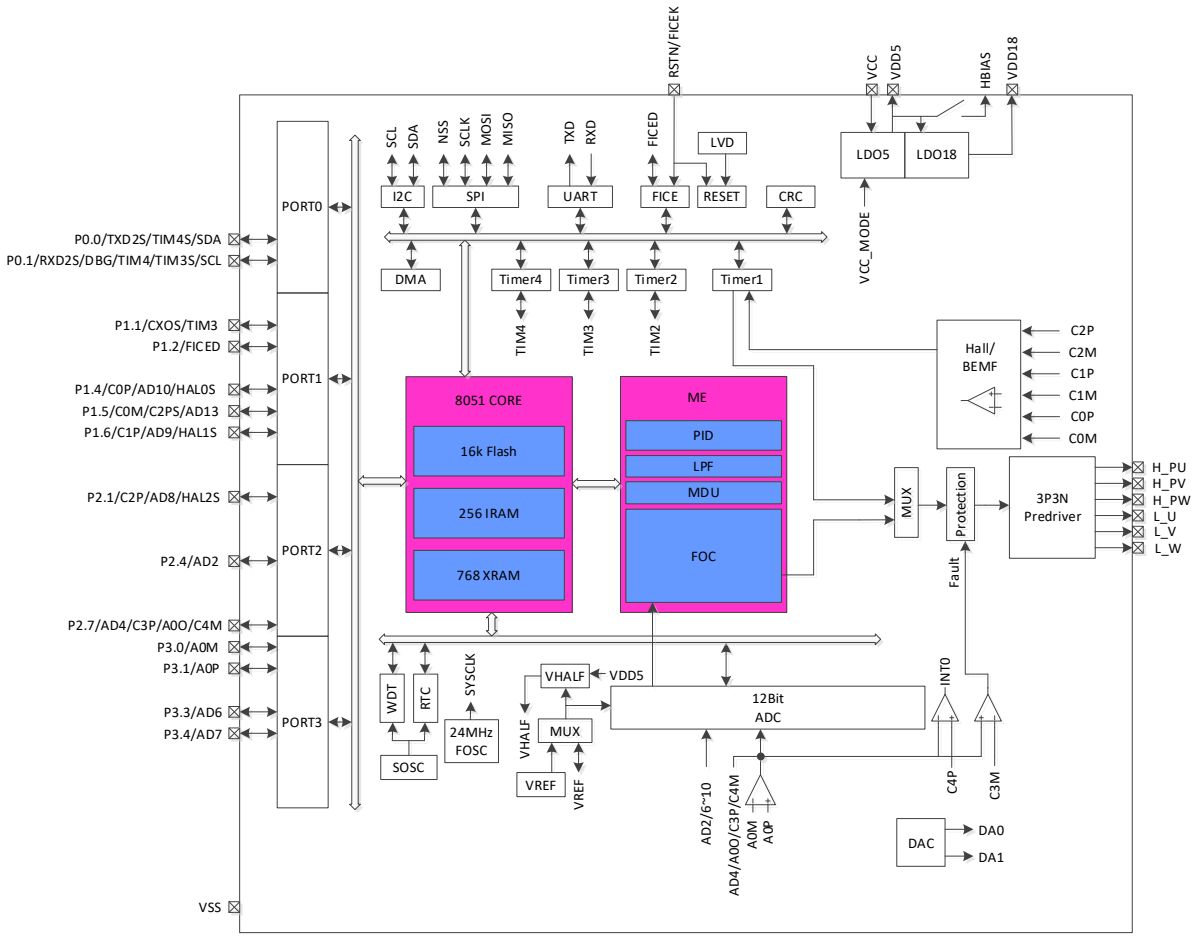


Figure 1-2 Functional Block Diagram of FU6832N

1.4.3 Functional Block Diagram of FU6832S

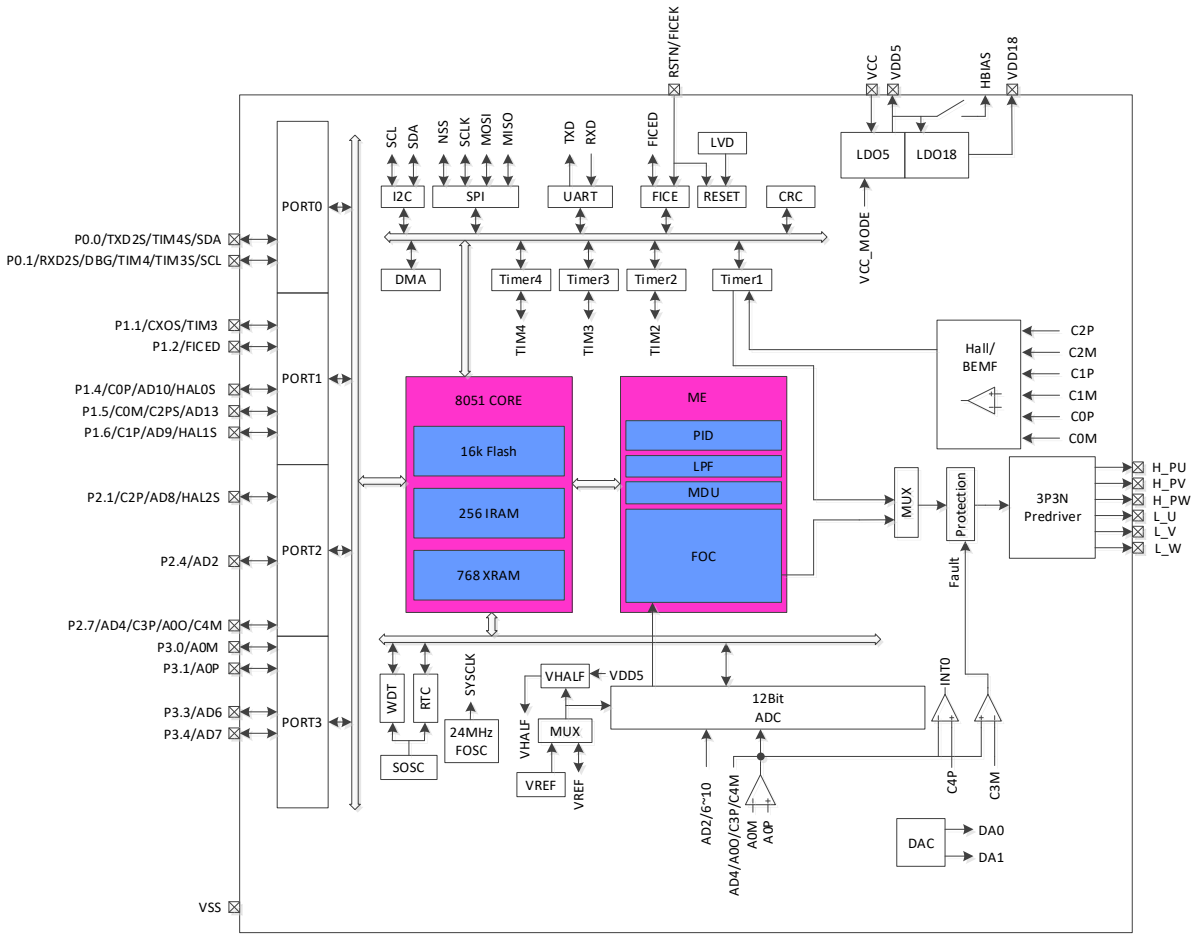


Figure 1-3 Functional Block Diagram of FU6832S

1.4.4 Functional Block Diagram of FU6832F

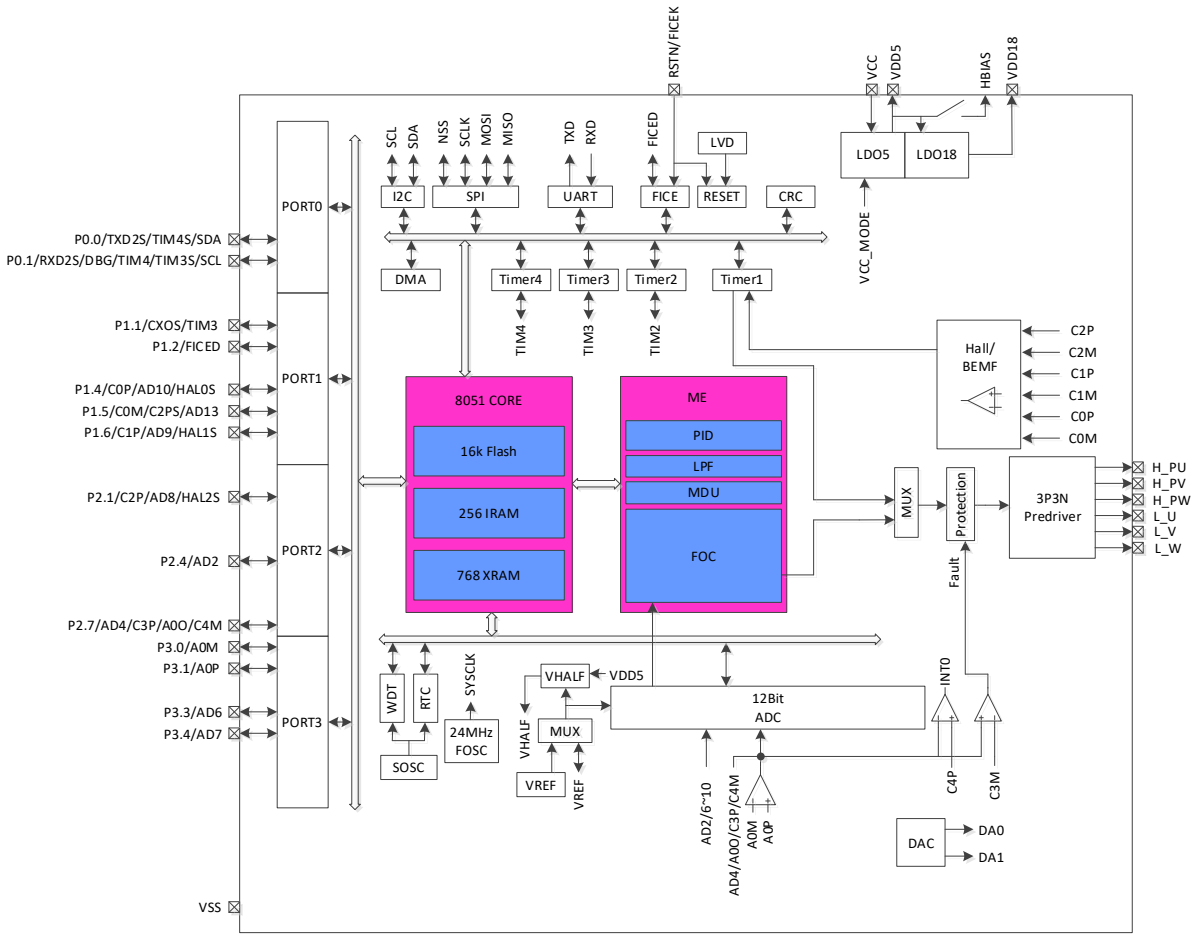


Figure 1-4 Functional Block Diagram of FU6832F

1.5 Memory Organization

The internal storage space is divided into Program Memory and Data Memory, which are independently addressed.

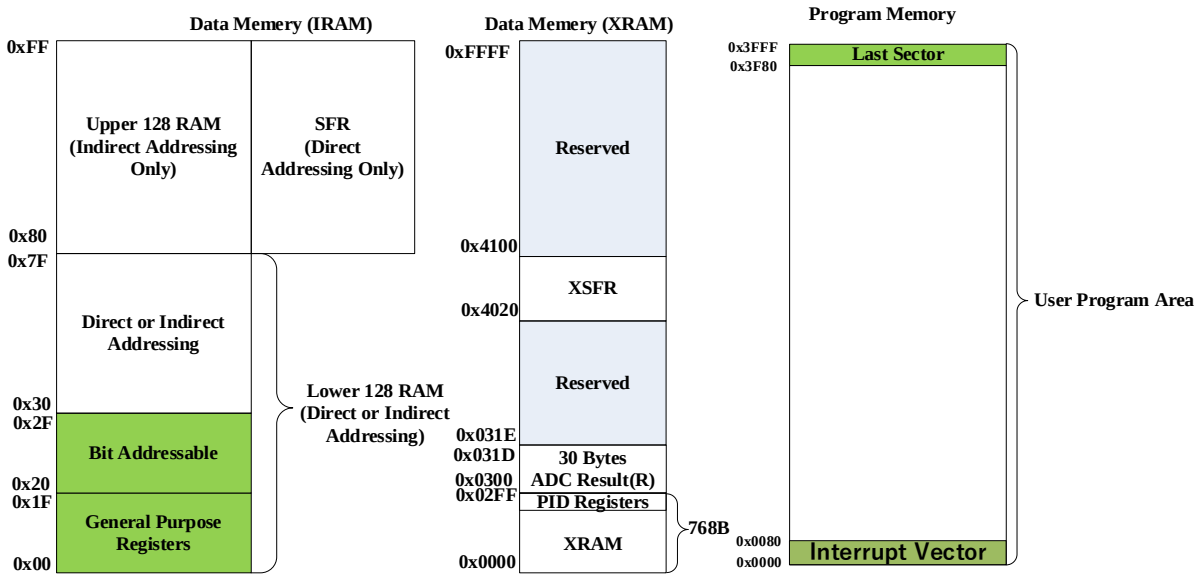


Figure 1-5 Memory Organization

1.5.1 Program Memory

The instruction space is addressed in the range of 0x0000 ~ 0x3FFF. The instruction space storage medium is Flash, which is used to store the control program.

The first sector (0x0000 ~ 0x007F) is the interrupt vector address area, which is used to store the start address of each interrupt subroutine. The last sector (0x3F80 ~ 0x3FFF) contains chip internal control bits.

1.5.2 Data Memory

The data space is divided into External Data Memory and Internal Data Memory as shown in Figure 1-5.

The External Data Memory is addressed in the range of 0x0000 ~ 0xFFFF and can only be accessed via the MOVX instruction. It includes XRAM (0x0000 ~ 0x02A7), extend control register area (0x02A8 ~ 0x02EF, 0x4020 ~ 0x40FF) and the ADC conversion results memory area (0x0300 ~ 0x031D).

The Internal Data Memory is addressed in the range of 0x00 ~ 0xFF. 0x00 ~ 0x1F is the general register space, containing 4 groups of 8 each, for a total of 32 general registers. 0x20 ~ 0x7F is the low RAM space, supporting direct and indirect addressing access. 0x20 ~ 0x2F support bit addressing operations. When 0x80 ~ 0xFF is accessed in indirect addressing, it points to RAM. When 0x80 ~ 0xFF is accessed in direct addressing, it points to SFR.

1.5.3 SFR

Table 1-1 SFR Address Mapping

Addr	0(8)	1(9)	2(A)	3(B)	4(C)	5(D)	6(E)	7(F)
0xF8	DRV_OUT	PI_CR			P0_OE	P1_OE	P2_OE	P3_OE
0xF0	B							
0xE8	P4	P4_OE						
0xE0	ACC	CMP_CR4						
0xD8	IP3	EVT_FILT	CMP_CR2	LVSR	CMP_CR3			
0xD0	PSW	P1_IE	P1_IF	P2_IE	P2_IF	CMP_CR0	CMP_CR1	CMP_SR
0xC8	IP2	RST_SR	MDU_MD	MDU_D				
0xC0	IP1	MDU_CR	MDU_CL	MDU_CH	MDU_BL	MDU_BH	MDU_AL	MDU_AH
0xB8	IP0							
0xB0	P3							
0xA8	IE	TIM2_CR1	TIM2__CNTRL	TIM2__CNTRH	TIM2__DRL	TIM2__DRH	TIM2__ARRL	TIM2__ARRH
0xA0	P2	TIM2_CR0	TIM3__CNTRL	TIM3__CNTRH	TIM3__DRL	TIM3__DRH	TIM3__ARRL	TIM3__ARRH
0x98	UT_CR	UT_DR	UT_BAUDL	UT_BAUDH	TIM3_CR0	TIM3_CR1	TIM4_CR0	TIM4_CR1
0x90	P1		TIM4__CNTRL	TIM4__CNTRH	TIM4__DRL	TIM4__DRH	TIM4__ARRL	TIM4__ARRH
0x88	TCON	UT2_DR	UT2_CR					
0x80	P0	SP	DPL	DPH	FLA_KEY	FLA_CR		PCON

Notes:

- The registers with __ are 16-bit snapshot register. The snapshot registers are dynamic register which should be read by a variable. The value will be incorrect when the register is read directly
- 8-bit MCU should read a 16-bit register twice to get the value, high 8 bits and low 8 bits respectively. When the low 8 bits value of register change after MCU has read high 8 bits of register, the result will be incorrect. Therefore, when high 8 bit of snapshot register is read by MCU, the corresponding low 8 bits will be stored and read.
- Snapshot register must be read as a whole, with high 8 bit first and low 8 bits followed

1.5.4 XSFR

Table 1-2 XSFR Address Mapping

Addr	0(8)	1(9)	2(A)	3(B)	4(C)	5(D)	6(E)	7(F)
0x40e0	LIN_CR	LIN_SR	LIN_CSR	LIN_ID	LIN_SIZE	LIN_BAUDH	LIN_BAUDL	
0x40d8	FOC__POWH	FOC__POWL	FOC__IAMAXH	FOC__IAMAXL	FOC__IBMAXH	FOC__IBMAXL	FOC__ICMAXH	FOC__ICMAXL
	FOC_EOMEKLPF							
0x40d0	FOC__EALPH	FOC__EALPL	FOC__EBETH	FOC__EBETL	FOC__EOMEH	FOC__EOMEL	FOC__UQEXH	FOC__UQEXL
							FOC_KFGH	FOC_KFGL
0x40c8	FOC__IBH	FOC__IBL	FOC__IAH	FOC__IAL	FOC__THETAH	FOC__THETAL	FOC__ETHETAH	FOC__ETHETAL
0x40c0	FOC__IBETH	FOC__IBETL	FOC__VBETH	FOC__VBETL	FOC__VALPH	FOC__VALPL	FOC__ICH	FOC__ICL
			FOC_UDCPSH	FOC_UDCPSL	FOC_UQCPSH	FOC_UQCPSL		
0x40b8	FOC__UDH	FOC__UDL	FOC__UQH	FOC__UQL	FOC__IDH	FOC__IDL	FOC__IQH	FOC__IQL
0x40b0	FOC_DMAXH	FOC_DMAXL	FOC_DMINH	FOC_DMINL	FOC_QMAXH	FOC_QMAXL	FOC_QMINH	FOC_QMINL
0x40a8	FOC__RTHESTEPH	FOC__RTHESTEPL	FOC_RTHEACCH	FOC_RTHEACCL	FOC_RTHECNT	FOC_THECOR	FOC_THECOMPH	FOC_THECOMPL
			FOC_EOMELPFH	FOC_EOMELPFL		CMP_SAMR		
0x40a0	FOC_CR1	FOC_CR2	FOC_TSMIN	FOC_TGLI	FOC_TBLO	FOC_TRGDLY	FOC_CSOH	FOC_CSOL
0x4098	FOC__UDCFLTH	FOC__UDCFLTL						FOC_CR0
	TIM1__ITRIPH	TIM1__ITRIPL						
0x4090	FOC_IDREFH	FOC_IDREFL	FOC_IQREFH	FOC_IQREFL	FOC_DQKPH	FOC_DQKPL	FOC_DQKIH	FOC_DQKIL
	TIM1__URESH	TIM1__URESL	TIM1__UIGNH	TIM1__UIGNL	TIM1__KFH	TIM1__KFL	TIM1__KRH	TIM1__KRL
0x4088	FOC_EK3H	FOC_EK3L	FOC_EK4H	FOC_EK4L	FOC_EK1H	FOC_EK1L	FOC_EK2H	FOC_EK2L
	TIM1__RARRH	TIM1__RARRL	TIM1__RCNTRH	TIM1__RCNTRL	TIM1__UCOPH	TIM1__UCOPL	TIM1__UFLPH	TIM1__UFLPL
0x4080	FOC_FBASEH	FOC_FBASEL	FOC_EFREQACCH	FOC_EFREQACCL	FOC_EFREQMINH	FOC_EFRQMINL	FOC_EFREQHOLDH	FOC_EFREQHOLDL
	TIM1_DBR7H	TIM1_DBR7L	TIM1__BCNTRH	TIM1__BCNTRL	TIM1__BCCRH	TIM1__BCCRL	TIM1__BARRH	TIM1__BARRL
0x4078	FOC_KSLIDEH	FOC_KSLIDEL	FOC_EKLPFMINH	FOC_EKLPFMINL	FOC_EBMFKH	FOC_EBMFKL	FOC_OMEKLPFH	FOC_OMEKLPFL
	TIM1_DBR3H	TIM1_DBR3L	TIM1_DBR4H	TIM1_DBR4L	TIM1_DBR5H	TIM1_DBR5L	TIM1_DBR6H	TIM1_DBR6L
0x4070	TIM1_BCORH	TIM1_BCORL			FOC_EKPH	FOC_EKPL	FOC_EKIH	FOC_EKIL
					TIM1_DBR1H	TIM1_DBR1L	TIM1_DBR2H	TIM1_DBR2L

Addr	0(8)	1(9)	2(A)	3(B)	4(C)	5(D)	6(E)	7(F)
0x4068	TIM1_CR0	TIM1_CR1	TIM1_CR2	TIM1_CR3	TIM1_CR4	TIM1_ IER	TIM1_SR	
0x4060	DRV_DTR	DRV_SR	DRV_CR		SYST_ARRH	SYST_ARRL	DRV__CNTRH	DRV__CNTRL
0x4058	DRV_DRH	DRV_DRL	DRV_COMRH	DRV_COMRL	DRV_CMRH	DRV_CMRL	DRV_ARRH	DRV_ARRL
0x4050	P1_AN	P2_AN	P3_AN	P0_PU	P1_PU	P2_PU	P3_PU	P4_PU
0x4048			DAC1_DR	DAC_DR	PH_SEL	PH_SEL1	AMP_CR	VREF_VHALF_CR
0x4040	DMA1_BAH	DMA1_BAL	UT2_BAUDL	UT2_BAUDH	CAL_CR0	CAL_CR1		
0x4038	ADC_SCYC	ADC_CR	DMA0_CR0	DMA1_CR0	DMA0_LEN	DMA1_LEN	DMA0_BAH	DMA1_BAL
0x4030	SPI_CR0	SPI_CR1	SPI_CLK	SPI_DR	AMP0_GAIN	DAC_CR	ADC_MASK_SYSCH	ADC_MASK_SYSCL
0x4028	I2C_CR	I2C_ID	I2C_DR	I2C_SR	RTC_TMH	RTC_TML	RTC_STA	TSD_CR
0x4020		CRC_DIN	CRC_CR	CRC_DR	CRC_BEG	CRC_CNT	WDT_CR	WDT_ARR
0x4018								
0x4010								
0x4008								
0x4000								
0x0318	AD12_DRH	AD12_DRL	AD13_DRH	AD13_DRL	AD14_DRH	AD14_DRH	—	—
0x0310	AD8_DRH	AD8_DRL	AD9_DRH	AD9_DRL	AD10_DRH	AD10_DRL	AD11_DRH	AD11_DRL
0x0308	AD4_DRH	AD4_DRL	AD5_DRH	AD5_DRL	AD6_DRH	AD6_DRL	AD7_DRH	AD7_DRL
0x0300	AD0_DRH	AD0_DRL	AD1_DRH	AD1_DRL	AD2_DRH	AD2_DRL	AD3_DRH	AD3_DRL
0x02f8								
0x02f0								
0x02e8	PI0_EK1		PI0_EK		PI0_UKH		PI0_UKL	
0x02e0	PI0_KP		PI0_KI		PI0_UKMAX		PI0_UKMIN	
0x02d8	PI1_EK1		PI1_EK		PI1_UKH		PI1_UKL	
0x02d0	PI1_KP		PI1_KI		PI1_UKMAX		PI1_UKMIN	
0x02c8	PI2_UKH		PI2_UKL		PI2_KD		PI2_EK2	
0x02c0	PI2_UKMAX		PI2_UKMIN		PI2_EK1		PI2_EK	
0x02b8	PI3_KD		PI3_EK2		PI2_KP		PI2_KI	
0x02b0	PI3_EK1		PI3_EK		PI3_UKH		PI3_UKL	
0x02a8	PI3_KP		PI3_KI		PI3_UKMAX		PI3_UKMIN	

Notes:

- The registers with __ are 16-bit snapshot register. The snapshot registers are dynamic register which should be read by a variable. The value will be incorrect when the register is read directly
- 8-bit MCU should read a 16-bit register twice to get the value, high 8 bits and low 8 bits respectively. When the low 8 bit value of register change after MCU has read high 8 bits of register, the result will be incorrect. Therefore, when high 8 bits of snapshot register is read by MCU, the corresponding low 8 bit will be stored and read.
- Snapshot register must be read as a whole, with high 8 bits first and low 8 bits followed

2 Pin Definitions

2.1 FU6832L LQFP48 Pin List

Table 2-1 FU6832L LQFP48 Pin List

Pin	FU6832 LQFP48	IO Type	Description
P2.2/ C2M/ A2M	1	DB/ AI/ AI	GPIO, configurable as input for INT1 Negative input of CMP2 Negative input of AMP2
P2.3/ AD1/ A2O/ C4P/ DA1	2	DB/ AI/ AO/ AI/ AO	GPIO, configurable as input for INT1 Input of ADC channel 1 Output of AMP2 Positive input of CMP4 Output of DAC1, without buffer output
P2.4/ AD2	3	DB/ AI	GPIO, configurable as input for INT1 Input of ADC channel 2, used for bus voltage sampling
P2.5/ AD3	4	DB/ AI	GPIO, configurable as input for INT1 Input of ADC channel 3
P2.6/ C3M/ DA0/ AD11	5	DB/ AI/ AO/ AI	GPIO, configurable as input for INT1 Negative input of CMP3 Output of DAC0, without buffer output Input of ADC channel 11
P2.7/ AD4/ C3P/ A0O/ C4M	6	DB/ AI/ AI/ AO/ AI	GPIO, configurable as input for INT1 Input of ADC channel 4, used for bus current sampling Positive input of CMP3 Output of AMP0 Negative input of CMP4
P3.0/ A0M	7	DB/ AI	GPIO Negative input of AMP0
P3.1/ A0P	8	DB/ AI	GPIO Positive input of AMP0
P3.2/ AD5/ VHALF	9	DB/ AI/ AO	GPIO Input of ADC channel 5 Output of VREF/2 voltage. Shunt a 1μF capacitance to ground
P3.3/ AD6	10	DB/ AI	GPIO Input of ADC channel 6
P3.4/ AD7	11	DB AI	GPIO Input of ADC channel 7
P3.5/ VREF	12	DB/ AI	GPIO Input of external ADC reference voltage, or output of internal VREF. Shunt a 1μF ~ 4.7μF capacitance to ground
P4.4	13	DB	GPIO
P4.5	14	DB	GPIO
P3.6/ HAL2/ RXD2	15	DB/ DI/ DI	GPIO Digital input of Hall-IC2 UART2 RXD or LIN RXD
P3.7/ HALL/ TXD2	16	DB/ DI/ DO	GPIO Digital input of Hall-IC1 UART2 TXD or LIN TXD
P0.0/ TIM4S/ TXD2S/ SDA	17	DB/ DB/ DO/ DB	GPIO, configurable as input for INT0 Input or output of Timer4 after function transfer UART2 TXD or LIN TXD after function transfer I2C SDA, with configurable collector open drain output

Pin	FU6832 LQFP48	IO Type	Description
P0.1/ RXD2S/ DBG/ TIM4/ TIM3S/ SCL	18	DB/ DI/ DO/ DB/ DB/ DB	GPIO, configurable as input for INT0 UART2 RXD or LIN RXD after function transfer Output of Debug signal Input or output of Timer4 Input or output of Timer3 after function transfer I2C SCL, with configurable collector open drain output
P0.2/ HAL0	19	DB/ DI	GPIO, configurable as input for INT0 Digital input of Hall-IC0
P0.3	20	DB	GPIO, configurable as input for INT0
P0.4/ NSS	21	DB/ DB	GPIO, configurable as input for INT0 SPI NSS
P0.5/ TXD/ SCLK	22	DB/ DO/ DB	GPIO, configurable as input for INT0 UART1 TXD SPI SCLK
P0.6/ RXD/ MOSI	23	DB/ DI/ DB	GPIO, configurable as input for INT0 UART1 RXD SPI MOSI
P0.7/ MISO/ CXO/ TIM2S/ QEPA	24	DB/ DB/ DO/ DB/ DI	GPIO SPI MISO Output of comparator test signal Input or output of Timer2 after function transfer An input of QEP
P1.0/ TIM2/ QEPB	25	DB/ DB// DI	GPIO, configurable as input for INT1 Input or output of Timer2 B input of QEP
P1.1/ TIM3	26	DB/ DB	GPIO, configurable as input for INT1 Input or output of Timer3
P4.2	27	DB	GPIO
H_PU	28	DO	Pre-driver U phase high side output, with built-in 50kΩ pull-up resistor
H_PV	29	DO	Pre-driver V phase high side output, with built-in 50kΩ pull-up resistor
H_PW	30	DO	Pre-driver W phase high side output, with built-in 50kΩ pull-up resistor
L_U	31	DO	Pre-driver U phase low side output, with built-in 25kΩ pull-down resistor
L_V	32	DO	Pre-driver V phase low side output, with built-in 25kΩ pull-down resistor
L_W	33	DO	Pre-driver W phase low side output, with built-in 25kΩ pull-down resistor
NC	34		NC
VCC	35	P	Power supply. The input voltage range is decided by the power supply mode. Shunt a 10μF capacitance to ground ■ Single power supply high voltage mode: VCC_MODE = 0, VCC input range is 5V ~ 28V, VDD5 is generated by internal LDO ■ Single power supply low voltage mode: VCC_MODE = 1, VDD5 input range is 3V ~ 5.5V and short with VCC ■ Dual power supply mode VCC_MODE = 1, VCC input range is 5V ~ 36V, and VDD5 input is 5V
VSS	36	P	Ground
VDD5	37	P	Input of power supply or output of the internal 5V LDO. Shunt a 1μF ~ 4.7μF capacitance to ground Please refer to the VCC pin description
NC	38		NC
RSTN/ FICEK	39	DI/ DI	Input of external reset, with built-in pull-up resistor Clock line of FICE

Pin	FU6832 LQFP48	IO Type	Description
VDD18	40	P	Output of 1.8V LDO. Shunt a 1μF ~ 4.7μF capacitance to ground
P1.2/ FICED	41	DB/ DB	GPIO, configurable as input for INT1 Data line of FICE
P1.3/ HBIAS/ C1PS/ AD12	42	DB/ DO/ AI/ AI	GPIO, configurable as input for INT1 Hall bias power supply, internally connected to VDD5 via a switch, allowing for large current output Positive input of CMP1 after function transfer Input of ADC channel 12
P1.4/ C0P/ AD10/ HAL0S	43	DB/ AI/ AI/ DI	GPIO, configurable as input for INT1 Positive input of CMP0 Input of ADC channel 10 Digital input of Hall-IC0 after function transfer
P1.5/ C0M/ C2PS/ AD13	44	DB/ AI/ AI/ AI	GPIO, configurable as input for INT1 Negative input of CMP0 Positive input of CMP2 after function transfer Input of ADC channel 13
P1.6/ C1P/ A1P/ AD9/ HAL1S	45	DB/ AI/ AI/ AI/ DI	GPIO, configurable as input for INT1 Positive input of CMP1 Positive input of AMP1 Input of ADC channel 9 Digital input of Hall-IC1 after function transfer
P1.7/ C1M/ A1M	46	DB/ AI/ AI	GPIO, configurable as input for INT1 Negative input of CMP1 Negative input of AMP1
P2.0/ AD0/ A1O	47	DB/ AI/ AO	GPIO, configurable as input for INT1 Input of ADC channel 0 Output of AMP1
P2.1/ C2P/ A2P/ AD8/ HAL2S	48	DB/ AI/ AI/ AI/ DI	GPIO, configurable as input for INT1 Positive input of CMP2 Positive input of AMP2 Input of ADC channel 8 Digital input of Hall-IC2 after function transfer

Notes:

- DI = Digital Input
- DO = Digital Output
- DB = Digital Bidirectional
- AI = Analogue Input
- AO = Analogue Output
- P = Power Supply

2.2 FU6832L LQFP48 Pinout

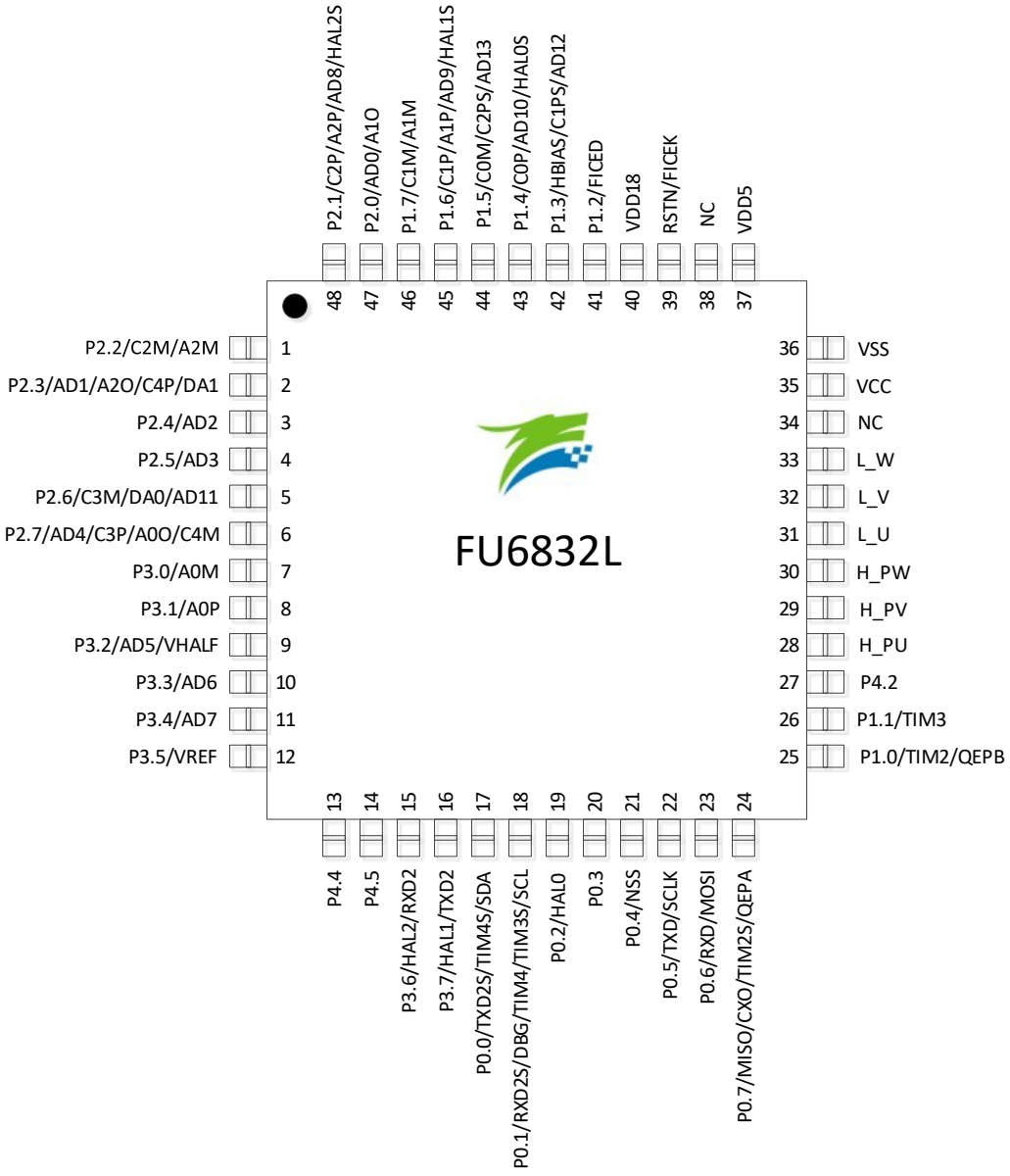


Figure 2-1 FU6832L LQFP48 Pinout

2.3 FU6832N QFN32 Pin List

Table 2-2 FU6832N QFN32 Pin List

Pin	FU6832 QFN32	IO Type	Description
P2.1/ C2P/ A2P/ AD8/ HAL2S	1	DB/ AI/ AI/ AI/ DI	GPIO, configurable as input for INT1 Positive input of CMP2 Positive input of AMP2 Input of ADC channel 8 Digital input of Hall-IC2 after function transfer
P2.2/ C2M/ A2M	2	DB/ AI/ AI	GPIO, configurable as input for INT1 Negative input of CMP2 Negative input of AMP2
P2.3/ AD1/ A2O/ C4P/ DA1	3	DB/ AI/ AO/ AI/ AO	GPIO, configurable as input for INT1 Input of ADC channel 1 Output of AMP2 Positive input of CMP4 Output of DAC1, without buffer output
P2.4/ AD2	4	DB/ AI	GPIO, configurable as input for INT1 Input of ADC channel 2, used for bus voltage sampling
P2.7/ AD4/ C3P/ A0O/ C4M	5	DB/ AI/ AI/ AO/ AI	GPIO, configurable as input for INT1 Input of ADC channel 4, used for bus current sampling Positive input of CMP3 Output of AMP0 Negative input of CMP4
P3.0/ A0M	6	DB/ AI	GPIO Negative input of AMP0
P3.1/ A0P	7	DB/ AI	GPIO Positive input of AMP0
P3.2/ AD5/ VHALF	8	DB/ AI/ AO	GPIO Input of ADC channel 5 Output of VREF/2 voltage. Shunt a 1μF capacitance to ground
P3.4/ AD7	9	DB AI	GPIO Input of ADC channel 7
P0.0/ TIM4S/ TXD2S/ SDA/	10	DB/ DB/ DO/ DB/	GPIO, configurable as input for INT0 Input or output of Timer4 after function transfer UART2 TXD or LIN TXD after function transfer I2C SDA, with configurable collector open drain output
P0.1/ RXD2S/ DBG/ TIM4/ TIM3S/ SCL/	11	DB/ DI/ DO/ DB/ DB/ DB/	GPIO, configurable as input for INT0 UART2 RXD or LIN RXD after function transfer Output of Debug signal Input or output of Timer4 Input or output of Timer3 after function transfer I2C SCL, with configurable collector open drain output
P0.5/ TXD/ SCLK	12	DB/ DO/ DB	GPIO, configurable as input for INT0 UART1 TXD SPI SCLK
P0.6/ RXD/ MOSI	13	DB/ DI/ DB	GPIO, configurable as input for INT0 UART1 RXD SPI MOSI
P0.7/ MISO/ CXO/ TIM2S/ P1.1/ TIM3	14	DB/ DB/ DO/ DB/ DB/ DB	GPIO SPI MISO Output of comparator test signal Input or output of Timer2 after function transfer GPIO, configurable as input for INT1 Input or output of Timer3

Pin	FU6832 QFN32	IO Type	Description
H_PU	15	DO	Pre-driver U phase high side output, with built-in 50kΩ pull-up resistor
H_PV	16	DO	Pre-driver V phase high side output, with built-in 50kΩ pull-up resistor
H_PW	17	DO	Pre-driver W phase high side output, with built-in 50kΩ pull-up resistor
L_U	18	DO	Pre-driver U phase low side output, with built-in 25kΩ pull-down resistor
L_V	19	DO	Pre-driver V phase low side output, with built-in 25kΩ pull-down resistor
L_W	20	DO	Pre-driver W phase low side output, with built-in 25kΩ pull-down resistor
VCC	21	P	Power supply. The input voltage range is decided by the power supply mode. Shunt a 10μF capacitance to ground ■ Single power supply high voltage mode: VCC_MODE = 0, VCC input range is 5V ~ 28V, VDD5 is generated by internal LDO ■ Single power supply low voltage mode: VCC_MODE = 1, VDD5 input range is 3V ~ 5.5V and short with VCC ■ Dual power supply mode VCC_MODE = 1, VCC input range is 5V ~ 36V, and VDD5 input is 5V
VSS	22	P	Ground
VDD5	23	P	Input of power supply or output of the internal 5V LDO. Shunt a 1μF ~ 4.7μF capacitance to ground Please refer to the VCC pin description
RSTN/ FICEK	24	DI/ DI	Input of external reset, with built-in pull-up resistor Clock line of FICE
VDD18	25	P	Output of 1.8V LDO. Shunt a 1μF ~ 4.7μF capacitance to ground
P1.2/ FICED	26	DB/ DB	GPIO, configurable as input for INT1 Data line of FICE
P1.3/ HBIAS/ C1PS/ AD12	27	DB/ DO/ AI/ AI	GPIO, configurable as input for INT1 Hall bias power supply, internally connected to VDD5 via a switch, allowing for large current output Positive input of CMP1 after function transfer Input of ADC channel 12
P1.4/ C0P/ AD10/ HAL0S	28	DB/ AI/ AI/ DI	GPIO, configurable as input for INT1 Positive input of CMP0 Input of ADC channel 10 Digital input of Hall-IC0 after function transfer
P1.5/ C0M/ C2PS/ AD13	29	DB/ AI/ AI/ AI	GPIO, configurable as input for INT1 Negative input of CMP0 Positive input of CMP2 after function transfer Input of ADC channel 13
P1.6/ C1P/ A1P/ AD9/ HAL1S	30	DB/ AI/ AI/ AI/ DI	GPIO, configurable as input for INT1 Positive input of CMP1 Positive input of AMP1 Input of ADC channel 9 Digital input of Hall-IC1 after function transfer
P1.7/ C1M/ A1M	31	DB/ AI/ AI	GPIO, configurable as input for INT1 Negative input of CMP1 Negative input of AMP1
P2.0/ AD0/ A1O	32	DB/ AO/ AI	GPIO, configurable as input for INT1 Input of ADC channel 0 Output of AMP1

2.4 FU6832N QFN32 Pinout

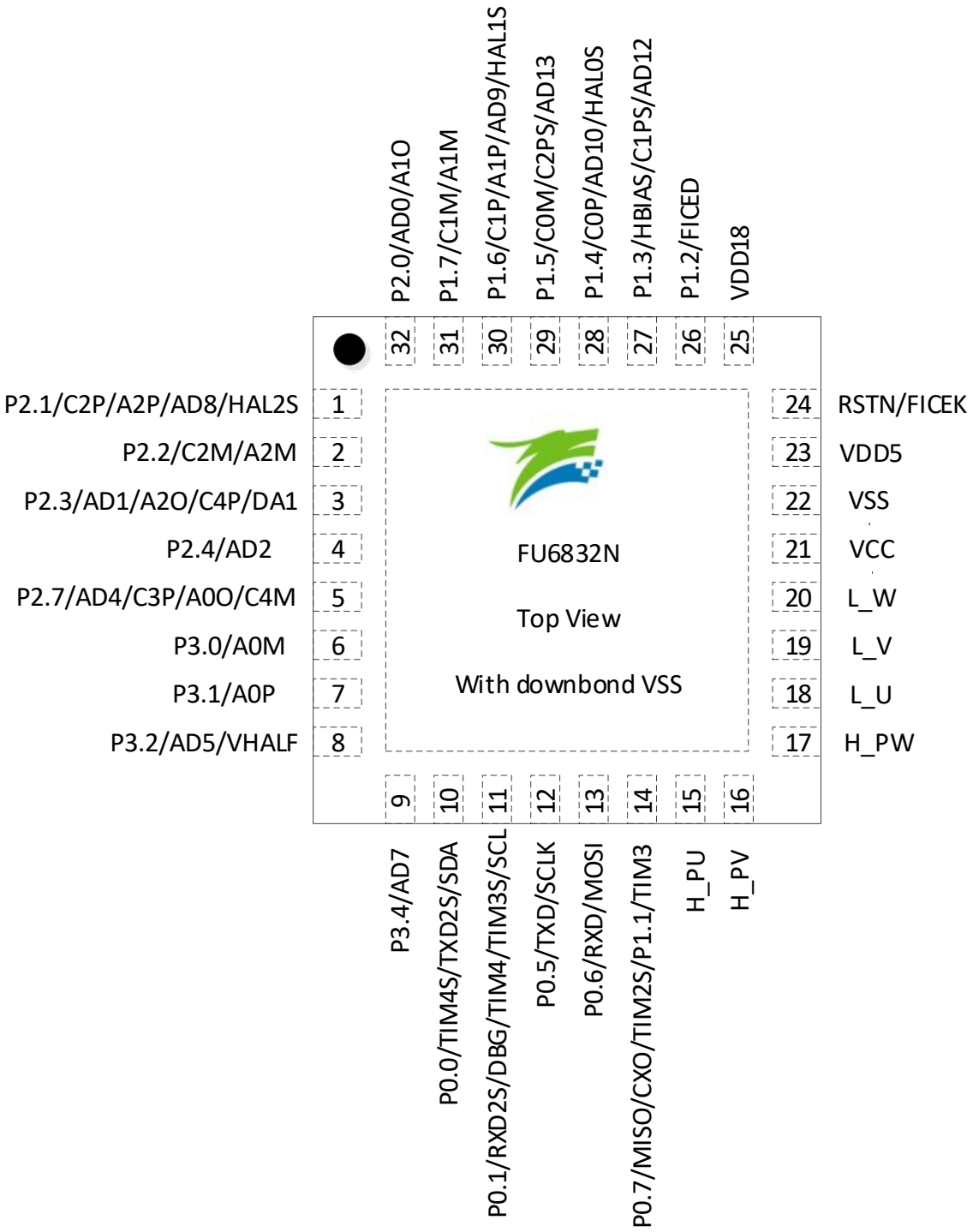


Figure 2-2 FU6832N QFN32 Pinout

2.5 FU6832S SSOP24 Pin List

Table 2-3 FU6832S SSOP24 Pin List

Pin	FU6832 SSOP24	IO Type	Description
P1.1/ CXOS/ TIM3	1	DB/ DO/ DB	GPIO, configurable as input for INT1 Output of comparator test signal after function transfer Input or output of Timer3
H_PU	2	DO	Pre-driver U phase high side output, with built-in 50kΩ pull-up resistor
H_PV	3	DO	Pre-driver V phase high side output, with built-in 50kΩ pull-up resistor
H_PW	4	DO	Pre-driver W phase high side output, with built-in 50kΩ pull-up resistor
L_U	5	DO	Pre-driver U phase low side output, with built-in 25kΩ pull-down resistor
L_V	6	DO	Pre-driver V phase low side output, with built-in 25kΩ pull-down resistor
L_W	7	DO	Pre-driver W phase low side output, with built-in 25kΩ pull-down resistor
VCC	8	P	Power supply. The input voltage range is decided by the power supply mode. Shunt a 10μF capacitance to ground ■ Single power supply high voltage mode: VCC_MODE = 0, VCC input range is 5V ~ 28V, VDD5 is generated by internal LDO ■ Single power supply low voltage mode: VCC_MODE = 1, VDD5 input range is 3V ~ 5.5V and short with VCC ■ Dual power supply mode VCC_MODE = 1, VCC input range is 5V ~ 36V, and VDD5 input is 5V
VSS	9	P	Ground
VDD5	10	P	Input of power supply or output of the internal 5V LDO. Shunt a 1μF ~ 4.7μF capacitance to ground Please refer to the VCC pin description
RSTN/ FICEK	11	DI/ DI	Input of external reset, with built-in pull-up resistor Clock line of FICE
VDD18	12	P	Output of 1.8V LDO. Shunt a 1μF ~ 4.7μF capacitance to ground
P1.2/ FICED	13	DB/ DB	GPIO, configurable as input for INT1 Data line of FICE
P1.4/ C0P/ AD10/ HAL0S	14	DB/ AI/ AI/ DI	GPIO, configurable as input for INT1 Positive input of CMP0 Input of ADC channel 10 Digital input of Hall-IC0 after function transfer
P1.6/ C1P/ AD9/ HAL1S	15	DB/ AI/ AI/ DI	GPIO, configurable as input for INT1 Positive input of CMP1 Input of ADC channel 9 Digital input of Hall-IC1 after function transfer
P2.1/ C2P/ AD8/ HAL2S	16	DB/ AI/ AI/ DI	GPIO, configurable as input for INT1 Positive input of CMP2 Input of ADC channel 8 Digital input of Hall-IC2 after function transfer
P2.4/ AD2	17	DB/ AI	GPIO, configurable as input for INT1 Input of ADC channel 2, used for bus voltage sampling
P2.7/ AD4/ C3P/ A0O/ C4M	18	DB/ AI/ AI/ AO/ AI	GPIO, configurable as input for INT1 Input of ADC channel 4, used for bus current sampling Positive input of CMP3 Output of AMP0 Negative input of CMP4

Pin	FU6832 SSOP24	IO Type	Description
P3.0/ A0M	19	DB/ AI	GPIO Negative input of AMP0
P3.1/ A0P	20	DB/ AI	GPIO Positive input of AMP0
P3.3/ AD6	21	DB/ AI	GPIO Input of ADC channel 6
P3.4/ AD7	22	DB AI	GPIO Input of ADC channel 7
P0.0/ TIM4S/ TXD2S/ SDA/	23	DB/ DB/ DO/ DB/	GPIO, configurable as input for INT0 Input or output of Timer4 after function transfer UART2 TXD or LIN TXD after function transfer I2C SDA, with configurable collector open drain output
P0.1/ RXD2S/ DBG/ TIM4/ TIM3S/ SCL/	24	DB/ DI/ DO/ DB/ DB/ DB/	GPIO, configurable as input for INT0 UART2 RXD or LIN RXD after function transfer Output of Debug signal Input or output of Timer4 Input or output of Timer3 after function transfer I2C SCL, with configurable collector open drain output

2.6 FU6832S SSOP24 Pinout

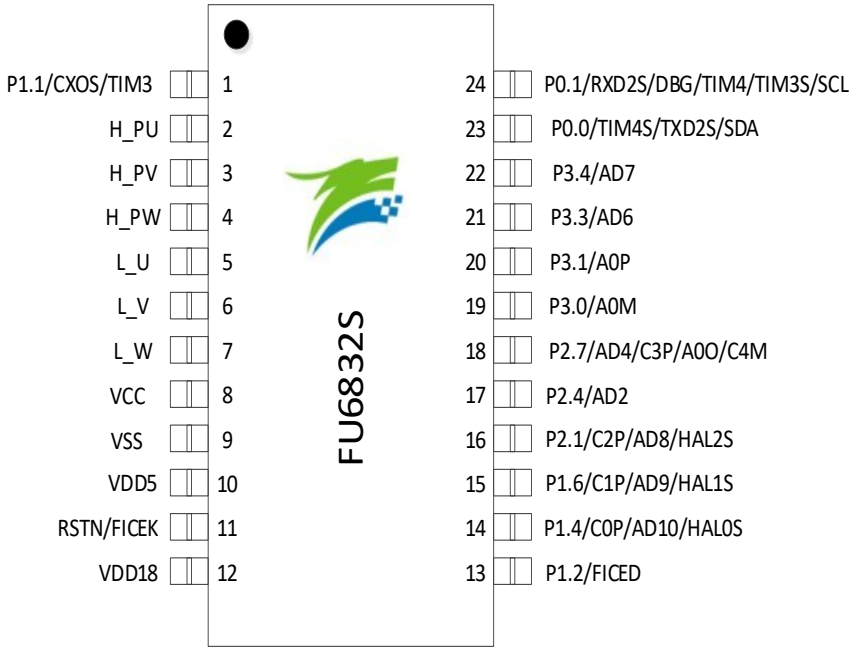


Figure 2-3 FU6832S SSOP24 Pinout

2.7 FU6832F QFN24 Pin List

Table 2-4 FU6832F QFN24 Pin List

Pin	FU6832 QFN24	IO Type	Description
P2.1/ C2P/ AD8/ HAL2S	1	DB/ AI/ AI/ DI	GPIO, configurable as input for INT1 Positive input of CMP2 Input of ADC channel 8 Digital input of Hall-IC2 after function transfer
P2.4/ AD2	2	DB/ AI	GPIO, configurable as input for INT1 Input of ADC channel 2, used for bus voltage sampling
P2.7/ AD4/ C3P/ A0O/ C4M	3	DB/ AI/ AI/ AO/ AI	GPIO, configurable as input for INT1 Input of ADC channel 4, used for bus current sampling Positive input of CMP3 Output of AMP0 Negative input of CMP4
P3.0/ A0M	4	DB/ AI	GPIO Negative input of AMP0
P3.1/ A0P	5	DB/ AI	GPIO Positive input of AMP0
P3.3/ AD6	6	DB/ AI	GPIO Input of ADC channel 6
P3.4/ AD7	7	DB AI	GPIO Input of ADC channel 7
P0.0/ TIM4S/ TXD2S/ SDA/	8	DB/ DB/ DO/ DB/	GPIO, configurable as input for INT0 Input or output of Timer4 after function transfer UART2 TXD or LIN TXD after function transfer I2C SDA, with configurable collector open drain output
P0.1/ RXD2S/ DBG/ TIM4/ TIM3S/ SCL/	9	DB/ DI/ DO/ DB/ DB/ DB/	GPIO, configurable as input for INT0 UART2 RXD or LIN RXD after function transfer Output of Debug signal Input or output of Timer4 Input or output of Timer3 after function transfer I2C SCL, with configurable collector open drain output
P1.1/ CXOS/ TIM3	10	DB/ DO/ DB	GPIO, configurable as input for INT1 Output of comparator test signal after function transfer Input or output of Timer3
H_PU	11	DO	Pre-driver U phase high side output, with built-in 50kΩ pull-up resistor
H_PV	12	DO	Pre-driver V phase high side output, with built-in 50kΩ pull-up resistor
H_PW	13	DO	Pre-driver W phase high side output, with built-in 50kΩ pull-up resistor
L_U	14	DO	Pre-driver U phase low side output, with built-in 25kΩ pull-down resistor
L_V	15	DO	Pre-driver V phase low side output, with built-in 25kΩ pull-down resistor
L_W	16	DO	Pre-driver W phase low side output, with built-in 25kΩ pull-down resistor
VCC	17	P	Power supply. The input voltage range is decided by the power supply mode. Shunt a 10μF capacitance to ground ■ Single power supply high voltage mode: VCC_MODE = 0, VCC input range is 5V ~ 28V, VDD5 is generated by internal LDO ■ Single power supply low voltage mode: VCC_MODE = 1, VDD5 input range is 3V ~ 5.5V and short with VCC ■ Dual power supply mode VCC_MODE = 1, VCC input range is 5V ~ 36V, and VDD5 input is 5V
VSS	18	P	Ground

Pin	FU6832 QFN24	IO Type	Description
VDD5	19	P	Input of power supply or output of the internal 5V LDO. Shunt a 1μF ~ 4.7μF capacitance to ground Please refer to the VCC pin description
RSTN/ FICEK	20	DI/ DI	Input of external reset, with built-in pull-up resistor Clock line of FICE
VDD18	21	P	Output of 1.8V LDO. Shunt a 1μF ~ 4.7μF capacitance to ground
P1.2/ FICED	22	DB/ DB	GPIO, configurable as input for INT1 Data line of FICE
P1.4/ C0P/ AD10/ HAL0S	23	DB/ AI/ AI/ DI	GPIO, configurable as input for INT1 Positive input of CMP0 Input of ADC channel 10 Digital input of Hall-IC0 after function transfer
P1.6/ C1P/ AD9/ HAL1S	24	DB/ AI/ AI/ DI	GPIO, configurable as input for INT1 Positive input of CMP1 Input of ADC channel 9 Digital input of Hall-IC1 after function transfer

2.8 FU6832F QFN24 Pinout

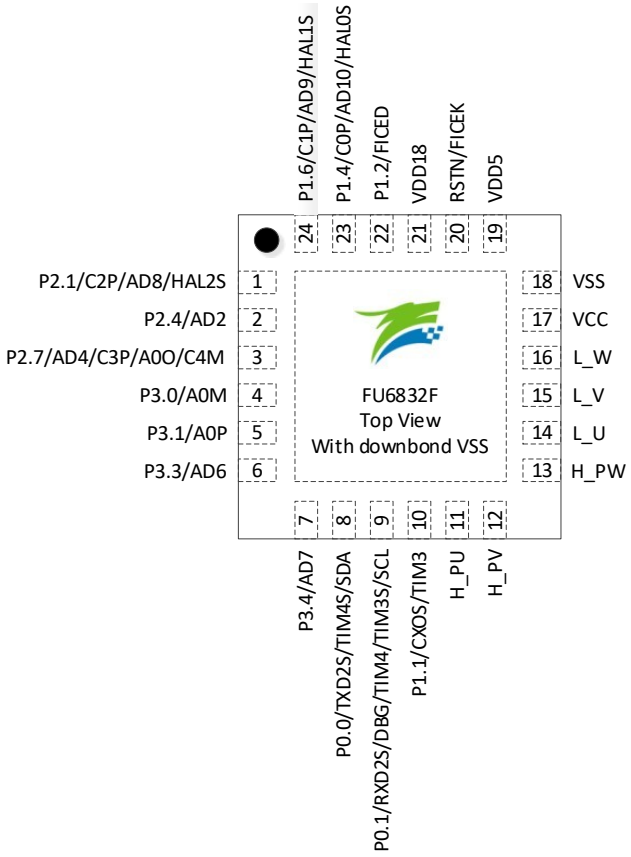


Figure 2-4 FU6832F QFN24 Pinout

2.9 Function and Pin Comparison between FU6832L/N/S/F

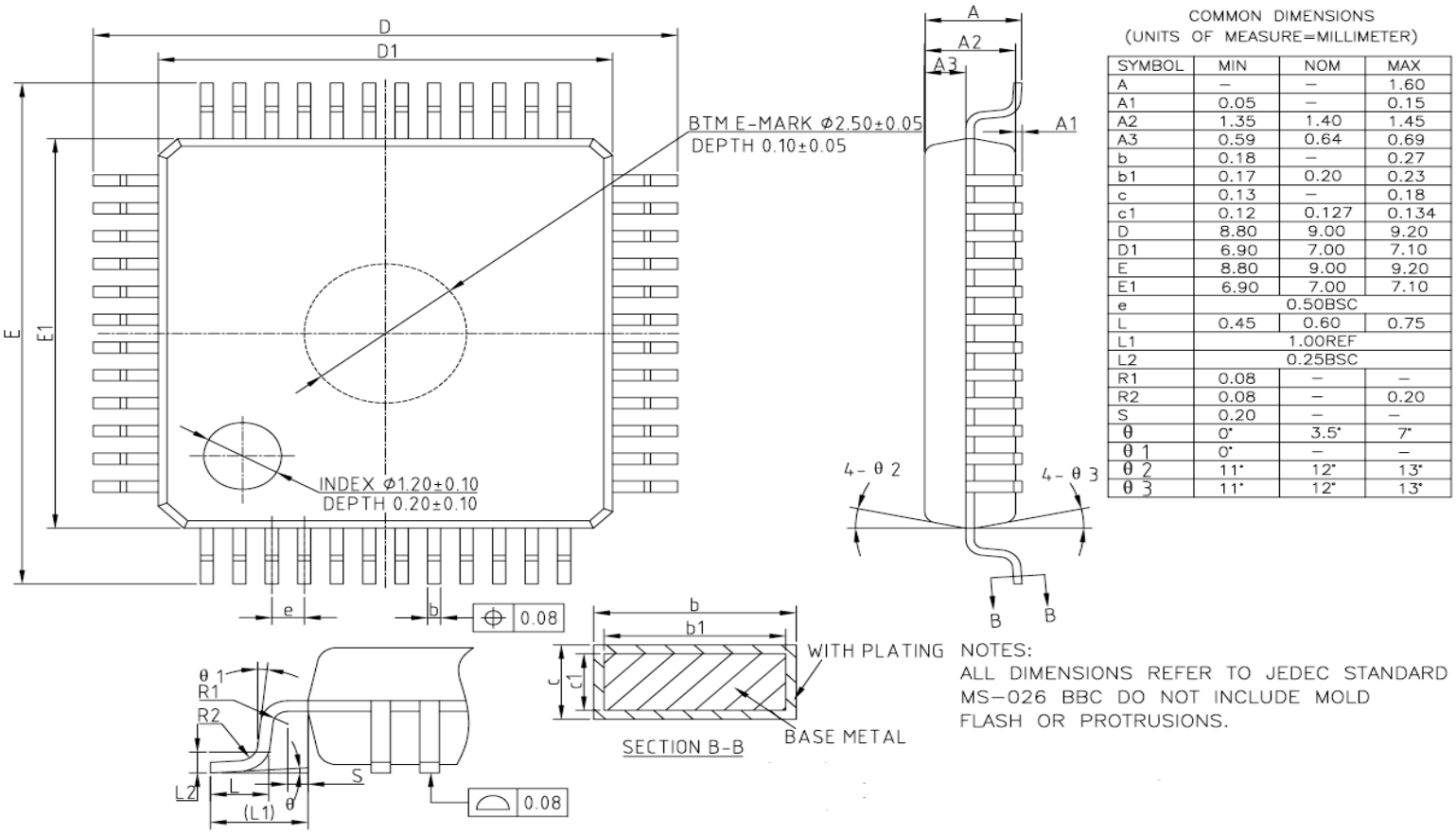
Table 2-5 Table of Function and Pin Comparison between FU6832L/N/S/F

Function	Pin	FU6832L GPIO	FU6832N GPIO	FU6832S GPIO	FU6832F GPIO
CMPG	C0P	P1.4	P1.4	P1.4	P1.4
	C0M	P1.5	P1.5	—	—
	C1P	P1.6	P1.6	P1.6	P1.6
	C1PS	P1.3	P1.3	—	—
	C1M	P1.7	P1.7	—	—
	C2P	P2.1	P2.1	P2.1	P2.1
	C2PS	P1.5	P1.5	—	—
	C2M	P2.2	P2.2	—	—
CMP3	C3P	P2.0	P2.0	—	—
		P2.3	P2.3	—	—
		P2,7	P2,7	P2,7	P2,7
	C3M	P2.6	—	—	—
CMP4	C4P	P2.3	P2.3	—	—
	C4M	P2.7	P2.7	P2.7	P2.7
AMP0	A0P	P3.1	P3.1	P3.1	P3.1
	A0M	P3.0	P3,0	P3,0	P3,0
	A0O	P2.7	P2.7	P2.7	P2.7
AMP1	A1P	P1.6	P1.6	P1.6	P1.6
	A1M	P1.7	P1.7	—	—
	A1O	P2.0	P2.0	—	—
AMP2	A1P	P2.1	P2.1	P2.1	P2.1
	A2M	P2.2	P2.2	—	—
	A2O	P2.3	P2.3	—	—
DAC	DAC0	P2.6	—	—	—
	DAC1	P2.7	P2.7	P2.7	P2.7
Bus voltage sampling	AD2	P2.4	P2.4	P2.4	P2.4
	AD14	—	—	—	—
Bus current sampling	AD4	P2.7	P2.7	P2.7	P2.7
U/V phase current sampling	AD0	P2.0	P2.0	—	—
	AD1	P2.1	P2.1	P2.1	P2.1
Hall-IC input	HAL0	P0.2	—	—	—
	HAL0S	P1.4	P1.4	P1.4	P1.4
	HAL1	P3.7	—	—	—
	HAL1S	P1.6	P1.6	P1.6	P1.6
	HAL2	P3.6	—	—	—
	HAL2S	P2.1	P2.1	P2.1	P2.1
Hall bias power supply	HBIAS	P1.3	P1.3	—	—
Timer2	TIM2	P0.7	P0.7	—	—
		P1.0	—	—	—
Timer3	TIM3	P1.1	P1.1	P1.1	P1.1
	TIM3S	P0.1	P0.1	P0.1	P0.1
Timer4	TIM4	P0.1	P0.1	P0.1	P0.1
	TIM4S	P0.0	P0.0	P0.0	P0.0
UART1	TXD	P0.5	P0.5	—	—
	RXD	P0.6	P0.6	—	—
UART2/LIN	TXD2	P3.7	—	—	—
	TXD2S	P0.0	P0.0	P0.0	P0.0
	RXD2	P3.6	—	—	—
	RXD2S	P0.1	P0.1	P0.1	P0.1

Function	Pin	FU6832L GPIO	FU6832N GPIO	FU6832S GPIO	FU6832F GPIO
I2C	SDA	P0.0	P0.0	P0.0	P0.0
	SCL	P0.1	P0.1	P0.1	P0.1
SPI	SCLK	P0.5	P0.5	—	—
	MOSI	P0.6	P0.6	—	—
	MISO	P0.7	P0.7	—	—
	NSS	P0.4	—	—	—
Voltage reference output	VREF	P3.5	—	—	—
Half voltage reference output	VHALF	P3.2	P3.2	—	—
Comparator test output	CXO	P0.7	P0.7	—	—
	CXOS	P1.1	P1.1	P1.1	P1.1
Debug signal	DBG	P0.1	P0.1	P0.1	P0.1
FICE interface	FICEK	RSTN	RSTN	RSTN	RSTN
	FICED	P1.2	P1.2	P1.2	P1.2

3 Package Size

3.1 LQFP48_7X7

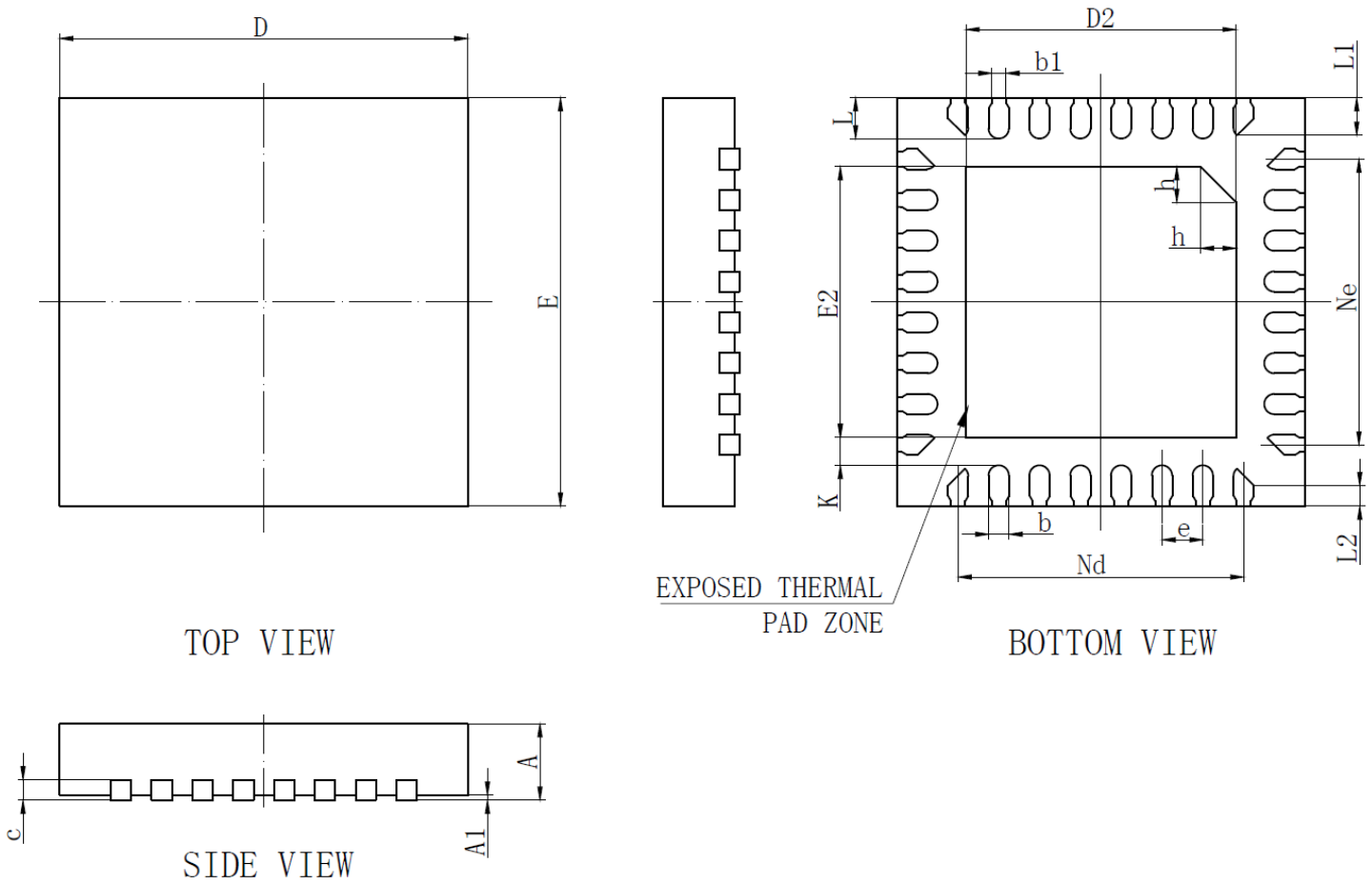


SYMBOL	MIN	NOM	MAX
A	–	–	1.60
A1	0.05	–	0.15
A2	1.35	1.40	1.45
A3	0.59	0.64	0.69
b	0.18	–	0.27
b1	0.17	0.20	0.23
c	0.13	–	0.18
c1	0.12	0.127	0.134
D	8.80	9.00	9.20
D1	6.90	7.00	7.10
E	8.80	9.00	9.20
E1	6.90	7.00	7.10
e		0.50BSC	
L	0.45	0.60	0.75
L1		1.00REF	
L2		0.25BSC	
R1	0.08	–	–
R2	0.08	–	0.20
S	0.20	–	–
θ	0°	3.5°	7°
θ1	0°	–	–
θ2	11°	12°	13°
θ3	11°	12°	13°

NOTES:
ALL DIMENSIONS REFER TO JEDEC STANDARD MS-026 BBC DO NOT INCLUDE MOLD FLASH OR PROTRUSIONS.

Figure 3-1 Package Size Diagram of LQFP48_7X7

3.2 QFN32_4X4



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.70	0.75	0.80
A1	0	0.02	0.05
b	0.15	0.20	0.25
b1	0.14REF		
c	0.18	0.20	0.25
D	3.90	4.00	4.10
D2	2.60	2.65	2.70
e	0.40BSC		
Nd	2.80BSC		
E	3.90	4.00	4.10
E2	2.60	2.65	2.70
Ne	2.80BSC		
K	0.20	-	-
L	0.35	0.40	0.45
L1	0.30	0.35	0.40
L2	0.15	0.20	0.25
h	0.30	0.35	0.40

Figure 3-2 Package Size Diagram of QFB32_4X4

3.3 SSOP24_8.65X3.9

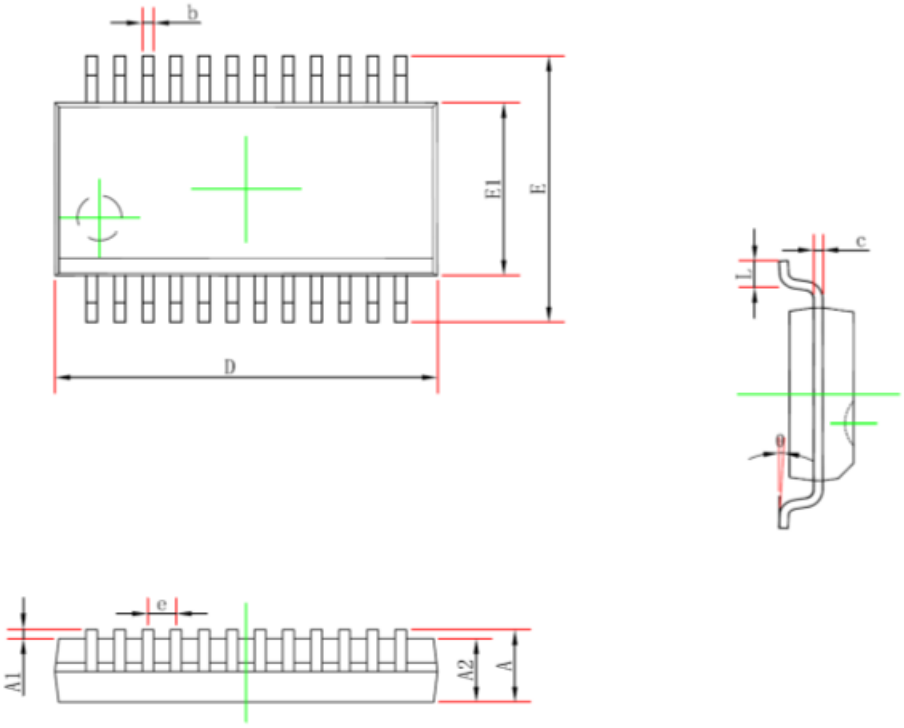


Figure 3-3 Package Size Diagram of SSOP24_8.65X3.9

Table 3-1 Package Size Table of SSOP24_8.65X3.9

Symbol	Dimensions In Millimeter		Dimensions In Inches	
	Min	Max	Min	Max
A	—	1.750	—	0.069
A1	0.100	0.250	0.004	0.010
A2	1.250	—	0.049	—
b	0.203	0.305	0.008	0.012
c	0.102	0.254	0.004	0.010
D	8.450	8.850	0.333	0.348
E1	3.800	4.000	0.150	0.157
E	5.800	6.200	0.228	0.244
e	0.635(BSC)		0025(BSC)	
L	0.400	1.270	0.016	0.050
θ	0°	8°	0°	8°

3.4 QFN24_4X4

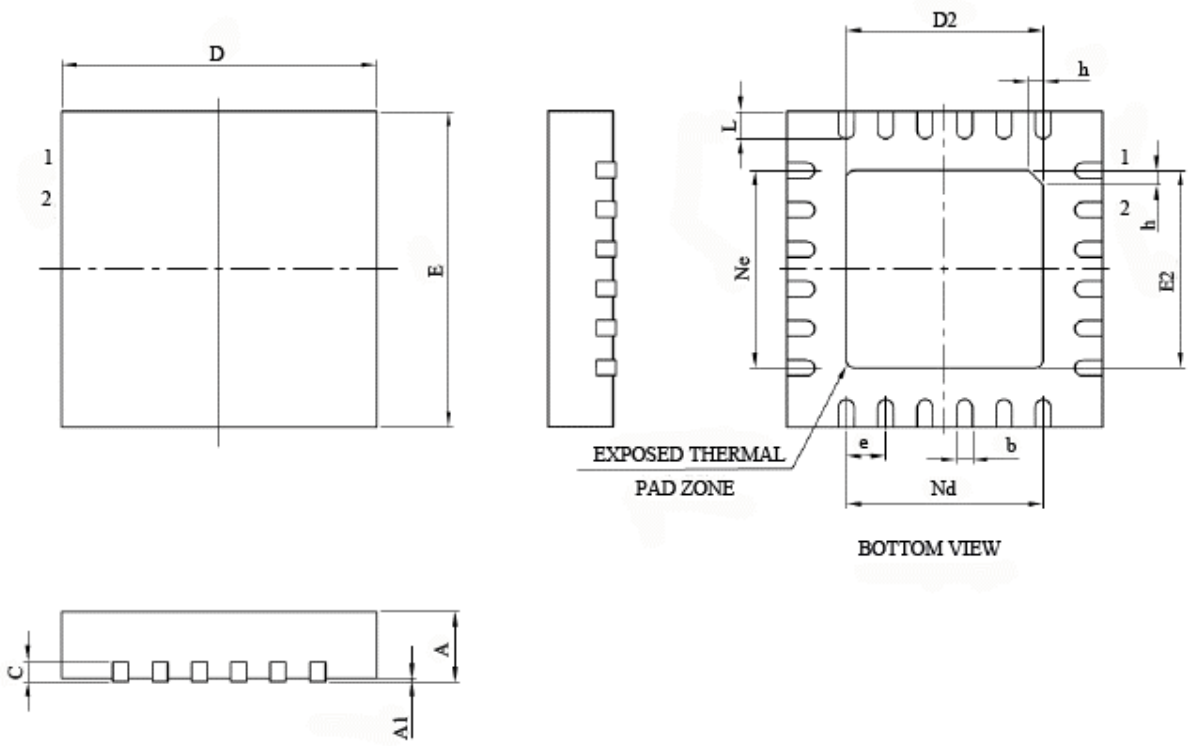


Figure 3-4 Package Size Diagram of QFN24_4X4

Table 3-2 Package Size Table of QFN24_4X4

Symbol	Dimensions In Millimeter		
	Min	Nom	Max
A	0.70	0.75	0.80
A1	—	0.02	0.05
b	0.18	0.25	0.30
c	0.18	0.20	0.25
D	3.90	4.00	4.10
D2	2.35	2.50	2.65
e	0.50BSC		
Ne	2.50BSC		
Nd	2.50BSC		
E	3.90	4.00	4.10
E2	2.35	2.50	2.65
L	0.35	0.40	0.45
h	0.30	0.35	0.40
N	Pin Number = 24		

4 Ordering Information

Table 4-1 Package Type Information

Type	MIPS(Peak)	Flash(kByte)	XRAM(Byte)	Clock circuits		3P3N pre-driver	Drive method		I2C/UART/SPI/LIN	DMA	GPIO	Timer	Analogue peripherals								Leadfree	Package
													ADC			DAC		VREF	Operational amplifier	Comparator		
				Internal fast clock	Internal slow clock		Square wave	FOC					Number	Channel	Bit	Number	Bit					
FU6832L	24	16	768	√	√	√	√	√	√	√	35	6	1	14	12	2	9/6	√	3	4	√	LQFP48 (7x7 mm)
FU6832N	24	16	768	√	√	√	√	√	√	√	22	6	1	11	12	2	9/6	√	3	4	√	QFN32 (4x4mm)
FU6832S	24	16	768	√	√	√	√	√	√	√	13	6	1	7	12	2	9/6	√	1	3	√	SSOP24 (8.65x3.9mm)
FU6832F	24	16	768	√	√	√	√	√	√	√	13	6	1	7	12	2	9/6	√	1	4	√	QFN24 (4x4mm)

5 Electrical Characteristics

5.1 Absolute Maximum Ratings

Table 5-1 Absolute Maximum Ratings

Parameter	Conditions	Min	Typ	Max	Unit
Operating ambient temperature T_A		–40	—	85	°C
Operating ambient temperature T_A	Dual power supply mode, VCC_MODE = 1, VDD5 = 5V, VCC < 28V	–40	—	105	°C
	Single power supply mode, VCC_MODE = 0, VCC < 15V	–40	—	105	°C
Operating junction temperature T_J		–40	—	150	°C
Storage temperature		–55	—	150	°C
VCC to VSS		–0.3	—	36	V
VDD5 to VSS		–0.3	—	6.5	V
RSTN, GPIO to VSS		–0.3	—	VDD5 + 0.3	V

Note: Exceeding the values listed in Table 5-1 may permanently damage the chip. These are stress ratings only, and functional operation of the device at those or any other conditions beyond those indicated in the operation listings of this specification is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

5.2 Global Electrical Characteristics

Table 5-2 Global Electrical Characteristics

(Unless otherwise specified, T_A = 25°C, VCC = 5V ~ 28V)

Parameter	Conditions	Min	Typ	Max	Unit
VCC operating voltage [1]	Single power supply high voltage Mode	5	—	28	V
	Dual power supply mode[2] VCC_MODE = 1, VCC ≥ VDD5	5	—	36	V
VDD5 operating voltage	VCC connected withVDD5[2]	3	—	5.5	V
I_{VCC} operating current [3]		—	20	40	mA
I_{VCC} standby current [3]		—	6	12	mA
I_{VCC} sleep current		—	50	150	μA

Notes:

[1] VCC voltage rise rate ranges from 0.5V/μs ~ 0.1V/s depending on different sample batchs

[2] VDD5 must be maintained at 5 ~ 5.5V when Flash is written or erased

[3] Characteristics may vary for different configurations

5.3 GPIO Electrical Characteristics

Table 5-3 GPIO Electrical Characteristics

(Unless otherwise specified, T_A = 25°C, VCC = 5V ~ 28V)

Parameter	Conditions	Min	Typ	Max	Unit
Rising time of output	50pF Load, from 10% to 90%, T_A = 25°C	—	15	—	ns
Falling time of output	50pF Load, from 90% to 10%, T_A = 25°C	—	13	—	ns
V_{OH} high level output voltage	I_{OH} = 4mA	VDD5 - 0.7	—	—	V
V_{OL} low level output voltage	I_{OL} = 8mA	—	—	VSS + 0.7	V
V_{IH} high level input voltage [1]		0.7*VDD5	—	—	V
V_{IL} low level input voltage		—	—	0.2*VDD5	V
Pull-up resistor [2]		—	33	—	kΩ
Pull-up resistor [3]		—	5.6	—	kΩ
Pull-down resistor [4]		—	10	—	kΩ

Notes:

[1] When VDD5 = 5V, the minimum value of V_{IH} is 0.6*VDD5

[2] GPIO expect P0[2:0], P1[6:3], P2[1] and P3[7:6]

[3] P0[2:0], P1[6:3], P2[1] and P3[7:6]

[4] P0[1] and P1[1]

5.4 Pre-driver IO Electrical Characteristics

Table 5-4 Pre-driver IO Electrical Characteristics

(Unless otherwise specified, T_A = 25°C, VCC = 15V, VCC_MODE = 0)

Parameter	Conditions	Min	Typ	Max	Unit
High side output source current		—	150	—	mA
High side output sink current		—	90	—	mA
Low side output source current		—	150	—	mA
Low side output sink current		—	180	—	mA
Rising time of high side output	1nF Load, from 10% to 90%	—	25	—	ns
Falling time of high side output	1nF Load, from 90% to 10%	—	90	—	ns
Rising time of low side output	1nF Load, from 10% to 90%	—	115	—	ns

Falling time of low side output	1nF Load, from 90% to 10%	—	60	—	ns

5.5 ADC Electrical Characteristics

Table 5-5 ADC Electrical Characteristics

(Unless otherwise specified, T_A = 25°C, VCC = 5V ~ 28V)

Parameter	Conditions	Min	Typ	Max	Unit
Integral nonlinearity	12 bit mode	—	2	—	LSB
Differential nonlinearity	12 bit mode	—	1.5	—	LSB
Offset error	12 bit mode	—	6	—	LSB
Signal to noise rate	f_{IN} = 350kHz	—	70.8	—	dB
Effective number of bits	f_{IN} = 350kHz	—	10.5	—	Bit
Spurious free dynamic range	f_{IN} = 350kHz	—	68.2	—	dB
Total harmonic distortion	f_{IN} = 350kHz	—	67	—	dB
R_{IN} input resistance		—	800	—	Ω
C_{IN} input capacitance		—	30	—	pF
Conversion time		—	13	—	ADCLK[1]
Sampling time		3	—	63	ADCLK[1]

Note: ADCLK = 12MHz

5.6 Voltage Reference Electrical Characteristics

Table 5-6 Voltage Reference Electrical Characteristics

(Unless otherwise specified, T_A = –40°C ~ 85°C, VCC = 5V ~ 28V)

Parameter	Conditions	Min	Typ	Max	Unit
VREF	VREF_CR[VREFVSEL]= 00	4.3	4.5	4.7	V
VHALF		VREF/2 - 0.2	VREF/2	VREF/2 + 0.2	V

5.7 Operational Amplifier Electrical Characteristics

Table 5-7 Operational Amplifier Electrical Characteristics

(Unless otherwise specified, T_A = 25°C, VCC = 5V ~ 28V)

Parameter	Conditions	Min	Typ	Max	Unit
V_{ICMR} input common mode voltage range		0	—	VDD5 - 1.5	V
V_{OS} input offset voltage	T_A = 25°C	—	5	10	mV

A_{OL} open loop gain	R_L = 100kΩ	—	80	—	dB
Unit gain bandwidth	C_L = 40pF	6	10	—	MHz
Slew rate	C_L = 40pF	10	15	—	V/μS

5.8 BEMF Electrical Characteristics

Table 5-8 BEMF Electrical Characteristics

(Unless otherwise specified, T_A = 25°C, VCC = 5V ~ 28V, VCC_MODE = 0)

Parameter	Conditions	Min	Typ	Max	Unit
Built-in resistance		5.4	6.8	8.2	kΩ
Relative accuracy between built-in resistors		—	1	—	%

5.9 OSC Electrical Characteristics

Table 5-9 OSC Electrical Characteristics

(Unless otherwise specified, T_A = –40°C ~ 85°C, VCC = 5V ~ 28V, VCC_MODE = 0)

Parameter	Conditions	Min	Typ	Max	Unit
System clock frequency		23.5	24	24.5	MHz
Low speed clock frequency		29	32.8	37	kHz

Note: System clock frequency is SYSCLK, System clock period is T. Unless otherwise specified, The System clock frequency of chip is 24MHz, T = 1/SYSCLK = 41.67ns.

5.10 Reset Electrical Characteristics

Table 5-10 Reset Electrical Characteristics

(Unless otherwise specified, T_A = 25°C, VCC = 5V ~ 28V, VCC_MODE = 0)

Parameter	Conditions	Min	Typ	Max	Unit
Minimum RSTN low time to generate a reset		50	—	—	μs
VDD5 reset threshold	Reset voltage LVR = 3.0V	2.8	3.0	3.2	V

5.11 LDO Electrical Characteristics

Table 5-11 LDO Electrical Characteristics

(Unless otherwise specified, T_A = 25°C, VCC = 5V ~ 28V, VCC_MODE = 0)

Parameter	Conditions	Min	Typ	Max	Unit
VDD5 to VSS	VCC = 7V ~ 28V, VCC_MODE = 0	4.7	5	5.3	V
VDD18 to VSS		1.65	1.85	2.0	V

5.12 Thermal Characteristics

Table 5-12 LQFP48 Thermal Characteristics

Parameter	Conditions	Value	Unit
θ_{JA} Thermal resistance junction to ambient [1]	JEDEC standard, 2S2P PCB	52.4	°C/W
	JEDEC standard, 1S0P PCB	72.2	°C/W
θ_{JC} Thermal resistance junction to case [1]	JEDEC standard, 1S0P PCB	17	°C/W

Note:

[1] Test results may vary depending on the actual conditions

Table 5-13 QFN32 Thermal Characteristics

Parameter	Conditions	Value	Unit
θ_{JA} Thermal resistance junction to ambient [1]	JEDEC standard, 2S2P PCB	47	°C/W
	JEDEC standard, 1S0P PCB	74	°C/W
θ_{JC} Thermal resistance junction to case [1]	JEDEC standard, 2S2P PCB	20	°C/W

Note:

[1] Test results may vary depending on the actual conditions

Table 5-14 SSOP24 Thermal Characteristics

Parameter	Conditions	Value	Unit
θ_{JA} Thermal resistance junction to ambient [1]	JEDEC standard, 2S2P PCB	75	°C/W

Note:

[1] Test results may vary depending on the actual conditions

Table 5-15 QFN24 Thermal Characteristics

Parameter	Conditions	Value	Unit
θ_{JA} Thermal resistance junction to ambient [1]	JEDEC standard, 2S2P PCB	50	°C/W
θ_{JC} Thermal resistance junction to case [1]	JEDEC standard, 2S2P PCB	25	°C/W

Note:

[1] Test results may vary depending on the actual conditions

6 Reset

6.1 Sources of Reset (RST_SR)

The chip has 7 reset sources:

- Power on reset (RSTPOW)
- External pin reset (RSTEXT)
- Low voltage reset (RSTLVD)
- Watchdog reset (RSTWDT)
- Flash error detection reset (RSTFED)
- Debug reset (RSTDBG)
- Soft reset (SOFTR)

The reset flag is queryable and recorded in register RST_SR. The last reset source will set the relevant flag bit to 1 and other bits will be cleared. To clear reset flag, set RST_SR[RSTCLR] to 1 to clear RST_SR[7:3] & RST_SR[0]. After reset, MCU restart the program from address 0.

6.2 Reset Enable

Please refer to the corresponding control registers.

6.3 External Pin Reset and Power on Reset

When RSTN pin is low for 50μs, the chip reset.

When the chip power on and VDD5 is higher than the reset voltage threshold, the chip reset.

6.4 Low Voltage Protection Reset

The chip internal circuitry will monitor VDD. When VDD voltage drops to the reset voltage threshold, the internal monitor circuitry will send the corresponding reset signal to reset the chip.

The enable of VDD monitor circuitry and the low voltage threshold are configurable by the corresponding register.

6.5 Watchdog Reset

After enable WDT, software must periodically write 1 to WDT_CR[WDTRF] which will initialize watchdog up counter. When watchdog up counter reaches its maximum value, WDT generates an output pulse to reset chip, which can be used to prevent software from running out of control.

6.6 Flash Error Detection Reset

The Flash manipulation module provides MOVX instruction to allow software to write, erase and read Flash. If a Flash erase is attempted to the last sector (0x3F80 ~ 0x3FFF) or a Flash write is attempted to the last byte (0x3FFF), the Flash error detection reset is generated. The Flash error detection reset is always enabled and cannot be disabled.

6.7 Debug Reset

When the chip is on the debug state, click reset button of IDE to send a Debug reset signal.

6.8 Soft Reset

Setting RST_SR[SOFTR] to 1, the chip will be forced to reset immediately. After reset, the flag bit RST_SR[SOFTR] will be set to 1.

6.9 Reset Registers

6.9.1 RST_SR (0xC9)

Bit	7	6	5	4	3	2	1	0
Name	RSTPOW/ RSTCLR	RSTEXT	RSTLVD	RSV	RSTWDT	RSTFED	RSTDBG	SOFTR
Type	R/W1	R	R	—	R	R	R	R/W1
Reset	—	—	—	—	—	—	—	—

Bit	Name	Description
[7]	RSTPOW/ RSTCLR	Power on reset flag Read: 0: Last reset was not from a power on reset 1: Last reset was from a power on reset Write: 0: No meaning 1: Clear RST_SR[7:3] & RST_SR[0] to 0
[6]	RSTEXT	External pin reset flag 0: Last reset was not from an external pin reset 1: Last reset was from an external pin reset
[5]	RSTLVD	Low voltage reset flag 0: Last reset was not from a low voltage reset 1: Last reset was from a low voltage reset
[4]	RSV	Reserved
[3]	RSTWDT	Watchdog reset flag 0: Last reset was not from a watchdog reset 1: Last reset was from a watchdog reset
[2]	RSTFED	Flash error detection reset flag 0: Last reset was not from a Flash error detection reset 1: Last reset was from a Flash error detection reset
[1]	RSTDBG	Debug reset flag 0: last reset not from a Debug reset 1: Last reset from a Debug reset
[0]	SOFTR	Soft reset Read: 0: Last reset was not a reset owing to this bit being written 1 1: Last reset was a reset owing to this bit being written 1 Write: 0: No meaning 1: Generates a soft reset

7 Interrupt

7.1 Interrupt Introduction

There are 16 interrupt sources inside the chip. Each interrupt source has four levels of priority, which are configured via registers IP0 ~ IP3. Interrupt flag bits are addressed in SFR or XSFR. The corresponding interrupt flag bit is set to 1 by hardware when the interrupt conditions are met by internal circuitry or external signals. If the enable bits of global interrupt and individual interrupt, with interrupt flag 1, are set to 1, an interrupt request is sent to CPU. If no higher priority interrupt subroutine is being executed, the interrupt is entered to start the execution of subroutine.

Except reset, each interrupt source can be set a priority level. The subroutine of a lower priority interrupt can be interrupted by a higher priority interrupt during its execution. The lower priority interrupt subroutine will be continued after the higher priority interrupt subroutine has finished. New interrupts cannot interrupt the interrupt subroutine with the same priority level. Priority of each interrupt can be set individually. The values 0 ~ 3 indicate the priority in order from lower to higher with the default value being 0. If two interrupt requests generate at the same time, the interrupt with the higher priority will be entered first. If two interrupt sources have the same priority, they are arbitrated in a default priority order. See Table 7-2 for the default priority order, the lower the mark the higher the priority level.

7.2 Interrupt Enable

IE[EA] is the global interrupt enable bit. When EA = 0, CPU does not response any interrupt request.

Each interrupt can be individually enabled or disabled by the configuration of the corresponding interrupt enable bit in SFR or XSFR. When the enable bit of the global interrupt or an interrupt is cleared, the interrupt flag bit that is set to 1 will be pending. The MCU will immediately enter the interrupt subroutine once the enable bit is set to 1. Therefore, clear corresponding interrupt flag bit before enable the interrupt.

7.3 External Interrupt

The external interrupt has 2 interrupt sources, INT0 and INT1.

The digital input signals of P0.0 ~ P0.6, and the output signal of CMP4, can be used as the trigger source of INT0, which is configured by LVSR[EXT0CFG]. These interrupt sources share one interrupt entry, one interrupt flag bit TCON[IF0] and one interrupt enable bit IE[EX0]. The interrupt trigger level is selected by TCON[IT0]. The priority is configured in IP0[PX0].

The digital input signals of P1.0 ~ 1.7 and P2.0 ~ 2.7 can be used as the trigger source of INT1. The interrupt flag registers are P1IF, P2IF and the interrupt enable registers are P1IE, P2IE. Each trigger source has a corresponding interrupt flag bit and an interrupt enable bit. INT1 can select multiple trigger sources, and the trigger source can be identified in the interrupt subroutine by P1IF and P2IF. These 16 interrupt trigger sources share one interrupt

entry, one interrupt enable bit IE[EX1]. To enable INT1, first set IE[EX1] to 1 and then configure the corresponding enable bit. The interrupt trigger level is selected by TCON[IT1]. The priority is configured in IP0[PX1].

Table 7-1 nterrupt Flag Register & Interrupt Enable Register of External Interrupt INT1 I

SFR Addr	Bit	Name	Description	R/W	Reset
0xD1	[7:0]	P1IE[7:0]	Port1 each port INT1 function enable 0: Disable 1: Enable	R/W	0x00
0xD2	[7:0]	P1IF[7:0]	Port1 each port INT1 flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear the corresponding bit to 0 1: No meaning	R/W0	0x00
0xD3	[7:0]	P2IE[7:0]	Port2 each port INT1 function enable 0: Disable 1: Enable	R/W	0x00
0xD4	[7:0]	P2IF[7:0]	Port2 each port INT1 flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear the corresponding bit to 0 1: No meaning	R/W0	0x00

7.4 Interrupt Summary

Table 7-2 Interrupt Summary

Interrupt source	Priority Order	Vector Address	Interrupt Flag	Software Clear?	Interrupt Enable Bit	Priority Configuration
Reset	Highest	0x0000	None	No	Always enable	Highest
LVW interrupt TSD interrupt	0	0x0003	LVSR[0] TCON[5]	Yes	CCFG1[6] IE[1]	IP0[1:0]
INT0	1	0x000B	TCON[2]	Yes	IE[0]	IP0[3:2]
INT1	2	0x0013	P1IF[7:0] P2IF[7:0]	Yes	IE[2]	IP0[5:4]
FG interrupt DRV compare match interrupt	3	0x001B	DRV_SR[5:4]	Yes	DRV_SR[3] DRV_SR[2:0]	IP0[7:6]
Timer2 interrupt	4	0x0023	TIM2_CR1[7:5]	Yes	TIM2_CR1[4:3] TIM2_CR0[3]	IP1[1:0]
Timer1 interrupt	5	0x002B	TIM1_SR[4:0]	Yes	TIM1_IER[4:0]	IP1[3:2]
ADC interrupt	6	0x0033	ADC_CR[0]	Yes	ADC_CR[1]	IP1[5:4]
CMP0/1/2 interrupt	7	0x003B	CMP_SR[6:4]	Yes	CMP_CR0[5:0]	IP1[7:6]
RTC interrupt	8	0x0043	RTC_STA[6]	Yes	IE[6]	IP2[1:0]
Timer3 interrupt	9	0x004B	TIM3_CR1[7:5]	Yes	TIM3_CR1[4:3] TIM3_CR0[3]	IP2[3:2]
Systick interrupt	10	0x0053	DRV_SR[7]	Yes	DRV_SR[6]	IP2[5:4]
Timer4 interrupt	11	0x005B	TIM4_CR1[7:5]	Yes	TIM4_CR1[4:3] TIM4_CR0[3]	IP2[7:6]
CMP3 interrupt	12	0x0063	CMP_SR[7]	Yes	CMP_CR0[7:6]	IP3[1:0]
I2C interrupt UART1 interrupt	13	0x006B	I2C_SR[0] UT_CR[1:0]	Yes	I2C_CR[0] IE[4]	IP3[3:2]
SPI interrupt UART2 interrupt LIN interrupt	14	0x0073	SPI_CR1[7] UT2_CR[1:0] LIN_SR[7:5] LIN_SR[2:0] LIN_CSR[3]	Yes	IE[3] UT2_BAUDH[5] LIN_CR[3]	IP3[5:4]
DMA interrupt	15	0x007B	DMA0_CR0[0] DMA1_CR0[0]	Yes	DMA0_CR0[2]	IP3[7:6]

Note:

- For UT_CR[RI], UT_CR[TI], DMA0_CR0[DMAIF] and DMA0_CR1[DMAIF], software writes 1 to those interrupt flags and MCU will generate the corresponding interrupt request. Except those flags, other flags cannot be pended by software.
- For registers containing serval interrupt flags, to prevent the error clearing of interrupt flags, software writes 1 to interrupt flag bits, which are not required to be cleared, during a clear flag operation. E.g., use code DRV_SR = (DRV_SR & 0x7F) | 0x30 to clear DRV_SR[SYSTIF].

7.5 Interrupt Register

7.5.1 IE (0xA8)

Bit	7	6	5	4	3	2	1	0
Name	EA	RTCIE	RSV	ES0	SPIIE	EX1	TSDIE	EX0
Type	R/W	R/W	—	R/W	R/W	R/W	R/W	R/W
Reset	0	0	—	0	0	0	0	0

Bit	Name	Description
[7]	EA	Global interrupt enable 0: Disable 1: Enable
[6]	RTCIE	RTC interrupt enable 0: Disable 1: Enable
[5]	RSV	Reserved
[4]	ES0	UART1 interrupt enable 0: Disable 1: Enable
[3]	SPIIE	SPI interrupt enable 0: Disable 1: Enable
[2]	EX1	INT1 interrupt enable 0: Disable 1: Enable
[1]	TSDIE	TSD interrupt enable 0: Disable 1: Enable
[0]	EX0	INT0 interrupt enable 0: Disable 1: Enable

7.5.2 IP0 (0xB8)

Bit	7	6	5	4	3	2	1	0
Name	PDRV		PX1		PX0		PLVW_TSD	
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:6]	PDRV	FG/DRV comparison match interrupt priority setting
[5:4]	PX1	INT1 interrupt priority setting
[3:2]	PX0	INT0 interrupt priority setting
[1:0]	PLVW_TSD	LVW/TSD interrupt priority setting

Note: The values 0 ~ 3 indicate the priority in order from lower to higher with totally 4 levels.

7.5.3 IP1 (0xC0)

Bit	7	6	5	4	3	2	1	0
Name	PCMP		PADC		PTIM1		PTIM2	
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:6]	PCMP	CMP0/1/2 interrupt priority setting
[5:4]	PADC	ADC interrupt priority setting
[3:2]	PTIM1	Timer1 interrupt priority setting
[1:0]	PTIM2	Timer2 interrupt priority setting

Note: The values 0 ~ 3 indicate the priority in order from lower to higher with totally 4 levels.

7.5.4 IP2 (0xC8)

Bit	7	6	5	4	3	2	1	0
Name	PTIM4		PSYSTICK		PTIM3		PRTC	
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:6]	PTIM4	Timer4 interrupt priority setting
[5:4]	PSYSTICK	Systick interrupt priority setting
[3:2]	PTIM3	Timer3 interrupt priority setting
[1:0]	PRTC	RTC interrupt priority setting

Note: The values 0 ~ 3 indicate the priority in order from lower to higher with totally 4 levels.

7.5.5 IP3 (0xD8)

Bit	7	6	5	4	3	2	1	0
Name	PDMA		PSPI_UT2		PI2C_UT1		PCMP3	
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:6]	PDMA	DMA interrupt priority setting
[5:4]	PSPI_UT2	SPI/UART2/LIN interrupt priority setting
[3:2]	PI2C_UT1	I2C/UART1 interrupt priority setting
[1:0]	PCMP3	CMP3 interrupt priority setting

Note: The values 0 ~ 3 indicate the priority in order from lower to higher with totally 4 levels.

7.5.6 TCON (0x88)

Bit	7	6	5	4	3	2	1	0
Name	RSV		TSDIF	IT1		IF0	IT0	
Type	—	—	R/W0	R/W	R/W	R/W0	R/W	R/W
Reset	—	—	0	0	0	0	0	0

Bit	Name	Description
[7:6]	RSV	Reserved
[5]	TSDIF	TSD interrupt flag TSDIF reflects that an ovetemperature event has occurred Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning Note: This flag is often used with the overtemperature status bit LVSR[TSDF]
[4:3]	IT1	INT1 trigger level selection 00: Rising edge triggers interrupt 01: Falling edge triggers interrupt 1X: Level change (rising or falling) triggers interrupt
[2]	IF0	INT0 interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[1:0]	IT0	INT0 trigger level selection 00: Rising edge triggers interrupt 01: Falling edge triggers interrupt 1X: Level change (rising or falling) triggers interrupt

8 I2C

8.1 I2C Introduction

The I2C module provides an industry standard two-wire serial interface and is a simple bi-directional synchronous serial bus for communication between MCU and external I2C devices as shown in Figure 8-1. The bus consists of two serial lines: SDA and SCL. P0.0 is the SDA and P0.1 is the SCL. P0.0 and P0.1 will be automatically open-drain when I2C is enabled.

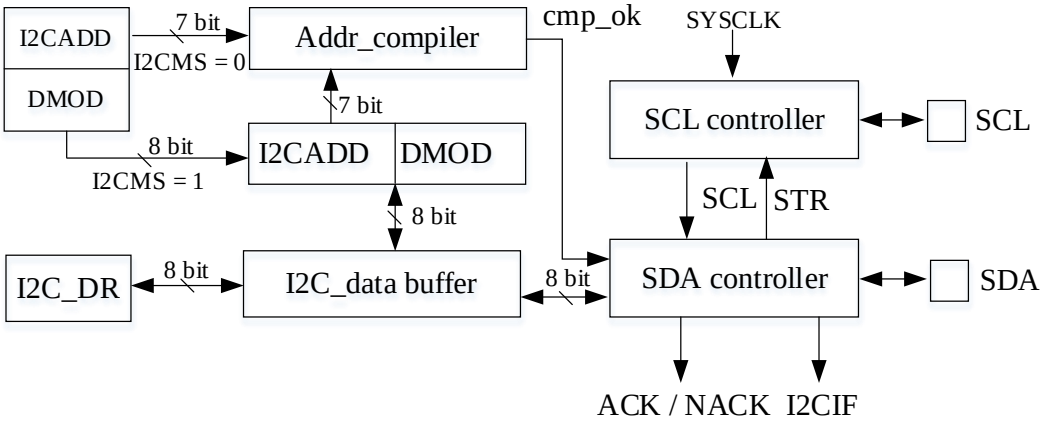


Figure 8-1 Block Diagram of I2C

Features:

- Supports standard mode (up to 100 kHz), fast mode (up to 400 kHz) and fast plus mode (up to 1 MHz)
- Supports master mode and slave mode
- Supports 7-bit address mode and general call address mode
- Supports DMA data transfer

SDA and SCL are high level when the bus is idle, which is the only basis for detecting whether the bus is idle or not. There is only one master device and at least one slave device active on the bus during the transmission. When the bus is occupied, other devices must wait for the bus idle before controlling the bus to start an I2C communication. The master start the bus to transfer data. Clock signal is sent to all devices via SCL and the slave address and read/write mode are sent via SDA. When a device on the bus matches the address, it will act as a slave. The relationships between masters and slaves or data transfer direction on the bus are not constant. The process of the master sending data to the slave is as follows: The master first addresses the slave device and waits for the slave response. After the slave response, sends data to the slave. Finally, the master terminates the data transmission. The communication process is shown in Figure 8-2. The process of the master receiving data from the slave is as follows: The master first addresses the slave and waits for the slave response. After the slave response, the master receives the data sent by the slave. Finally the master terminates the receiving process. The communication process is shown in Figure 8-3. In this case, the master generates the timing clock and stops the data transmission.

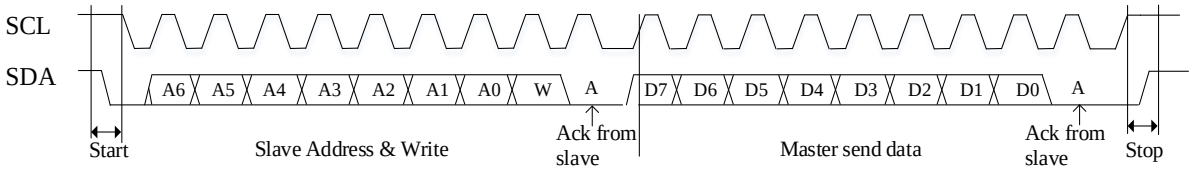


Figure 8-2 Master Sends Data to Slave

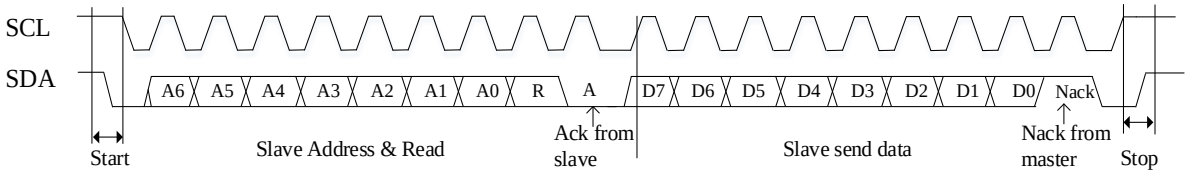


Figure 8-3 Master Receives Data from Slave

8.2 I2C Instructions

8.2.1 Master Mode

1. Set I2C_CR[I2CMS] to 1 to select master mode;
2. Configure I2C_CR [I2CSPD] to set the clock frequency of SCL;
3. Configure I2C_ID[I2CADD] to set the slave address;
4. Configure I2C_SR[DMOD] to set the read/write direction;
5. Set I2C_CR[I2CEN] to 1 to enable I2C;
6. Set I2C_SR[I2CSTA] to 1 to send START and address. After receiving ACK/NACK, I2C_SR[STR] will be set to 1 by hardware and SCL will be forced to low by the master;
7. Sending data: Write the sent data to I2C_DR register. After resetting I2C_SR[STR] and SCL is released, the master starts to send data. When the data is sent and ACK/NACK is received, I2C_SR[STR] will be set to 1 by hardware and SCL will be forced to low by the master;
8. Receiving data: After resetting I2C_SR[STR] and SCL is released, the master starts to receive data. When the data is received, I2C_SR[STR] will be set to 1 by hardware and SCL will be forced to low by the master. After setting ACK/NACK via I2C_SR[NACK], clearing I2C_SR[STR] to release SCL to send ACK/NACK signal. When new data is received, I2C_SR[STR] will be set to 1 by hardware and SCL will be forced to low by the master;
9. Stop communication: Set I2C_SR[I2CSTP] to 1 when I2C_SR[STR] is 1 and stop signal will be sent after I2C_SR[STR] is cleared.

8.2.2 Slave Mode

1. Set I2C_CR[I2CMS] to 0 to select slave mode;
2. Configure I2C_ID[I2CADD] to set the slave address or set I2C_ID[GC] to 1 to enable general call mode;
3. Set I2C_CR[I2CEN] to 1 to enable I2C;
4. After receiving the START signal and the correct address, I2C_SR[I2CSTA] and I2C_SR[STR] will be set to 1 by hardware and SCL will be forced to low by the slave. Set ACK/NACK via I2C_SR[NACK] and confirm whether this communication is receiving data or sending data via I2C_SR[DMOD];
5. Sending data: Write the sent data to I2C_DR register. Clear I2C_SR[STR] to release SCL and send data after sending ACK/NACK. When data sending is completed and ACK/NACK is received from the master, I2C_SR[STR] will be set to 1 by hardware and SCL will be forced to low by the slave;
6. Receiving data: Clear I2C_SR[STR] to release SCL to start receiving data. When data receiving is completed, I2C_SR[STR] will be set to 1 by hardware and SCL will be forced to low by the slave. After setting ACK/NACK via I2C_SR[NACK], reset I2C_SR[STR] to release SCL and send ACK/NACK. When a new data is received, I2C_SR[STR] will be set to 1 by hardware and SCL will be forced to low by the slave;
7. RESTART function: When the slave receives the START signal in the busy state, it will stop the current work and wait for receiving address.

8.2.3 I2C Interrupt Sources

The interrupt sources of I2C include:

- I2C_SR[STR] = 1 generates interrupt. This interrupt source is valid in both master and slave modes
- I2C_SR[I2CSTP] = 1 generates interrupt. This interrupt source is only valid in slave mode

8.3 I2C Registers

8.3.1 I2C_CR (0x4028)

Bit	7	6	5	4	3	2	1	0
Name	I2CEN	I2CMS	RSV			I2CSPD		I2CIE
Type	R/W	R/W	—	—	—	R/W	R/W	R/W
Reset	0	0	—	—	—	0	0	0

Bit	Name	Description
[7]	I2CEN	I2C enable Enable the corresponding GPIO to switch to I2C mode with collector open-drain output. Whether the I2C pull-up is enabled or not is determined by the pull-up setting 0: Disable 1: Enable
[6]	I2CMS	Master/slave mode selection 0: Slave mode 1: Master mode
[5:3]	RSV	Reserved
[2:1]	I2CSPD	I2C transfer rate configuration, valid only in master mode

		00: 100kHz 01: 400kHz 10: 1MHz 11: Reserved
[0]	I2CIE	I2C interrupt enable 0: Disable 1: Enable

8.3.2 I2C_ID (0x4029)

Bit	7	6	5	4	3	2	1	0
Name	I2CADD							GC
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	0	1	0	1	0	1	0

Bit	Name	Description
[7:1]	I2CADD	Slave address
[0]	GC	General call mode, valid only in slave mode 0: General call is disabled 1: General call is enabled, i.e., address 0x00 will also be responded

8.3.3 I2C_DR (0x402A)

Bit	7	6	5	4	3	2	1	0
Name	I2C_DR							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	I2C_DR	I2C Data Register Read: Data to be sent or data received Write: Data to be sent

8.3.4 I2C_SR (0x402B)

Bit	7	6	5	4	3	2	1	0
Name	I2CBSY	DMOD	RSV	I2CSTA	I2CSTP	STR	NACK	I2CIF
Type	R	R/W	—	R/W	R/W	R/W0	R/W	R
Reset	0	0	—	0	0	0	0	0

Bit	Name	Description
[7]	I2CBSY	I2C busy flag I2C_SR[I2CBSY] will be cleared by hardware when I2C_CR[I2CEN] is 0 Master mode: This bit will be set to 1 by hardware after sending START and be cleared by hardware after sending STOP Slave mode: This bit will be set to 1 by hardware when START is received and address matches, and will be cleared by hardware when STOP is received
[6]	DMOD	I2C read/write flag 0: Write mode (master sends data, slave receives data) 1: Read mode (master receives data, slave sends data) Note: Slave mode is read only

<table>
<tr><td>[5]</td><td>RSV</td><td>Reserved</td></tr>
<tr><td>[4]</td><td>I2CSTA</td><td>Master mode:
Written to 1 by software and start sending START and address after both SCL and SDA are confirmed high by hardware. This bit will be cleared by hardware automatically when the transmission is finished, and I2C_SR[I2CSTA] writing is forbidden during the process of sending or receiving data. After the data is sent or received, set I2C_SR[I2CSTA] to 1 to send RESTART.
0: Not START and address
1: Send START or RESTART and address

Slave mode:
Be set to 1 after hardware receives START and address matches, cleared by software

Table 8-1 Relationship between I2C_SR[I2CSTA/I2CSTP] and the I2C Data Type
<table>
<tr><th>I2CSTA</th><th>I2CSTP</th><th>I2C Data Type</th></tr>
<tr><td>0</td><td>0</td><td>Data</td></tr>
<tr><td>0</td><td>1</td><td>STOP</td></tr>
<tr><td>1</td><td>0</td><td>START + address</td></tr>
<tr><td>1</td><td>1</td><td>STOP received first, then START + address</td></tr>
</table>
Note: When I2C_SR[I2CEN] is 0, I2C_SR[I2CSTA] will be automatically cleared</td></tr>
<tr><td>[3]</td><td>I2CSTP</td><td>Master mode:
This bit cannot be written to 1 by software unless I2C_SR[I2CBSY] is 1. Clear I2C_SR[STR] to release SCL to start sending STOP. This bit will be cleared automatically after sending STOP. When I2CSTA and I2CSTP are written to 1 at the same time and I2C_SR[I2CBSY] is 1, I2C will send STOP first, then START and address. I2C_SR[STR] will be set to 1 by hardware after START and address byte are sent. I2C_SR[I2CSTP] writing is forbidden during the process of sending or receiving data.
0: STOP will not be sent
1: STOP will be sent

Slave mode:
This bit will be set to 1 by hardware after receiving STOP and cleared by software
Refer to Table 8-1 for status flags

Note: When I2C_CR[I2CEN] is 0, I2CSTP will be cleared by hardware</td></tr>
<tr><td>[2]</td><td>STR</td><td>I2C bus suspend flag
Master mode:
After hardware sends START and address or DATA byte, I2C_SR[STR] will be set to 1 by hardware while SCL is forced to low. SCL will be released after I2C_SR[STR] is cleared by software
When I2C_SR[I2CSTA] and I2C_SR[I2CSTP] are both 1, I2C_SR[STR] will be set to 1 only after hardware sends STOP and START and address

Slave mode:
After hardware receives START and address matched or DATA byte, I2C_SR[STR] will be set to 1 while SCL is forced to low. SCL will be released after I2C_SR[STR] is cleared by software

Note: This bit is set to 1 by hardware and cleared by software. When I2C_SR[I2CEN] is 0, I2C_SR[STR] will be cleared automatically</td></tr>
<tr><td>[1]</td><td>NACK</td><td>Feedback from the receiver to the sender after a byte has been transferred. This bit will be automatically cleared when I2C_SR[I2CEN] is 0</td></tr>
</table>

		0: ACK, indicates that the receiver can continue to receive data 1: NACK, indicates that the receiver wishes to stop data transmission When the device is in read mode, configure I2C_SR[NACK] to send ACK/NACK after receiving the 8th bit of data 0: Bit9 sends ACK 1: Bit9 sends NACK When the device is in write mode, read I2C_SR[NACK] after sending the 8th bit of data to receive ACK/NACK 0: Bit9 receives ACK 1: Bit9 receives NACK
[0]	I2CIF	I2C interrupt flag 0: No interrupt pending 1: Interrupt pending In master and slave mode, interrupt generates when I2C_SR[STR] = 1 In slave mode, interrupt generates when I2C_SR[I2CSTP] = 1

9 SPI

9.1 SPI Introduction

SPI is a high-speed, full-duplex synchronous serial bus and the schematic block diagram is shown in Figure 9-1. SPI supports both master/slave modes and 3/4-wire transmission, allowing multiple masters and slaves on the bus.

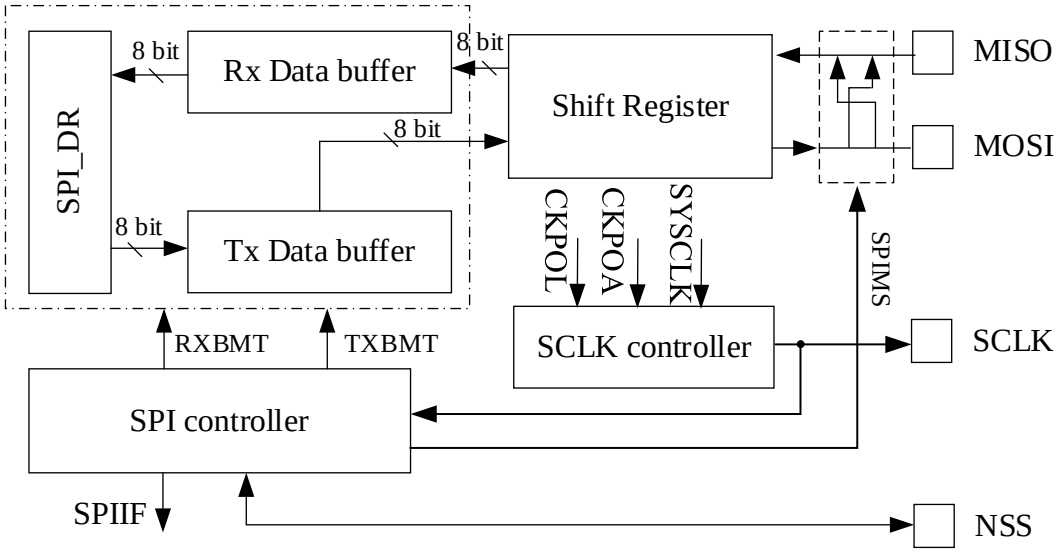


Figure 9-1 Schematic Block Diagram of SPI

9.2 SPI Instructions

9.2.1 Signal Instructions

The four signals for SPI are MOSI, MISO, SCLK, and NSS.

9.2.1.1 MOSI

The MOSI signal is the output of the master and the input of the slave for the data transfer from the master to the slave. When the data is transmitted, the highest bit is first, i.e., the MSB of the shift register is serially sent to the MOSI pin in sequence.

9.2.1.2 MISO

The MISO signal is the output of the slave and the input of the master. When SPI is disabled or operating in 4-wire slave mode, which is not selected, the MISO pin is placed in the high-impedance state. It is used for data transfer from slave to master when SPI is operating as slave in 3-wire mode or 4-wire slave mode, which is selected. When the data is transmitting, the highest bit is first, i.e., the MSB of the shift register is serially sent to the MISO pin in sequence.

9.2.1.3 SCLK

The SCLK signal is the output of the master and the input of the slave and is used to synchronize serial data transfers between the master and slave. SCLK is generated by the master. In 4-wire slave mode, when the slave device is not selected (NSS = 1), SCLK is ignored.

9.2.1.4 NSS

The function of NSS is selected by the configuration of SPI_CR1[NSSMOD], i.e., the SPI operating mode. SPI operating modes include 3-wire mode, 4-wire slave/multi-master mode, and 4-wire single master mode. When the device is operating in 4-wire slave/multi-master mode, NSS is configured as an input to select the slave, or to disable a particular master on the bus in multi-master mode to avoid SPI bus conflicts when more than two masters try to transfer data at the same time. When the device is operating in 4-wire single master mode, NSS is configured as a chip select output. When the device is operating in 3-wire mode, NSS is disabled. In master mode, other slaves can be selected by GPIO pins.

Set SPI_CR1[NSSMOD] = 00 to configure SPI working in 3-wire mode. NSS port is not necessary in this mode and there is only one master and one slave on the SPI bus. The connection is as shown in Figure 9-2.

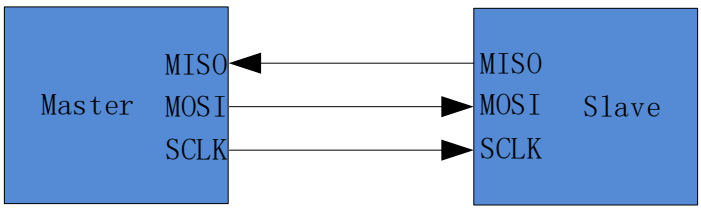


Figure 9-2 Connection Diagram of 3-Wire SPI Mode

Set SPI_CR1[NSSMOD] = 01 to configure SPI working in 4-wire slave mode or multi-master mode. In this mode, the NSS of the devices on the SPI bus are all configured as inputs, waiting to be addressed by the master. Set SPI_CR0[SPIMS] = 0 to configure SPI for 4-wire slave mode. When NSS is low, the slave is selected. when NSS is high, the slave is not selected. Set SPI_CR0[SPIMS] = 1 to configure SPI for master mode, which defaults to multi-master mode. In multi-master mode, the NSS input is used to disable the SPI master mode. When the NSS signal of the master on the bus is low level, SPI_CR0[SPIMS] is cleared by hardware and switched from master mode to slave mode, while SPI_CR1[MODF] is set to 1. In this mode, the SPI bus allows multiple masters to participate in communication. The connection is shown in Figure 9-3.

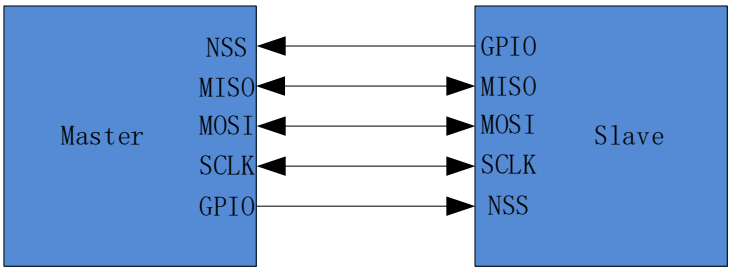


Figure 9-3 Connection Diagram of 4-Wire Multi-Master Mode

Set SPI_CR1[NSSMOD] = 1X to configure SPI working in 4-wire single master mode. In this mode, the master NSS is configured as output and the slave NSS is input. The master configures SPI_CR1[NSSMOD0] to determine the output level of NSS as a chip selected signal of a slave. The chip select of other slaves can be selected through GPIO. The connection is shown in Figure 9-4.

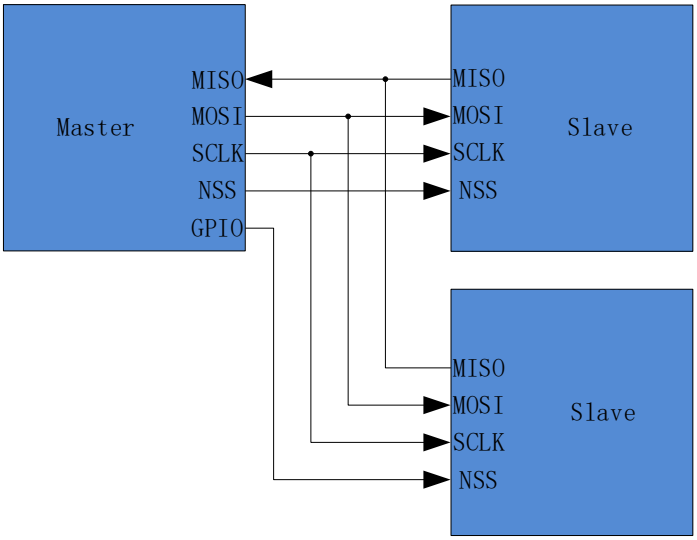


Figure 9-4 Connection Diagram of 4-Wire Single Master Mode

9.2.2 Master Mode of SPI

Set SPI_CR0[SPIMS] = 1 and SPI works in master mode, which provides SCLK signal for the bus. When writing data to SPI_DR, the data is first written to the transmit buffer and SPI_CR1[TXBMT] will be cleared. If the shift register is empty, then the data in the transmit buffer will be transferred to the shift register and the transfer starts. The master SPI serially sends the MSB of shift register to MOSI. After the transfer, SPI_CR1[SPIIF] and SPI_CR1[TXBMT] will be set to 1. In full-duplex operation, when the SPI master sends data to the slave on the MOSI line, the addressed SPI slave can simultaneously send the data of shift register to the master on the MISO line. Therefore, SPI_CR1[SPIIF] serves as both the transmit complete flag and the receive complete flag, and the data in the shift register is the data received by MISO, which is transferred to the receive buffer. If data is written to SPI_DR when SPI_CR1[TXBMT] is 0, the write conflict flag bit SPI_CR1[WCOL] will be set to 1 and the data in the transmit buffer will be unchanged.

9.2.2.1 Master Mode Configuration

1. Configure SPI_CR1[NSSMOD] to set the SPI working mode;
2. Configure SPI_CR0[CPOL] to set the clock polarity;
3. Configure SPI_CR0[CPHA] to set the clock phase;
4. Set SPI_CR0[SPIMS] to 1 to select master mode;
5. Configuring SPI_CLK to set the SCLK frequency;
6. Set SPI_CR1[SPIEN] to 1 to enable SPI;
7. Writing the data to SPI_DR. SPI will transmit data for each write;
8. After SPI_CR1[SPIIF] is set to 1, read SPI_DR to get the received data.

9.2.3 Slave Mode of SPI

Set SPI_CR0[SPIMS] = 0 and SPI works in slave mode. In this mode, SCLK is provided by the master SPI. The data is shifted in from MOSI and shifted out from MISO. When the signal of SCLK is not input, the shift register of slave is in the stop state. When the signal of SCLK is input, the shift register of slave starts to act and begins to receive and send data through MOSI and MISO. The slave device cannot start data transfer. The data sent to the master is preloaded to the transmit buffer by writing SPI_DR. If the shift register is empty, the data in the transmit buffer will be transferred to the shift register. When the transmission is completed, SPI_CR1[SPIIF] and SPI_CR1[TXBMT] will be set to 1. The received data is transferred to receive buffer and the receive buffer empty flag bit SPI_CR0[RXBMT] will be cleared, indicating the unread data. If SPI_CR0[RXBMT] is 0 and there is new data ready to be sent to the receive buffer. SPI_CR1[RXOVR] will be set to 1 and the data in the receive buffer will be unchanged. When data is written to SPI_DR, SPI_CR1[TXBMT] will be cleared. If a data is written in this case, the write conflict flag bit SPI_CR1[WCOL] will be set to 1 and the data in the transmit buffer will be unchanged.

9.2.3.1 Slave Mode Configuration

1. Configure SPI_CR1[NSSMOD] to set the SPI working mode;
2. Configure SPI_CR0[CPOL] to set the clock polarity;
3. Configure SPI_CR0[CPHA] to set the clock phase;
4. Set SPI_CR0[SPIMS] to 0 to select slave mode;
5. Set SPI_CR1[SPIEN] to 1 to enable SPI;
6. Write data to SPI_DR and wait for the master to send the clock signal.

9.2.4 SPI Interrupt Source

The interrupt sources of SPI include:

- SPI interrupt flag SPI_CR1[SPIIF] will be set to 1 after each byte transfer finished
- When the SPI_DR is written while the data in the transmit buffer has not been transferred to the shift register, the write conflict flag SPI_CR1[WCOL] will be set to 1 and the write operation will be ignored.
- When the SPI is configured to work as a master in a multi-master system and NSS is forced to low, the mode error flag SPI_CR1[MODF] will be set to 1. When a mode error occurs, the SPI_CR0[SPIMS] and SPI_CR1[SPIEN] will be cleared. SPI is forbidden to allow another master to control the bus.
- The receive overflow flag SPI_CR1[RXOVR] will be set to 1 when SPI works in slave mode and a transmission is completed with the last transmitted data in the receive buffer unread. And the last received data will not be transferred to the receive buffer.

9.2.5 Serial Clock Timing

Configuring the phase and idle level selection bits in SPI_CR0 can perform four combinations of serial clock phase and idle level. SPI_CR0[CPHA] selects the clock phase (the edge of the SCLK signal when the shift register data jumps). SPI_CR0[CPOL]selects the idle level. The configuration of the master and the slave must be the same. When configure the clock phase and idle level, SPI should be disabled (SPI_CR1[SPIEN] = 0). The timing relationship of the clock and data lines at different combinations of clock phase and idle levels are shown in Figure 9-5 and Figure 9-6.

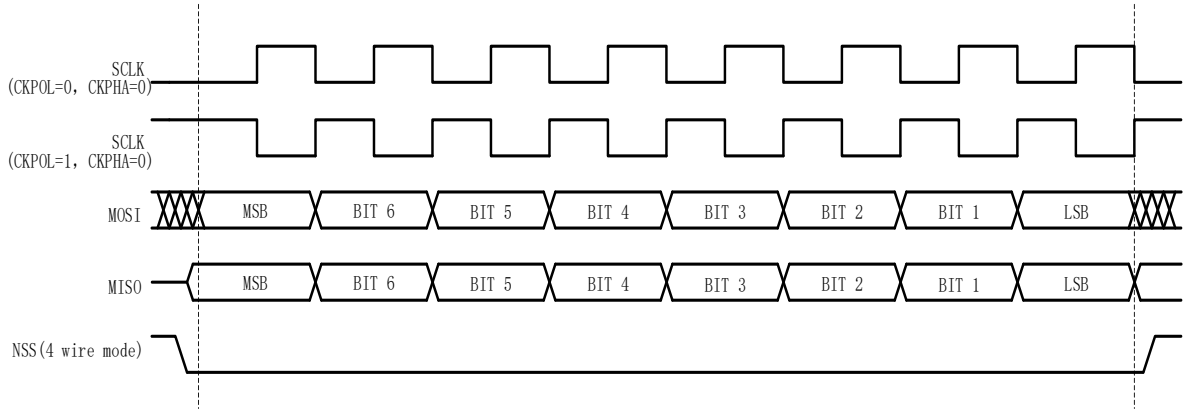


Figure 9-5 Data/Clock Line Timing Diagram (SPI_CR0[CPHA] = 0)

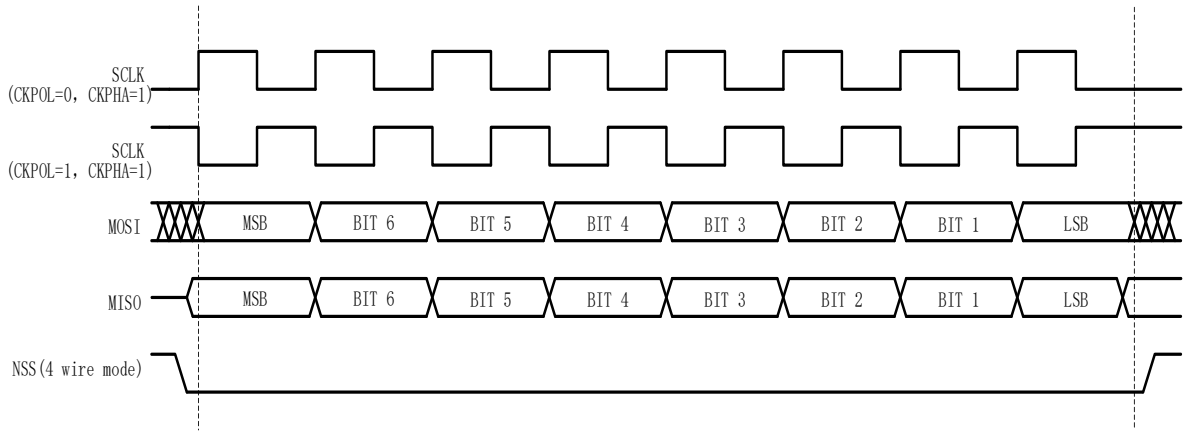


Figure 9-6 Data/Clock Line Timing Diagram (SPI_CR0[CPHA] = 1)

9.3 SPI Registers

9.3.1 SPI_CR0 (0x4030)

Bit	7	6	5	4	3	2	1	0
Name	SPIBSY	SPIMS	CPHA	CPOL	SLVSEL	NSSIN	SRMT	RXBMT
Type	R	R/W	R/W	R/W	R	R	R	R
Reset	0	0	0	0	0	1	1	1

Bit	Name	Description
[7]	SPIBSY	Busy flag 0: No SPI transfer 1: SPI is transferring
[6]	SPIMS	Master/slave mode selection 0: Slave 1: Master
[5]	CPHA	Clock phase 0: Receive data at the first active edge of SCLK, send data at the second edge 1: Send data at the first active edge of SCLK, receive data at the second edge
[4]	CPOL	Clock idle level 0: Low level for idle level 1: High level for idle level
[3]	SLVSEL	NSS chip select flag This bit will be set to 1 when the filtered signal of NSS is low, indicating that the device is selected as slave. When NSS is high, this bit will be cleared, indicating that the device is not selected as slave. 0: Not selected as slave 1: Selected as slave
[2]	NSSIN	NSS real-time signal, unfiltered
[1]	SRMT	Shift register empty flag (valid only in slave mode) 0: Data transfer from the transmit buffer to the shift register or SCLK has changed 1: No data in the shift register and no data in the transmit and receive buffers Note: SPI_CR0[SRMT] is 1 in master mode
[0]	RXBMT	Receive buffer empty flag (valid only in slave mode) 0: There is a new data that has not been read 1: Data is read and there is no new data

		Note: SPI_CR0[RXBMT] is 1 in the master mode

Note: Clock phase idle level mode SPI_CR0[CPHA:CPOL]:

- 00: Receive on rising edge, send on falling edge, idle level is low
- 01: Send on rising edge, receive on falling edge, idle level is high
- 10: Send on rising edge, receive on falling edge, idle level is low
- 11: Receive on rising edge, send on falling edge, idle level is high

9.3.2 SPI_CR1 (0x4031)

Bit	7	6	5	4	3	2	1	0
Name	SPIIF	WCOL	MODF	RXOVRN	NSSMOD1	NSSMOD0	TXBMT	SPIEN
Type	R/W0	R/W0	R/W0	R/W0	R/W	R/W	R	R/W
Reset	0	0	0	0	0	0	1	0

Bit	Name	Description
[7]	SPIIF	SPI Interrupt Flag This bit will be set to 1 by hardware after each data (8-bit) transfer Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[6]	WCOL	Write conflict interrupt flag When TXBMT is 0, write to SPI_DR will set this bit to 1 This bit can only be cleared to 0 by software Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[5]	MODF	Master mode error interrupt flag This bit will be set to 1 when a master mode conflict is detected (SPI_CR0[NSSIN] = 0, SPI_CR0[SPIMS] = 1 and SPI_CR1[NSSMOD] = 01) This bit can only be cleared by software Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[4]	RXOVRN	Receive overflow interrupt flag (valid only in slave mode) This bit will be set to 1 by hardware (and generates a SPI interrupt) when the last bit of the current transfer has been shifted into the shift register and the receive buffer still holds unread data from the previous transfer. This bit will not be automatically cleared by hardware and can only be cleared by software. Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[3:2]	NSSMOD	SPI mode selection 00: 3-wire slave mode or 3-wire master mode. NSS signal not connected to the port 01: 4-wire slave or multi-master mode (default value). NSS is input port 1X: 4-wire single master mode, NSS is output port and outputs the value of SPI_CR1[2]
[1]	TXBMT	Transmit buffer empty flag This bit will be cleared when new data is written to the transmit buffer, and will be set to 1 when the data in the transmit buffer is transferred to the SPI shift register, indicating that new data can be written to the transmit buffer. 0: There is a new data written to the transmit buffer 1: Data in the transmit buffer has been transferred to the shift register

[0]	SPIEN	SPI Enable 0: Disable 1: Enable

9.3.3 SPI_CLK (0x4032)

Bit	7	6	5	4	3	2	1	0
Name	SPI_CLK							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	SPI_CLK	SPI Baud Rate Setting Valid in master mode, writable only when SPI_CR1[SPIEN] = 0 Baud rate = SYSCLK/2/(SPI_CLK + 1) E.g., Baud rate 2400kHz, then SPI_CLK = (24M/2/2400k) - 1 = 4, i.e., 0x04

9.3.4 SPI_DR (0x4033)

Bit	7	6	5	4	3	2	1	0
Name	SPI_DR							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	SPI_DR	SPI Data Registers SPI_DR register is used to send and receive SPI data Read: Data in the receive buffer Write: Write data to the transmit buffer and start transmitting

10 UART

10.1 UART Introduction

UART is a full-duplex or half-duplex serial data exchange interface. The transmission baud rate is configurable and supports DMA.

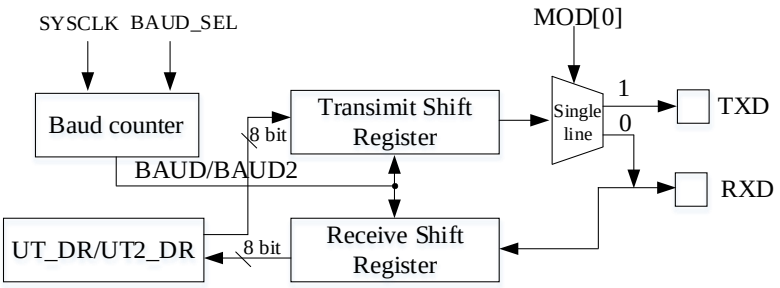


Figure 10-1 Block Diagram of UART

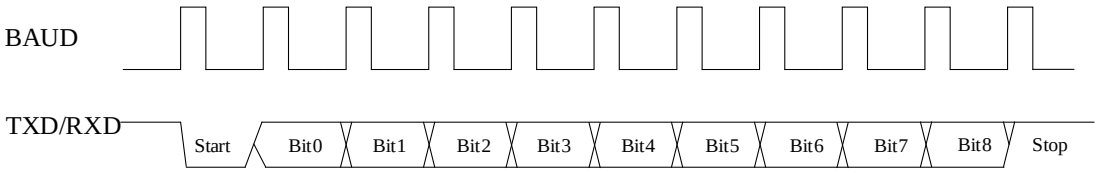


Figure 10-2 Communication Timing Diagram of UART

10.2 UART Instructions

10.2.1 UART1

10.2.1.1 UART1 Mode0

Mode 0 works in single-wire half-duplex mode. RXD is both the transmit data bus and the receive data bus. The data, sending or receiving, is 10 bits (1-bit start, 8-bit data, 1-bit stop) and the baud rate is configured by UT_BAUD[BAUD].

Sending data: Write the sent data to UT_DR and clear UT_CR[TI]. RXD will output 10-bit data. UT_CR[TI] will be set to 1 when the transmission is completed.

Receiving data: Set UT_CR[REN] to 1 to start receiving and clear UT_CR[RI]. The data is received via RXD. After the reception is completed, UT_CR[RI] will be set to 1 and read UT_DR to get the received data.

10.2.1.2 UART1 Mode1

Mode 1 works in full/half duplex mode. TXD is the transmit data bus, RXD is the receive data bus. The data, sending or receiving, is 10 bits (1-bit start, 8-bit data, 1-bit stop) and the baud rate is configured by UT_BAUD.

Sending data: Write the sent data to UT_DR and clear UT_CR[TI]. TXD will output 10-bit data. UT_CR[TI] will be set to 1 after the transmission is completed.

Receiving data: Set UT_CR[REN] to 1 to start receiving data and clear UT_CR[RI]. The data is received via RXD. After the reception is completed, UT_CR[RI] will be set to 1 and read UT_DR to get the received data.

10.2.1.3 UART1 Mode2

Mode 2 works in single-wire half-duplex mode. RXD is both transmit data bus and receive data bus. The data, sending or receiving, is 11 bits (1-bit start, 9-bit data, 1-bit stop) and baud rate is configured by UT_BAUD[BAUD].

Sending data: Write the low 8 bits of sent data to UT_DR, the 9th bit to UT_CR[TB8] and clear UT_CR[TI]. TXD will output 11-bit data. UT_CR[TI] will be set to 1 after the transmission is completed.

Receiving data: Set UT_CR[REN] to 1 to start receiving data and clear UT_CR[RI]. The data is received via RXD. When reception is completed, UT_CR[RI] will be set to 1. UT_CR[RB8] stores the 9th bit of data, and UT_DR stores the low 8 bits of data.

10.2.1.4 UART1 Mode3

Mode 3 works in full/half duplex mode. TXD is the transmit data bus, RXD is the receive data bus. The data, sending or receiving, is 11 bits (1-bit start, 9-bit data, 1-bit stop) and the baud rate is configured by UT_BAUD.

Sending data: Write the low 8 bits of the sent data to UT_DR, the 9th bit to UT_CR[TB8] and clear UT_CR[TI]. TXD will output 11-bit data, and UT_CR[TI] will be set to 1 after the transmission is completed.

Receiving data: Set UT_CR[REN] to 1 to start receiving and clear UT_CR[RI]. The data is received via RXD. When reception is completed, UT_CR[RI] will be set to 1. UT_CR[RB8] stores the 9th bit of data, and UT_DR stores the low 8 bits of data.

10.2.1.5 UART1 Interrupt

UART1 interrupt include:

- After UART1 sends data, UT_CR[TI] will be set to 1 by hardware
- After UART1 receives data and STOP, UT_CR[RI] will be set to 1 by hardware

10.2.2 UART2

10.2.2.1 UART2 Mode0

Mode 0 works in single-wire half-duplex mode. RXD is both the transmit data bus and the receive data bus. The data, sending or receiving, is 10 bits (1-bit start, 8-bit data, 1-bit stop) and the baud rate is configured by UT2_BAUD[BAUD2].

Sending data: Write the sent data to UT2_DR and clear UT2_CR[UT2TI]. RXD will output 10-bit data. UT2_CR[UT2TI] will be set to 1 when the transmission is completed.

Receiving data: Set UT2_CR[UT2REN] to 1 to start receiving data and clear UT2_CR[UT2RI]. The data is

received via RXD. After the reception is completed, UT2_CR[UT2RI] will be set to 1 and read UT_DR to get the received data.

10.2.2.2 UART2 Mode1

Mode 1 works in full/half duplex mode. TXD is the transmit data bus, RXD is the receive data bus. The data, sending or receiving, is 10 bits (1-bit start, 8-bit data, 1-bit stop) and the baud rate is configured by UT2_BAUD[BAUD2].

Sending data: Write the sent data to UT2_DR and clear UT2_CR[UT2TI]. TXD will output 10-bit data. UT2_CR[UT2TI] will be set to 1 after the transmission is completed.

Receiving data: Set UT2_CR[UT2REN] to 1 to start receiving data and clear UT2_CR[UT2RI]. The data is received via RXD. After the reception is completed, UT2_CR[UT2RI] will be set to 1 and read UT2_DR to get the received data.

10.2.2.3 UART2 Mode2

Mode 2 works in single-wire half-duplex mode. RXD is both transmit data bus and receive data bus. The data, sending or receiving, is 11 bits (1-bit start, 9-bit data, 1-bit stop), baud rate is configured by UT2_BAUD[BAUD2].

Sending data: Write the low 8 bits of sent data to UT2_DR, the 9th bit to UT2_CR[UT2TB8] and clear UT2_CR[UT2TI]. TXD will output 11-bit data. UT2_CR[UT2TI] will be set to 1 after the transmission is completed.

Receiving data: Set UT2_CR[UT2REN] to 1 to start receiving data and clear UT2_CR[UT2RI]. The data is received via RXD. When reception is completed, UT2_CR[UT2RI] will be set to 1. UT2_CR[UT2RB8] stores the 9th bit of data, and UT_DR stores the low 8 bits of data.

10.2.2.4 UART2 Mode3

Mode 3 works in full/half duplex mode. TXD is the transmit data bus, RXD is the receive data bus. The data, sending or receiving, is 11 bits (1-bit start, 9-bit data, 1-bit stop) and the baud rate is configured by UT2_BAUD[BAUD2].

Sending data: Write the low 8 bits of the sending data to UT2_DR, write the 9th bit to UT2_CR[UT2TB8] and clear UT2_CR[UT2TI]. TXD will output 11-bit data, and UT2_CR[UT2TI] will be set to 1 after the transmission is completed.

Receiving data: Set UT2_CR[UT2REN] to 1 to start receiving and clear UT2_CR[UT2RI]. The data is received via RXD. When reception is completed, UT2_CR[UT2RI] will be set to 1. UT2_CR[UT2RB8] stores the 9th bit of data, and UT2_DR stores the low 8 bits of data.

10.2.2.5 UART2 Interrupt

UART interrupt include:

- After the UART2 sends data, UT2_CR[UT2TI] will be set to 1 by hardware
- After the UART2 receives data and STOP, UT2_CR[UT2RI] will be set to 1 by hardware

10.3 UART1 Registers

10.3.1 UT_CR (0x98)

Bit	7	6	5	4	3	2	1	0
Name	MOD		SM2	REN	TB8	RB8	TI	RI
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:6]	MOD	Mode selection 00: Mode 0 01: Mode 1 10: Mode 2 11: Mode 3
[5]	SM2	Communication mode 0: Single-device communication 1: Multi-device communication
[4]	REN	Serial input enable 0: Disable 1: Enable
[3]	TB8	Bit9 of the sent data in mode 2 and mode 3
[2]	RB8	Bit9 of the received data in mode 2 and mode 3
[1]	TI	Data sending completed interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: Generate sending completed interrupt
[0]	RI	Data reception completed interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: Generate reception completed interrupt

10.3.2 UT_DR (0x99)

Bit	7	6	5	4	3	2	1	0
Name	UT_DR							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	UT_DR	Send/receive data

		Read: Data received Write: Data to be sent Note: The data buffer of UART1 contains two independent receive and send buffers, which can send and receive data at the same time. The send buffer can only be written but not read and the receive buffer can only be read but not written. Both buffers share a same address.

10.3.3 UT_BAUD (0x9A, 0x9B)

UT_BAUDH(0x9B)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	BAUD_SEL	RSV			BAUD[11:8]			
Type	R/W	—	—	—	R/W	R/W	R/W	R/W
Reset	0	—	—	—	0	0	0	0
UT_BAUDL(0x9A)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	BAUD[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	0	0	1	1	0	1	1

Bit	**Name**	**Description**
[15]	BAUD_SEL	Frequency multiplier enable 0: Disable 1: Enable
[14:12]	RSV	Reserved
[11:0]	BAUD	Baud rate setting Baud rate = SYSCLK/(16/(1 + UT_BAUD[BAUD_SEL]))/(UT_BAUD[BAUD] + 1) E.g., Baud rate = 9600, UT_BAUD[BAUD_SEL] = 0, UT_BAUD[BAUD] = (24M/16/9600/(1 + 0)) - 1 = 155 (0x9B)

10.4 UART2 Registers

10.4.1 UT2_CR (0x8A)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	UT2MOD		UT2SM2	UT2REN	UT2TB8	UT2RB8	UT2TI	UT2RI
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W0	R/W0
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[7:6]	UT2MOD	Mode selection 00: Mode 0 01: Mode 1 10: Mode 2 11: Mode 3
[5]	UT2SM2	Communication mode 0: Single-device communication 1: Multi-device communication
[4]	UT2REN	Serial input enable 0: Disable 1: Enable
[3]	UT2TB8	Bit 9 of the sent data in mode 2 and mode 3
[2]	UT2RB8	Bit 9 of the received data in mode 2 and mode 3

[1]	UT2TI	Data sending completed interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[0]	UT2RI	Data reception completed interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning

10.4.2 UT2_DR (0x89)

Bit	7	6	5	4	3	2	1	0
Name	UT2_DR							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	UT2_DR	Send/receive Data Read: Data received Write: Data to be sent Note: The data buffer of UART2 contains two independent receive and send buffers, which can send and receive data at the same time. The send buffer can only be written but not read and the receive buffer can only be read but not written. Both buffers share a same address.

10.4.3 UT2_BAUD (0x4042, 0x4043)

UT2_BAUDH(0x4042)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	BAUD2_SEL	UART2CH	UART2IEN	RSV	BAUD2[11:8]			
Type	R/W	R/W	R/W	—	R/W	R/W	R/W	R/W
Reset	0	0	0	—	0	0	0	0
UT2_BAUDL(0x4043)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	BAUD2[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	0	0	1	1	0	1	1

Bit	**Name**	**Description**
[15]	BAUD2_SEL	Frequency multiplier enable 0: Disable 1: Enable
[14]	UART2CH	UART2 function transfer enable 0: UART2 port function not transferred, P3.6 for RXD; P3.7 for TXD 1: UART2 port function transferred, P0.1 for RXD; P0.0 for TXD
[13]	UART2IEN	UART2 interrupt enable 0: Disable 1: Enable
[12]	RSV	Reserved
[11:0]	BAUD2	Baud rate setting Baud rate = SYSCLK/(16/(1 + UT2_BAUD[BAUD_SEL]))/(UT2_BAUD[BAUD2] + 1) E.g., Baud rate = 9600, UT2_BAUD[BAUD_SEL] = 0; UT2_BAUD[BAUD2] = (24M/16/9600/(1 + 0)) - 1 = 155 (0x9B)

11 LIN

11.1 LIN Introduction

LIN is an asynchronous serial communication interface, mainly used in automotive network. LIN controller complies with the 2.2 standard (backward compatible). As shown in Figure 11-1, the controller has a complete LIN hardware interface, which works in slave mode, and supports baud rate adaption. The pin is multiplexed with UART2.

For more information and standard on the LIN protocol, please refer to the LIN Association (http://www.lin-subbus.org).

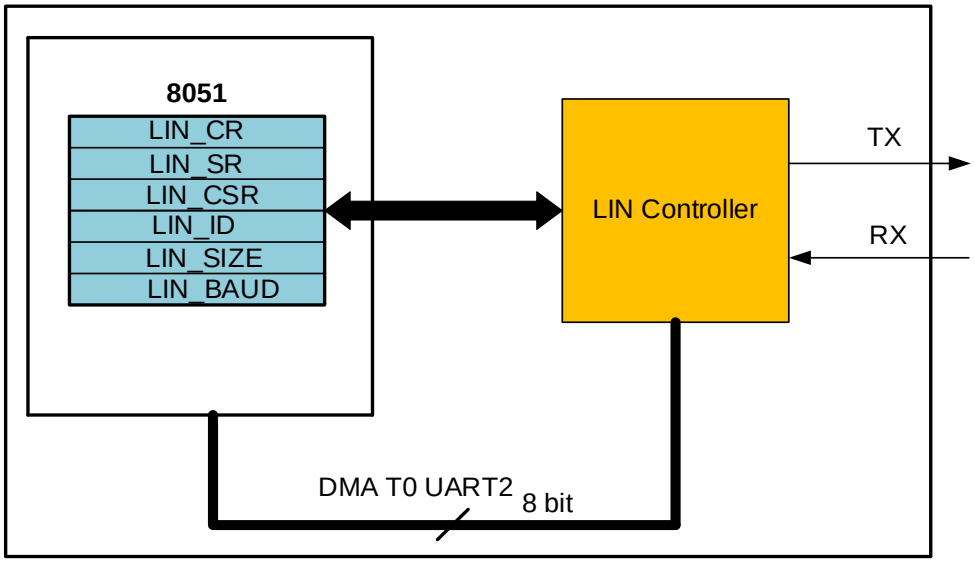


Figure 11-1 Block Diagram of LIN

As shown in Figure 11-1, the LIN controller contains three parts:

- LIN control register: Provides the interface between CPU and LIN controller via internal XSFR of 8051
- LIN data caching: Data can be sent and received by configuring DMA 0/1
- LIN controller: Used to handle data transfer and status control of the LIN bus

11.2 Instructions of the Slave LIN

When LIN works in slave node, it must wait for the master to send commands.

When the LIN interrupt is enabled, the interrupt will be triggered in five cases as shown in Table 11-1.

Table 11-1 Interrupt Source and Description of LIN

Interrupt source	Description	Related flags	Clear flags
Bus idle	Bus idle for more than 4s	LIN_SR[LINIDLE]	Write 0 to LIN_SR[LINIDLE]

External wake-up	Wake-up signal received	LIN_CSR[LINWAKUP]	Write 0 to LIN_CSR[LINWAKUP]
Frame header received	When the frame header is received and the ID check is correct	LIN_SR[LINREQ]	Write 1 to LIN_SR[LINACK]/ Write 0 to LIN_SR[LINREQ]/ New frame header found
Sending/receiving data completed	The slave receiving or sending data is completed	LIN_SR[LINDONE]	Write 0 to LIN_SR[LINDONE] / New frame header found
Error occurred	Error request received: Synchronization error/ ID check error/ Data check error	LIN_SR[ERRSYNC] LIN_SR[ERRPRTY] LIN_SR[ERRCHK]	Write 0 to LIN_CSR[CLRERR]/ New frame header found

Data transmission and reception process of the slave:

1. LIN controller detects the headers (interval and synchronization signals) of the message frames sent by the master on the LIN bus. The baud rate of the data is automatically identified by the synchronization signal. When the slave recognizes the ID, LIN_SR[LINREQ] will be set to 1 if the ID checksum is correct. When the ID checksum is incorrect, an ID checksum error occurs and LIN_SR[ERRPRTY] will be set to 1;
2. Slave sending operation: Set LIN_CR[LINRW] to 1 to load data length LIN_SIZE and DMA buffer data. Set LIN_CSR[LINACK] to 1 and frame header transfer data to the master;
3. Slave receiving operation: Clear LIN_CR[LINRW], set LIN_CSR[LINACK] to 1, and frame header will receive the data sent by the master;
4. LIN_SR[LINDONE] will be set to 1 when the receiving or sending process of the slave has finished.

11.3 Sleep and Wake up

To reduce system power consumption, the LIN bus protocol defines sleep mode.

After the slave receives the sleep request command from the master and analyzes it correctly, LIN_CSR[LINSLP] will be set to 1 by software to enter sleep mode.

LIN_SR[IDLE] will be set to 1 when the bus is idle for more than 4s with the slave LIN not in sleep mode. In this case, assume the LIN bus in sleep mode and set LIN_CSR[LINSLP] to 1 to enter sleep mode.

When a wake-up signal is detected from the master or other slave on the LIN bus (LIN_CSR[LINWAKUP] is set to 1), the sleep mode is automatically exited. The slave can also send a wake-up signal (Set LIN_CSR[LINWAKUP] to 1) to wake up the master or other slaves.

11.4 Error Detection

When the slave detects an error, LIN_CSR[CLRERR] will be set to 1. LIN generate an error interrupt request and stop the process of the current frame. The error, synchronization error, data check error or ID check error, can be read by LIN_SR[ERRSYNC], LIN_SR[ERRCHK] and LIN_SR[ERRPRTY]. Clear the flag (LIN_CSR[CLRERR]) after the error handling is completed.

11.5 Other Matters

When the slave LIN is enabled, new frame headers (including synchronic interval, synchronic field and PID) can be detected when the device is not in sleep mode.

Set LIN_CSR[LINSTOP] to 1 to terminate the current frame during receiving/sending process of slave. LIN_SR[ABORT] will be set to 1.

11.6 LIN Registers

11.6.1 LIN_CR (0x40E0)

Bit	7	6	5	4	3	2	1	0
Name	RSV				LINIE	CHKMOD	LINRW	AUTOSIZE
Type	—	—	—	—	R/W	R/W	R/W	R/W
Reset	—	—	—	—	0	0	0	0

Bit	Name	Description
[7:4]	RSV	Reserved
[3]	LINIE	LIN interrupt enable 0: Disable 1: Enable
[2]	CHKMOD	Check mode selection 0: Enhanced check 1: Standard check
[1]	LINRW	Sending/receiving selection 0: Current frame is a receiving operation 1: Current frame is a sending operation
[0]	AUTOSIZE	Use the ID bits (LIN_ID[5:4]) to configure the data length enable 0: Disable 1: Enable The relationship between LIN_ID[5:4] and data length is: 0X: 2 bytes 10: 4 bytes 11: 8 bytes

11.6.2 LIN_SR (0x40E1)

Bit	7	6	5	4	3	2	1	0
Name	ERRSYNC	ERRCHK	ERRPRTY	ABORT	LINACT	LINIDLE	LINDONE	LINREQ
Type	R	R	R	R	R	R/W0	R/W0	R/W0
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	ERRSYNC	Synchronization error (synchronization timeout or synchronization too fast), this bit will be cleared by hardware when a new frame arrives or LIN[CLRERR] is cleared 0: No synchronization error occurred 1: Synchronization error occurred
[6]	ERRCHK	Data check error, this bit will be cleared by hardware when a new frame arrives or LIN_CSR[CLRERR] is cleared 0: No data check error occurred 1: Data check error occurred
[5]	ERRPRTY	ID check error, this bit will be cleared by hardware when a new frame arrives or LIN_CSR[CLRERR] is cleared 0: No ID check error occurred 1: ID check error occurred
[4]	ABORT	Transmission abort flag This bit will be set to 1 when the frame header is received or LIN_CSR[LINSTOP] is set to 1 during transmission 0: No transmission abort occurs 1: Transmission abort occurs
[3]	LINACT	LIN bus in transmission flag

		0: No data transmitting on the bus 1: There is data transmitting on the bus
[2]	LINIDLE	LIN bus idle interrupt flag This bit will be set to 1 when the bus is idle for more than 4s Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[1]	LINDONE	Transmission completed interrupt flag This bit will be set to 1 when the data receiving or sending of the slave is completed and cleared when new frame arrives or software write 0 to LIN_SR[LINDONE] Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[0]	LINREQ	Frame received interrupt flag This bit will be set to 1 when frame header is received and ID check is correct. This bit is cleared when new frame arrives, LIN_CSR[LINACK] written to 1 by software or software write 0 to LIN_SR[LINREQ]. Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning

11.6.3 LIN_CSR (0x40E2)

Bit	7	6	5	4	3	2	1	0
Name	RSV		LINSLP	CLRERR	LINWAKUP	LINACK	LINSTOP	LINEN
Type	—	—	R/W	R/W0	R/W	W1	W1	R/W
Reset	—	—	0	0	0	0	0	0

Bit	Name	Description
[7:6]	RSV	Reserved
[5]	LINSLP	LIN sleep state enable Read: 0: LIN is active 1: LIN is in sleep state Write: 0: LIN waked up from sleep state 1: LIN enter sleep state
[4]	CLRERR	LIN fault status clearing Read: 0: No error occurred 1: Error occurred Write: 0: Error flag cleared 1: No meaning
[3]	LINWAKUP	LIN Wake-Up Read: 0: No wake-up signal received 1: Wake-up signal received

		Write: 0: No meaning 1: Send wake-up signal
[2]	LINACK	Used to response the frame header. The check mode, read/write mode, data, data length must be configured before setting this bit to 1 0: No meaning 1: Response frame header
[1]	LINSTOP	Stop signal. LIN will stop the current data sending or receiving and wait for a new frame header after this bit is set to 1. LIN_SR[ABORT] will be set to 1 0: No meaning 1: Stop current data sending or receiving and wait for a new frame header
[0]	LINEN	LIN Enable 0: Disable 1: Enable

11.6.4 LIN_ID (0x40E3)

Bit	7	6	5	4	3	2	1	0
Name	RSV		LIN_ID					
Type	—	—	R	R	R	R	R	R
Reset	—	—	0	0	0	0	0	0

Bit	Name	Description
[7:6]	RSV	Reserved
[5:0]	LIN_ID	ID received by LIN

11.6.5 LIN_SIZE (0x40E4)

Bit	7	6	5	4	3	2	1	0
Name	RSV				LIN_SIZE			
Type	—	—	—	—	R/W	R/W	R/W	R/W
Reset	—	—	—	—	0	0	0	0

Bit	Name	Description
[7:4]	RSV	Reserved
[3:0]	LIN_SIZE	Received/sent data length

11.6.6 LIN_BAUD (0x40E5, 0x40E6)

LIN_BAUDH(0x40E5)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	LIN_BAUD[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	1	0	0
LIN_BAUDL(0x40E6)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name			LIN_BAUD[7:0]					
Type	R	R	R	R	R	R	R	R
Reset	1	0	1	0	1	1	1	1

Bit	Name	Description
[15:0]	LIN_BAUD	Baud rate configuration Baud rate = SYSCLK/(LIN_BAUD + 1)

12 MDU

12.1 MDU Introduction

MDU is a computational co-processing unit, which can assist the CPU to process complex operations efficiently. MDU provides multiplication, division, trigonometric functions and low-pass filter. MDU module can be called multiple times in different interrupt programs and main programs.

12.2 MDU Features

- Can be used in nested interrupts
- Hardware speedup, saving CPU load
- Supports the following modes:
  - 16-bit signed multiplication
  - 16-bit signed multiplication (result shifted 1 bit left)
  - 16-bit unsigned multiplication
  - 32-bit/16-bit unsigned division
  - Low-pass filter
  - Coordinate transformation
  - Arctangent

12.3 MDU Instructions

12.3.1 Operation Process

MDU operation process:

1. Set MDU_CR[MDURUN] to 1;
2. Configure MUD_MD to select the operation mode;
3. Write data to MDU_A, MDU_B, MDU_C and MDU_D. The operation will be started when detecting MDU_C[7:0] is written to;
4. Wait for MDU_CR[MDUBUSY] to be cleared by hardware;
5. Set MDU_CR[MDUDONE] to 1.

Notes:

- Before using MDU, MDU_CR[MDURUN] must be set to 1. After operation finished, MDU_CR[MDUDONE] must be set to 1. These two steps ensure the data of MDU will not be affected by nested calls of different interrupts and the main function.
- Make sure the operation mode and other register have been written before writing to MDU_C[7:0]

12.3.2 16-Bit Signed Multiplication with the Result Shifted Left by 1 Bit

When MDU_MD[2:0] = 000, the MDU works in 16-bit signed multiplication mode with the result shifted left by 1 bit. As shown in Table 12-1, write 16-bit signed data to MDU_A and MDU_C as the multiplied number and multiplier respectively. The result is the 32-bit signed data obtained by the product shifting left by 1 bit. The high 16 bits of this data is read by MDU_A, and the low 16 bits is read by MDU_B.

Table 12-1 Meaning of the Register in 16-Bit Signed Multiplication with the Result Shifted Left by 1 Bit Mode

Data Register	Input Data	Output Data
MDU_A	Multiplied number	High 16 bits of the product
MDU_B	—	Low 16 bits of the product
MDU_C	Multiplier	—
MDU_D	—	—

12.3.3 16-Bit Signed Multiplication

When MDU_MD[2:0] = 001, the MDU works in 16-bit signed multiplication mode. As shown in Table 12-2, write 16-bit signed data to MDU_A and MDU_C as the multiplied number and multiplier, respectively. The result is 31-bit signed data. The high 16 bits of the data is read by MDU_A, and the low 16 bits is read by MDU_B.

Table 12-2 Meaning of the Register in 16-Bit Signed Multiplication Mode

Data Register	Input Data	Output Data
MDU_A	Multiplied number	High 16 bits of the product
MDU_B	—	Low 16 bits of the product
MDU_C	Multiplier	—
MDU_D	—	—

12.3.4 16-Bit Unsigned Multiplication

When MDU_MD[2:0] = 010, the MDU works in 16-bit unsigned multiplication mode. As shown in Table 12-3, write 16-bit unsigned data to MDU_A and MDU_C as the multiplied number and multiplier, respectively. The result is 32-bit unsigned data. The high 16 bits of the data is read by MDU_A, and the low 16 bits is read by MDU_B.

Table 12-3 Meaning of the Register in 16-Bit Unsigned Multiplication Mode

Data Register	Input Data	Output Data
MDU_A	Multiplied number	High 16 bits of the product
MDU_B	—	Low 16 bits of the product
MDU_C	Multiplier	—
MDU_D	—	—

12.3.5 32-Bit/16-Bit Unsigned Division

When MDU_MD[2:0] = 011, the MDU works in 32-bit/16-bit unsigned division mode. As shown in Table 12-4, the dividend is 32-bit unsigned data, and the divisor is 16-bit unsigned data. High 16 bits of the dividend is written to MDU_A, low 16 bits of the dividend is written to MDU_B, and divisor is written to MDU_C. The result is 32-bit unsigned quotient with 16-bit unsigned remainder. The high 16 bits of the quotient is read by MDU_A, the low 16 bits is read by MDU_B, and the remainder is read by MDU_C.

Table 12-4 Meaning of the Register in Unsigned Division Mode

Data Register	Input Data	Output Data
MDU_A	High 16 bits of the dividend	High 16 bits of the quotient
MDU_B	Low 16 bits of the dividend	Low 16 bits of the quotient
MDU_C	Divisor	Remainder
MDU_D	—	—

12.3.6 Low-Pass Filter

When MDU_MD[2:0] = 110, MDU works in LPF mode.

The calculation formula of LPF is:

$$Y_k = Y_{k-1} + K \times (X_k - Y_{k-1})$$

Where,

Y_k: Filtered value

Y_{k-1}: Last filtered value

K: Filter coefficient

X_k: Value to be filtered

As shown in Table 12-5, Y_k and Y_{k-1} are 32-bit signed data, X_k is 16-bit signed data, and K is 8-bit unsigned data. High 16 bits of Y_{k-1} is written to MDU_B, low 16 bits of Y_{k-1} is written to MDU_C, K is written to MDU_D and X_k is written to MDU_A. The result is Y_k. High 16 bits of Y_k is read by MDU_B, and low 16 bits is read by MDU_C.

Table 12-5 Meaning of the Register in LPF Mode

Data Register	Input Data	Output Data
MDU_A	X_k	—
MDU_B	Y_{k-1}[31:16]	Y_k[31:16]
MDU_C	Y_{k-1}[15:0]	Y_k[15:0]
MDU_D	K	—

12.3.7 Coordinate Transformation

When MDU_MD[2:0] = 100, MDU works in coordinate transformation mode. As shown in Figure 12-1, the

coordinate transformation converts the components cos_i, sin_i of vector A under the x-y axis to the components cos_o, sin_o under the x'-y' axis, with the x'-y' axis lagging the x-y axis by θ.

The formula for coordinate transformation is:

$$cos_o = cos_i \times cos\,\theta - sin_i \times sin\,\theta$$

$$sin_o = cos_i \times sin\,\theta + sin_i \times cos\,\theta$$

In particular, when sin_i is 0, the coordinate transformation is a sine and cosine calculation with cos_i as the amplitude, calculated as:

$$cos_o = cos_i \times cos\,\theta$$

$$sin_o = cos_i \times sin\,\theta$$

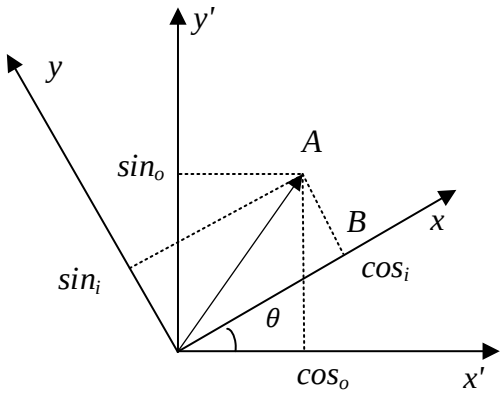


Figure 12-1 Coordinate Transformation

As shown in Table 12-6, cos_i, sin_i, θ, cos_o and sin_o are all 16-bit signed data. Write cos_i to MDU_A, θ to MDU_B and sin_i to MDU_C to calculate cos_o and sin_o. cos_o is read by MDU_A and sin_o by MDU_C.

Table 12-6 Meaning of the Register in Coordinate Transformation Mode

Data Register	Input Data	Output Data
MDU_A	cos_i	cos_o
MDU_B	θ	—
MDU_C	sin_i	sin_o
MDU_D	—	—

12.3.8 Arctangent

When MDU_MD[2:0] = 101, the MDU works in arctangent function mode.

Arctangent function calculates the magnitude and angle of the vector based on the input sine and cosine values. The calculation formula is:

$$U = \sqrt{(U \sin\theta)^2 + (U \cos\theta)^2}$$

$$\theta = tan^{-1}\left(\frac{U sin\,\theta}{U cos\,\theta}\right)$$

Where,

Usinθ: Sine component of vector

Ucosθ: Cosine component of vector

θ: Calculated vector angle

U: Calculated vector magnitude

As shown in Table 12-7, $Ucos_\theta$, $Usin_\theta$, *U* and *θ* are 16-bit signed data. Write $Ucos_\theta$ to MDU_A and $Usin_\theta$ to MDU_C to calculate *U* and *θ*. *U* is read by MDU_A, and *θ* is read by MDU_B.

Table 12-7 Meaning of the Register in Arctangent Mode

Data Register	Input Data	Output Data
MDU_A	*Ucosθ*	*U*
MDU_B	—	*θ*
MDU_C	*Usinθ*	—
MDU_D	—	—

12.4 MDU Registers

12.4.1 MDU_CR (0xC1)

Bit	7	6	5	4	3	2	1	0
Name	MDUBUSY	MDUDONE	MDURUN	RSV				
Type	R	W1	W1	—	—	—	—	—
Reset	0	0	0	—	—	—	—	—

Bit	Name	Description
[7]	MDUBUSY	MDU busy flag When MDU_C[7:0] is written, MDU will start 0: MDU idle 1: MDU operating
[6]	MDUDONE	MDU end operation bit 0: No meaning 1: After MDU operation is finished, software set this bit to 1. This operation ensures that MDU calculation is correct when it is called multiple times in nested interrupt
[5]	MDURUN	MDU start operation bit 0: No meaning 1: Before MDU operation starts, software set this bit to 1. This operation ensures that the MDU calculation is correct when it is called multiple times in nested interrupt
[4:0]	RSV	Reserved

12.4.2 MDU_MD (0xCA)

Bit	7	6	5	4	3	2	1	0
Name	RSV					MDUMOD		

Type	—	—	—	—	—	R/W	R/W	R/W
Reset	—	—	—	—	—	0	0	0

Bit	Name	Description
[7:3]	RSV	Reserved
[2:0]	MDUMOD	MDU mode selection 000: 16-bit signed multiplication with the result shifted left by 1 bit 001: 16-bit signed multiplication 010: 16-bit unsigned multiplication 011: 32-bit/16-bit unsigned division 100: Coordinate transformation (sin/cos calculation) 101: Arctangent function 110: Low-pass filter 111: Reserved

12.4.3 MDU_A (0xC7, 0xC6)

MDU_AH(0xC7)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	MDU_A[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
MDU_AL(0xC6)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	MDU_A[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	MDU_A	A data register of MDU, the contents of this register in different modes are shown in Table 12-8

Table 12-8 MDU_A contents in Different Modes

MDU_MD[2:0]	Write	Read
000	Multiplied number	High 16 bits of the product
001	Multiplied number	High 16 bits of the product
010	Multiplied number	High 16 bits of the product
011	High 16 bits of dividend	cos_o
100	cos_i	High 16 bits of quotient
101	$Ucos\theta$	U
110	X_k	—

12.4.4 MDU_B (0xC5, 0xC4)

MDU_BH(0xC5)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	MDU_B[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
MDU_BL(0xC4)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	MDU_B[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	MDU_B	B data register of MDU, the contents of this register in different modes are shown in Table 12-9

Table 12-9 MDU_B contents in Different Modes

MDU_MD[2:0]	Write	Read
000	—	Low 16 bits of the product
001	—	Low 16 bits of the product
010	—	Low 16 bits of the product
011	Low 16 bits of the dividend	Low 16 bits of the quotient
100	θ	—
101	—	θ
110	Y_{k-1}[31:16]	Y_k[31:16]

12.4.5 MDU_C (0xC3, 0xC2)

MDU_CH(0xC3)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	MDU_C[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
MDU_CL(0xC2)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	MDU_C[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	MDU_C	C data register of MDU, the contents of this register in different modes are in Table 12-10

Table 12-10 MDU_C contents in Different Modes

MDU_MD[2:0]	Write	Read
000	Multiplier	—
001	Multiplier	—
010	Multiplier	—
011	Divisor	Remainder
100	sin_i	sin_o
101	$Usin\theta$	—
110	Y_{k-1}[15:0]	Y_k[15:0]

12.4.6 MDU_D (0xCB)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	MDU_D							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	MDU_D	D data register of MDU MDU_MD[2:0] = 110: *K* of LPF

13 PI/PID

13.1 PI/PID Introduction

PI/PID regulator is a linear controller. The output is generated by the linear combination of the proportionality, integration and differentiation of the deviation. Then the output is implemented by an actuator. In motor control systems, it can be used for speed and position control.

PI formula is:

$$U_k = U_{k-1} + Kp \times (E_k - E_{k-1}) + Ki \times E_k$$

PID formula is:

$$U_k = U_{k-1} + Kp \times (E_k - E_{k-1}) + Ki \times E_k + Kd \times (E_k - 2 \times E_{k-1} + E_{k-2})$$

Where,

U_k: output for round k calculation

U_{k-1}: output for round k - 1 calculation

E_k: deviation for round k input

E_{k-1}, E_{k-2}: deviations for round k - 1 and round k -2 input

Kp, Ki, Kd: Proportional, integral and differential coefficients of regulator

The maximum value of U_k is PIx_UKMAX (x = 0 ~ 3) and the minimum value is PIx_UKMIN

13.2 PI/PID Features

- Parameter range is configurable
- Support multiple calls, not support nested calls
- The operation result PIx_UK is 32-bit
- Read the result after the busy flag is 0

13.3 PI/PID Instructions

1. Initialization is required before PI/PID operation. Configure Kp, Ki, Kd and the maximum, minimum values of output;
2. Set PI_CR[PIxSTA] = 1 to start PI/PID operation, and the busy flag PI_CR[PIBSY] will be automatically set to 1;
3. Read PI_CR[PIBSY] by software. PI_CR[PIBSY] with value of 0 indicates that the calculation is completed and calculation result PIx_UK data is updated;
4. Read PIx_UK to obtain the output.

Notes:

- The data format of PI_KP is Q12 and the data format of other registers is Q15
- PIx_UK and PIx_EK1 default to the last calculated UK and Ek. Writing to PIx_EK1 and PIx_UK can change

the related values

- For multiple calls, save the parameters after PI operation and initialize parameters before operation. The initialization operation is as follows:

```
PIx_KP = KP;                    // Initialize Kp
PIx_KI = KI;                    // Initialize Ki
PIx_KD = KD;                    // Initialize Kd
PIx_UKMAX = UKMAX;              // Initialize maximum output
PIx_UKMIN = UKMIN;              // Initialize minimum output
PIx_EK1 = X;                    // Initialize Ek-1
PIx_UKH = Y1;                   // Initialize high 16 bits of Uk-1
PIx_UKL = Y2;                   // Initialize low 16 bits of Uk-1
```

13.4 PI/PID Registers

13.4.1 PI_CR (0xF9)

Bit	7	6	5	4	3	2	1	0
Name	T2TSS	RSV		PIBSY	PI3STA	PI2STA	PI1STA	PI0STA
Type	R/W	—	—	W	W	W	W	W
Reset	0	—	—	0	0	0	0	0

Bit	Name	Description
[7]	T2TSS	Input mode selection for Timer2 step mode 0: P1.0 is the direction line, P0.7 is the pulse counting line 1: P1.0 is reverse pulse counting line, P0.7 is forward pulse counting line
[6:5]	RSV	Reserved
[4]	PIBSY	PI busy flag 0: PI idle 1: PI in operation
[3]	PI3STA	PI3 enable 0: Disable 1: Enable
[2]	PI2STA	PI2 enable 0: Disable 1: Enable
[1]	PI1STA	PI1 enable 0: Disable 1: Enable
[0]	PI0STA	PI0 enable 0: Disable 1: Enable

13.4.2 PI0_KP (0x02E0, 0x02E1)

PI0_KPH(0x02E0)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI0_KP[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI0_KPL(0x02E1)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI0_KP[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI0_KP	Proportional coefficient of PI0

13.4.3 PI0_KI (0x02E2, 0x02E3)

PI0_KIH(0x02E2)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI0_KI[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI0_KIL(0x02E3)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI0_KI[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI0_KI	Integral coefficient of PI0

13.4.4 PI0_UKMAX (0x02E4, 0x02E5)

PI0_UKMAXH(0x02E4)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI0_UKMAX[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI0_UKMAXL(0x02E5)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI0_UKMAX[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI0_UKMAX	Maximum output of PI0

13.4.5 PI0_UKMIN (0x02E6, 0x02E7)

PI0_UKMINH(0x02E6)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI0_UKMIN[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI0_UKMINL(0x02E7)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI0_UKMIN[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI0_UKMIN	Minimum output of PI0

13.4.6 PI0_EK1 (0x02E8, 0x02E9)

PI0_EK1H(0x02E8)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI0_EK1[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI0_EK1L(0x02E9)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI0_EK1[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI0_EK1	Last deviation of PI0

13.4.7 PI0_EK (0x02EA, 0x02EB)

PI0_EKH(0x02EA)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI0_EK[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI0_EKL(0x02EB)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI0_EK[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI0_EK	Current deviation of PI0

13.4.8 PI0_UKH (0x02EC, 0x02ED)

PI0_UKHH(0x02EC)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI0_UKH[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI0_UKHL(0x02ED)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI0_UKH[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI0_UKH	High 16 bits of output of PI0

13.4.9 PI0_UKL (0x02EE, 0x02EF)

PI0_UKLH(0x02EE)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI0_UKL[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI0_UKLL(0x02EF)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI0_UKL[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI0_UKL	Low 16 bits of output of PI0

13.4.10 PI1_KP (0x02D0, 0x02D1)

PI1_KPH(0x02D0)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI1_KP[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI1_KPL(0x02D1)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI1_KP[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI1_KP	Proportional coefficient of PI1

13.4.11 PI1_KI (0x02D2, 0x02D3)

PI1_KIH(0x02D2)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI1_KI[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI1_KIL(0x02D3)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI1_KI[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI1_KI	Integral coefficient of PI1

13.4.12 PI1_UKMAX (0x02D4, 0x02D5)

PI1_UKMAXH(0x02D4)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI1_UKMAX[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI1_UKMAXL(0x02D5)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI1_UKMAX[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI1_UKMAX	Maximum output of PI1

13.4.13 PI1_UKMIN (0x02D6, 0x02D7)

PI1_UKMINH(0x02D6)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI1_UKMIN[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI1_UKMINL(0x02D7)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI1_UKMIN[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI1_UKMIN	Minimum output of PI1

13.4.14 PI1_EK1 (0x02D8, 0x02D9)

PI1_EK1H(0x02D8)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI1_EK1[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI1_EK1L(0x02D9)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI1_EK1[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI1_EK1	Last deviation of PI1

13.4.15 PI1_EK (0x02DA, 0x02DB)

PI1_EKH(0x02DA)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI1_EK[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI1_EKL(0x02DB)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI1_EK[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI1_EK	Current deviation of PI1

13.4.16 PI1_UKH (0x02DC, 0x02DD)

PI1_UKHH(0x02DC)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI1_UKH[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI1_UKHL(0x02DD)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI1_UKH[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI1_UKH	High 16 bits of output of PI1

13.4.17 PI1_UKL (0x02DE, 0x02DF)

PI1_UKLH(0x02DE)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI1_UKL[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI1_UKLL(0x02DF)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI1_UKL[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI1_UKL	Low 16 bits of output of PI1

13.4.18 PI2_KP (0x02BC, 0x02BD)

PI2_KPH(0x02BC)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI2_KP[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI2_KPL(0x02BD)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI2_KP[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI2_KP	Proportional coefficient of PI2

13.4.19 PI2_KI (0x02BE, 0x02BF)

PI2_KIH(0x02BE)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI2_KI[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI2_KIL(0x02BF)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI2_KI[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI2_KI	Integral coefficient of PI2

13.4.20 PI2_UKMAX (0x02C0, 0x02C1)

PI2_UKMAXH(0x02C0)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI2_UKMAX[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI2_UKMAXL(0x02C1)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI2_UKMAX[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI2_UKMAX	Maximum output of PI2

13.4.21 PI2_UKMIN (0x02C2, 0x02C3)

PI2_UKMINH(0x02C2)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI2_UKMIN[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI2_UKMINL(0x02C3)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI2_UKMIN[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI2_UKMIN	Minimum output of PI2

13.4.22 PI2_EK1 (0x02C4, 0x02C5)

PI2_EK1H(0x02C4)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI2_EK1[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI2_EK1L(0x02C5)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI2_EK1[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI2_EK1	Last deviation of PI2

13.4.23 PI2_EK (0x02C6, 0x02C7)

PI2_EKH(0x02C6)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI2_EK[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI2_EKL(0x02C7)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI2_EK[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	**Name**	**Description**						
[15:0]	PI2_EK	Current deviation of PI2						

13.4.24 PI2_UKH (0x02C8, 0x02C9)

PI2_UKHH(0x02C8)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI2_UKH[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI2_UKHL(0x02C9)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI2_UKH[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	**Name**	**Description**						
[15:0]	PI2_UKH	High 16 bits of output of PI2						

13.4.25 PI2_UKL (0x02CA, 0x02CB)

PI2_UKLH(0x02CA)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI2_UKL[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI2_UKLL(0x02CB)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI2_UKL[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
Bit	**Name**	**Description**						
[15:0]	PI2_UKL	Low 16 bits of output of PI2						

13.4.26 PI2_KD (0x02CC, 0x02CD)

PI2_KDH(0x02CC)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI2_KD[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI2_KDL(0x02CD)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI2_KD[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI2_KD	Differential coefficient of PI2

13.4.27 PI2_EK2 (0x02CE, 0x02CF)

PI2_EK2H(0x02CE)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI2_EK2[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI2_EK2L(0x02CF)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI2_EK2[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI2_EK2	Deviation before last deviation of PI2

13.4.28 PI3_KP (0x02A8, 0x02A9)

PI3_KPH(0x02A8)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI3_KP[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI3_KPL(0x02A9)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI3_KP[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI3_KP	Proportional coefficient of PI3

13.4.29 PI3_KI (0x02AA, 0x02AB)

PI3_KIH(0x02AA)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI3_KI[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI3_KIL(0x02AB)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI3_KI[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI3_KI	Integral coefficient of PI3

13.4.30 PI3_UKMAX (0x02AC, 0x02AD)

PI3_UKMAXH(0x02AC)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI3_UKMAX[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI3_UKMAXL(0x02AD)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI3_UKMAX[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI3_UKMAX	Maximum output of PI3

13.4.31 PI3_UKMIN (0x02AE, 0x02AF)

PI3_UKMINH(0x02AE)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI3_UKMIN[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI3_UKMINL(0x02AF)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI3_UKMIN[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI3_UKMIN	Minimum output of PI3

13.4.32 PI3_EK1 (0x02B0, 0x02B1)

PI3_EK1H(0x02B0)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI3_EK1[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI3_EK1L(0x02B1)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI3_EK1[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI3_EK1	Last deviation of PI3

13.4.33 PI3_EK (0x02B2, 0x02B3)

PI3_EKH(0x02B2)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI3_EK[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI3_EKL(0x02B3)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI3_EK[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI3_EK	Current deviation of PI3

13.4.34 PI3_UKH (0x02B4, 0x02B5)

PI3_UKHH(0x02B4)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI3_UKH[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI3_UKHL(0x02B5)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI3_UKH[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI3_UKH	High 16 bits of output of PI3

13.4.35 PI3_UKL (0x02B6, 0x02B7)

PI3_UKLH(0x02B6)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI3_UKL[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI3_UKLL(0x02B7)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI3_UKL[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI3_UKL	Low 16 bits of output of PI3

13.4.36 PI3_KD (0x02B8, 0x02B9)

PI3_KDH(0x02B8)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	**15**							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI3_KDL(0x02B9)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI3_KD[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI3_KD	Differential coefficient of PI3

13.4.37 PI3_EK2 (0x02BA, 0x02BB)

PI3_EK2H(0x02BA)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	PI3_EK2[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
PI3_EK2L(0x02BB)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	PI3_EK2[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	PI3_EK2	Deviation before last deviation of PI2

14 FOC

14.1 FOC Instructions

14.1.1 FOC Introduction

FOC module is used for motor drive based on sensorless FOC or sensor-based FOC. It can also be used for motor control based on SVPWM. When DRV_CR[FOC_EN] = 0, FOC module does not work and FOC clock is stopped. The relevant registers are in reset state and cannot be written to.

FOC module contains angle estimator, PI controller, coordinate transformation module, current sampling module and PWM output module, which can implement current closed loop in hardware. The angle estimator uses the sampling motor current to estimate the rotor position and sensorless FOC is performed. MCU can also process position sensor signal to obtain the rotor position and sensor-based FOC is performed.

- Sensorless FOC: Angle for coordinate transformation is obtained by angle estimator, and the motor speed is estimated for speed closed-loop control
- Sensor-based FOC: FOC module provides the angle input interface. MCU sample position sensor signal and calculate the electrical angle of motor. Software sends the result to FOC module for coordinate transformation

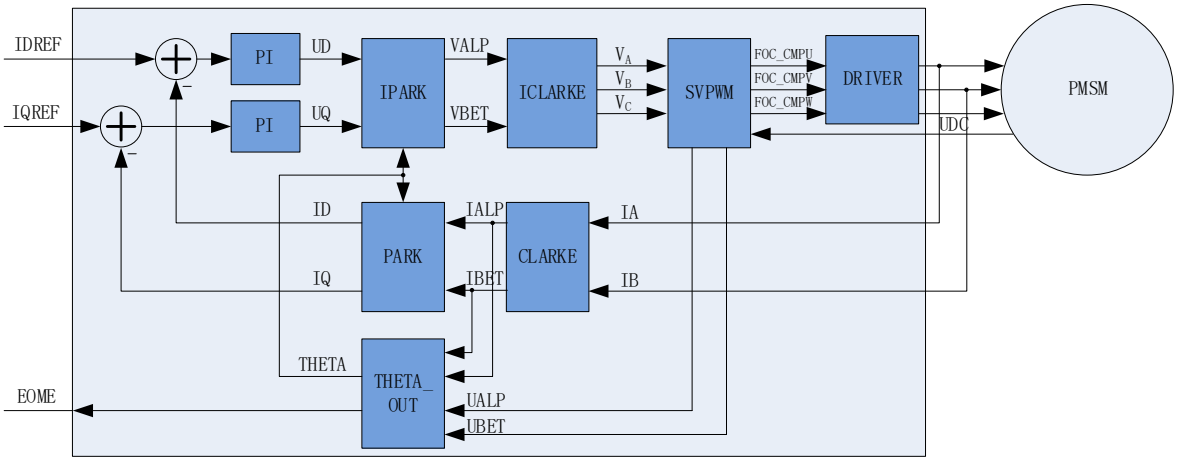


Figure 14-1 Schematic Diagram of FOC

14.1.2 Reference Input

The current loop of FOC module uses the d-axis current reference value FOC_IDREF and the q-axis current reference value FOC_IQREF as the reference, and uses the d-axis current sampling value FOC__ID and the q-axis current sampling value FOC__IQ as the feedback. FOC module provides real-time estimated motor speed FOC__EOME. MCU can use FOC__EOME as the feedback to build speed loop and sends the output of speed loop to FOC_IQREF to implement the speed-current dual closed loop control.

14.1.3 PI Controller

FOC module contains 2 PI controllers:

1. Flux control: PI controller of d-axis current, current reference FOC_IDREF minus feedback current FOC__ID as the error input. Proportional coefficient FOC_DQKP and the integral coefficient FOC_DQKI for adjustment of PI performance. FOC_DMAX and FOC_DMIN for limiting of the output amplitude. The output is voltage reference of d-axis FOC__UD;
2. Torque control: PI controller of q-axis current, current reference FOC_IQREF minus feedback current FOC__IQ as the error input. Proportional coefficient FOC_DQKP and the integral coefficient FOC_DQKI for adjustment of PI performance. FOC_QMAX and FOC_QMIN for limiting of the output amplitude. The output is voltage reference of q-axis FOC__UQ.

14.1.4 Coordinate Transformations

14.1.4.1 Inverse Park Transformation

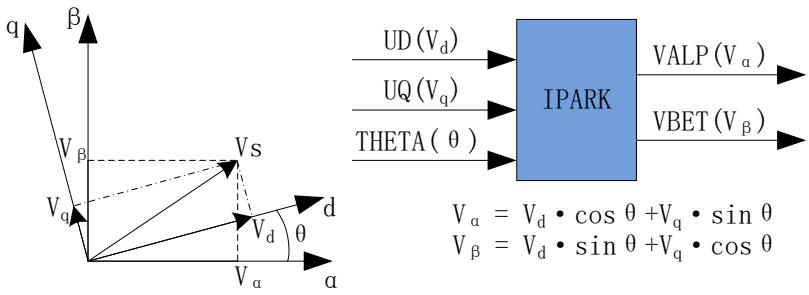


Figure 14-2 Inverse Park Transformation

Inverse Park transformation is used to transform two voltage vector obtained by PI controller, FOC__UD and FOC__UQ, from dq-axis coordinate to αβ-axis coordinate.

14.1.4.2 Inverse Clarke Transformation

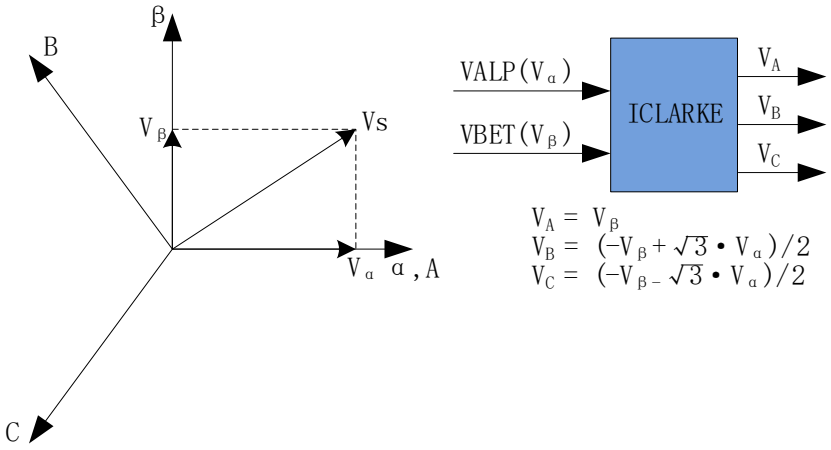


Figure 14-3 Inverse Clarke Transformation

Inverse Clarke transformation is used to transform voltage vector from αβ-axis coordinate to 3-phase stationary coordinate.

14.1.4.3 Clarke Transformation

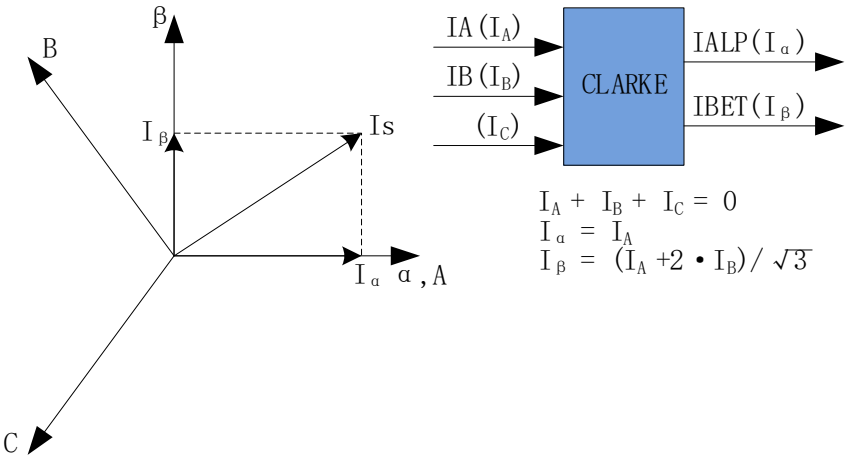


Figure 14-4 Clarke Transformation

Clarke transformation is used to transform the sampled current from 3-phase stationary coordinate to αβ-axis coordinate.

14.1.4.4 Park Transformation

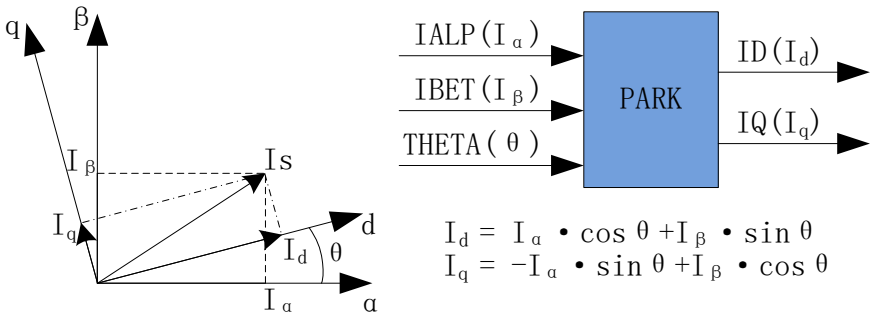


Figure 14-5 Park Transformation

Park transformation is used to transform the current vector, obtained after Clarke transformation, from αβ-axis coordinate to dq-axis coordinate to get the sampled dq-axis current FOC__ID and FOC__IQ.

14.1.5 SVPWM

SVPWM algorithm is an important part of FOC. The main idea is to use the switch of inverter space voltage vector to obtain the quasi-circular rotating magnetic field. This method can decrease the harmonic components of phase current, the harmonic losses of the motor and the torque ripple, and have a high voltage utilization.

SVPWM generate pulse-width modulation signals for the 3-phase motor voltage control. The process of generating the pulse width is reduced to a few simple equations. Since the power device high side and the low side of a phase cannot be turned on simultaneously, there are two states, phase connected to bus voltage (represented by 1) or phase connected to ground (represented by 0). Therefore, voltage vector output of inverter has totally 2^3 = 8 possible states. $X_C X_B X_A$ represents the voltage vector, where X_C represents the state of C phase, X_B represents the state of B phase and X_A represents the state of A phase. E.g., 100 represents the state that C phase voltage is connected to bus voltage and A, B phase are connected to ground. When the states of 3-phase are all 1 or 0, there is no voltage drop between 2 phase and the state is called inactive state or zero voltage vector. The other 6 states which have voltage output are active voltage vectors with an adjacent state rotation offset of 60 degrees.

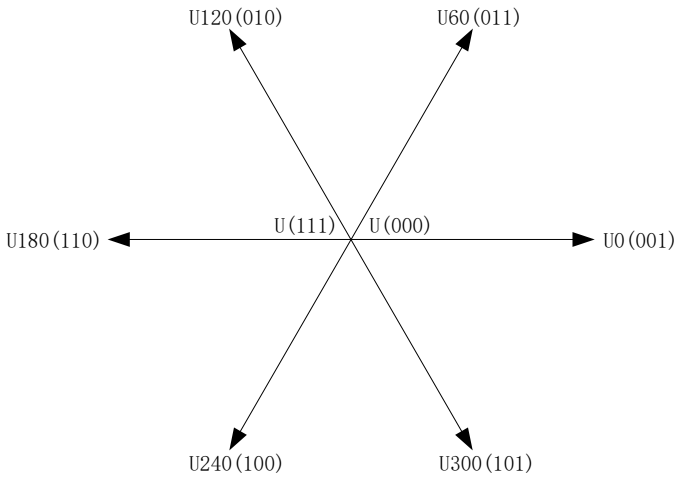


Figure 14-6 SVPWM Voltage Vector

SVPWM uses the sum of two adjacent vectors to generate any voltage vector located in the voltage vector space. As shown in Figure 14-7, supposing that U_{OUT} is the desired vector and it is in the sector between U60 and U0. Based on the principle of equal impulse, the effect, U0 applied 2*T1 time and U1 applied 2*T2 time, is equivalent to the U_{OUT}. The rest of time (T0) is applied by zero voltage vector.

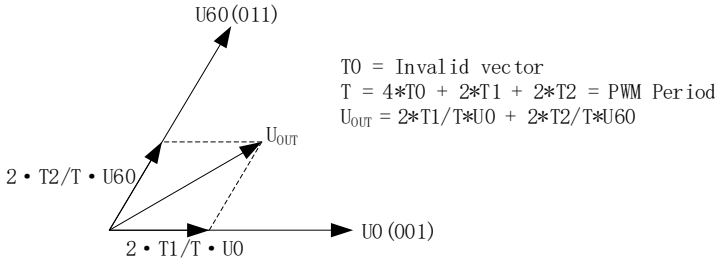


Figure 14-7 Voltage Vector Synthesis

Table 14-1 Inverter States of SVPWM

C phase	B phase	A phase	U_{ALP}	U_{BET}	State
0	0	0	0	0	000
0	0	1	$2/3*U_{DC}$	0	001
0	1	1	$1/3*U_{DC}$	$1/3*U_{DC}$	011
0	1	0	$-1/3*U_{DC}$	$1/3*U_{DC}$	010
1	1	0	$-2/3*U_{DC}$	0	110
1	0	0	$-1/3*U_{DC}$	$-1/3*U_{DC}$	100
1	0	1	$1/3*U_{DC}$	$-1/3*U_{DC}$	101
1	1	1	0	0	111

14.1.5.1 Continuous SVPWM

In the single-resistance current sampling mode, Continuous SVPWM is always used. In the dual/triple-resistances current sampling mode, set FOC_CR2[F5SEG] to 0 to select Continuous SVPWM as the output mode.

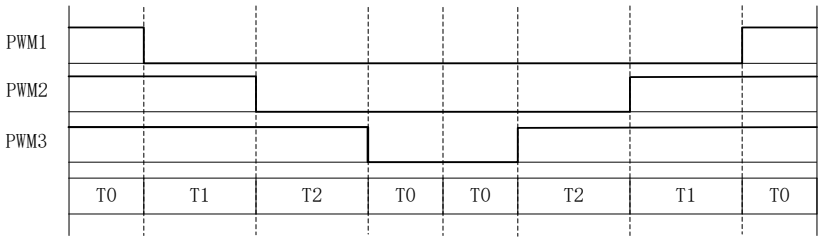


Figure 14-8 Continuous SVPWM Output Waveform

14.1.5.2 Discontinuous SVPWM

Discontinuous SVPWM is available in the dual/triple-resistances current sampling mode. Set FOC_CR2[F5SEG] to 1 to enable this mode.

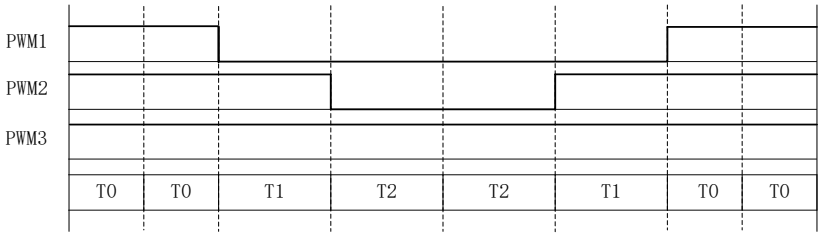


Figure 14-9 Discontinuous SVPWM Output Waveform

14.1.6 Overmodulation

Overmodulation is available in single/dual/triple-resistance current sampling mode. Set FOC_CR1[OVMDL] to 1 to enable overmodulation function. The output, FOC__UD and FOC__UQ, and limit value, FOC_DMAX and FOC_QMIN, will be multiplied by 1.15 in this mode.

14.1.7 Deadtime Compensation

Deadtime compensation is available in dual/triple-resistances current sampling mode. The compensation value of deadtime is configured by FOC_TSMIN. This mode can improve the quality of phase current at low speed.

14.1.8 Current and Voltage Sampling

In FOC mode, bus voltage and phase current will be sampled by hardware automatically. To start this function, enable ADC and operational amplifier and configure the corresponding control registers. The configuration of ADC channel and mode is not required. Current sampling mode, Single/dual/triple-resistance, is selected by setting FOC_CR1[CSM]. In single-resistance mode, the default sampling channel of the bus current (itrip) is ADC

channel 4. In dual-resistance mode, the default sampling channels of A phase current (ia) and B phase current (ib) are ADC channel 0 and channel 1. In triple-resistance mode, the default sampling channels of ia, ib and C phase current (ic) are ADC channel 0, channel 1 and channel 4. Channel 14, with bulit-in divider resistor and configurable dividing factor, or channel 2 can be selected for bus voltage sampling.

14.1.8.1 Single-Resistances Current Sampling Mode

Set FOC_CR1[CSM] to 00 to select the single-resistance current sampling mode. In this mode, FOC module samples itrip twice during the DRV counter counting-up operation, and samples the bus voltage after FOC algorithm calculation completed.

Since deadtime affects the accuracy of current sampling, FOC module sample within T1' and T2', which is the active voltage vector applied time with deatime removed. FOC_TRGDLY is the register which moves the current sampling time, and this register should be configured reasonably to ensure sampling is completed within T1' and T2'. E.g., FOC_TRGDLY = 5, the sampling time is delayed for 5*T = 208ns; FOC_TRGDLY = 0xFB(–5), the sampling time is advanced for 5*T = 208ns.

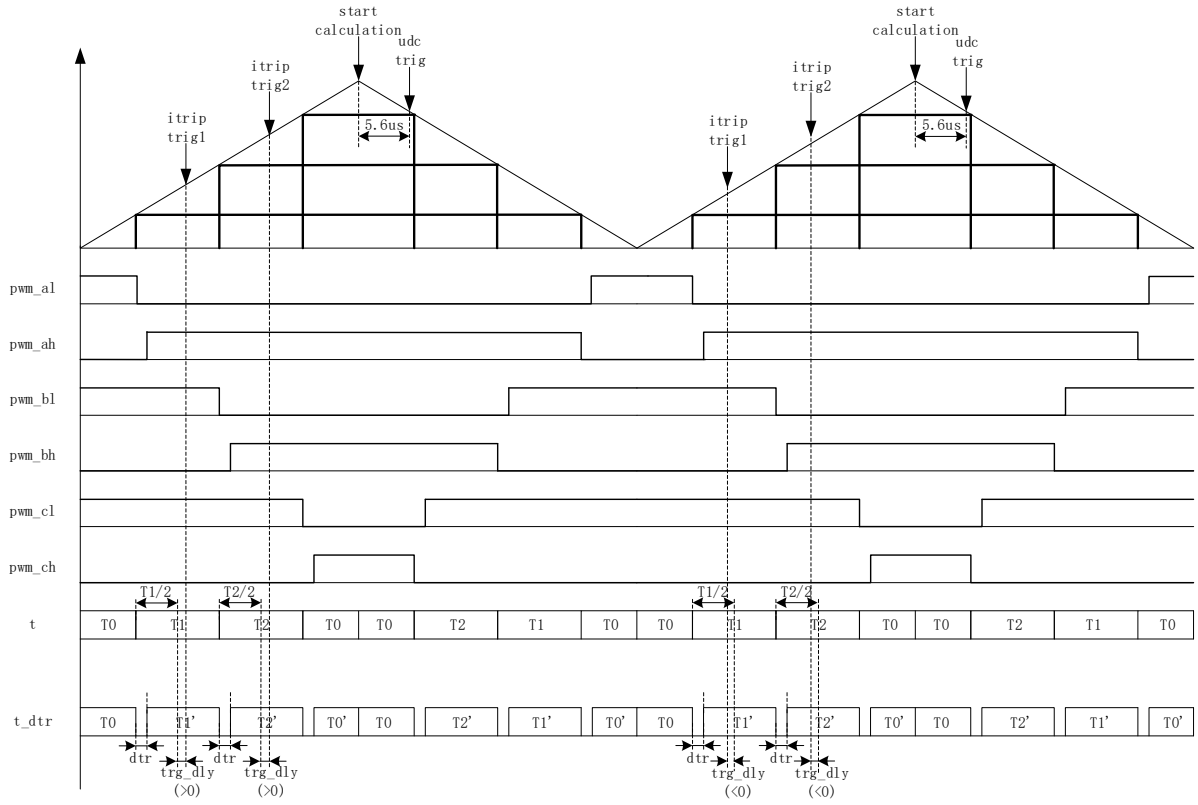


Figure 14-10 Single Resistances Current Sampling Timing Diagram

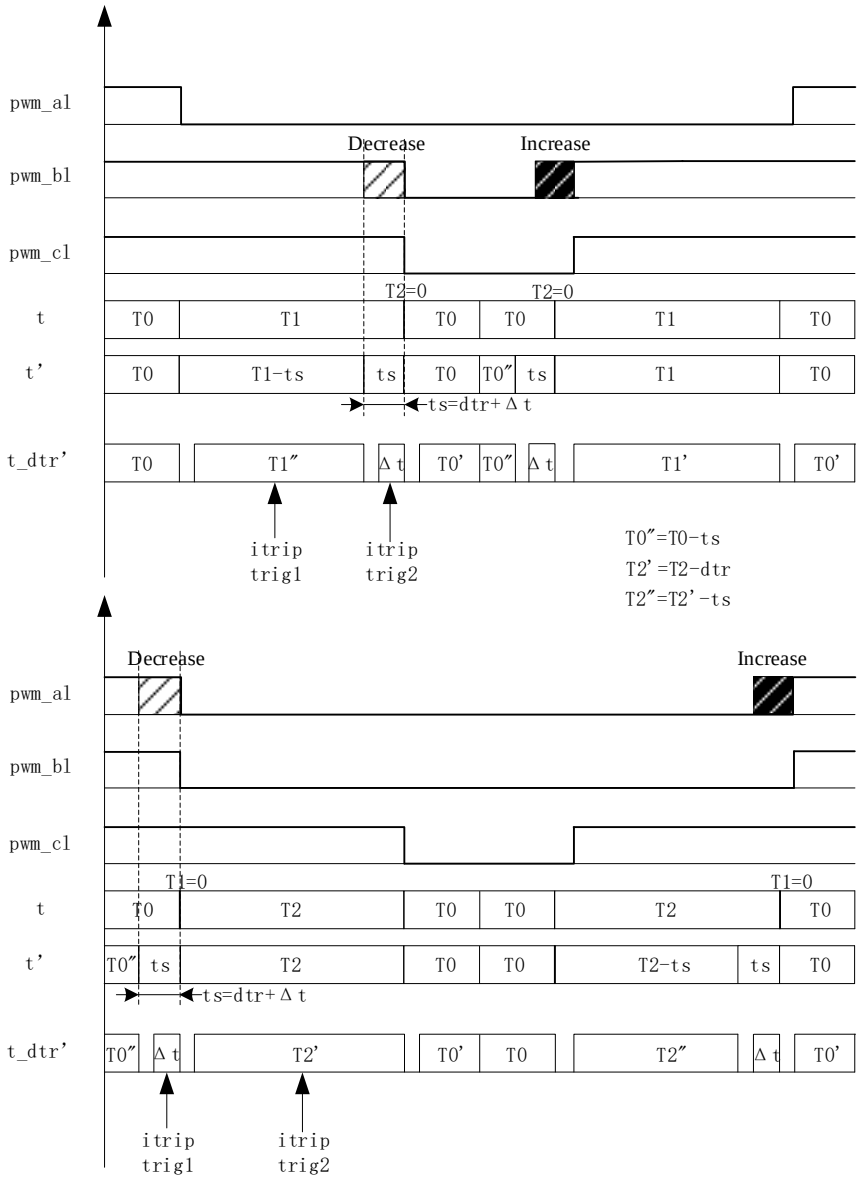


Figure 14-11 PWM Wave Shift in Single Resistances Current Sampling Mode

The time of sampling window could be not enough to sample the current in low modulation index and sector switching area. PWM waveform should be adjusted to ensure the minimum samping window required in the case. FOC_TSMIN (FOC_TSMIN = minimum sampling window + deadtime) is used to configure the minimum active voltage vector applied time, and FOC module will adjust the PWM waveform automatically.

14.1.8.2 Dual/Triple-Resistances Current Sampling Mode

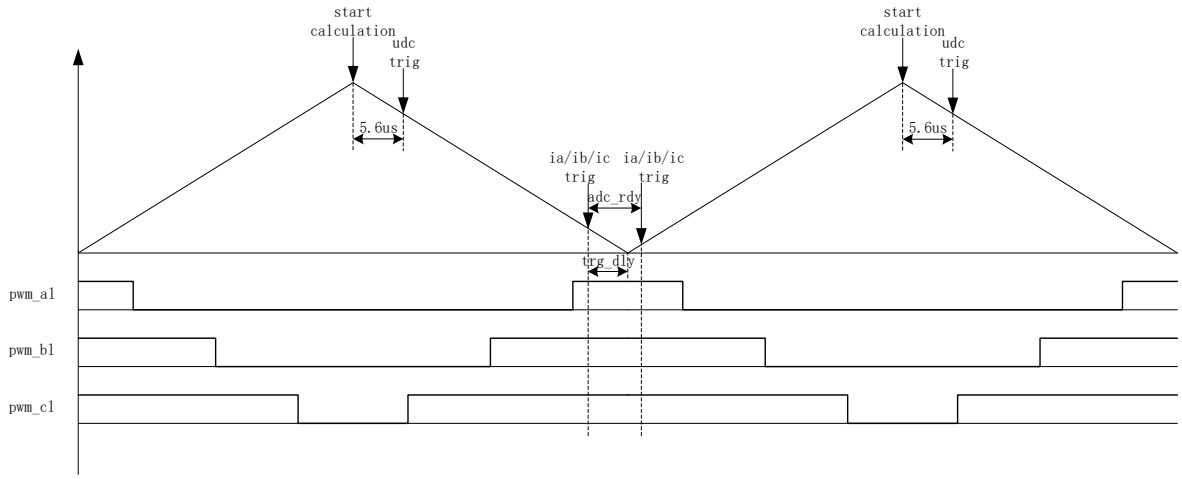


Figure 14-12 Dual/Triple-Resistances Concurrent Current Sampling Mode

Set FOC_CR1[CSM] to 10/11 and FOC_CR2[DSS] to 0 to select dual/triple-resistances concurrent current sampling mode. In triple-resistances mode, FOC_TRGDLY is used to configure the sampling time of a phase current (decided by the reference voltage), and the other phase will be sampled at the end of the previous sampling. In dual-resistances mode, FOC_TRGDLY is used to configure the sampling time of ia, and ib will be sampled at the end of ia sampling. TRG_DLY should be configured reasonably to ensure the both current sampling time on zero voltage vector (000). E.g., FOC_TRGDLY = 0xB2, the sampling time of ia is 50*T = 2.08μs advanced DRV counter underflow event and ib will sampled at the end of ia sampling.

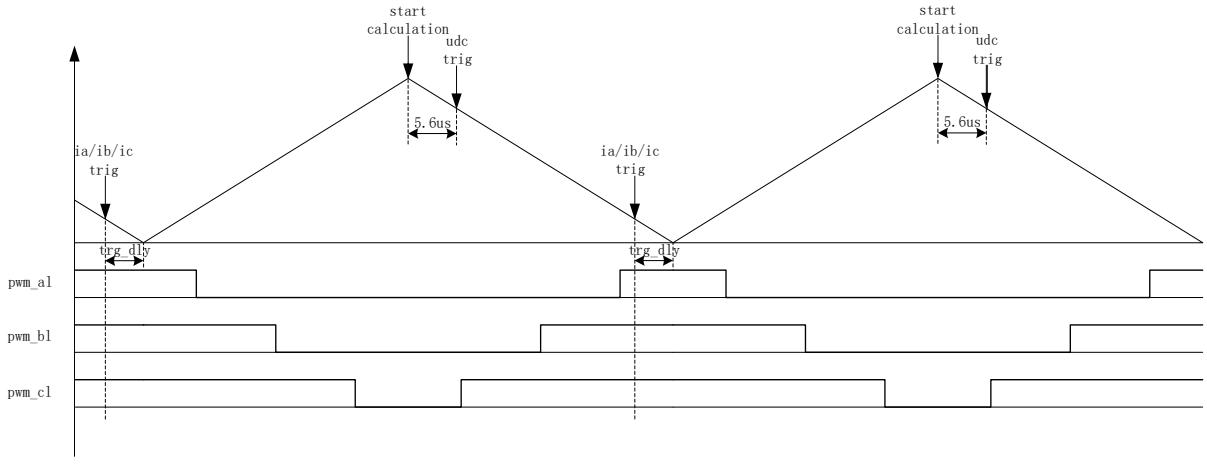


Figure 14-13 Dual/Triple-Resistances Alternating Current Sampling Mode

Set FOC_CR1[CSM] to 10/11 and FOC_CR2[DSS] to 1 to select dual/triple-resistances alternating current sampling mode. In this mode, FOC algorithm will be calculated in every PWM period. However, only one phase current is sampled at each PWM period (i.e., a phase current is sampled in the first PWM period and the other in

the second period). TRG_DLY should be configured reasonably to ensure the current, one of ia, ib or ic, sampling time on zero voltage vector (000). E.g., FOC_TRGDLY = 0xB2, the sampling time of phase current is 50*T = 2.08μs advanced DRV counter underflow event.

In dual/triple-resistances current sampling mode, bus voltage is sampled after FOC algorithm calculation completed.

14.1.8.3 Current Sampling Offset

The current sampling offset voltage should be added to sample full range of current due to the existence of the positive and negative phase current. When phase current is 0, ADC result is the offset value. ADC result minus this value, 0x4000 default, is the sampling current. Since the ADC reference voltage and hardware are nonideal, there is a deviation between the default value and the real value. Therefore, it is necessary to calibrate the offset. The calibration procedure is as following: When there is no current in three phases, MCU starts to sample the corresponding channel and the results, averaging all the sampled value, is written to FOC_CSO. Providing the ADC sampling range is 0 ~ 5V and the offset is 2.5V, FOC_CSO = 2.5V/5V*32768= 16384 (0x4000).

- FOC_CHC[CSOC] = 00/11, setting FOC_CSO is write the offset of itrip and ic
- FOC_CHC[CSOC] = 01, setting FOC_CSO is write the offset of ia
- FOC_CHC[CSOC] = 10, setting FOC_CSO is write the offset of ib

14.1.9 Angle Mode

Angle module includes angle estimation module, ramping module and estimated angle smooth switching module. The sources of angle are as follows:

- Forced ramping angle
- Forced pulling angle
- Estimator estimated angle
- Estimator forced angle

Table 14-2 Sources of Angle

FOC_CR1[RFAE]	FOC_CR1[ANGM]	FOC_CR1[EFAE]	Sources of Angle
1	X	X	Forced ramping angle
0	0	X	Forced pulling angle
0	1	0	Estimator estimated angle
0	1	1	omega > FOC_EFREQMIN: Estimator estimated angle omega < FOC_EFREQMIN: Estimator forced angle

14.1.9.1 Forced Ramping Angle

Forced ramping angle is controlled by angle FOC_THETA, speed FOC_RTHESTEP, acceleration FOC_RTHEACC and ramping counter FOC_RTHECNT. The formula is:

FOC__RTHESTEP (32-bit) = FOC__RTHESTEP (32-bit) + FOC_RTHEACC (low 16 bits)

THETA_OL (16-bit) = THETA_OL (16-bit) + FOC__RTHESTEP (high 16 bits)

Where, THETA_OL is a chip internal variable. In forced ramping angle mode, THETA_OL is written to FOC_THETA as the used angle. Software writing a value to FOC__THETA will also update this value to THETA_OL.

Forced ramping angle has a highest priority. Set FOC_CR1[RFAE] to 1 to enable the ramping function. Ramping module makes a ramping operation in every PWM cycle. When the value of the counter reaches the set value, set by FOC_RTHECNT, FOC_CR1[RFAE] is cleared by hardware, and then the ramping is over. Thereafter, according to the value of FOC_CR1[ANGM], the angle comes from estimator (FOC_CR1[ANGM] = 1) or forced pulling angle (FOC_CR1[ANGM] = 0).

14.1.9.2 Forced Pulling Angle

Forced ramping angle is controlled by angle FOC_THETA, speed FOC_RTHESTEP. The formula is:

THETA_OL (16-bit) = THETA_OL (16-bit) + FOC__RTHESTEP (high 16 bits)

Where, THETA_OL is a chip internal variable. In forced pulling angle mode, THETA_OL is written to FOC_THETA as the used angle. Software writing a value to FOC__THETA will also update this value to THETA_OL.

- Setting FOC_CR1[RFAE] to 1 and FOC_CR1[ANGM] to 0, forced pulling mode is started after ramping function. The speed is the cumulative result after ramping. This mode implement a forced uniform speed control
- Setting FOC_CR1[RFAE] to 0 and FOC_CR1[ANGM] to 0, angle is forced pulling angle directly. The speed RTHESTEP is the initial speed written by software. Set RTHESTEP to 0 to implement a pre-position function. Set motor speed to RTHESTEP to implement a sensor-based FOC. (Sensor-based FOC principle: Software writes the angle and speed to FOC__THETA and FOC__RTHESTEP to make modifications. FOC module will generate angle in each PWM period based on the written angle and speed.)

14.1.9.3 Estimator Angle

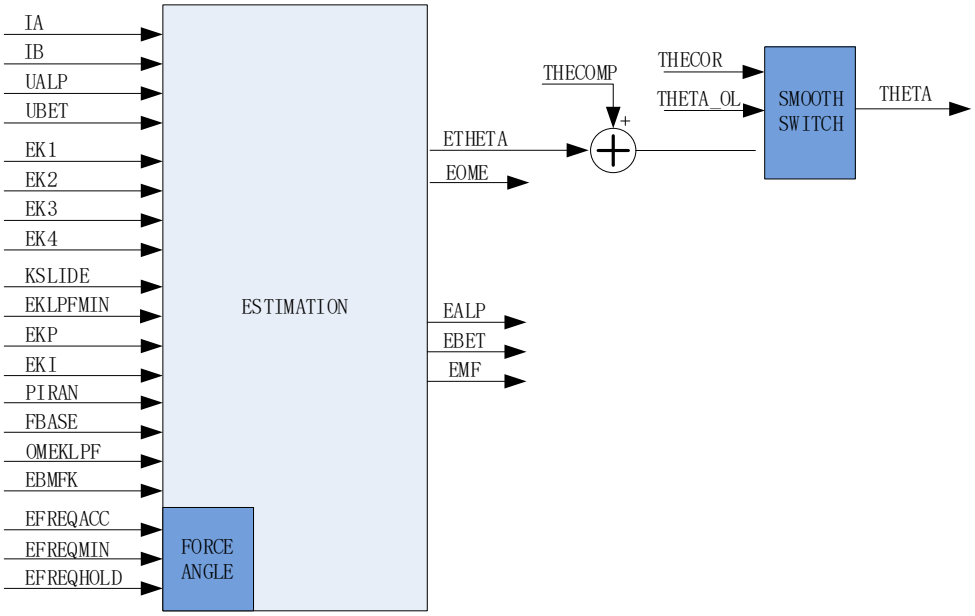


Figure 14-14 Principle Block Diagram of Angle Estimator

14.1.9.3.1 Estimator Estimated Angle

The estimator builds the motor model based on the motor parameters and control parameters, and output the estimated angle based on the sampling current and output voltage. The estimator can be selected as the PLL mode or SMO mode by configuring FOC_CR2[ESEL].

14.1.9.3.2 Estimator Forced Angle

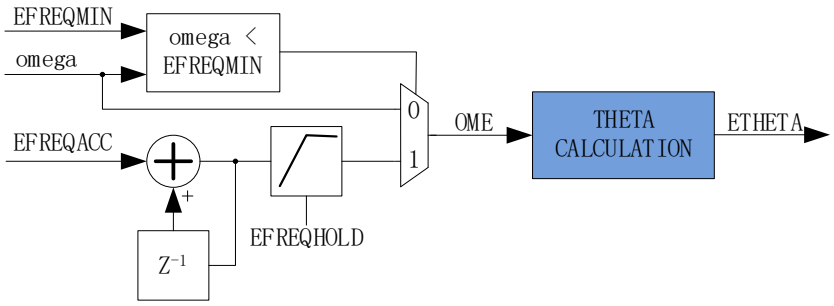


Figure 14-15 Principle Block Diagram of Estimator Forced angle

This function is similar as the ramping function. Due to the low speed at motor starting process, there can be a deviation in angle and speed estimation with the small effective signal. This may result in the launch failure. In this case, the estimator outputs the forced angle to make the motor start successfully.

Set FOC_CR1[RFAE] to 0, FOC_CR1[ANGM] to 1, FOC_CR1[EFAE] to 1 to enable estimator forced angle function. As shown in Figure 14-15, the estimator compares the value of real time estimated speed (omega) and FOC_EFREQMIN to select omega or forced speed as the used speed (OME). When omega < FOC_EFREQMIN, the forced speed is used as OME. The forced speed will start at 0 and increase by FOC_EFREQACC in each PWM period, with the maximum value FOC_EFREQHOLD. When omega ≥ FOC_EFREQMIN, omega is used as OME.

Estimator estimated speed FOC__EOME is the low pass filtering result of OME with the coefficient set by FOC_OMEKLPF.

14.1.9.3.3 Angle Smooth Switching

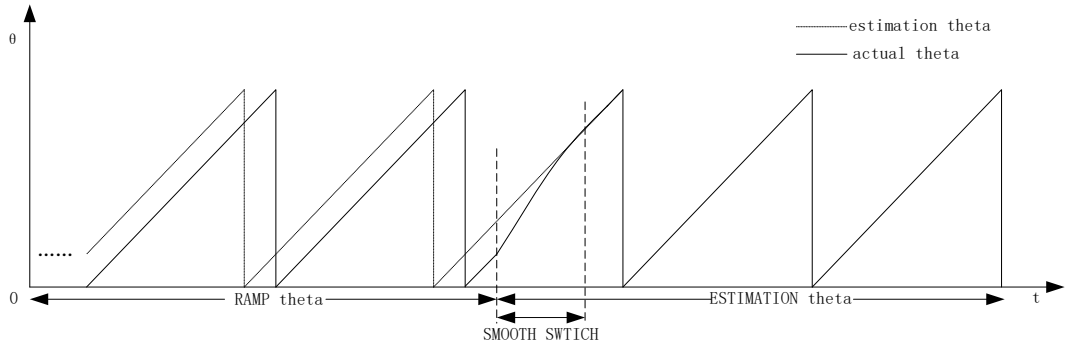


Figure 14-16 Angle Smooth Switching Waveform

Setting FOC_CR1[RFAE] to 1 and FOC_CR1[ANGM] to 1, the motor starts with the ramping function. At the end of the ramping, angle will switch to estimator angle mode. Unfortunately, there is usually a deviation between the estimated angle and the forced ramping angle. If the angle is switched from forced ramping angle to estimated angle directly, there could be motor jitter due to the inappropriate angle difference. To deal with this problem, a smooth switching of angle is preferred.

At the end of the ramping, if the deviation between estimated angle FOC_ETHETA and the forced ramping angle THETA_OL is less than or equal to FOC_THECOR, FOC_ETHETA is used as the output angle. If the deviation is larger than FOC_THECOR, THETA_OL will be modified smoothly to close FOC_ETHETA with the step of FOC_THECOR at every PWM cycle. Until the deviation is less than THECOR, FOC_ETHETA is used as the output angle.

14.1.9.3.4 Angle Compensation

Angle compensation register FOC_THECOMP is used as a compensation of the estimated angle FOC__ETHETA and the result is FOC__THETA. When FOC_THECOMP is a negative value, FOC__THETA is lagged than FOC__ETHETA. When FOC_THECOMP is a positive value, FOC__THETA is advanced than FOC__ETHETA.

14.1.10 Motor Real Time Monitor

MCU could monitor the state of motor using the following real time variables provided by FOC module:

- Used angle FOC__THETA
- Estimated angle FOC__ETHETA, Estimated speed FOC__EOME
- d-aixs voltage FOC__UD, q-aixs voltage FOC__UQ
- d-aixs current FOC__ID, q-aixs current FOC__IQ
- α-aixs voltage FOC__VALP, β-aixs voltage FOC__VBET
- Bus voltage FOC__UDCFLT
- Phase current FOC__IA, FOC__IB, FOC__IC and maximum phase current FOC__IAMAX, FOC__IBMAX, FOC__ICMAX
- α-aixs current (equal to FOC__IA), β-aixs current FOC__IBET
- α-aixs BEMF FOC__EALP, β-aixs BEMF FOC__EBET
- The magnitude of BEMF FOC__EMF
- Motor power FOC__POW

14.1.10.1 RSD

FOC provides RSD (the detection of tail wind and contrary wind) function. Drive motor with FOC_IDREF set to 0, FOC_IQREF set to 0 and FOC_CR0[ESCMS] set to 1, and rotor state could be detected by FOC__ETHETA and FOC__EOME. If FOC__ETHETA decreases or FOC__EOME is a negative value, the motor is in the contrary wind state. It is necessary to brake first and then start motor with the ramping forced angle mode. If FOC__ETHETA increases or FOC__EOME is a positive value, the motor is in the tail wind state. The motor can be started using estimated angle directly.

14.1.10.2 BEMF Detection

Estimator estimates α-axis BEMF FOC__EALP and β-axis BEMF FOC__EBET with the motor parameters, and outputs the magnitude of BEMF (FOC__EMF). This value can be used to implement protections, such as lock protection or phase unconnected protection.

14.1.10.3 Motor Power

Based on the sampling current, modulation index of SVPWM and bus voltage, FOC calculates motor power.

14.1.11 FG Generation

The FG signal is generated by FOC module and Timer4 working together. FOC module calculates a FG value based on the frequency base fbase, low-pass filtered speed FOC_EOMELPF and FG coefficient FOC_KFG in every PWM cycle. The result will update to TIM4_ARR and half value of result will update to TIM4_DR by

hardware. It should be noted that Timer4 must work at output mode and the clock division factor of Timer4 should be configured according to the motor maximum speed. FOC_KFG is set as: FOC_KFG = SYSCLK/(2^ TIM4_CR0[T4PSC]*fbase*x), where, x is the expected number of FG signal in one electric cycle. If the result exceeds 65535, Timer4 clock division factor TIM4_CR0[T4PSC] should be adjusted.

When FOC_KFG = 0, this function is disable and TIM4__ARR, TIM4__DR will not be updated.

14.2 FOC Registers

14.2.1 FOC_CR0 (0x409F)

Bit	7	6	5	4	3	2	1	0
Name	OMIF	OMAF	MERRS		UCSEL	OMAS	ESCMS	EDIS
Type	R	R	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	OMIF	omega < FOC_EFREQMIN flag. This bit is available even if FOC_CR1[EFAE] = 0 0: omega ≥ FOC_EFREQMIN 1: omega < FOC_EFREQMIN
[6]	OMAF	omega > FOC_EFREQMIN flag 0: omega ≤ FOC_EFREQMIN 1: omega > FOC_EFREQMIN
[5:4]	MERRS	The maximum error in SMO 00: 0.5 01: 0.25 10: 0.125 11: 1.0
[3]	UCSEL	Bus voltage (UDC) sampling channel selection In FOC mode, enable Driver counter and the bus voltage will be sampled automatically by hardware. This bit selects the ADC channel for bus voltage sampling 0: ADC channel 2 1: ADC channel 14 ADC channel 14 is internal bus voltage sampling channel, and the configuration of ADC_CR[ADCRATIO] selects the voltage division ratio. ADC channel 2 is the external bus voltage sampling channel, which use external voltage resistor division Note: The configuration of ADC_MASK is not required
[2]	OMAS	Output selection when omega is too large When omega[15:8] is larger than FOC_EFREQMAX, OME is set as: 0: FOC_EFREQMAX*256 1: FOC_EFREQHOLD
[1]	ESCMS	Angle output mode selection 0: Test mode 1: Recommended mode
[0]	EDIS	FOC__EALP, FOC__EBET calculation forbid 0: Not forbid 1: Forbid

14.2.2 FOC_CR1 (0x40A0)

Bit	7	6	5	4	3	2	1	0
Name	OVMDL	EFAE	RFAE	ANGM	CSM		RSV	SVPWMEN
Type	R/W	R/W	R/W	R/W	R/W	R/W	—	R/W
Reset	0	0	0	0	0	0	—	0

Bit	Name	Description
[7]	OVMDL	Overmodulation enable 0: Disable 1: Enable
[6]	EFAE	Estimator forced angle enable Enable estimator forced mode, and angle will switch to estimated angle automatically 0: Disable 1: Enable
[5]	RFAE	Forced ramping angle enable Enable forced ramping mode. After ramping, the angle mode switches to estimated mode or forced pulling mode and FOC_CR1[RFAE] is cleared by hardware 0: Disable 1: Enable
[4]	ANGM	Angle Mode FOC_CR1[RFAE] = 0: Select the angle mode FOC_CR1[RFAE] = 1: Select the angle mode after ramping 0: Forced pulling mode 1: Estimator mode
[3:2]	CSM	Current sampling mode 00: Single-resistance current sampling mode 01: Dual-resistance current sampling mode 10: Reserved 11: Triple-resistance current sampling mode
[1]	RSV	Reserved
[0]	SVPWMEN	SVPWM enable 0: Disable 1: Enable

14.2.3 FOC_CR2 (0x40A1)

Bit	7	6	5	4	3	2	1	0
Name	ESEL	ICLR	F5SEG	DSS	CSOC		UQD	UDD
Type	R/W	R/W1	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	ESEL	Angle estimator mode selection 0: SMO 1: PLL, FOC_KSILDE register is KP of PLL PI controller and the FOC_KLPFMIN register is KI of PLL PI controller
[6]	ICLR	Clear FOC_IAMAX/FOC_IBMAX/FOC_ICMAX to 0 0: No meaning 1: Clear FOC_IAMAX/ FOC_IBMAX/ FOC_ICMAX to 0 and this bit will be cleared by hardware
[5]	F5SEG	SVPWM mode selection 0: Continuous SVPWM 1: Discontinuous SVPWM (cannot be selected in single-resistance current sampling mode)
[4]	DSS	Current sampling mode selection in dual/triple resistances mode 0: Concurrent sampling mode 1: Alternating sampling mode, FOC algorithm is calculated in every PWM period and only one phase current is sampled at each PWM period
[3:2]	CSOC	Current sampling offset calibration selection Write this bit to select the offset of FOC_CSO. In single-resistance sampling, write 00 or 11 to select itrip offset. In dual-resistances sampling, write 01 to select ia offset and

		write 10 to select ib offset. In triple-resistances sampling, write 01 to select ia offset, write 10 to select ib offset and write 00 or 11 to select ic offset. 00: itrip and ic 01: ia 10: ib 11: itrip and ic
[1]	UQD	q-Axis PI controller forbid. When FOC_CR2[UQD] is set to 1, FOC__UQ is not updated by PI controller 0: Not forbid 1: Forbid
[0]	UDD	d-Axis PI controller forbid. When FOC_CR2[UQD] is set to 1, FOC__UQ is not updated by PI controller 0: Not forbid 1: Forbid

14.2.4 FOC_TSMIN (0x40A2)

Bit	7	6	5	4	3	2	1	0
Name	FOC_TSMIN							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	FOC_TSMIN	Single resistance sampling mode: Minimum window for sampling Dual/triple resistances sampling mode: Deadtime compensation time Range: [0,255] TSMIN = sampling window T_{window} + deadtime T_{DT} Providing T_{window} = 1μs, T_{DT} = 1μs, TSMIN = 2μs, Ts = 62.5μs, FOC_TSMIN = (1 + 1)/62.5*4096 = 131

14.2.5 FOC_TGLI (0x40A3)

Bit	7	6	5	4	3	2	1	0
Name	FOC_TGLI							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	FOC_TGLI	High side of bridge narrow pulse elimination This function is used in high voltage applications. The high side of bridge must be longer than a certain time. Set this register to configure a value, and high side of bridge will not turn on if the conducting time is less than this value. Range: [0,255] Providing to eliminate pulse narrower than 1μs, T_{DT} = 1μs, Ts = 62.5μs FOC_TGLI = (1 + 1)/62.5*4096 = 131

14.2.6 FOC_TBLO (0x40A4)

Bit	7	6	5	4	3	2	1	0
Name	FOC_TBLO							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description

[7:0]	FOC_TBLO	Sampling masking time in triple-resistances sampling mode When the low side of bridge conducting time is less than FOC_TBLO, the current of this phase is not sampled and obtained by special process Range: [0,255] Providing the low side of bridge conducting time less than 1μs will not be sampled, FOC_TBLO = 1000ns/41.67ns = 24

14.2.7 FOC_TRGDLY (0x40A5)

Bit	7	6	5	4	3	2	1	0
Name	FOC_TRGDLY							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	FOC_TRGDLY	Current sampling time configuration When FOC_TRGDLY is set to 0, FOC module will sample current at Single-resistance mode: Midpoint of deadtime add active voltage vector applied time Dual/triple-resistance mode: Midpoint of voltage vector 000 (minima of Driver counter) Range: [–128,127] Single-resistance mode: FOC_TRGDLY = 5, the sampling time delay 5*T = 208ns; FOC_TRGDLY = –5, the sampling time advance 5*T=208ns Dual/triple-resistance mode: FOC_TRGDLY = 0x85, sample current 5*T = 208.3ns before Driver counter minima; FOC_TRGDLY = 0x05, sample current 5*T = 208.3ns after Driver counter minima

14.2.8 FOC_CSO (0x40A6, 0x40A7)

FOC_CSOH(0x40A6)								
Bit	15	14	13	12	11	10	9	8
Name	FOC_CSO[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	1	0	0	0	0	0	0
FOC_CSOL(0x40A7)								
Bit	7	6	5	4	3	2	1	0
Name	FOC_CSO[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_CSO	Current sampling offset Set FOC_CR2[CSOC] to select the corresponding current, write FOC_CSO to calibrate the current sampling offset of itrip in single-resistance mode, ia, ib in dual-resistances mode and ia, ib and ic in triple-resistances mode Range: [0,32767], MSB is always 0 Providing ADC sampling voltage range is 0V ~ 5V and offset is 2.5V, FOC_CSO = 2.5V/5V*32768 = 16384 (0x4000)

14.2.9 FOC__RTHESTEP (0x40A8, 0x40A9)

FOC__RTHESTEPH(0x40A8)								
Bit	15	14	13	12	11	10	9	8
Name	FOC__RTHESTEP[15:8]							

Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC__RTHESTEPL(0x40A9)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__RTHESTEP[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__RTHESTEP	Speed of ramping module, FOC__RTHESTEP is an internal 32-bit variable. MSB is sign bit. Software writes high 16 bits Range: [–32768,32767] FOC__RTHESTEP (32-bit) = FOC__RTHESTEP (32-bit) + FOC_RTHEACC (low 16 bits) THETA_OL (16-bit) = THETA_OL (16-bit) + FOC__RTHESTEP (high 16 bits)

14.2.10 FOC_RTHEACC (0x40AA, 0x40AB)

FOC_RTHEACCH(0x40AA)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_RTHEACC[15:8]							
Type	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0
FOC_RTHEACCL(0x40AB)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_RTHEACC[7:0]							
Type	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_RTHEACC	Ramping acceleration, FOC_RTHEACC is an internal 32-bit variable. MSB is sign bit. Software writes low 16 bits and high 16 bits are always 0 Range: [–32768,32767] FOC__RTHESTEP (32-bit) = FOC__RTHESTEP (32-bit) + FOC_RTHEACC (low 16 bits) THETA_OL (16-bit) = THETA_OL (16-bit) + FOC__RTHESTEP (high 16 bits)

14.2.11 FOC_EOMELPF (0x40AA, 0x40AB)

FOC_EOMELPFH(0x40AA)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_EOMELPF[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC_EOMELPFL(0x40AB)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_EOMELPF[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_EOMELPF	Filtered estimated motor speed

		The filter coefficient is FOC_EOMEKLPF and filtered speed is calculated in each PWM period Range: [–32768,32767]

14.2.12 FOC_RTHECNT (0x40AC)

Bit	7	6	5	4	3	2	1	0
Name	FOC_RTHECNT							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	FOC_RTHECNT	Ramping times = RTHECNT*256 When ramping function is enabled, the ramping angle will be increased in each PWM period. After RTHECNT*256 times, ramping function is finished

14.2.13 FOC_THECOR (0x40AD)

Bit	7	6	5	4	3	2	1	0
Name	FOC_THECOR							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R
Reset	0	0	0	0	0	0	0	1

Bit	Name	Description
[7:0]	FOC_THECOR	Step value of angle smooth switching The step value of angle smooth switching after ramping function. The format is the same as FOC__THETA Range: [0,255]

14.2.14 FOC__EMF (0x40AE, 0x40AF)

FOC__EMFH(0x40AE)								
Bit	15	14	13	12	11	10	9	8
Name	FOC__EMF[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__EMFL(0x40AF)								
Bit	7	6	5	4	3	2	1	0
Name	FOC__EMF[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__EMF	The BEMF magnitude of estimator The value is the root of sum of square of FOC__EALP and square of FOC__ EBET Range: [0,32767]

14.2.15 FOC_THECOMP (0x40AE, 0x40AF)

FOC_THECOMPH(0x40AE)								
Bit	15	14	13	12	11	10	9	8
Name	FOC_THECOMP[15:8]							
Type	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0

FOC_THECOMPL(0x40AF)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_THECOMP[7:0]							
Type	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_THECOMP	Angle compensation value (write only) The estimated angle (FOC__ETHETA) plus the compensation value is the output angle of estimator (FOC__THETA). The format is the same as FOC__THETA Range: [–32768,32767]

14.2.16 FOC_DMAX (0x40B0, 0x40B1)

FOC_DMAXH(0x40B0)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_DMAX[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_DMAXL(0x40B1)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_DMAX[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_DMAX	Maximum limit of d-axis PI controller output Range: [–32768,32767]

14.2.17 FOC_DMIN (0x40B2, 0x40B3)

FOC_DMINH(0x40B2)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_DMIN[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_DMINL(0x40B3)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_DMIN[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_DMIN	Minimum limit of d-axis PI controller output Range: [–32768,32767]

14.2.18 FOC_QMAX (0x40B4, 0x40B5)

FOC_QMAXH(0x40B4)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_QMAX[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

FOC_QMAXL(0x40B5)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_QMAX[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_QMAX	Maximum limit of q-axis PI controller output Range: [–32768,32767]

14.2.19 FOC_QMIN (0x40B6, 0x40B7)

FOC_QMINH(0x40B6)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_QMIN[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_QMINL(0x40B7)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_QMIN[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_QMIN	Minimum limit of q-axis PI controller output Range: [–32768,32767]

14.2.20 FOC__UD (0x40B8, 0x40B9)

FOC__UDH(0x40B8)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__UD[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC__UDL(0x40B9)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__UD[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__UD	The output of d-axis PI controller Range: [–32768,32767]

14.2.21 FOC__UQ (0x40BA, 0x40BB)

FOC__UQH(0x40BA)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__UQ[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC__UQL(0x40BB)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__UQ[7:0]							

Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__UQ	The output of q-axis PI controller Range: [–32768,32767]

14.2.22 FOC__ID (0x40BC, 0x40BD)

FOC__IDH(0x40BC)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__ID[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__IDL(0x40BD)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__ID[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__ID	The d-axis current from Park transformation Range: [–32768,32767]

14.2.23 FOC__IQ (0x40BE, 0x40BF)

FOC__IQH(0x40BE)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__IQ[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__IQL(0x40BF)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__IQ[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__IQ	The q-axis current from Park transformation Range: [–32768,32767]

14.2.24 FOC__IBET (0x40C0, 0x40C1)

FOC__IBETH(0x40C0)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__IBET[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__IBETL(0x40C1)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__IBET[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description

[15:0]	FOC__IBET	The β-axis current from Clarke transformation Range: [–32768,32767]

14.2.25 FOC__VBET (0x40C2, 0x40C3)

FOC__VBETH(0x40C2)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__VBET[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__VBETL(0x40C3)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__VBET[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__VBET	Output β-axis voltgage of FOC module Range: [–32768,32767]

14.2.26 FOC__VALP (0x40C4, 0x40C5)

FOC__VALPH(0x40C4)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__VALP[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__VALPL(0x40C5)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__VALP[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__VALP	Output α-axis voltgage of FOC module Range: [–32768,32767]

14.2.27 FOC_UDCPS (0x40C2, 0x40C3)

FOC_UDCPSH(0x40C2)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_UDCPS[15:8]							
Type	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0
FOC_UDCPSL(0x40C3)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_UDCPS[7:0]							
Type	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_UDCPS	d-Axis voltage compensation value The result of d-axis PI controller adding FOC_UDCPS is used for inverse Park transformation Range: [–32768,32767]

14.2.28 FOC_UQCPS (0x40C4, 0x40C5)

FOC_UQCPSH(0x40C4)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_UQCPS[15:8]							
Type	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0
FOC_UQCPSL(0x40C5)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_UQCPS[7:0]							
Type	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_UQCPS	q-Axis voltage compensation value The result of q-axis PI controller adding FOC_UQCPS is used for inverse Park transformation Range: [–32768,32767]

14.2.29 FOC__IC (0x40C6, 0x40C7)

FOC__ICH(0x40C6)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__IC[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__ICL(0x40C7)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__IC[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__IC	The sampling C phase current Range: [–32768,32767]

14.2.30 FOC__IB (0x40C8, 0x40C9)

FOC__IBH(0x40C8)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__IB[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__IBL(0x40C9)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__IB[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__IB	The sampling B phase current Range: [–32768,32767]

14.2.31 FOC__IA (0x40CA, 0x40CB)

FOC__IAH(0x40CA)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__IA[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__IAL(0x40CB)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__IA[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__IA	The sampling A phase current Range: [–32768,32767]

14.2.32 FOC__THETA (0x40CC, 0x40CD)

FOC__THETAH(0x40CC)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__THETA[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC__THETAL(0x40CD)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__THETA[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__THETA	Angle used for inverse Park transformation Range: [–32768,32767] –32768 ~ 32767 mapping –180°~ 180° Providing FOC__THETA = 8192, the angle is 8192/32768*180° = 45°

14.2.33 FOC__ETHETA (0x40CE, 0x40CF)

FOC__ETHETAH(0x40CE)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__ETHETA[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC__ETHETAL(0x40CF)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__ETHETA[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__ETHETA	Read: The angle estimated by estimator (before compensation), the format is the same as FOC__THETA Write: The initial angle in estimator Range: [–32768,32767]

14.2.34 FOC__EALP (0x40D0, 0x40D1)

FOC__EALPH(0x40D0)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__EALP[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__EALPL(0x40D1)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__EALP[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__EALP	The α-axis estimated BEMF Range: [–32768,32767]

14.2.35 FOC__EBET (0x40D2, 0x40D3)

FOC__EBETH(0x40D2)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__EBET[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__EBETL(0x40D3)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__EBET[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__EBET	The β-axis estimated BEMF Range: [–32768,32767]

14.2.36 FOC__EOME (0x40D4, 0x40D5)

FOC__EOMEH(0x40D4)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__EOME[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC__EOMEL(0x40D5)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__EOME[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__EOME	Estimator output speed obtained by filtering OME Range: [–32768,32767]

14.2.37 FOC__UQEX (0x40D6, 0x40D7)

FOC__UQEXH(0x40D6)

Bit	15	14	13	12	11	10	9	8
Name	FOC__UQEX[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__UQEXL(0x40D7)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__UQEX[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__UQEX	Overflow value of q-axis PI controller. FOC_UQEX = FOC__UQ – FOC_QMAX When FOC_UQ > FOC_QMAX, FOC_UQEX is positive When FOC_UQ < FOC_QMAX, FOC_UQEX is negative FOC_UQEX can be used to as the input of field weakening controller. Range: [–32768,32767]

14.2.38 FOC_KFG (0x40D6, 0x40D7)

FOC_KFGH(0x40D6)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_KFG[15:8]							
Type	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0
FOC_KFGL(0x40D7)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_KFG[7:0]							
Type	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_KFG	The coefficient of FG calculation FOC module calculates a result value based on FOC_EOMELPF and FOC_KFG in each PWM period. The result will update to TIM4__ARR and half value of the result will update to TIM4__DR by hardware More details please refer to FG Generation Note: The frequency division coefficients of Timer4 should be configured according to the motor maximum speed. When FOC_KFG = 0, this function is disabled. Range: [0,65535]

14.2.39 FOC__POW (0x40D8, 0x40D9)

FOC__POWH(0x40D8)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__POW[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__POWL(0x40D9)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__POW[7:0]							

Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__POW	Motor power Range: [–32768,32767]

14.2.40 FOC_EOMEKLPF (0x40D8)

Bit	7	6	5	4	3	2	1	0
Name	FOC_EOMEKLPF							
Type	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	FOC_EOMEKLPF	The coefficient of filtered estimated motor speed This LPF is calculated in every PWM period Range 1 ~ 255 mapping 1/32768 ~ 255/32768

14.2.41 FOC__IAMAX (0x40DA, 0x40DB)

FOC__IAMAXH(0x40DA)								
Bit	15	14	13	12	11	10	9	8
Name	FOC__IAMAX[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__IAMAXL(0x40DB)								
Bit	7	6	5	4	3	2	1	0
Name	FOC__IAMAX[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__IAMAX	A phase current maximum value Recorded maximum value of A phase current. This value could be unreliable unless motor rotate a full electrical period The current maximum value will not be cleared automatically. Set FOC_CR2[ICLR] to 1 to clear this register Range: [–32768,32767]

14.2.42 FOC__IBMAX (0x40DC, 0x40DD)

FOC__IBMAXH(0x40DC)								
Bit	15	14	13	12	11	10	9	8
Name	FOC__IBMAX[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__IBMAXL(0x40DD)								
Bit	7	6	5	4	3	2	1	0
Name	FOC__IBMAX[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__IBMAX	B phase current maximum value Recorded maximum value of B phase current. This value could be unreliable unless motor rotate a full electrical period The current maximum value will not be cleared automatically. Set FOC_CR2[ICLR] to 1 to clear this register Range: [–32768,32767]

14.2.43 FOC__ICMAX (0x40DE, 0x40DF)

FOC__ICMAXH(0x40DE)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__ICMAX [15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__ICMAXL(0x40DF)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__ICMAX [7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__ICMAX	C phase current maximum value Recorded maximum value of C phase current. This value could be unreliable unless motor rotate a full electrical period The current maximum value will not be cleared automatically. Set FOC_CR2[ICLR] to 1 to clear this register Range: [–32768,32767]

14.2.44 FOC_EFREQMAX (0x406F)

Bit	7	6	5	4	3	2	1	0
Name	FOC_EFREQMAX[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	1	1	1	1	1	1	1

Bit	Name	Description
[7:0]	FOC_EFREQMAX	The maximum of omega When omega[15:8] is bigger than FOC_EFREQMAX, the output speed OME is : FOC_CR0[OMAS] = 0: FOC_EFREQMAX*256 FOC_CR0[OMAS] = 1: FOC_EFREQHOLD Range: [0,127] 0 ~ 127 mapping omega 0 ~ 32767 Note: This function will be disabled when MSB of FOC_EFREQMAX is 1

14.2.45 FOC_EKP (0x4074, 0x4075)

FOC_EKPH(0x4074)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**

Name	FOC_EKP[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_EKPL(0x4075)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_EKP[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_EKP	PI controller KP coefficient of angle estimator. MSB is always 0, Q12 format Range: [0,32767]

14.2.46 FOC_EKI (0x4076, 0x4077)

FOC_EKIH(0x4076)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_EKI[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_EKIL(0x4077)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_EKI[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_EKI	PI controller KI coefficient of angle estimator. MSB is always 0, Q15 format Range: [0,32767]

14.2.47 FOC_EBMFK (0x407C, 0x407D)

FOC_EBMFKH(0x407C)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_EBMFK[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_EBMFKL(0x407D)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_EBMFK[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_EBMFK	LPF coefficient (EKLPF) of BEMF low pass filter setting, Q15 format Range: [0,32767] EKLPF = FOC_EBMFK*FOC__EOME FOC_EBMFK = 2*π*fbase*Ts

14.2.48 FOC_KSLIDE (0x4078, 0x4079)

FOC_KSLIDEH(0x4078)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_KSLIDE/FOC_PLLKP[15:8]							

Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_KSLIDEL(0x4079)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_KSLIDE/FOC_PLLKP [7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_KSLIDE /FOC_PLLKP	FOC_CR2[ESEL] = 0: Gain coefficient of SMO, Q15 format FOC_CR2[ESEL] = 1: PI controller KP coefficient of PLL, Q12 format Range: [0,32767], MSB is always 0

14.2.49 FOC_EKLPFMIN (0x407A, 0x407B)

FOC_EKLPFMINH(0x407A)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_EKLPFMIN/FOC_PLLKPI[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_EKLPFMINH(0x407B)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_EKLPFMIN/FOC_PLLKPI[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_EKLPFMIN /FOC_PLLKI	FOC_CR2[ESEL] = 0: The minimum value of BEMF low pass filter factor. EKLPF will be forced to this value when it is lower than this value. Q15 format FOC_CR2[ESEL] = 1: PI controller KI coefficient of PLL, Q12 format Range: [0,32767], MSB is always 0

14.2.50 FOC_OMEKLPF (0x407E, 0x407F)

FOC_OMEKLPFH(0x407E)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_OMEKLPF[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_OMEKLPFL(0x407F)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_OMEKLPF[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_OMEKLPF	The LPF coefficient of FOC__EOME, MSB is always 0, Q15 format Range: [0,32767]

14.2.51 FOC_FBASE (0x4080, 0x4081)

FOC_FBASEH(0x4080)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**

Name	FOC_FBASE[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_FBASEL(0x4081)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_FBASE[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_FBASE	Frequency base of estimator Range: [0,32767] FOC_FBASE = fbase*Ts*32768 Providing fbase = 200Hz, Ts = 62.5μs, FOC_FBASE = 200*0.0000625*32768 = 409(0x199)

14.2.52 FOC_EFREQACC (0x4082, 0x4083)

FOC_EFREQACCH(0x4082)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_EFREQACC[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_EFREQACCL(0x4083)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_EFREQACC[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_EFREQACC	Speed increment of the forced angle mode. FOC_EFREQACC is an internal 24-bit variable and MSB is sign bit. Software writes low 16 bits Range: [0,65535] Providing fbase = 200Hz, pole pairs (pp) = 4, speed_base = 60*fbase/pp = 3000rpm. To set speed increment to 3rpm, FOC_EFREQACC = 3rpm/speed_base*32768*256 = 8388 (0x20C4)

14.2.53 FOC_EFREQMIN (0x4084, 0x4085)

FOC_EFREQMINH(0x4084)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_EFREQMIN[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_EFREQMINL(0x4085)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_EFREQMIN[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_EFREQMIN	Switch speed of estimator estimated angle mode, FOC_EFREQMIN is an internal 24–bit variable, MSB is sign bit. Software write low high 16 bits When omega is bigger than or equal to FOC_EFREQMIN, omega is

		used as OME Range: [–32768,32767] Providing fbase = 200Hz, pole pairs pp = 4, speed_base = 60*fbase/pp = 3000rpm. To set switch speed to 30rpm, FOC_EFREQMIN = 30rpm/speed_base*32768 = 327 (0x147)

14.2.54 FOC_EFREQHOLD (0x4086, 0x4087)

FOC_EFREQHOLDH(0x4086)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_EFREQHOLD[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_EFREQHOLDL(0x4087)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_EFREQHOLD[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	FOC_EFREQHOLD	The maximum value of estimator forced speed, FOC_EFREQHOLD is an internal 24-bit variable, MSB is sign bit. Software write high 16 bits Estimator forced speed will increase to the speed set by this register and stops increment Range: [–32768,32767] Providing fbase = 200Hz, pole pairs pp = 4, speed_base = 60*fbase/pp = 3000rpm. To set maximum forced speed to 60rpm, FOC_EFREQHOLD = 60rpm/speed_base*32768 = 655 (0x028F)

14.2.55 FOC_EK3 (0x4088, 0x4089)

FOC_EK3H(0x4088)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_EK3[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_EK3L(0x4089)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_EK3[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	FOC_EK3	The 3rd coefficient of the current model in estimator, MSB is always 0, Q15 format Range: [–32768,32767]

14.2.56 FOC_EK4 (0x408A, 0x408B)

FOC_EK4H(0x408A)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_EK4[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

FOC_EK4L(0x408B)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_EK4[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_EK4	The 4th coefficient of the current model in estimator, Q15 format Range: [–32768,32767]

14.2.57 FOC_EK1 (0x408C, 0x408D)

FOC_EK1H(0x408C)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_EK1[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_EK1L(0x408D)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_EK1[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_EK1	The 1st coefficient of the current model in estimator, MSB is always 0, Q15 format Range: [0,32767]

14.2.58 FOC_EK2 (0x408E, 0x408F)

FOC_EK2H(0x408E)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_EK2[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_EK2L(0x408F)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC_EK2[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_EK2	The 2nd coefficient of the current model in estimator, MSB is always 0, Q15 format Range: [0,32767]

14.2.59 FOC_IDREF (0x4090, 0x4091)

FOC_IDREFH(0x4090)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC_IDREF[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_IDREFL(0x4091)								

Bit	7	6	5	4	3	2	1	0
Name	FOC_IDREF[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_IDREF	Reference of d-axis current PI controller Range: [–32768,32767]

14.2.60 FOC_IQREF (0x4092, 0x4093)

FOC_IQREFH(0x4092)								
Bit	15	14	13	12	11	10	9	8
Name	FOC_IQREF[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_IQREFL(0x4093)								
Bit	7	6	5	4	3	2	1	0
Name	FOC_IQREF[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_IQREF	Reference of q-axis current PI controller Range: [–32768,32767]

14.2.61 FOC_DQKP (0x4094, 0x4095)

FOC_DQKPH(0x4094)								
Bit	15	14	13	12	11	10	9	8
Name	FOC_DQKP[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_DQKPL(0x4095)								
Bit	7	6	5	4	3	2	1	0
Name	FOC_DQKP[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_DQKP	KP coefficient of dq-axis PI controller, MSB is always 0, Q12 format Range: [0,32767]

14.2.62 FOC_DQKI (0x4096, 0x4097)

FOC_DQKIH(0x4096)								
Bit	15	14	13	12	11	10	9	8
Name	FOC_DQKI[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
FOC_DQKIL(0x4097)								
Bit	7	6	5	4	3	2	1	0
Name	FOC_DQKI[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC_DQKI	KI coefficient of dq-axis PI controller, MSB is always 0, Q15 format Range: [0,32767]

14.2.63 FOC__UDCFLT (0x4098, 0x4099)

FOC__UDCFLTH(0x4098)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	FOC__UDCFLT[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
FOC__UDCFLTL(0x4099)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	FOC__UDCFLT[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	FOC__UDCFLT	Filtered bus voltage The FOC module samples the bus voltage and filters it for motor control. ADC channel 2 (external voltage resistor division) or channel 14 (internal voltage resistor division) can be selected Range: [0,32767] Providing the sampled bus voltage is divided by 6 and ADC voltage range is 0V ~ 5V, the sampling range of bus voltage is 0V ~ 30V. Providing FOC_UDCFLT is 19661(0x4CCD), the bus voltage is 19661/32768*5V*6 = 18V.

15 Timer1

15.1 Timer1 Instructions

Timer1 consists of a 16-bit up counting base counter and a 16-bit up counting reload counter. Timer1 can be used in the applications of square wave controlled BLDC motor drive. Timer1 features as following:

- The 16-bit up counting base counter is used to record the time between two position detected events or two phases commutations (60 degree time) and also can be used to forced commutation control when phase detection is failure.
- The 16-bit up counting reload counter is used to control the time from position detected to phase commutation, as well as diode freewheeling masking time, in which position detection is disabled
- The 3-bit programmable frequency prescaler divides the system clock. The divided clock is used as the clock source of the two counters
- Configurable input filtering and sampling time
- Position detection module generates the position signal required for phase commutation according to the input signal
- 7 groups state register control comparator and pre-driver output
- 6 Interrupt sources

The internal structure of Timer1 is shown in Figure 15-1.

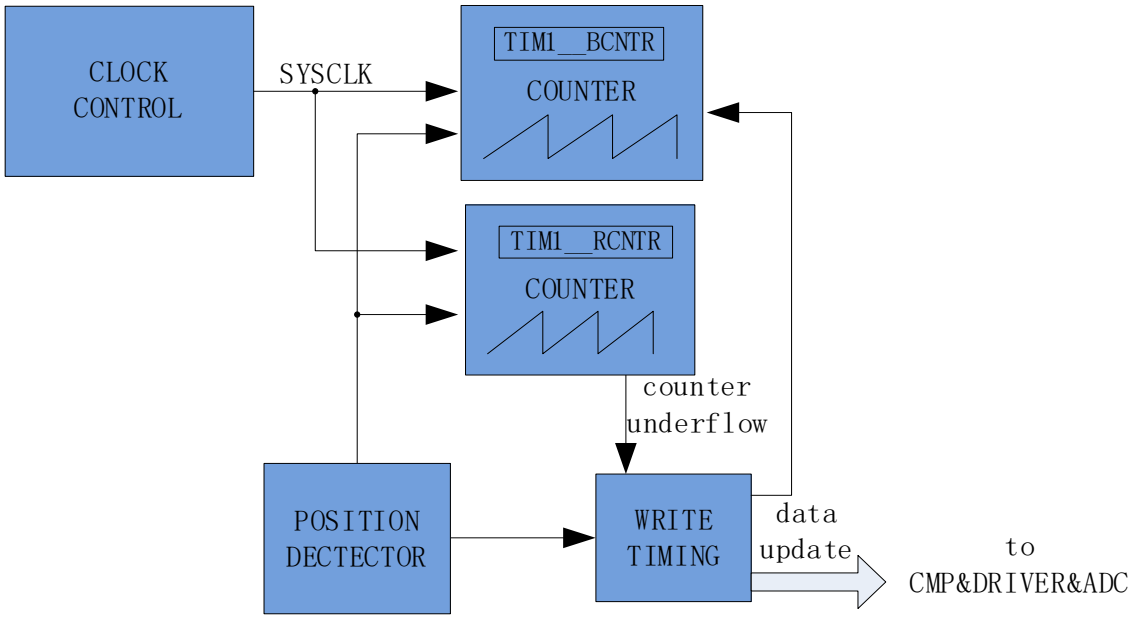


Figure 15-1 Internal Structure of Timer1

15.1.1 Timer1 Counting Units

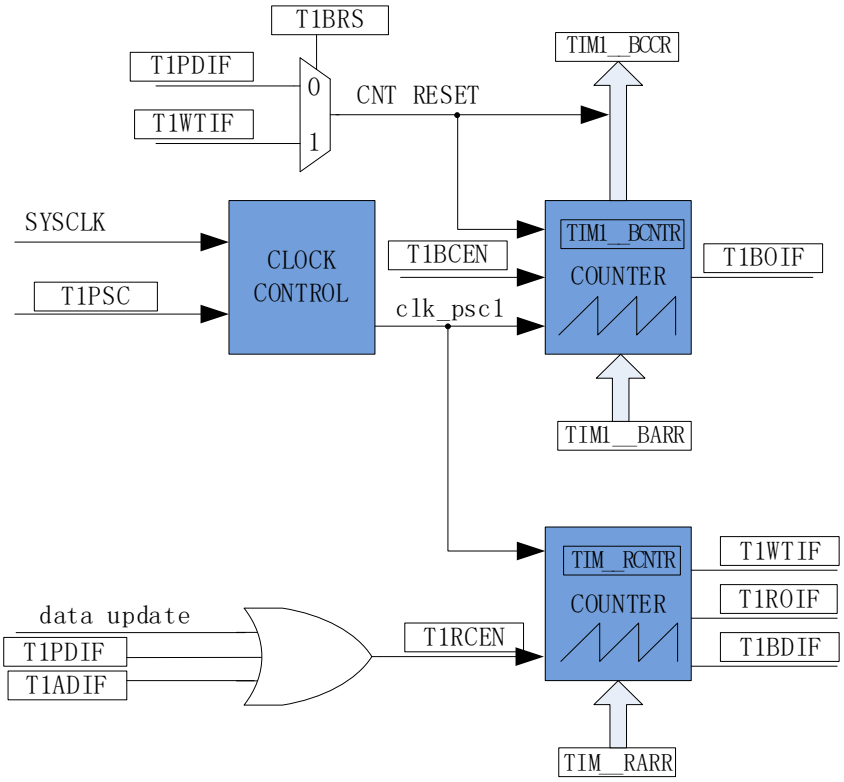


Figure 15-2 Timer Counting Units

Timer1 consists of a frequency prescaler, an up counting base counter and an up counting reload counter.

15.1.1.1 Frequency Prescaler

Frequency prescaler is used to divide the system clock and generate clock source of base counter and reload counter. 8 frequency division coefficients of prescaler are available and can be selected by TIM1_CR3[T1PSC]. Since this register has no buffer, the clock source frequency will be changed immediately after TIM1_CR3[T1PSC] has been written. Therefore, the frequency division coefficients should be configured when both the base counter and reload counter are not working. The clock source frequency formula is: clk_psc1 = SYSCLK/(2^T1PSC). The clock frequency corresponding to TIM1_CR3[T1PSC] is shown in Table 15-1.

Table 15-1 The Clock Frequency Corresponding to TIM1_CR3[T1PSC]

TIM1_CR3[T1PSC]	Coefficient	clk_psc1(Hz)	TIM1_CR3[T1PSC]	Coefficient	clk_psc1(Hz)
000	1	24M	100	16	1.5M
001	2	12M	101	32	750k
010	4	6M	110	64	375k
011	8	3M	111	128	187.5k

15.1.1.2 Base Counter

The base counter is a 16-bit up counter, and the count value is TIM1__BCNTR. When a position detected interrupt (TIM1_SR[T1PDIF]) or a writing sequence interrupt (TIM1_SR[T1WTIF]), selected by TIM1_CR2[T1BRS], is generated, TIM1__BCNTR is recorded into capture register TIM1__BCCR and TIM1__BCNTR restarts from 0. TIM1__BCCR captures the time between twice position detected interrupt or twice writing sequence interrupt, i.e., motor 60 degree. This value of time could be averaged by the previous value, factor configurable set by TIM1_CR0[T1CFLT]. The result is recorded as 60 degree reference time sent into TIM1_BCOR. Set TIM1_CR1[BAPE] to 1 to enable TIM1__BARR auto load function, TIM1__BARR will load the value of TIM1_BCOR by hardware. When the count value of TIM1__BCNTR increases to TIM1__BARR, the base counter overflow interrupt flag TIM1_SR[T1BOIF] is set to 1. If forced phase commutation is enabled, writing sequence event will be generated and the value of base timer will be cleared. If forced phase commutation is disabled, the base counter will increase to 0xFFFF and be cleared.

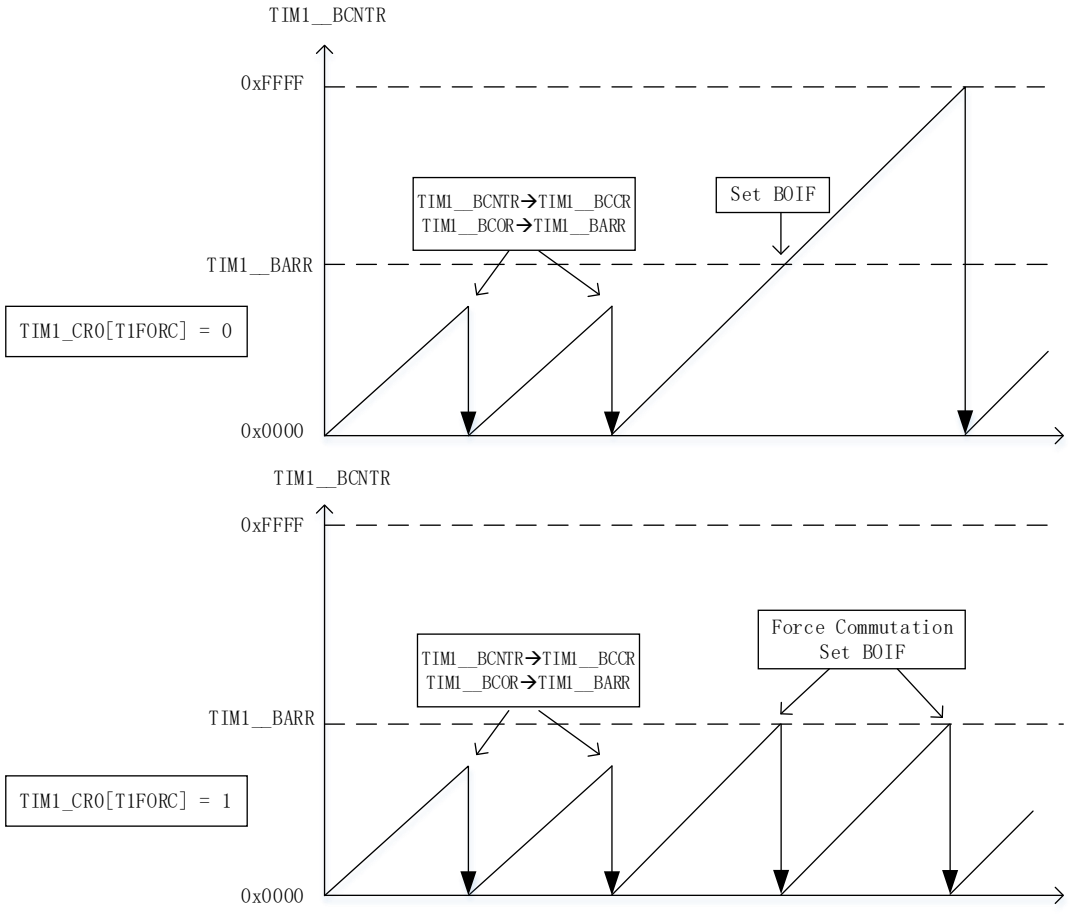


Figure 15-3 Waveform of Base Counter

When software mode is enable (TIM1_IER[T1MAME] = 1), TIM1__BCNTR is cleared by base counter overflow event instead of TIM1_CR2[T1BRS].

15.1.1.3 Reload Counter

The reload counter is a 16-bit up counter, and the count value is TIM1__RCNTR. When the counter value of TIM1__RCNTR equals to TIM1__RARR, overflow event is generated. The reload counter overflow interrupt flag TIM1_SR[T1ROIF] is set to 1. TIM1__RCNTR and TIM1_CR0[T1RCEN] are cleared and counting will not be restarted until TIM1_CR0[T1RCEN] is set to 1. TIM1_CR0[T1RCEN] could be set to 1 to restart reload counter counting when position detected interrupt or writing sequence interrupt is generated.

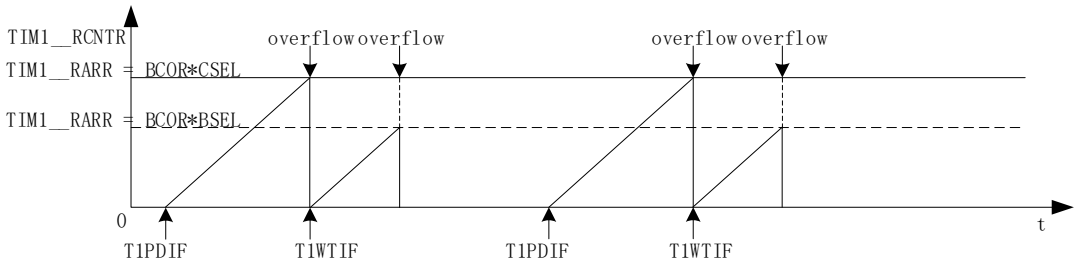


Figure 15-4 Waveform of Reload Counter

15.1.2 Position Detection

15.1.2.1 Position Detection Signal

TIM1_CR3[T1TIS] select position detection signal from CMP0/1/2 (CMP based position detection), GPIO (Hall effected position detection) or ADC (ADC based position detection). The configuration of CMP_CR1[HALLSEL] selects GPIO from P1.4/P1.6/P2.1 (after function transfer) or P0.2/P3.7/P3.6. The configuration of TIM1_CR3[T1INM] select whether the signal of CMP/GPIO is filtered. When a predefined signal trigger source is detected, the position detected interrupt is generated. Position detected event has 2 kinds of event, CMP/GPIO position detected event and ADC position detected event.

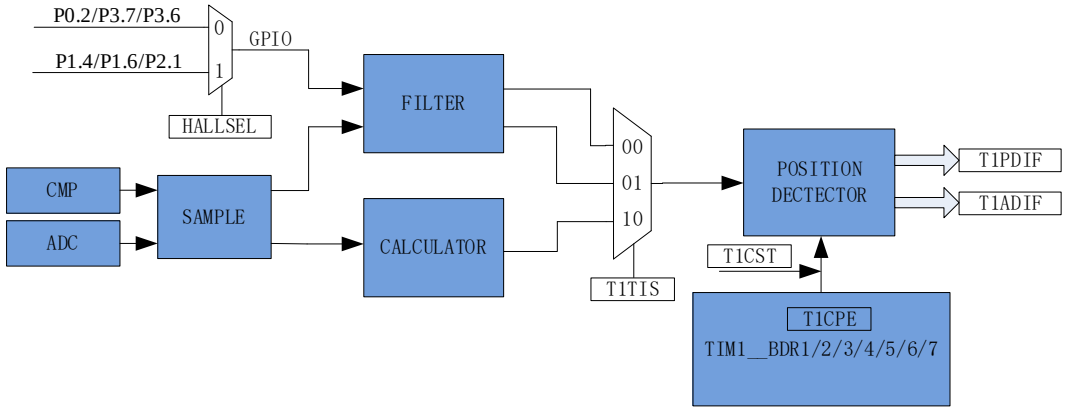


Figure 15-5 Function Block Diagram of Position Detection

15.1.2.2 CMP/GPIO Position Detected Event

Configure the register group TIM1_DBR1/2/3/4/5/6/7[T1CPE] to select the active edge group of position detection signal. When the active edge of CMP/GPIO position detection signal is detected, the position of motor is detected and CMP/GPIO position detected interrupt flag (TIM1_SR[T1PDIF]) is set to 1. The sequence of TIM1_DBR1/2/3/4/5/6/7[T1CPE] is controlled by TIM1_CR4[T1CST].

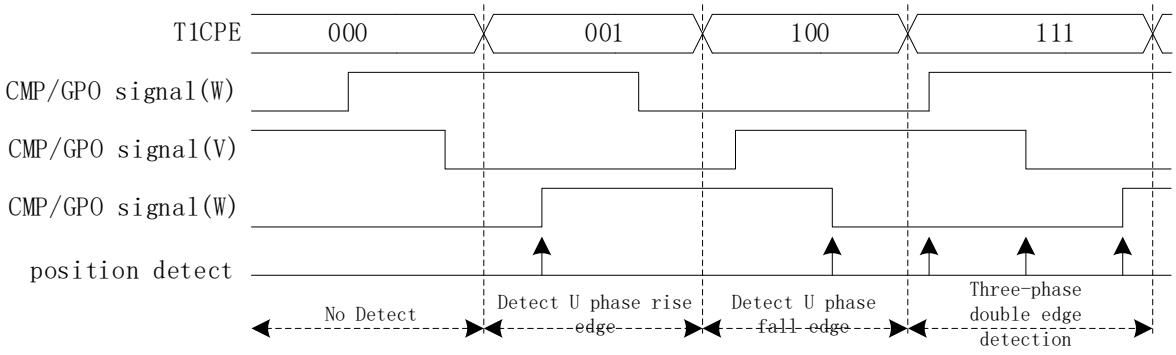


Figure 15-6 Timing Diagram of CMP/GPIO Position Detection

The relation between active edge and TIM1_DBR1/2/3/4/5/6/7[T1CPE] is as shown in Table 15-2.

Table 15-2 The Relation between Active Edge and TIM1_DBR1/2/3/4/5/6/7[T1CPE]

T1CPE	Description	T1CPE	Description
000	0	100	Detect Falling edge of U phase and U phase corresponding comparator is enabled
001	Detect Rising edge of U phase and U phase corresponding comparator is enabled	101	Detect Rising edge of W phase and W phase corresponding comparator is enabled
010	Detect Falling edge of W phase and W phase corresponding comparator is enabled	110	Detect Falling edge of V phase and V phase corresponding comparator is enabled
011	Detect Rising edge of V phase and V phase corresponding comparator is enabled	111	Detect 3-phase both edge and 3-phase corresponding comparators are enabled

15.1.2.3 ADC Position Detected Event

Set TIM1_CR3[T1TIS] = 10 to select the position detection signal from ADC. Timer1 control ADC to sample the voltage of active phase and floating phase. To get the position information, Timer1 calculate format as following:

$$\mathrm{TIM1_URES} = K \times \mathrm{TIM1_UCOP} - \mathrm{TIM1_UFLP}$$

where,

K: ADC position detection factor

TIM1__UCOP: ADC sampling value of active phase

TIM1__UFLP: ADC sampling value of floating phase

TIM1_DBR1/2/3/4/5/6/7[T1CPE] control the specific variables represented by K, TIM1__UCOP and TIM1__UFLP. The Relation between TIM1_DBR1/2/3/4/5/6/7[T1CPE] and K, TIM1__UCOP, TIM1__UFLP is shown in Table 15-3.

Table 15-3 The Relation between TIM1_DBR1/2/3/4/5/6/7[T1CPE] and K, TIM1__UCOP, TIM1__UFLP

T1CPE	Description
000	Reserved
001	K is TIM1_KR, TIM1__UCOP is W phase voltage, TIM1__UFLP is U phase voltage
010	K is TIM1_KF, TIM1__UCOP is U phase voltage, TIM1__UFLP is W phase voltage
011	K is TIM1_KR, TIM1__UCOP is U phase voltage, TIM1__UFLP is V phase voltage
100	K is TIM1_KF, TIM1__UCOP is V phase voltage, TIM1__UFLP is U phase voltage
101	K is TIM1_KR, TIM1__UCOP is V phase voltage, TIM1__UFLP is W phase voltage
110	K is TIM1_KF, TIM1__UCOP is W phase voltage, TIM1__UFLP is V phase voltage
111	Reserved

When TIM1__URES has a negative step or a positive step, ADC position detected event is generated and ADC position detected interrupt flag (TIM1_SR[T1ADIF]) is set to 1. The time of ADC position detected event is controllable by setting factor K. In this method, the phase commutation degree can be controlled flexibly.

15.1.2.4 Sampling

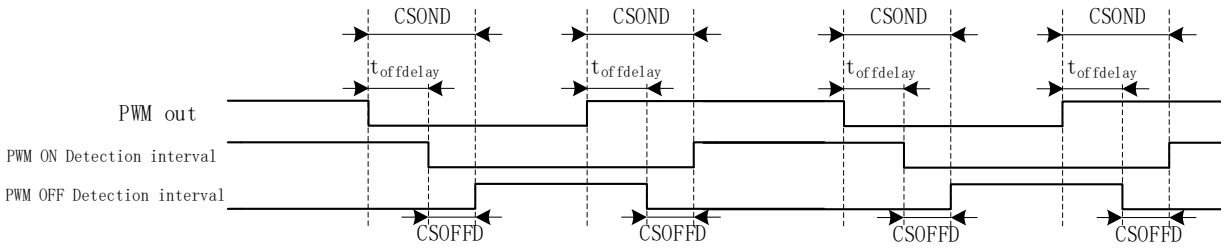


Figure 15-7 Time Diagram of Sampling

Due to the influence of the inverter nonlinearity, the BEMF signal lags the PWM output of chip. CMP_SAMR[CSOFFD], CMP_SAMR [CSOND] and CMP_CR4[FAEN] should be set reasonably to adjust the sampling interval and obtain the effective position detection signal. When TIM1_CR3[T1TIS] = 01, Timer1 enables CMP0/1/2 to output the compare results between phase BEMF and neutral point. When TIM1_CR3[T1TIS] = 10, Timer1 triggers ADC to sample floating voltage.

More details please refer to Comparator Sampling.

15.1.2.5 Filtering

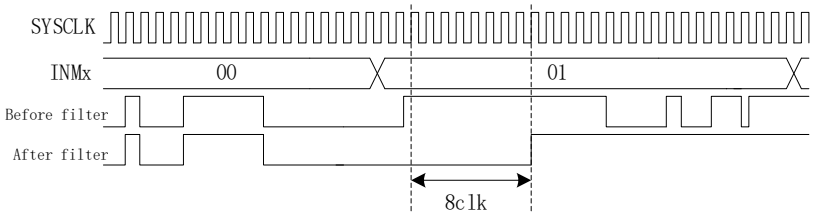


Figure 15-8 Timing Diagram of Filter Module

According to TIM1_CR3[T1INM] and CMP_CR4[FAEN], the filtering out pulse width of input noise can be selected as 8/16/24/32/64/96 system clocks. After enable the filtering, the signal will be lagged about 8/16/24/32/64/96 system clocks.

15.1.3 Writing Sequence Event

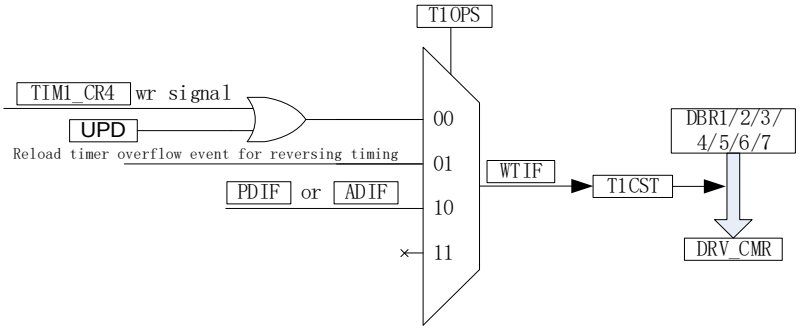


Figure 15-9 Block Diagram of Writing Sequence

When the control logic, predefined in TIM1_DBR1/2/3/4/5/6/7, is sent to driver register DRV_CMR, a writing sequence interrupt is generated. The triggered source is selected by the configuration of TIM1_CR0[T1OPS], and software, reload counter overflow event or position detected event can be selected. When a writing sequence interrupt is generated, writing sequence interrupt flag is set to 1. If the value of TIM1_CR4[T1CST] is in 001 ~ 110, TIM1_CR4[T1CST] will add 1 automatically.

15.1.4 Timer1 Interrupt

Timer1 has 6 interrupt sources:

- Base counter overflow interrupt
- Reload counter overflow interrupt
- Writing sequence interrupt
- Diode freewheeling end interrupt
- CMP/GPIO position detected interrupt
- ADC position detected interrupt

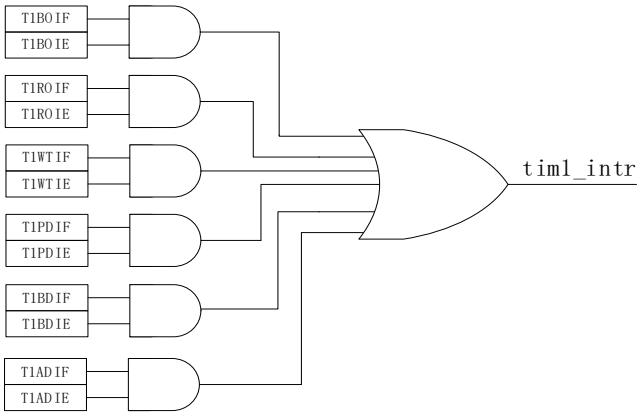


Figure 15-10 Timer1 Interrupt Sources

15.2 Square Wave Control based BLDC Motor

For BLDC motor square wave control application, Timer1 works with CMP0/1/2 and Driver module to achieve the following functions:

- Automatic record of 60 degree time, filtered as 60 degree reference time
- Automatic forced phase commutation when position signal is not detected
- Automatic diode freewheeling masking, i.e., stopping position detection during diode freewheeling time
- Automatic control of the time from position detected to phase commutation to achieve automatic commutation
- Take over CMP_CR2[CMP0SEL] to control CMP0/1/2 automatically
- Comparator signal can be set to avoid power device switching oscillation for sampling, and the signal can be configured to be filtered after sampling
- Take over DRV_CMR register to control 6 PWM outputs automatically

15.2.1 Six-Step Phase Commutation of Square Wave Control

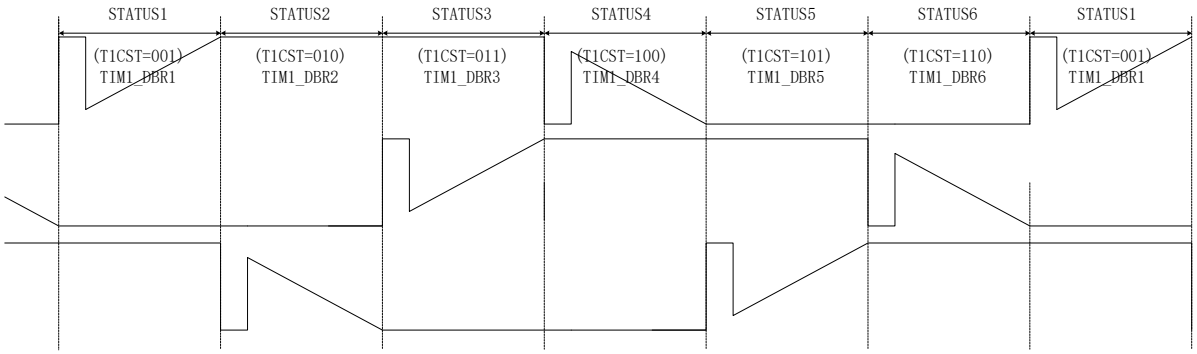


Figure 15-11 Diagram of Six-Step Phase Commutation of Square Wave Control

TIM1_CR4[T1CST] is the commutation state machine. Among them, state 0 is used to output off state, state 7 is customizable for braking, pre-charging, pre-positioning, starting and other functions. States 1 ~ 6 are used for six-step automatic commutation, and the state machine TIM1_CR4[T1CST] automatically adds 1 after the phase commutation.

The states 1 ~ 7 maps to the TIM1_DBR1 ~ 7. When the writing sequence interrupt occurs, the TIM1_DBRx corresponding to the current state will be automatically transferred to DRV_CMR and CMP_CR2[CMP0SEL] to achieve phase commutation and position detection.

15.2.2 Square Wave Control Working Principle

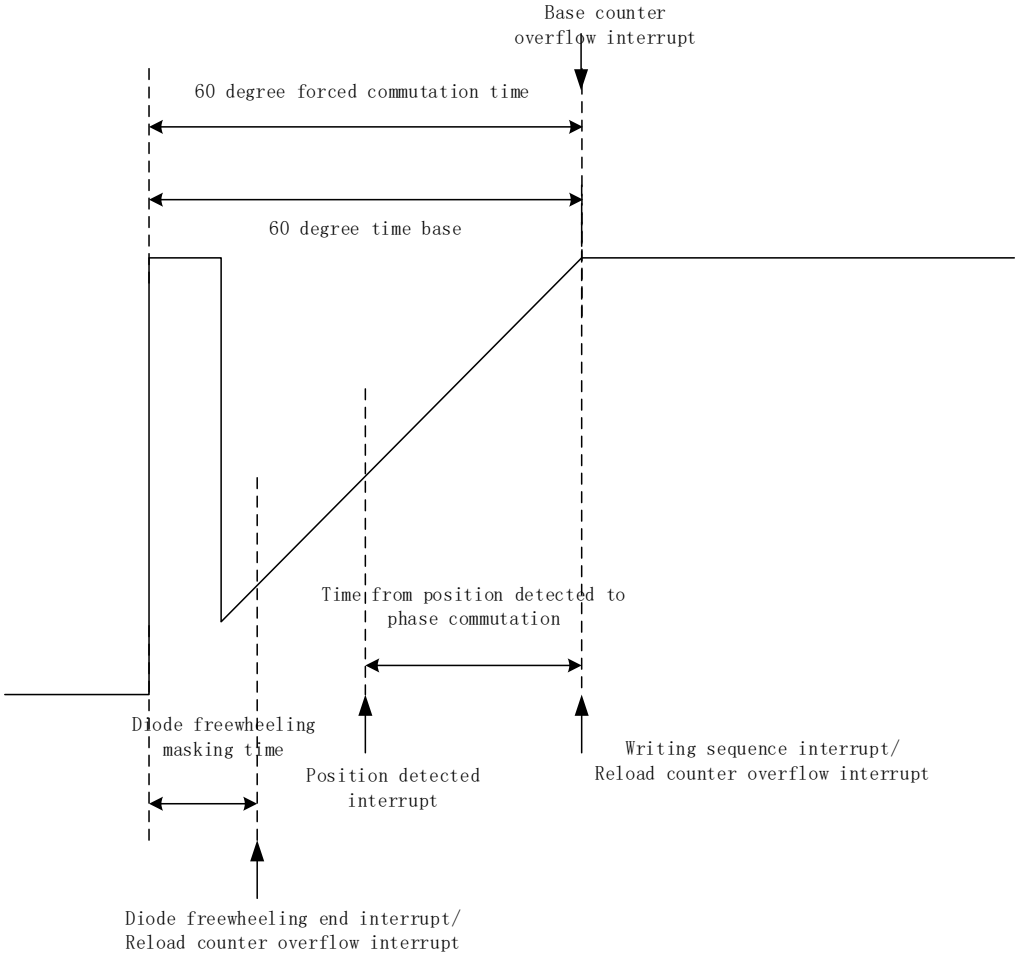


Figure 15-12 Diagram of Square Wave Control Working Principle

15.2.2.1 60 Degree Time Base

TIM1__BCCR captures the time of last 60 degree. Set TIM1_CR2[T1BRS] = 0 to capture the time between two writing sequence interrupts and set TIM1_CR2[T1BRS] = 1 to capture the time between two position detected interrupts.

TIM1_BCOR is the filtered 60 degree time, i.e., 60 degree time base. TIM1_CR0[CFLT] can select the last 1/2/4/8 TIM1__BCCR averaged to get TIM1_BCOR.

In the square wave control mode, the diode freewheeling masking time, the time from position detected to commutation, and the time to force commutation are determined according to the 60 degree time base TIM1_BCOR.

When the base timer is auto-load enabled (TIM1_CR1[T1BAPE] = 1) and the base timer is reset due to a position detection interrupt or a write timing interrupt, the value of TIM1_BCOR is transferred to TIM1_BARR to control the forced phase commutation.

15.2.2.2 60 Degree Forced Commutation

When the motor is rotating smoothly, the ZCP is generally detected after 30 degrees of rotation after a phase commutation and generates a position detection interrupt. If the ZCP is not detected in 60 degree after the phase commutation, the position detection fails and a forced commutation is required. Set TIM1_CR0[FORC] = 1 to enable the forced commutation function. At the previous commutation, the write timing interrupt clears the counter TIM1_BCNTR to 0 and restarts counting, while TIM1_BCCR captures the count value of TIM1_BCNTR, which is filtered and stored in TIM1_BCOR as the 60 degree time base. When the auto-load function is enabled (TIM1_CR1[T1BAPE] = 1), the value of TIM1_BCOR is loaded into TIM1_BARR when the base counter is cleared. When TIM1_BCNTR matches TIM1_BARR (no ZCP detected in 60 degree after commutation), the base counter overflow interrupt flag TIM1_SR[T1BOIF] is set to 1 for forced phase commutation, at this time the counter TIM1_BCNTR is cleared to 0, and will not start counting again. When an ZCP detected within 60 degrees after the phase commutation, even if TIM1__BCNTR > TIM1__BARR, the forced commutation will not be triggered and TIM1_SR[T1BOIF] will not be set to 1. When the forced commutation function is disabled (TIM1_CR0[FORC] = 0) and TIM1__BCNTR > TIM1__BARR, the interrupt flag TIM1_SR[T1BOIF] is set to 1 and no forced phase commutation is automatically performed. Manual phase commutation can be performed by the base counter overflow interrupt flag and the position detected interrupt flag.

15.2.2.3 Diode Freewheeling Masking

After the commutation, the inductance energy of the phase is released to the power supply or ground through the diode since the original active phase becomes a floating phase. During the diode freewheeling process, the floating phase BEMF signal cannot be measured. By masking the comparator signal or ADC sampling value during the diode freewheeling time, the wrong signal generated by the freewheeling is avoided to cause wrong commutation. At the end of freewheeling masking, the freewheeling masking end interrupt flag TIM1_SR[T1BDIF] will be generated.

Freewheeling masking time is set by TIM1_CR1[BSEL] with the formula: Masking angle = TIM1_CR1[BSEL]/128*60°.

15.2.2.4 Angle of Position Detected to Commutation

After the commutation, a ZCP is detected (generating a position detected interrupt) and the hardware starts counting according to the software-set time between ZCP and the commutation. After the counting ends, the hardware automatically implements phase commutation and generates the write sequence interrupt flag TIM1_SR[T1WTIF].

The time between ZCP and commutation is set by TIM1_CR2[CSEL] with the formula: Commutation angle = TIM1_CR2[CSEL]/128*60°.

15.2.2.5 Cycle-by-Cycle Current Limiting

Refer to 29.1.1.2 for cycle-by-cycle current limiting.

15.3 Timer1 Registers

15.3.1 TIM1_CR0 (0x4068)

Bit	7	6	5	4	3	2	1	0
Name	T1RWEN	T1CFLT		T1FORC	T1OPS		T1BCEN	T1RCEN
Type	W1	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	T1RWEN	TIM1_CR0[TIRCEN] write enable 0: No meaning 1: When operating TIM1_CR0, TIM1_CR0[T1RWEN] and TIM1_CR0[T1RCEN] should be configured simultaneously to enable or disable TIM1_CR0[T1RCEN]. Write 0x81 to TIM1_CR0 to enable TIM1_CR0[T1RCEN] and write 0x80 to disable TIM1_CR0[T1RCEN]
[6:5]	T1CFLT	60 degree time base filtering selection The average of previous x times 60 degree is used as the time base 00: 1 times 60 degree 01: 2 times 60 degree 10: 4 times 60 degree 11: 8 times 60 degree
[4]	T1FORC	60 degree forced phase commutation enable 0: Disable 1: Enable Note: If ZCP has been detected, forced phase commutation will not be implemented regardless of this bit
[3:2]	T1OPS	Phase commutation trigger signal selection Select the trigger signal of TIM1_DBRx data sent to DRV_CMR. 00: Write 1 to TIM1_IER[T1UPD] or writeTIM1_CR4[T1CST] 01: The overflow event of reload counter for phase commutation 10: Position detected event 11: Reserved
[1]	T1BCEN	Base counter enable 0: Disable 1: Enable
[0]	T1RCEN	Reload counter enable When operating TIM1_CR0, TIM1_CR0[T1RWEN] and TIM1_CR0[T1RCEN] must be operated simultaneously to enable or disable TIM1_CR0[T1RCEN]. Write 0x81 to TIM1_CR0 to enable TIM1_CR0[T1RCEN] and write 0x80 to disable TIM1_CR0[T1RCEN] Position detected interrupt and writing sequence interrupt can automatically enable TIM1_CR0[T1RCEN]. TIM1_CR0[T1RCEN] will be cleared by hardware when the Reload counter overflow interrupt is generated In manual mode, the hardware enable and disable functions of TIM1_CR0[T1RCEN] are invalid 0: Disable 1: Enable

15.3.2 TIM1_CR1 (0x4069)

Bit	7	6	5	4	3	2	1	0
Name	T1BAPE	BSEL						

Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	T1BAPE	TIM1_BARR auto-load enable. After enabling, when the base counter is cleared due to a position detected event or a writing sequence event, write TIM1__BCOR to TIM1_BARR. This function will not be affected by manual mode 0: Disable 1: Enable
[6:0]	BSEL	Diode freewheeling angle selection Angle of diode freewheeling after phase commutation. During diode freewheeling time, the position is not detected Diode freewheeling angle = TIM1_CR1[BSEL]/128*60° Note: In manual mode, the diode freewheeling angle function is invalid

15.3.3 TIM1_CR2 (0x406A)

Bit	7	6	5	4	3	2	1	0
Name	T1BRS	CSEL						
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	T1BRS	Base counter reset source selection In manual mode, this function is invalid, and the base counter TIM1_BCNTR can only be cleared by base counter overflow event 0: Writing sequence interrupt 1: Position detected interrupt
[6:0]	CSEL	Phase commutation angle selection After a position detected event, the phase commutation will be implemented after the degree configured by TIM1_CR2[CSEL] Commutation angle = TIM1_CR2[CSEL]/128*60°

15.3.4 TIM1_CR3 (0x406B)

Bit	7		6	5	4	3	2	1	0
Name	RSV		T1PSC			T1TIS		T1INM	
Type	—	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	—	0	0	0	0	1	0	0

Bit	Name	Description
[7]	RSV	Reserved
[6:4]	T1PSC	Counter clock source frequency selection These bits are used to divide the system clock as the clock source for the base counter and the reload counter. The clock source frequency of the two counters: 000: 24MHz 001: 12MHz 010: 6MHz 011: 3MHz 100: 1.5MHz 101: 750kHz 110: 375kHz 111: 187.5kHz
[3:2]	T1TIS	Position detection signal selection 00: GPIO (select P1.4, P1.6, P2.1 or P0.2, P3.7, P3.6 according to CMP_CR1[HALLSEL]) 01: Output signal of CMP0/1/2 10: Output signal of ADC

		11: Reserved
[1:0]	T1INM	Filtering pulse width of the position detection signal selection. When the pulse width of the input signal is less than the set value, it will be filtered as noise. The filtering time is multiplied by 4 times according to the setting of CMP_CR4[FAEN]. When CMP_CR4[FAEN] = 0: 00: No filtering 01: 8 system clock cycles 10: 16 system clock cycles 11: 24 system clock cycles When CMP_CR4[FAEN]= 1: 00: No filtering 01: 32 system clock cycles 10: 64 system clock cycles 11: 96 system clock cycles

15.3.5 TIM1_CR4 (0x406C)

Bit	7	6	5	4	3	2	1	0
Name	RSV					T1CST		
Type	—	—	—	—	—	R/W	R/W	R/W
Reset	—	—	—	—	—	0	0	0

Bit	Name	Description
[7:3]	RSV	Reserved
[2:0]	T1CST	Phase commutation state machine The state machine correspond to different TIM1_DBRx at different states When TIM1_CR4[T1CST] is in state 001 ~ 111, Timer1 will automatically take over the enable of CMP0/1/2 according to TIM1_DBRx[T1CPE] When TIM1_CR4[T1CST] is in state 001 ~ 110, Timer1 will automatically add 1 cyclically when the writing sequence interrupt is generated

TIM1_CR4[T1CST]	TIM1_DBRx	TIM1_CR4[T1CST]	TIM1_DBRx
000	0	100	TIM1_DBR4
001	TIM1_DBR1	101	TIM1_DBR5
010	TIM1_DBR2	110	TIM1_DBR6
011	TIM1_DBR3	111	TIM1_DBR7

15.3.6 TIM1_IER (0x406D)

Bit	7	6	5	4	3	2	1	0
Name	T1UPD	T1MAME	T1ADIE	T1BOIE	T1RUIE	T1WTIE	T1PDIE	T1BDIE
Type	W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	T1UPD	When TIM1_CR0[T1OPS] = 00, writing 1 to this bit trigger data transmission. This bit will be cleared by hardware after writing 1
[6]	T1MAME	Manual mode enable When enabled, the base counter and reload counter will act as separate counters. The details are: TIM1__BCNTR of the base counter will be cleared by counter overflow interrupt instead of TIM1_CR2[T1BRS] TIM1_CR0[T1RCEN] will not be cleared or set to 1 automatically, and only operated by software TIM1__RCNTR of the reload counter can only be cleared by the reload counter overflow event

		TIM1__RARR of the reload counter will not be updated automatically, and only operated by software 0: Disable 1: Enable
[5]	T1ADIE	ADC position detected interrupt enable 0: Disable 1: Enable
[4]	T1BOIE	Base counter overflow interrupt enable 0: Disable 1: Enable
[3]	T1ROIE	Reload counter overflow interrupt enable 0: Disable 1: Enable
[2]	T1WTIE	Writing sequence interrupt enable 0: Disable 1: Enable
[1]	T1PDIE	CMP/GPIO position detected interrupt enable 0: Disable 1: Enable
[0]	T1BDIE	Diode freewheeling end interrupt enable 0: Disable 1: Enable

15.3.7 TIM1_SR (0x406E)

Bit	7	6	5	4	3	2	1	0
Name	RSV		T1ADIF	T1BOIF	T1ROIF	T1WTIF	T1PDIF	T1BDIF
Type	—	—	R/W0	R/W0	R/W0	R/W	R/W0	R/W0
Reset	—	—	0	0	0	0	0	0

Bit	Name	Description
[7:6]	RSV	Reserved
[5]	T1ADIF	ADC position detected interrupt event flag Generate a position detected interrupt when the ADC position detection signal is matched TIM1_DBRx[T1CPE] 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[4]	T1BOIF	Base counter overflow interrupt flag When the base counter counts up and TIM1_BCNTR matches TIM1_BARR, an overflow event occurs Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[3]	T1ROIF	Reload counter overflow interrupt flag When TIM1__RCNTR matches TIM1__RARR, an overflow event occurs and TIM1__RCNTR is cleared Read: 0: No interrupt pending

		1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[2]	T1WTIF	Writing sequence interrupt flag Writing sequence interrupt is generated when TIM1_DBRx is transferred to DRV_CMR 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: Generate write sequence interrupt when IM1_CR0[T1OPS] = 00, otherwise no meaning
[1]	T1PDIF	CMP/GPIO position detected interrupt flag Generate a position detected interrupt when the CMP/GPIO position detection signal is matched TIM1_DBRx[T1CPE] Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[0]	T1BDIF	Diode freewheeling end interrupt flag After phase commutation, start diode freewheeling and generate interrupt at end Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning

15.3.8 TIM1__BCOR (0x4070, 0x4071)

TIM1__BCORH(0x4070)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM1__BCOR[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIM1__BCORL(0x4071)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM1__BCOR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	TIM1__BCOR	Filtering value of TIM1_BCCR (i.e., 60 degree time base).

15.3.9 TIM1_DBR1 (0x4074, 0x4075)

TIM1_DBR1H(0x4074)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV	T1CPE			T1WHP	T1WLP	T1VHP	T1VLP
Type	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	0	0	0	0	0	0	0

TIM1_DBR1L(0x4075)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	T1UHP	T1ULP	T1WHE	T1WLE	T1VHE	T1VLE	T1UHE	T1ULE
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15]	RSV	Reserved
[14:12]	T1CPE	Position detection signal input edge polarity and comparator enable selection These bits are used to define the active edge of the position detection input signal and control the enable of the corresponding comparator. The detected edge in input signal, corresponding to the configuration, generates a position detected interrupt. Refer to CMP/GPIO Position Detected Event and Table 15-2
[11]	T1WHP	W phase high side output polarity 0: High level active 1: Low level active
[10]	T1WLP	W phase low side output polarity 0: High level active 1: Low level active
[9]	T1VHP	V phase high side output polarity 0: High level active 1: Low level active
[8]	T1VLP	V phase low side output polarity 0: High level active 1: Low level active
[7]	T1UHP	U phase high side output polarity 0: High level active 1: Low level active
[6]	T1ULP	U phase low side output polarity 0: High level active 1: Low level active
[5]	T1WHE	W phase high side output enable 0: Disable 1: Enable
[4]	T1WLE	W phase low side output enable 0: Disable 1: Enable
[3]	T1VHE	V phase high side output enable 0: Disable 1: Enable
[2]	T1VLE	V phase low side output enable 0: Disable 1: Enable
[1]	T1UHE	U phase high side output enable 0: Disable 1: Enable
[0]	T1ULE	U phase low side output enable 0: Disable 1: Enable

Note: When TIM1_DBR1[T1WLE] and TIM1_DBR1[T1WHE], TIM1_DBR1[T1VLE] and TIM1_DBR1[T1VHE] or TIM1_DBR1[T1ULE] and TIM1_DBR1[T1UHE] are enabled at the same time, the 3-phase high side and low side of bridges are complementary output and the deadtime is automatically inserted.

15.3.10 TIM1_DBR2 (0x4076, 0x4077)

TIM1_DBR2H(0x4076)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV	T1CPE			T1WHP	T1WLP	T1VHP	T1VLP
Type	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	0	0	0	0	0	0	0
TIM1_DBR2L(0x4077)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	T1UHP	T1ULP	T1WHE	T1WLE	T1VHE	T1VLE	T1UHE	T1ULE
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15]	RSV	Reserved
[14:12]	T1CPE	Position detection signal input edge polarity and comparator enable selection These bits are used to define the active edge of the position detection input signal and control the enable of the corresponding comparator. The detected edge in input signal, corresponding to the configuration, generates a position detected interrupt. Refer to CMP/GPIO Position Detected Event and Table 15-2
[11]	T1WHP	W phase high side output polarity 0: High level active 1: Low level active
[10]	T1WLP	W phase low side output polarity 0: High level active 1: Low level active
[9]	T1VHP	V phase high side output polarity 0: High level active 1: Low level active
[8]	T1VLP	V phase low side output polarity 0: High level active 1: Low level active
[7]	T1UHP	U phase high side output polarity 0: High level active 1: Low level active
[6]	T1ULP	U phase low side output polarity 0: High level active 1: Low level active
[5]	T1WHE	W phase high side output enable 0: Disable 1: Enable
[4]	T1WLE	W phase low side output enable 0: Disable 1: Enable
[3]	T1VHE	V phase high side output enable 0: Disable 1: Enable
[2]	T1VLE	V phase low side output enable 0: Disable 1: Enable
[1]	T1UHE	U phase high side output enable 0: Disable 1: Enable
[0]	T1ULE	U phase low side output enable 0: Disable 1: Enable

15.3.11 TIM1_DBR3 (0x4078, 0x4079)

TIM1_DBR3H(0x4078)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV	T1CPE			T1WHP	T1WLP	T1VHP	T1VLP
Type	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	0	0	0	0	0	0	0
TIM1_DBR3L(0x4079)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	T1UHP	T1ULP	T1WHE	T1WLE	T1VHE	T1VLE	T1UHE	T1ULE
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15]	RSV	Reserved
[14:12]	T1CPE	Position detection signal input edge polarity and comparator enable selection These bits are used to define the active edge of the position detection input signal and control the enable of the corresponding comparator. The detected edge in input signal, corresponding to the configuration, generates a position detected interrupt. Refer to CMP/GPIO Position Detected Event and Table 15-2
[11]	T1WHP	W phase high side output polarity 0: High level active 1: Low level active
[10]	T1WLP	W phase low side output polarity 0: High level active 1: Low level active
[9]	T1VHP	V phase high side output polarity 0: High level active 1: Low level active
[8]	T1VLP	V phase low side output polarity 0: High level active 1: Low level active
[7]	T1UHP	U phase high side output polarity 0: High level active 1: Low level active
[6]	T1ULP	U phase low side output polarity 0: High level active 1: Low level active
[5]	T1WHE	W phase high side output enable 0: Disable 1: Enable
[4]	T1WLE	W phase low side output enable 0: Disable 1: Enable
[3]	T1VHE	V phase high side output enable 0: Disable 1: Enable
[2]	T1VLE	V phase low side output enable 0: Disable 1: Enable
[1]	T1UHE	U phase high side output enable 0: Disable 1: Enable
[0]	T1ULE	U phase low side output enable 0: Disable 1: Enable

15.3.12 TIM1_DBR4 (0x4080, 0x4081)

TIM1_DBR4H(0x4080)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV	T1CPE			T1WHP	T1WLP	T1VHP	T1VLP
Type	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	0	0	0	0	0	0	0
TIM1_DBR4L(0x4081)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	T1UHP	T1ULP	T1WHE	T1WLE	T1VHE	T1VLE	T1UHE	T1ULE
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15]	RSV	Reserved
[14:12]	T1CPE	Position detection signal input edge polarity and comparator enable selection These bits are used to define the active edge of the position detection input signal and control the enable of the corresponding comparator. The detected edge in input signal, corresponding to the configuration, generates a position detected interrupt. Refer to CMP/GPIO Position Detected Event and Table 15-2
[11]	T1WHP	W phase high side output polarity 0: High level active 1: Low level active
[10]	T1WLP	W phase low side output polarity 0: High level active 1: Low level active
[9]	T1VHP	V phase high side output polarity 0: High level active 1: Low level active
[8]	T1VLP	V phase low side output polarity 0: High level active 1: Low level active
[7]	T1UHP	U phase high side output polarity 0: High level active 1: Low level active
[6]	T1ULP	U phase low side output polarity 0: High level active 1: Low level active
[5]	T1WHE	W phase high side output enable 0: Disable 1: Enable
[4]	T1WLE	W phase low side output enable 0: Disable 1: Enable
[3]	T1VHE	V phase high side output enable 0: Disable 1: Enable
[2]	T1VLE	V phase low side output enable 0: Disable 1: Enable
[1]	T1UHE	U phase high side output enable 0: Disable 1: Enable
[0]	T1ULE	U phase low side output enable 0: Disable 1: Enable

15.3.13 TIM1_DBR5 (0x4082, 0x4083)

TIM1_DBR5H(0x4082)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV	T1CPE			T1WHP	T1WLP	T1VHP	T1VLP
Type	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	0	0	0	0	0	0	0
TIM1_DBR5L(0x4083)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	T1UHP	T1ULP	T1WHE	T1WLE	T1VHE	T1VLE	T1UHE	T1ULE
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15]	RSV	Reserved
[14:12]	T1CPE	Position detection signal input edge polarity and comparator enable selection These bits are used to define the active edge of the position detection input signal and control the enable of the corresponding comparator. The detected edge in input signal, corresponding to the configuration, generates a position detected interrupt. Refer to CMP/GPIO Position Detected Event and Table 15-2
[11]	T1WHP	W phase high side output polarity 0: High level active 1: Low level active
[10]	T1WLP	W phase low side output polarity 0: High level active 1: Low level active
[9]	T1VHP	V phase high side output polarity 0: High level active 1: Low level active
[8]	T1VLP	V phase low side output polarity 0: High level active 1: Low level active
[7]	T1UHP	U phase high side output polarity 0: High level active 1: Low level active
[6]	T1ULP	U phase low side output polarity 0: High level active 1: Low level active
[5]	T1WHE	W phase high side output enable 0: Disable 1: Enable
[4]	T1WLE	W phase low side output enable 0: Disable 1: Enable
[3]	T1VHE	V phase high side output enable 0: Disable 1: Enable
[2]	T1VLE	V phase low side output enable 0: Disable 1: Enable
[1]	T1UHE	U phase high side output enable 0: Disable 1: Enable
[0]	T1ULE	U phase low side output enable 0: Disable 1: Enable

15.3.14 TIM1_DBR6 (0x4084, 0x4085)

TIM1_DBR6H(0x4084)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV	T1CPE			T1WHP	T1WLP	T1VHP	T1VLP
Type	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	0	0	0	0	0	0	0
TIM1_DBR6L(0x4085)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
cName	T1UHP	T1ULP	T1WHE	T1WLE	T1VHE	T1VLE	T1UHE	T1ULE
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15]	RSV	Reserved
[14:12]	T1CPE	Position detection signal input edge polarity and comparator enable selection These bits are used to define the active edge of the position detection input signal and control the enable of the corresponding comparator. The detected edge in input signal, corresponding to the configuration, generates a position detected interrupt. Refer to CMP/GPIO Position Detected Event and Table 15-2
[11]	T1WHP	W phase high side output polarity 0: High level active 1: Low level active
[10]	T1WLP	W phase low side output polarity 0: High level active 1: Low level active
[9]	T1VHP	V phase high side output polarity 0: High level active 1: Low level active
[8]	T1VLP	V phase low side output polarity 0: High level active 1: Low level active
[7]	T1UHP	U phase high side output polarity 0: High level active 1: Low level active
[6]	T1ULP	U phase low side output polarity 0: High level active 1: Low level active
[5]	T1WHE	W phase high side output enable 0: Disable 1: Enable
[4]	T1WLE	W phase low side output enable 0: Disable 1: Enable
[3]	T1VHE	V phase high side output enable 0: Disable 1: Enable
[2]	T1VLE	V phase low side output enable 0: Disable 1: Enable
[1]	T1UHE	U phase high side output enable 0: Disable 1: Enable
[0]	T1ULE	U phase low side output enable 0: Disable 1: Enable

15.3.15 TIM1_DBR7 (0x4086, 0x4087)

TIM1_DBR7H(0x4086)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV	T1CPE			T1WHP	T1WLP	T1VHP	T1VLP
Type	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	0	0	0	0	0	0	0
TIM1_DBR7L(0x4087)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	T1UHP	T1ULP	T1WHE	T1WLE	T1VHE	T1VLE	T1UHE	T1ULE
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15]	RSV	Reserved
[14:12]	T1CPE	Position detection signal input edge polarity and comparator enable selection These bits are used to define the active edge of the position detection input signal and control the enable of the corresponding comparator. The detected edge in input signal, corresponding to the configuration, generates a position detected interrupt. Refer to CMP/GPIO Position Detected Event and Table 15-2
[11]	T1WHP	W phase high side output polarity 0: High level active 1: Low level active
[10]	T1WLP	W phase low side output polarity 0: High level active 1: Low level active
[9]	T1VHP	V phase high side output polarity 0: High level active 1: Low level active
[8]	T1VLP	V phase low side output polarity 0: High level active 1: Low level active
[7]	T1UHP	U phase high side output polarity 0: High level active 1: Low level active
[6]	T1ULP	U phase low side output polarity 0: High level active 1: Low level active
[5]	T1WHE	W phase high side output enable 0: Disable 1: Enable
[4]	T1WLE	W phase low side output enable 0: Disable 1: Enable
[3]	T1VHE	V phase high side output enable 0: Disable 1: Enable
[2]	T1VLE	V phase low side output enable 0: Disable 1: Enable
[1]	T1UHE	U phase high side output enable 0: Disable 1: Enable
[0]	T1ULE	U phase low side output enable 0: Disable 1: Enable

15.3.16 TIM1__BCNTR (0x4082, 0x4083)

TIM1__BCNTRH(0x4082)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM1__BCNTR[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	0	0	0	0	0	0	0
TIM1__BCNTRL(0x4083)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM1__BCNTR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM1__BCNTR	The count value of base counter, which is used to count 60 degree commutation time Auto mode: Selects the reset source of TIM1__BCNTR according to TIM1_CR2[T1BRS]. TIM1__BCNTR overflow interrupt will not make TIM1__BCNTR recount Manual mode: TIM1__BCNTR overflow interrupt will make TIM1__BCNTR recount

15.3.17 TIM1__BCCR (0x4084, 0x4085)

TIM1__BCCRH(0x4084)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM1__BCCR[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIM1__BCCRL(0x4085)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM1__BCCR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM1__BCCR	The count vaule of capture base counter Auto mode: When the base counter is cleared due to position detected interrupt or sequence writing interrupt, the count value before cleared is stored to TIM1__BCCR Manual mode: When the base timer is cleared due to the overflow interrupt, the count value before cleared is stored to TIM1__BCCR

15.3.18 TIM1__BARR (0x4086, 0x4087)

TIM1__BARRH(0x4086)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM1__BARR[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIM1__BARRL(0x4087)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM1__BARR[7:0]							

Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM1__BARR	Reload value of base counter When the count value of the base timer is equal to TIM1__BARR, an overflow interrupt is generated and the counter is cleared

15.3.19 TIM1__RARR (0x4088, 0x4089)

TIM1__RARRH(0x4088)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM1__RARR[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIM1__RARRL(0x4089)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM1__RARR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM1__RARR	Auto-reload value of reload counter When the count of reload counter is equal to TIM1__RARR, an overflow interrupt event is generated and the value of counter is cleared Auto mode: The value corresponding to diode freewheeling angle (set by TIM1_CR1[BSEL]) is updated to TIM1__RARR when writing sequence interrupt is generated. The value corresponding to commutation angle (set by TIM1_CR2[CSEL]) is updated to TIM1__RARR when the position detected interrupt is generated. Manual mode: TIM1__RARR is written by software

15.3.20 TIM1__RCNTR (0x408A, 0x408B)

TIM1__RCNTRH(0x408A)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM1__RCNTR[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1
TIM1__RCNTRL(0x408B)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM1__RCNTR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1

Bit	Name	Description
[15:0]	TIM1__RCNTR	Count value of the reload counter for counting time of diode freewheeling masking and ZCP to phase commutation Note: In Manual mode, TIM1__RCNTR is only cleared by the reload counter overflow interrupt

15.3.21 TIM1__UCOP (0x408C, 0x408D)

TIM1__UCOPH(0x408C)

Bit	15	14	13	12	11	10	9	8
Name	TIM1__UCOP[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIM1__UCOPL(0x408D)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM1__UCOP[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM1__UCOP	ADC sample value of active phase voltage (second-highest bit alignment)

15.3.22 TIM1__UFLP (0x408E, 0x408F)

TIM1__UFLPH(0x408E)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM1__UCOP[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIM1__UFLPL(0x408F)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM1__UCOP[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM1__UFLP	ADC sample value of floating phase voltage (second-highest bit alignment)

15.3.23 TIM1__URES (0x4090, 0x4091)

TIM1__URESH(0x4090)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM1__URES[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIM1__URESL(0x4091)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM1__URES[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM1__URES	Calculation result of ADC position detection formula, Q15 format

15.3.24 TIM1__UIGN (0x4092, 0x4093)

TIM1__UIGNH(0x4092)

Bit	15	14	13	12	11	10	9	8
Name	TIM1__UIGN[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIM1__UFLPL(0x4093)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM1__UIGN[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM1__UIGN	When the ADC sampling value of the active phase voltage is less than this value, no calculation will be performed

15.3.25 TIM1__KF (0x4094, 0x4095)

TIM1__KFH(0x4094)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM1__KF[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIM1__KFL(0x4095)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM1__KF[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM1__KF	ADC position detection coefficient when the floating phase voltage drops Range [0,32767]

15.3.26 TIM1__KR (0x4096, 0x4097)

TIM1__KRH(0x4096)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM1__KR[15:8]							
Type	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIM1__KRL(0x4097)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM1__KR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM1__KR	ADC position detection coefficient when the floating phase voltage rises Range [0,32767]

15.3.27 TIM1__ITRIP (0x4098, 0x4099)

TIM1__ITRIPH(0x4098)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM1__ITRIP[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
TIM1__ITRIPL(0x4099)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM1__ITRIP[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM1__ITRIP	Filtered bus current When DRV__CNTR = 0, the hardware module automatically samples the bus current and filter it for software using. The default channel is ADC channel 4 Range [0,32767] Note: This value is obtained by averaging the instantaneous current values of 8 samples

16 Timer2

16.1 Timer2 Instructions

Timer2 has 5 working modes:

- Output mode: PWM generation
- Input capture mode: Detect the duration of high and low level of input PWM
- Input counter mode: Detect the time of set PWM wave numbers
- QEP & RSD mode: Quadrature Encoder Pulse & Rotating State Detection (tail wind or contrary wind detection) mode
- Step mode: Detect the direction, position and speed of the step motor

Timer2 features:

- 3-bit programmable prescaler divides the system clock
- 16-bit up counting base counter, counting clock source is the output of prescaler
- 16-bit up/down counting special counter, used for input counter mode, QEP & RSD mode and step mode. The counting clock source is active edge of external input signal
- Input filter module
- Edge detection module
- PWM generation module
- Interrupt event

16.1.1 Prescaler

Frequency prescaler is used to divide the system clock and generate clock source of base counter. 8 frequency division coefficients of prescaler are available and can be selected by TIM2_CR0[T2PSC]. Since this register has no buffer, the clock source frequency will be changed immediately after TIM2_CR0[T2PSC] has been written. Therefore, the frequency division coefficients should be configured when base counter is not working. The clock source frequency formula is: clk_psc2 = SYSCLK/(2^TIM2_CR0[T2PSC]). The clock frequency corresponding to different TIM2_CR0[T2PSC] value as shown in Table 16-1

Table 16-1 The Clock Frequency Corresponding to TIM2_CR0[T2PSC]

TIM2_CR0[T2PSC]	Coefficient	clk_psc2 (Hz)	TIM2_CR0[T2PSC]	Coefficient	clk_psc2 (Hz)
000	1	24M	100	16	1.5M
001	2	12M	101	32	750K
010	4	6M	110	64	375K
011	8	3M	111	128	187.5K

16.1.2 Reading, Writing and Counting of TIM2__CNTR

TIM2_CR1[T2CEN] = 1, TIM2__CNTR start to count. Software write operation to TIM2__CNTR directly changes the value of the register, so the counter should be disabled before performing the write operation. When reading TIM2__CNTR, software reads the high 8 bits first, and the hardware will synchronously cache the low 8 bits. When reading the low byte, the cached data will be read.

16.1.3 Output Mode

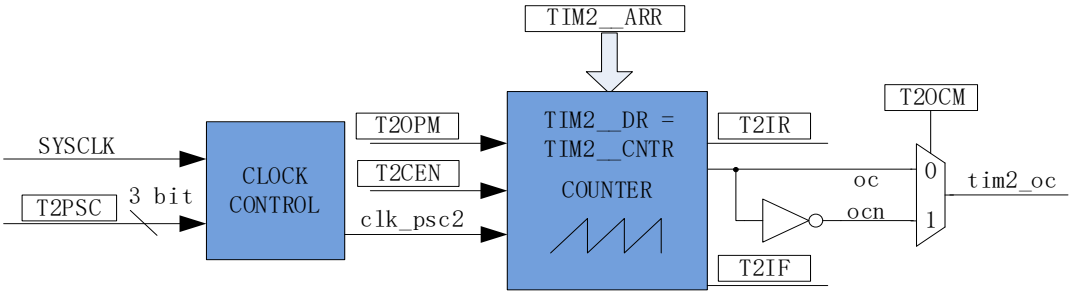


Figure 16-1 Principle Block Diagram of Output Mode

Counter output mode generate output signals according to TIM2_CR0[T2OCM] setting, and the comparison results between TIM2__CNTR and registers TIM2__DR, TIM2__ARR setting value. Meanwhile, corresponding interrupts is generated.

16.1.3.1 Reading and Writing of TIM2__ARR/TIM2__DR

In output mode, TIM2__ARR/TIM2__DR contains preload registers and shadow registers. When the software writes TIM2__ARR/TIM2__DR register, the data is saved in the preload register. When the overflow event TIM2_CR1[T2IF] is generated or the base counter stops working (TIM2_CR1[T2CEN] = 0), the set value is transferred to the shadow register.

TIM2__ARR/TIM2__DR is a 16-bit register, which requires to write the high byte first and then the low byte. The hardware ensures that the data in the preload register will not be transferred to the shadow register after the high byte is written and before the low byte is written.

E.g., TIM2__DR (preload register), DR_SH (shadow register) and TIM2__CNTR are used to generate PWM. When software writes TIM2__DR, TIM2__DR is not updated to DR_SH immediately, and is updated to TIM2__DR at the end of a PWM (TIM2__CNTR overflow event).

16.1.3.2 High/Low Level Output

When TIM2_CR0[T2OCM] = 0, if TIM2__DR > TIM2__ARR, the output signal is always low. When TIM2_CR0[T2OCM] = 1, if TIM2__DR > TIM2__ARR, the output signal is always high.

16.1.3.3 PWM Generation

In PWM output mode, TIM2__ARR determines PWM period, TIM2__DR determines duty cycle, and duty cycle = TIM2__DR/TIM2__ARR*100%. TIM2_CR0[T2OCM] = 0, the low level is output when TIM2__CNTR < TIM2__DR, and the high level is output when TIM2__CNTR ≥ TIM2__DR. TIM2_CR0[T2OCM] = 1, the high level is output when TIM2__CNTR < TIM2__DR, and the low level is output when TIM2__CNTR ≥ TIM2__DR. When TIM2__CNTR is increased to TIM2__ARR, the output signal is reversed.

16.1.3.4 Interrupt Event

- When TIM2__CNTR = TIM2__DR, a compare match event is generated and the interrupt flag bit TIM2_CR1[T2IR] is set to 1. The counter continues to count
- When TIM2__CNTR = TIM2__ARR, an overflow event is generated, and the interrupt flag bit TIM2_CR1[T2IF] is set to 1. The counter clears to 0 and restarts

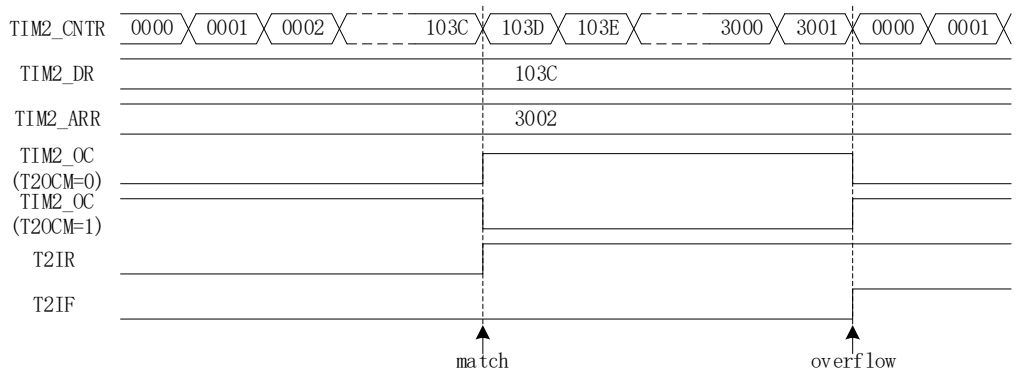


Figure 16-2 Output Mode Waveform

16.1.4 Input Signal Filtering and Edge Detection

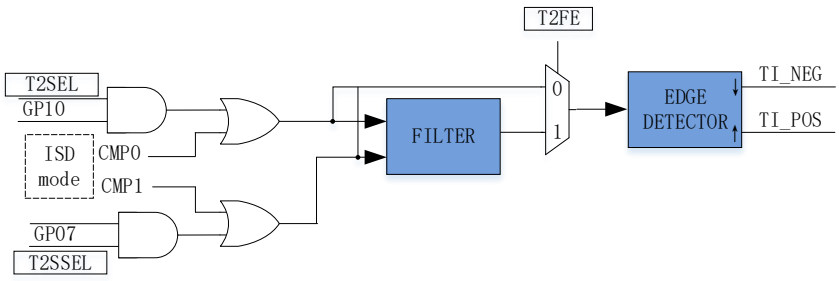


Figure 16-3 Block Diagram of Input Signal Filtering and Edge Detection

The input signal of Timer2 comes from P0.7 or P1.0, set by PH_SEL[T2SEL] and PH_SEL [T2SSEL] (refer to 22.3.14). The filter of input signal is optional. Configure TIM2_CR1[T2_FE] = 1 to enable the filtering function,

and the filtering circuit filters out the input noise below 4 system clock cycles. The filtered signal will be 4 clock cycles later than the signal before filtering. TIM2_CR0[T2CES] configures the active edge to count.

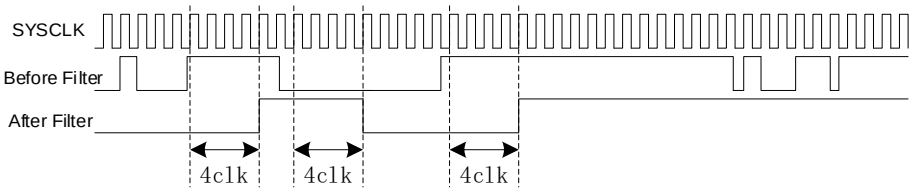


Figure 16-4 Timing Diagram of Filter Module

The edge detection module detects the filtered input signal and records the rising edge and falling edge for input capture mode or input counting mode.

16.1.5 Input Capture Mode

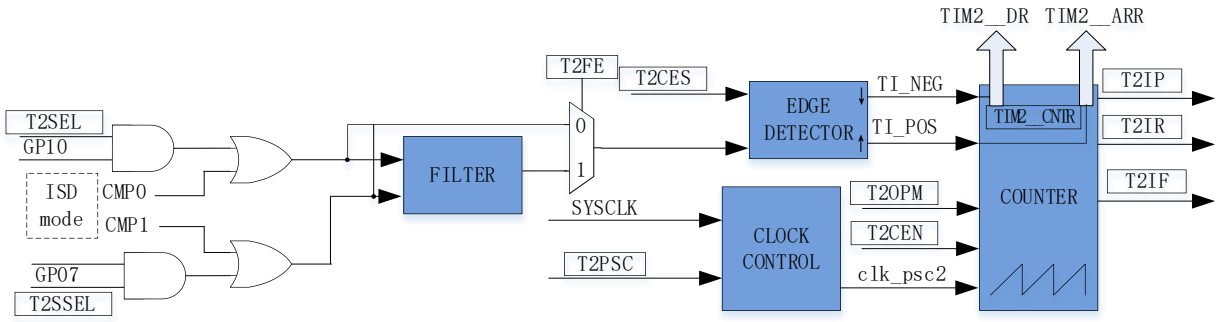


Figure 16-5 Principle Block Diagram of Input Capture Mode

The input capture mode detects the duty cycle and period of the PWM signal. When TIM_CR0[T2CES] = 0, the time between two adjacent rising edges is one cycle, and the time from rising edge to falling edge is pulse width (high-level pulse width). When TIM_CR0[T2CES] = 1, the time between two adjacent falling edges is one cycle, and the time from falling edge to rising edge is pulse width (low-level pulse width). When the predefined edge arrives, the count value TIM2__CNTR is stored in TIM2__DR and TIM2__ARR respectively to calculate the period and duty cycle of PWM waveform. The input signal can be filtered or not.

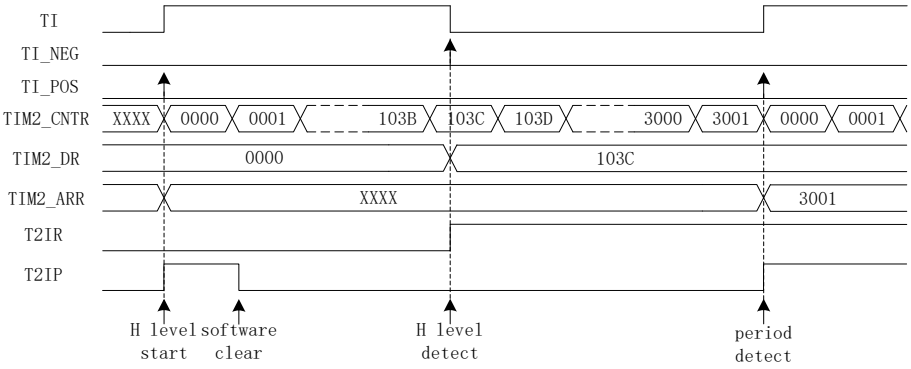


Figure 16-6 Timing Diagram of Input Capture Mode (T2CES = 0)

E.g., T2CES = 0, configure TIM2_CR1[T2CEN] = 1 to enable the counter, and the counter start counting. When the counter detects the first rising edge of the input (falling edge is invalid), TIM2__CNTR is cleared and recounts. When the falling edge of the input is detected, the value of TIM2__CNTR is stored in TIM2__DR, while the interrupt flag TIM2_CR1[T2IR] is set to 1, and TIM2__CNTR continues to count. When the second rising edge of input is detected, save the value of TIM2__CNTR into TIM2__ARR. The interrupt flag TIM2_CR1[T2IP] is set to 1, and TIM2__CNTR is cleared and restarts counting.

If the second rising edge of the input has not been detected by Timer2 and TIM2__CNTR reaches 0xFFFF, an overflow event occurs. The interrupt flag TIM2_CR1[T2IF] is set to 1, and the TIM2__CNTR is cleared and restarts counting.

16.1.6 Input Counter Mode

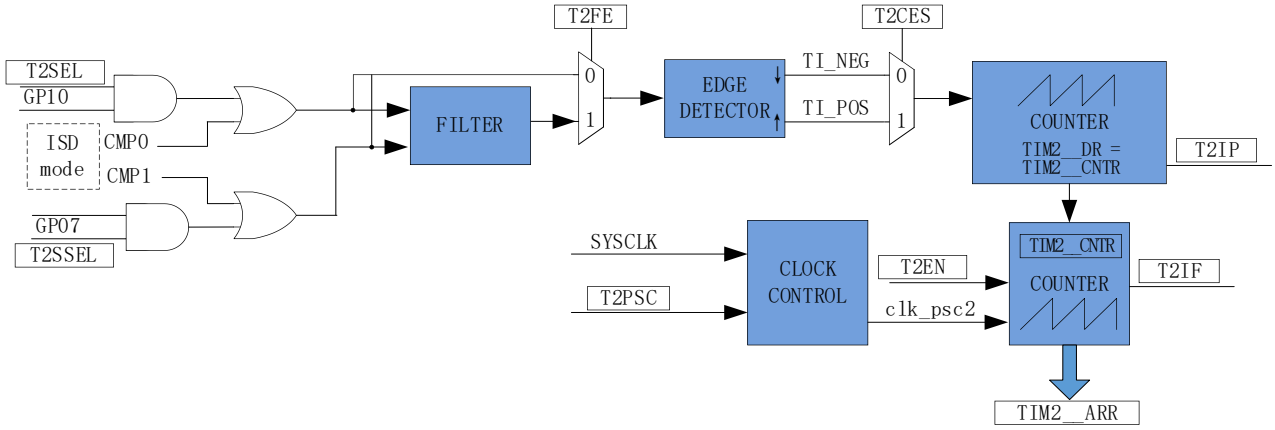


Figure 16-7 Principle Block Diagram of Input Counter Mode

In input counter mode, TIM2__DR includes preload register and shadow register. When software writes TIM2__DR register, the data is saved in the preload register first. It is sent to the shadow register in case of compare match event (TIM2_CR1[T2IP] = 1), overflow event (TIM2_CR1[T2IF] = 1) or counter disable

(TIM2_CR1[T2CEN] = 0). TIM2__DR is a 16-bit register. Software writes the high byte first and then the low byte. The hardware ensures that the data in the preload register will not be updated to the shadow register after the high byte is written and before the low byte is written.

The input counter mode is used to detect the time of a set PWM wave numbers. When the number of input PWM counted by the special counter CCNTR reaches the set value (TIM2_DR), TIM2__CNTR of the base counter is stored in TIM2__ARR. The input signal can be filtered or not. Configure TIM2_CR0[T2CES] = 1, the rising edge of the input PWM signal is used as the active counting edge of the special counter. Configure TIM2_CR0[T2CES] = 0, the falling edge of the input PWM signal is used as the active counting edge.

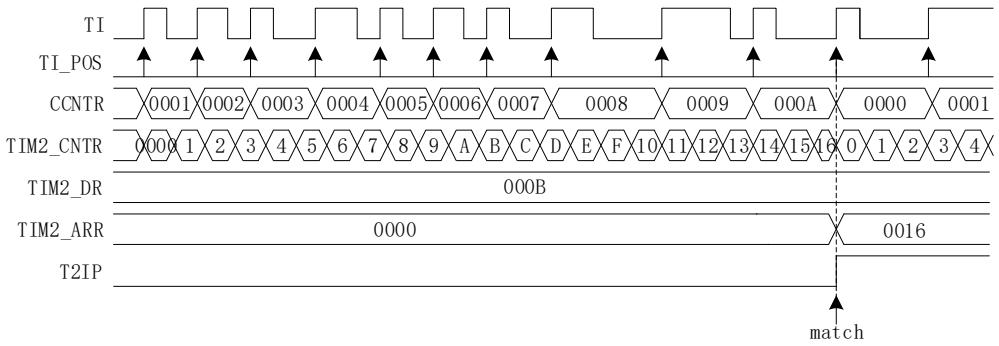


Figure 16-8 Timing Diagram of Input Counter Mode

TIM2_CR1[T2CEN] = 1, enable counter and the counter counts up. If the first active edge of the input signal is detected, TIM2__CNTR is cleared and restarts counting. Whenever the active edge of the input signal arrives, the count value of the special counter CCNTR is added by one. When the count value reaches TIM2__DR, TIM2__CNTR is stored in TIM2__ARR. TIM2_CR1[T2IP] is set to 1, TIM2__CNTR and CCNTR are cleared to 0 and restart counting.

When the number of input PWM has not reached the set value and TIM2__CNTR reaches 0xFFFF, an overflow event generates, and the interrupt flag TIM2_CR1[T2IF] is set to 1. TIM2__CNTR is cleared with CCNTR uncleared. TIM2__CNTR starts counting from 0, and CCNTR continues counting after the previous value.

16.1.7 QEP & RSD Mode

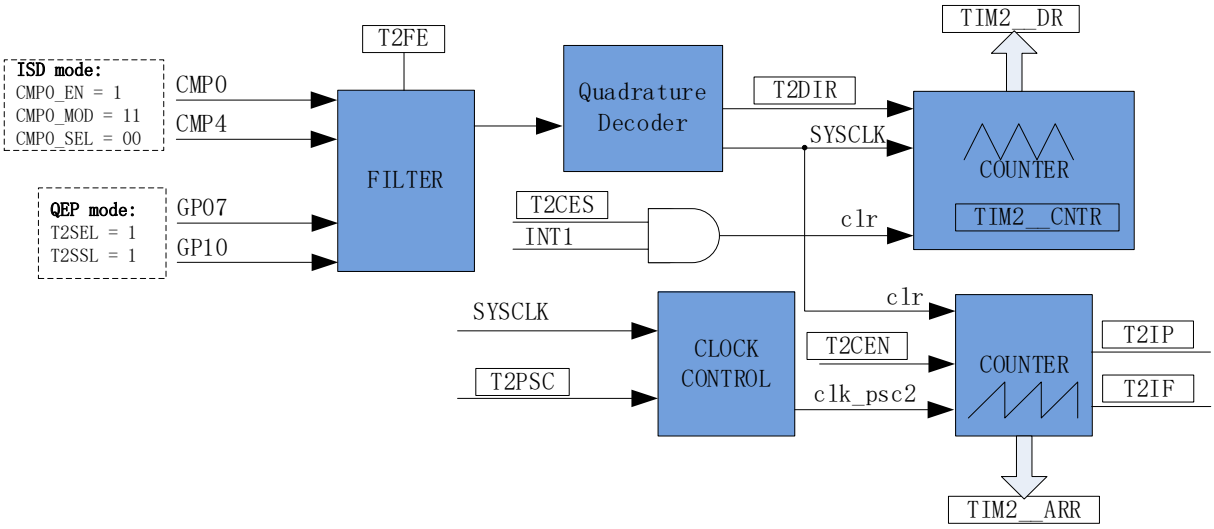


Figure 16-9 Principle Block Diagram of QEP & RSD Mode

QEP & RSD mode obtains the relative position, direction and speed of motor by detecting two channels orthogonal signal. P0. 7, P1. 0 (QEP mode) or CMP0, CMP1 (RSD mode) are the input signal sources, which are sent to the quadrature decoding module after filtering module to obtain the active edge and direction (TIM2_CR1[T2DIR]). Direction change will generate TIM2_CR1[T2IR] interrupt flag.

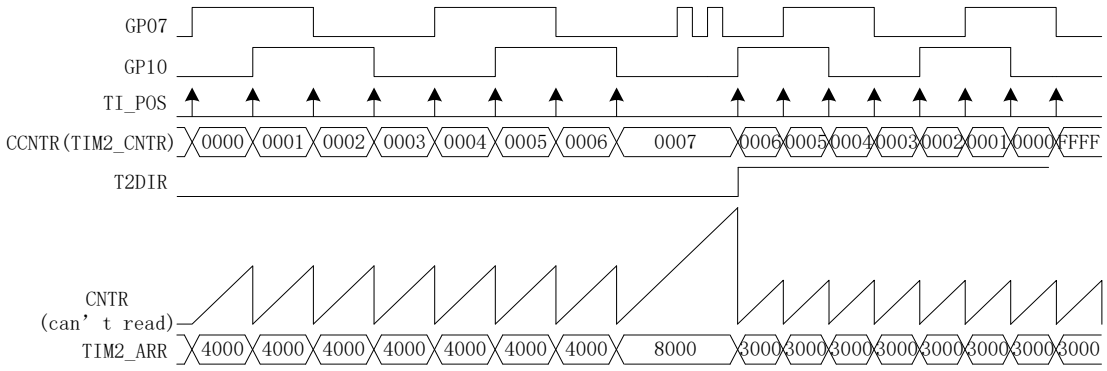


Figure 16-10 Timing Diagram of QEP & RSD Mode

The special counter is an up/down counter, and the signal source is the active edge of the orthogonal decoding module. TIM2_CR1[T2DIR] = 0, the direction is positive, and special counter count upward. When the active edge arrives, the counter is incremented by one. TIM2_CR1[T2DIR] = 1, the direction is reverse and special counter count down. When the active edge arrives, the counter decreases by one. The special counter can be cleared by INT1. Connect the mechanical zero point of the encoder to any port of INT1, enable INT1 and set TIM2_CR0[T2CES] = 1. When INT1 is generated, the current count value of the special counter is stored in

TIM2__DR and cleared. When the count value of the special counter reaches 65535 from 0, it will be automatically cleared to 0. When it decreases from 65535 to 0, it will be automatically set to 65535. Read TIM2__CNTR to get the value of special counter.

The base counter is an up counter used to record the time of two active counting edges. The clock source frequency can be divided. When the counting edge arrives, the current count value of the base counter is stored in the TIM2__ARR and cleared to 0. TIM2_CR1[T2IP] interrupt flag bit is set to 1. When the base counter counts to 0xFFFF, the count overflows and TIM2_CR1[T2IF] interrupt flag is generated.

16.1.7.1 RSD Comparator Sampling

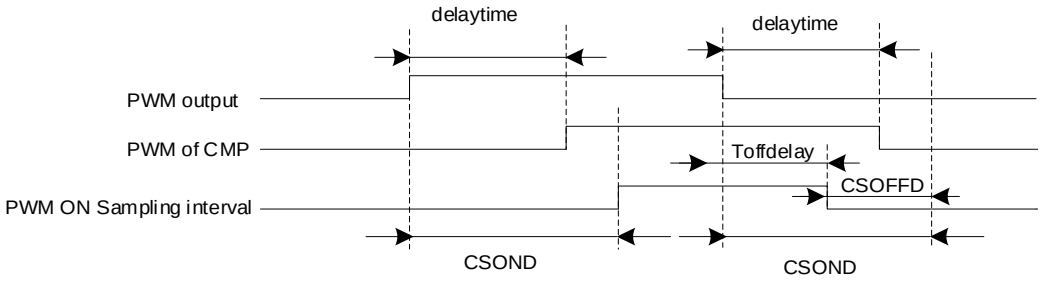


Figure 16-11 PWM ON Sampling mode

During RSD sampling, in order to obtain the accurate BEMF comparison signal, it is necessary to set the ON-delayed sampling time and OFF-advanced sampling time.

For details, please refer to Comparator Sampling.

16.1.8 Step Mode

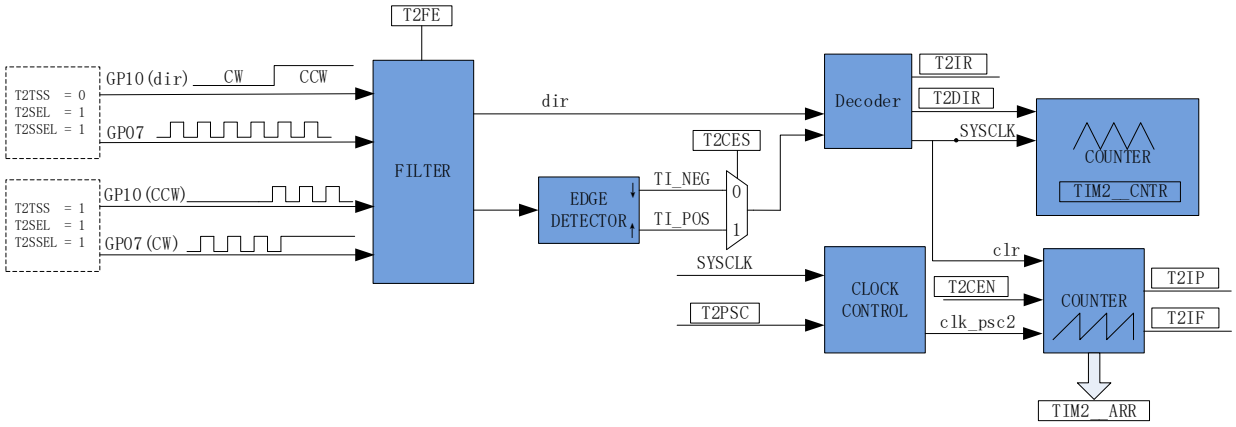


Figure 16-12 Principle Block Diagram of Step Mode

In the step mode, the relative position, direction and speed of the step motor are obtained by detecting two channel

inputs. P1.0 is direction input, P0.7 is pulse input. Select the rising edge or falling edge as the active edge by setting TIM2_CR0[T2CES]. Input signal, after filtering module, are sent to decoding module to obtain the active edge and direction TIM2_CR1[T2DIR]. The change of direction will generate TIM2_CR1[T2IR] interrupt.

Note: TIM2_CR1[T2DIR] and TIM2_CR1[T2IR] will not change unless P1.0 changes and P0.7 active edge arrives. To generate an interrupt immediately after P1.0 changes, use INT1.

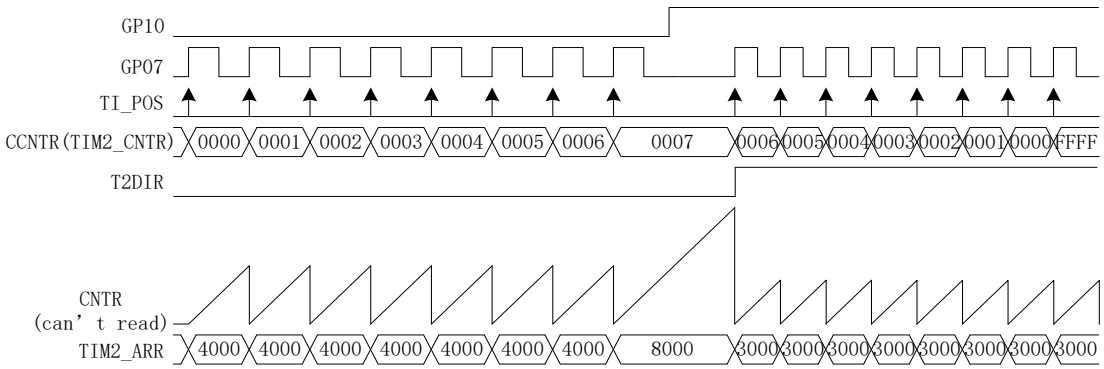


Figure 16-13 Timing Diagram of Step Mode

The special counter is an up/down counter, and the signal source is the active edge of the decoding module. When P1.0 = 0, TIM2_CR1[T2DIR] = 0, the direction is forward. When the active edge of P0.7 arrives, the special counter CCNTR increases by 1. When P1.0 = 1, TIM2_CR1[T2DIR] = 1 and the direction is reverse. When the active edge of P0.7 arrives, CCNTR decreases by 1. When the count value of the special counter reaches 65535 from 0, it will be cleared. When it decreases from 65535 to 0, it will be set to 65535. Read TIM2__CNTR to get the value of special counter.

The base counter is an up counter, which uses the output of prescaler as the clock source and is used to record the time between two active counting edges. When active counting edge arrives, the current count value of the base counter is stored in TIM2__ARR and cleared. The interrupt flag bit TIM2_CR1[T2IP] is set to 1. When the base counter counts to 0xFFFF, the count overflows and the interrupt (TIM2_CR1[T2IF]) is generated.

16.2 Timer2 Register

16.2.1 TIM2_CR0 (0xA1)

Bit	7	6	5	4	3	2	1	0
Name	T2PSC			T2OCM	T2IRE	T2CES	T2MOD	
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:5]	T2PSC	Base counter clock source frequency division selection Divide the system clock as the clock source of the base counter

		The clock source frequency after frequency division is: 000: 24MHz 001: 12MHz 010: 6MHz 011: 3MHz 100: 1.5MHz 101: 750kHz 110: 375kHz 111: 187.5kHz
[4]	T2OCM	Output mode: Output mode selection 0: TIM2__CNTR < TIM2__DR, output 0; TIM2__CNTR ≥ TIM2__DR, output 1 1: TIM2__CNTR < TIM2__DR, output 1; TIM2__CNTR ≥ TIM2__DR, output 0 Input capture mode: No meaning Input counter mode: No meaning QEP & RSD mode and step mode selection 0: QEP & RSD mode 1: Step mode
[3]	T2IRE	Output mode: Compare match interrupt enable Input capture mode: Pulse width detection interrupt enable Input counter mode: No meaning QEP & RSD mode: Direction change interrupt enable Step mode: Direction change interrupt enable 0: Disable 1: Enable
[2]	T2CES	Output mode: no meaning Input capture mode: Counting edge selection 0: the time between two adjacent rising edges is one cycle, and the time from rising edge to falling edge is pulse width (high-level pulse width) 1: The time between two adjacent falling edges is one cycle, and the time from falling edge to rising edge is pulse width (low-level pulse width) Input counter mode: Counting edge selection 0: Falling edge count 1: Rising edge count QEP & RSD mode: INT1 (zero point) clear special counter enable 0: Disable 1: Enable Step mode: Active counting edge selection 0: Falling edge count 1: Rising edge count
[1:0]	T2MOD	Mode selection 00: Input capture mode 01: Output mode 10: Input counter mode 11: QEP & RSD mode or step mode

16.2.2 TIM2_CR1 (0xA9)

Bit	7	6	5	4	3	2	1	0
Name	T2IR	T2IP	T2IF	T2IPE	T2IFE	T2FE	T2DIR	T2CEN
Type	R/W0	R/W0	R/W0	R/W	R/W	R/W	R	R/W
Reset	0	0	0	0	0	0	—	0

Bit	Name	Description

[7]	T2IR	Output mode: Compare match interrupt flag Input capture mode: Pulse width detection interrupt flag Input counter mode: No meaning Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[6]	T2IP	Output mode: No meaning Input capture mode: PWM cycle detected interrupt flag Input counter mode: Input PWM count match interrupt flag QEP & RSD mode & step mode: active edge detected interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[5]	T2IF	Output mode: Base counter overflow interrupt flag. This bit is set to 1 when the counter value TIM2__CNTR matches the comparison value TIM2__ARR Input capture mode: Base counter overflow interrupt flag. This bit is set to 1 when the counter value TIM2__CNTR accumulated to 0xFFFF and timer has not detected the input of a PWM cycle Input counter mode: Counter overflow interrupt flag. This bit is set to 1 when the number of input PWM has not reached the value of TIM2__DR and the counter value TIM2__CNTR is accumulated to 0xFFFF. QEP & RSD mode & step mode: Base counter overflow interrupt flag. When the base counter accumulates to 0xFFFF, this bit is set to 1 and the base counter is cleared to 0 Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[4]	T2IPE	Output mode: No meaning Input capture mode: PWM cycle detected interrupt enable Input counter mode: Input PWM count match interrupt enable 0: Disable 1: Enable
[3]	T2IFE	Output mode: Base counter overflow interrupt enable Input capture mode: Base counter overflow interrupt enable Input counter mode: Base counter overflow interrupt enable 0: Disable 1: Enable
[2]	T2FE	Input signal filtering enable 0: Disable 1: Enable
[1]	T2DIR	QEP & RSD: Motor rotation direction Indicate the rotation direction of the motor according to the phase relationship of the two input signals Step mode: Motor rotation direction Indicates the rotation direction of the motor according to the direction signal P1.0 0: Forward 1: Reverse
[0]	T2CEN	Timer enable 0: Disable 1: Enable

16.2.3 TIM2__CNTR (0xAA, 0xAB)

TIM2__CNTRH(0xAB)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM2__CNTR[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIM2__CNTRL(0xAA)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM2__CNTR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM2__CNTR	Output mode/input capture mode/input counter mode: Count value of the base counter QEP & RSD mode/step mode: Count value of the special counter

16.2.4 TIM2__DR (0xAC, 0xAD)

TIM2__DRH(0xAD)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM2__DR[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIM2__DRL(0xAC)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM2__DR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM2__DR	Output mode: Compare match values (written by software) Input capture mode: The count value of the detected input pulse width (written by hardware) Input counter mode: The number of PWM needs to be counted (written by software) QEP & RSD mode: When T2CES = 1 and INT1 (zero point) arrives, the value of the special counter (written by hardware) Step mode: No meaning

16.2.5 TIM2__ARR (0xAE, 0xAF)

TIM2__ARRH(0xAF)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIM2__ARR[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIM2__ARRL(0xAE)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIM2__ARR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIM2__ARR	Output mode: PWM waveform cycle (written by software) Input capture mode: The count value of the base counter of a detected PWM cycle (written by hardware) Input counter mode: The count value of the base counter when input PWM count matches (written by hardware) QEP & RSD mode & step mode: The count value of the base counter when the input edge is detected as an active edge (written by hardware)

17 Timer3/Timer4

17.1 Timer3/Timer4 Instructions

Timer3/Timer4 support output and input modes:

- Output mode: PWM generation
- Input capture mode: Detect the duration of high and low level of input PWM, which can be used to calculate PWM duty cycle

Timer3/Timer4 Features:

- 3-bit programmable prescaler divides the system clock as the clock source of base counter (the clock source of Timer3 can be doubled to 48MHz when works in input capture mode)
- 16-bit up counting base counter, the output of the prescaler is the counting clock source
- Input signal filtering
- Input signal edge detection
- Output PWM signal, single compare output
- Interrupt event

17.1.1 Prescaler

Frequency prescaler is used to divide the system clock and generate clock source of base counter. 8 frequency division coefficients of prescaler are available and can be selected by TIMx_CR0[TxPSC]. Since this register has no buffer, the clock source frequency will be changed immediately after TIMx_CR0[TxPSC] has been written. Therefore, the frequency division coefficients should be configured when base counter is not working. The clock source frequency formula is: clk_psc2 = SYSCLK/(2^TIM3_CR0[TxPSC]). The clock frequency corresponding to different TIM2_CR0[TxPSC] value as shown in Table 17-1.

Table 17-1 The Clock Frequency Corresponding to TxPSC

TIMx_CR0[TxPSC]	Coefficient	clk_pscx (Hz)	TIMx_CR0[TxPSC]	Coefficient	clk_pscx (Hz)
000	1	24M	100	16	1.5M
001	2	12M	101	32	750K
010	4	6M	110	64	375K
011	8	3M	111	128	187.5K

17.1.2 Reading, Writing and Counting of TIMx__CNTR

TIMx_CR1[TxEN] = 1, TIMx__CNTR starts count. Software write operation to TIMx__CNTR directly changes the value of the register, so it is required to disable the counter before performing the write operation. When software reads TIMx__CNTR, it reads the high byte first and then the low byte, and the hardware caches the low byte simultaneously. When the low byte is read, it reads the cached data.

17.1.3 Output Mode

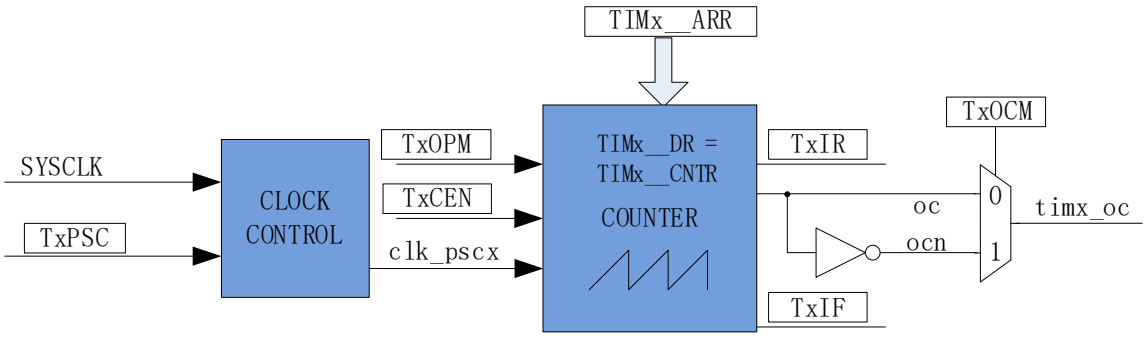


Figure 17-1 Principle Block Diagram of Output Mode

Counter output mode generate output signals according to TIMx_ CR0[TxOCM] setting, and the comparison results between TIMx__CNTR and registers TIMx__DR, TIMx__ARR setting value. Meanwhile, corresponding interrupts is generated.

17.1.3.1 High/Low Level Output

Set TIMx_CR0[TxOCM] = 0 and TIMx__DR > TIMx__ARR, the output signal is always low. Set TIMx_CR0[TxOCM] = 1 and TIMx__DR > TIMx__ARR, the output signal is always high.

17.1.3.2 PWM Generation

In PWM output mode, the set value of TIMx__ARR configures the PWM cycle, and the set value of TIMx__DR configures the duty cycle. Duty cycle = TIMx__DR/TIMx__ARR*100%. TIMx_CR0[Tx_OCM] = 0, if TIMx__CNTR < TIMx__DR, output low level, otherwise output high level. TIMx_CR0[Tx_OCM] = 1, if TIMx__CNTR < TIMx__DR, output high level, otherwise output low level. If TIMx__CNTR > TIMx_ARR, the output signal is reversed.

17.1.3.3 Interrupt Event

- When TIMx__CNTR = TIMx__DR, a compare match interrupt is generated. The interrupt flag TIMx_CR1[TxIR] is set to 1, and the counter continues to count
- When TIMx__CNTR = TIMx__ARR, an overflow event is generated. The interrupt flag TIMx_CR1[TxIF] is set to 1, the counter is cleared. Whether to recount is determined by TIMx_CR0[TxOPM]. TIMx_CR0[TxOPM]= 1, stop counting. TIMx_CR0[TxOPM]= 0, restart counting.

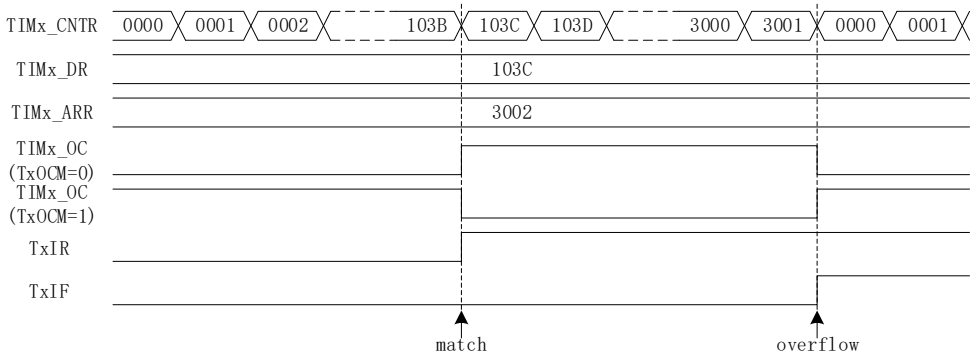


Figure 17-2 Output Waveform of Output Mode

17.1.4 Input Signal Filtering and Edge Detection

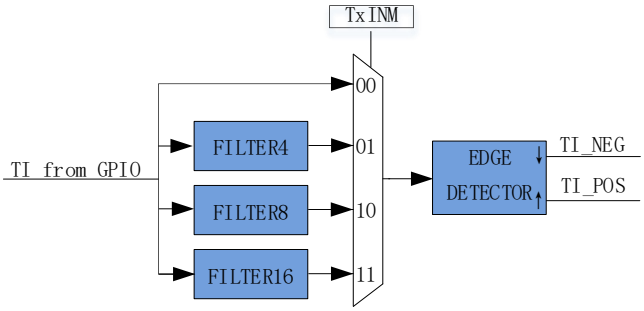


Figure 17-3 Block Diagram of Input Signal Filtering and Edge Detection

The input signal of Timer3/Timer4 comes from GPIO. Configure TIMx_CR1[TINM], the filtering circuit is disabled or filters out the input noise below 4/8/16 system clock cycles. The filtered signal is 4/8/16 system clock cycles delayed than the signal before filtering.

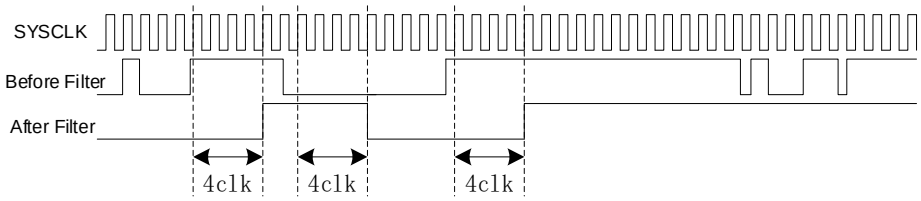


Figure 17-4 Timing Diagram of Filter Module

After input signal filtered by filtering module, the edge detection module detects the filtered input signal and records the rising edge and falling edge for input capture mode.

17.1.5 Input Capture Mode

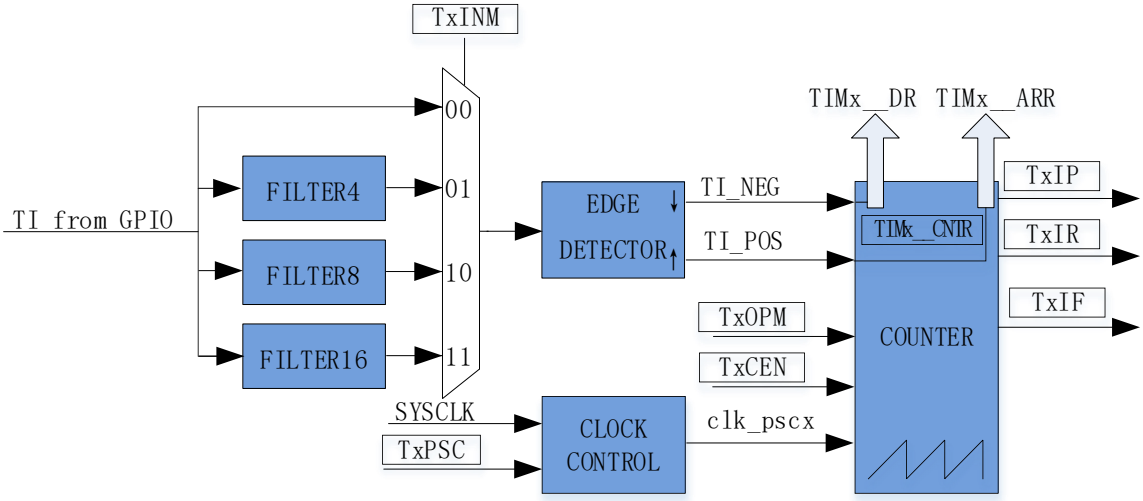


Figure 17-5 Principle Block Diagram of Input Capture Mode

The input capture mode is used to detect the pulse width and period of the input PWM. When TIMx_CR0[TxOCM] = 0, select the time between two adjacent rising edges as one cycle, and the time from rising edge to falling edge is pulse width (high-level pulse width). When TIMx_CR0[TxOCM] = 1, select the time between two adjacent falling edges as one cycle, and the time from falling edge to rising edge is pulse width (low-level pulse width). The count values TIMx__CNTR of pulse width and period are stored in TIMx__DR and TIMx__ARR respectively.

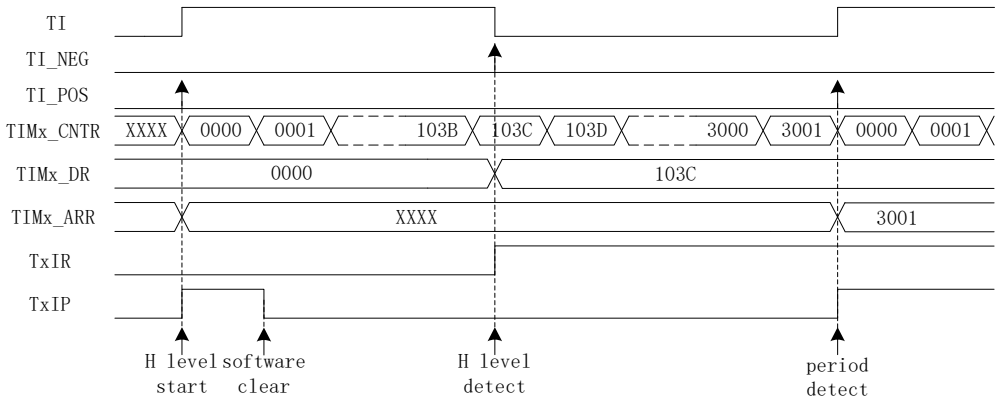


Figure 17-6 Timing Diagram of Input Capture Mode (TIMx_CR0[TxOCM] = 0)

E. g., TIMx_CR0[TxOCM]= 0, configure TIMx_CR1[TxEN] = 1 to enable the counter, and the counter counts up. When the first rising edge is detected, the counter is cleared and restarts counting. When the falling edge is detected, store the value of TIMx__CNTR in TIMx__DR. The interrupt flag TIMx_CR1[TxIR] is set to 1, and TIMx__CNTR continues to count. When the second rising edge is detected, save the value of TIMx__CNTR into

TIMx__ARR. The interrupt flag TIMx_CR1[TxIP] is set to 1 and TIMx__CNTR is cleared. TIMx_CR0[TxOPM] determines whether to restart counting. TIMx_CR0[TxOPM] = 1, stop counting; TIMx_CR0[TxOPM] = 0, restart counting.

When the second rising edge is not arrived and the count value TIMx__CNTR reaches 0xFFFF, an overflow event is generated. Interrupt flag TIMx_CR1[TxIF] is set to 1, and TIMx__CNTR is cleared. TIMx_CR0[TxOPM] determines whether to restart counting. TIMx_CR0[TxOPM] = 1, stop counting; TIMx_CR0[TxOPM] = 0, restart counting.

17.1.6 FG Output Mode of Timer4

Refer to FG Generation.

17.2 Timer3/Timer4 Register

17.2.1 TIMx_CR0 (0x9C/0x9E) (x = 3/4)

Bit	7	6	5	4	3	2	1	0
Name	TxPSC			TxOCM	TxIRE	RSV	TxOPM	TxMOD
Type	R/W	R/W	R/W	R/W	R/W	—	R/W	R/W
Reset	0	0	0	0	0	—	0	0

Bit	Name	Description
[7:5]	TxPSC	Base counter clock source frequency division selection Divide the system clock as the clock source of the base counter The clock source frequency after frequency division is: 000: 24MHz 001: 12MHz 010: 6MHz 011: 3MHz 100: 1.5MHz 101: 750kHz 110: 375kHz 111: 187.5kHz Note: In the input capture mode of Timer3, 111 corresponds to 48MHz
[4]	TxOCM	Output mode: Output mode selection 0: TIM2__CNTR < TIM2__DR, output 0; TIM2__CNTR ≥ TIM2__DR, output 1 1: TIM2__CNTR < TIM2__DR, output 1; TIM2__CNTR ≥ TIM2__DR, output 0 Input capture mode: Active edge selection 0: The time between two adjacent rising edges is one cycle, and the time from rising edge to falling edge is pulse width (high-level pulse width) 1: The time between two adjacent falling edges is one cycle, and the time from falling edge to rising edge is pulse width (low-level pulse width)
[3]	TxIRE	Output mode: Compare match interrupt enable Input capture mode: Pulse width detected interrupt enable 0: Disable 1: Enable
[2]	RSV	Reserved
[1]	TxOPM	Single mode The counter stops counting when the following events occur Output mode: Base counter overflow Input capture mode: PWM cycle detected or base counter overflow 0: Base counter does not stop 1: Base counter stop (TIMx_CR1[TxEN] cleared)
[0]	TxMOD	Working mode selection

		0: Input capture mode 1: Output mode

17.2.2 TIMx_CR1 (0x9D/0x9F) (x = 3/4)

Bit	7	6	5	4	3	2	1	0
Name	TxIR	TxIP	TxIF	TxIPE	TxIFE	TxINM		TxEN
Type	R/W0	R/W0	R/W0	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	TxIR	Output mode: Compare match interrupt flag Input capture mode: Pulse width detected interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[6]	TxIP	Output mode: No meaning Input capture mode: PWM cycle detected interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[5]	TxIF	Output mode: Base counter overflow interrupt flag. When TIMx__CNTR matches TIMx__ARR, this bit is set to 1 Input capture mode: Base counter overflow interrupt flag. When Timer has not detected the input PWM cycle and TIMx__CNTR accumulates to 0xFFFF, this bit is set to 1 Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[4]	TxIPE	Output mode: No meaning Input capture mode: PWM cycle detected interrupt enable. 0: Disable 1: Enable
[3]	TxIFE	Output mode: Base counter overflow interrupt enable Input capture mode: Base counter overflow interrupt enable 0: Disable 1: Enable
[2:1]	TxINM	Input signal filtering pulse width selection When the pulse width of the input signal is less than the set time, it is filtered as noise 00: No filtering 01: 4 system clock cycles, 4*41.67ns 10: 8 system clock cycles, 8*41.67ns 11: 16 system clock cycles, 16*41.67ns
[0]	TxEN	Base counter enable 0: Disable 1: Enable

17.2.3 TIMx__CNTR (0xA2, 0xA3/0x92, 0x93) (x = 3/4)

TIMx__CNTRH(0xA3/0x93)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIMx__CNTR[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIMx__CNTRL(0xA2/0x92)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIMx__CNTR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIMx__CNTR	Count value of base counter

17.2.4 TIMx__DR (0xA4, 0xA5/0x94, 0x95) (x = 3/4)

TIMx__DRH(0xA5/0x95)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIMx__DR[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIMx__DRL(0xA4/0x94)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIMx__DR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIMx__DR	Output mode: Compare match values (written by software) Input capture mode: Count value of PWM pulse width (written by hardware)

17.2.5 TIMx__ARR (0xA6, 0xA7/0x96, 0x97) (x = 3/4)

TIMx__ARRH(0xA7/0x97)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	TIMx__ARR[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0
TIMx__ARRL(0xA6/0x96)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	TIMx__ARR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	TIMx__ARR	Output mode: Reload value (written by software). FG mode refer to 14.1.11

		Input capture mode: Count value of one PWM cycle (written by hardware)

18 Systick

18.1 Systick Instructions

The chip can generate Systick interrupt with a fixed frequency, and the interrupt cycle is controlled by SYST_ARR.

Set DRV_SR[SYSTIE] = 1, enables Systick interrupt, and the interrupt entry is 10.

18.2 Systick Register

18.2.1 DRV_SR (0x4061)

Bit	7	6	5	4	3	2	1	0
Name	SYSTIF	SYSTIE	FGIF	DCIF	FGIE	DCIP	DCIM	
Type	R/W0	R/W	R/W0	R/W0	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	SYSTIF	Systick interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No Meaning
[6]	SYSTIE	Systick interrupt enable 0: Disable 1: Enable
[5]	FGIF	FG interrupt flag When FOC/square wave drives, an FGIF interrupt is generated in each rotation cycle (electrical cycle) Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No Meaning
[4]	DCIF	Driver compare match interrupt flag When the Driver counter value is equal to DRV_COMR, whether to generate an interrupt is determined according to the counting direction set by DRV_SR[DCIM] Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[3]	FGIE	FG interrupt enable After interrupt enable, one FG interrupt is generated in each electric cycle during FOC drive/square wave control 0: Disable 1: Enable
[2]	DCIP	Number of PWM cycles generating compare match interrupt 0: 1 interrupt in 1 PWM cycle 1: 1 interrupt in 2 PWM cycles
[1:0]	DCIM	Compare match interrupt mode selection

		When the count value is equal to DRV_COMR, whether to generate an interrupt request is determined according to the value of DRV_SR[DCIM] 00: No interrupt will be generated 01: Interrupt is generated when the counter counts up 10: Interrupt is generated when the counter counts down 11: Interrupt is generated when the counter counts up/down

18.2.2 SYST_ARR (0x4064, 0x4065)

SYST_ARRH(0x4064)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	SYST_ARR[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	1	0	1	1	1	0	1
SYST_ARRL(0x4065)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	SYST_ARR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	0	1	1	1	1	1	1

Bit	**Name**	**Description**
[15:0]	SYST_ARR	Systick reload value This value determines the period of systick interrupt. The default time is 1ms The calculation formula is: systick interrupt frequency = SYSCLK/(SYST_ARR[15:0] + 1) Range [0,65535]

19 Driver

19.1 Driver Instructions

19.1.1 Driver Introduction

The chip has built-in Pre-driver output.

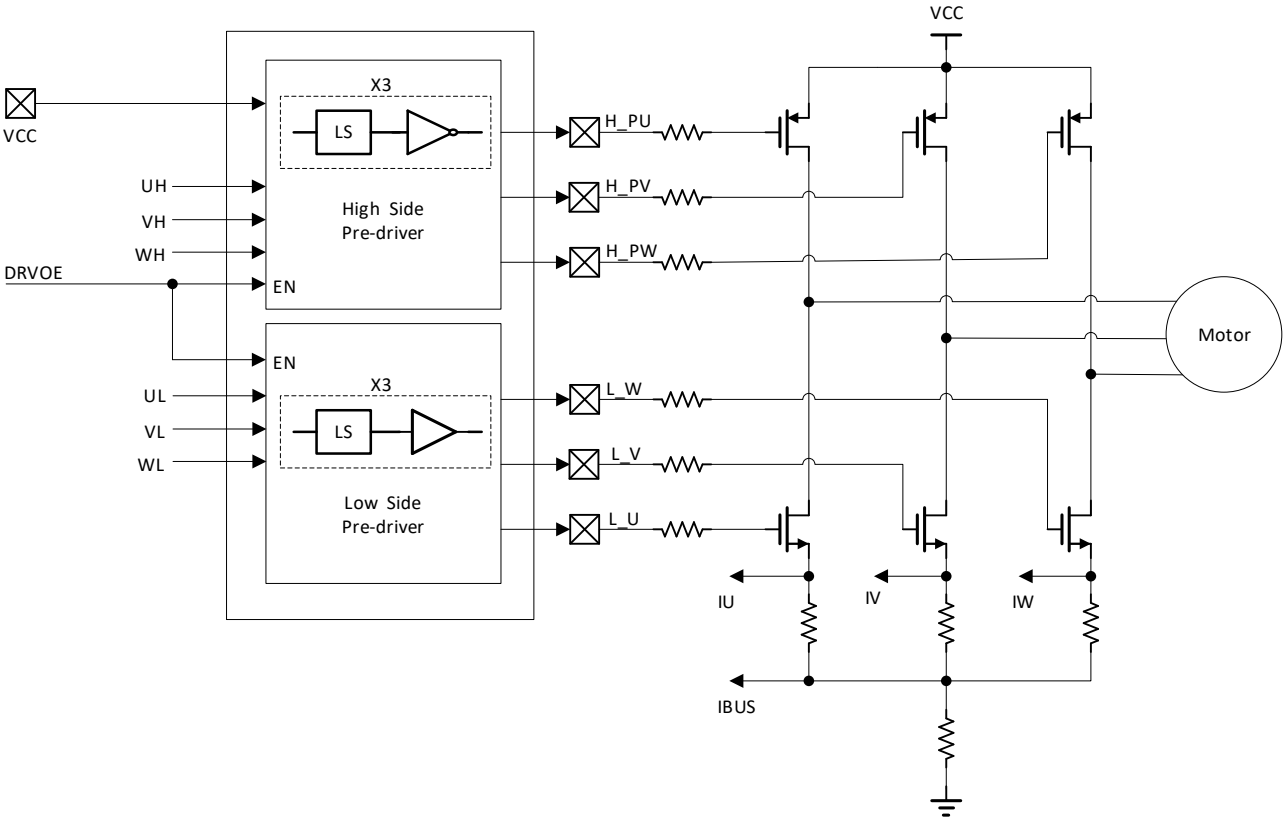


Figure 19-1 Block Diagram of Pre-driver Module

3P3N Pre-driver module is shown in Figure 19-1. UH/VH/WH and UL/VL/WL, 3-phase PWM signal, are the input signals of Pre-driver. H_PU/H_PV/H_PW and L_U/L_V/L_W are the output signals of Pre-driver. H_PU/H_PV/H_PW are reversely related to UH/VH/WH. DRV_CR[DRVOE] is the enable bit of Pre-driver.

Set RV_CR[DRVOE] = 1 to enable Pre-driver output, UH/VH/WH are reversely and sent to H_PU/H_PV/H_PW for driving gate of PMOS. UL/VL/WL are sent to L_U/L_V/L_W for driving gate of NMOS. PMOS and NMOS output voltages drive motors.

19.1.2 Output Control Module

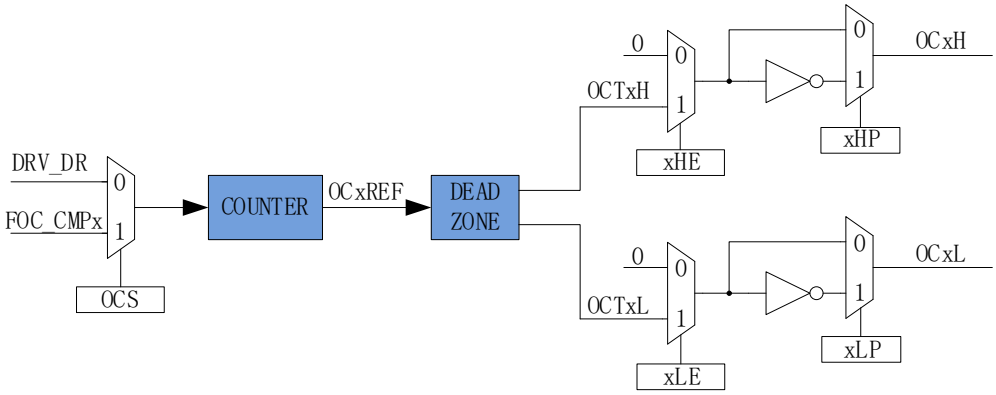


Figure 19-2 Block Diagram of Output Control Module Pre-stage

Before Driver module works, set DRV_CR[MESEL] to 1 to select FOC mode or set DRV_CR[MESEL] to 0 to select square wave control mode.

When DRV_CR[OCS] = 0, the comparison value of PWM comes from DRV_DR, and the referenced of output PWM signal is OCTxH. When OCxH and OCxL output at the same time, OCTxL output reverse signal. When DRV_CR[OCS] = 1, the comparison value of PWM comes from FOC module, and the referenced of output PWM signal is OCTxL. When OCxH and OCxL output at the same time, OCTxH output reverse signal.

19.1.2.1 Count and Compare Module

Set DRV_CR[OCS] to select the comparison value of the PWM from FOC_CMPU/V/W of the FOC module or DRV_DR set by software. The comparison value is sent to the counter for comparison to obtain the 3-phase original PWM signal OCxREF. When DRV_CNTR is smaller than the comparison value, OCxREF outputs high. When DRV_CNTR is larger than DRV_DR, OCxREF outputs low.

Configure DRV_CR[OCS] = 1, Driver module compares FOC_CMPU/V/W and the count value to generate the duty cycle OC1REF/OC2REF/OC3REF.

Configure DRV_CR[OCS] = 0, select DRV_DR set by software compare with the count value to generate OC1REF/OC2REF/OC3REF with the same duty cycle. Duty cycle = DRV_DR/DRV_ARR*100%.

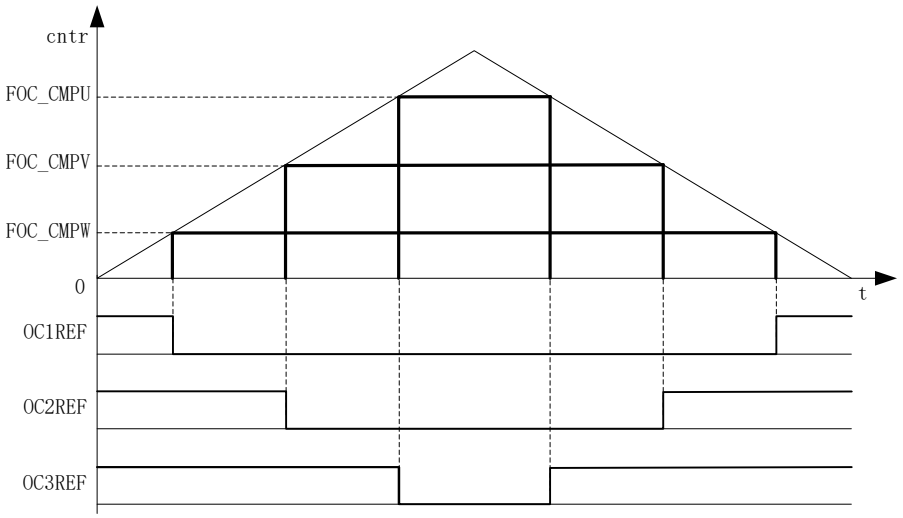


Figure 19-3 PWM Generation Diagram

19.1.2.2 Deadtime Module

OCxREF has hardware deadtime insertion. Each channel has an 8-bit deadtime generator, and three channels have the same dead time, which is set by DRV_DTR. In the rising edge of OCxREF, the actual output high level of OCxL is delayed by the time set by DRV_DTR than the rising edge of OCxREF. In the falling edge of OCxREF, the actual output high level of OCxH is delayed by the time set by DRV_DTR than the falling edge of OCxREF.

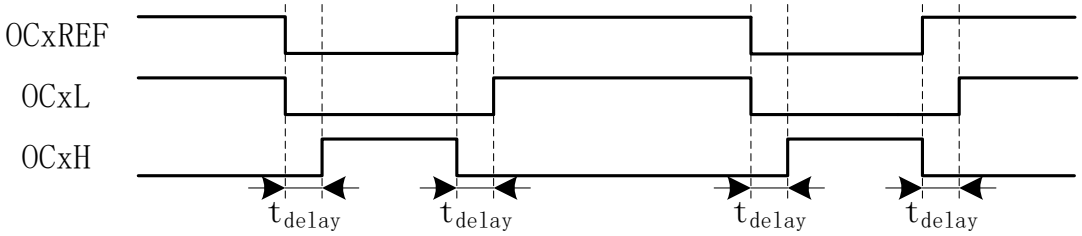


Figure 19-4 Complementary Output with Deadtime Insertion

19.1.2.3 Enable and Polarity of Output Signals

Software selects the output mode as idle level or complementary output with deadtime inserted by configuring DRV_CMR[xHE] and [xLE]. Software selects the polarity of output by configuring DRV_CMR[xHP] and [xLP]. In the application of square wave control, Timer1 automatically controls DRV_CMR to implement phase commutation function. Set DRV_CR[MESEL] = 0 to select square wave control mode. When Timer1 generates a writing sequence event, the corresponding TIM1_DBRx is transferred to DRV_CMR.

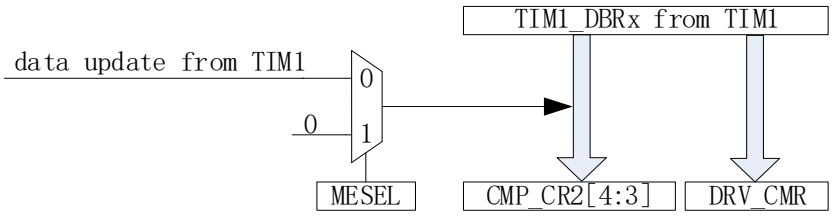


Figure 19-5 Timer1 Automatically Controls DRV_CMR and CMP_CR2[4:3]

Configure DRV_DR, DRV_ARR and DRV_CMR can implement pre-charging and brake, etc. DRV_DR, DRV_ARR control the duty cycle and frequency of PWM, and DRV_CMR[xHE], DRV_CMR[xLE] control the output modes.

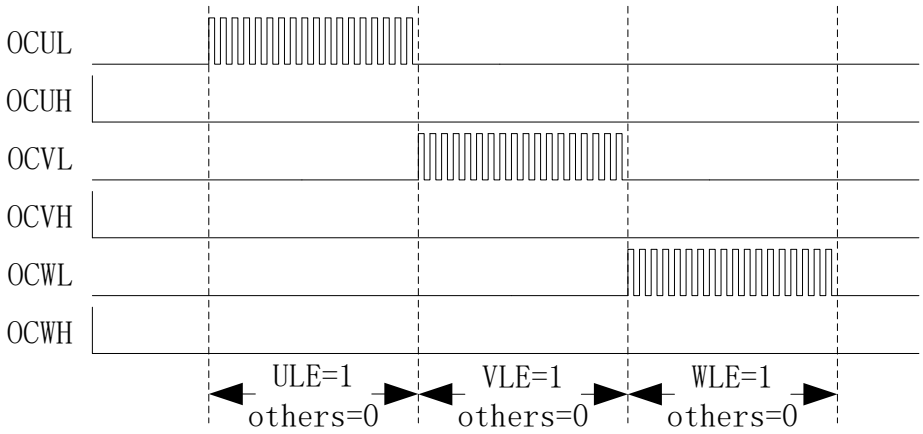


Figure 19-6 Waveform Diagram of Pre-charging

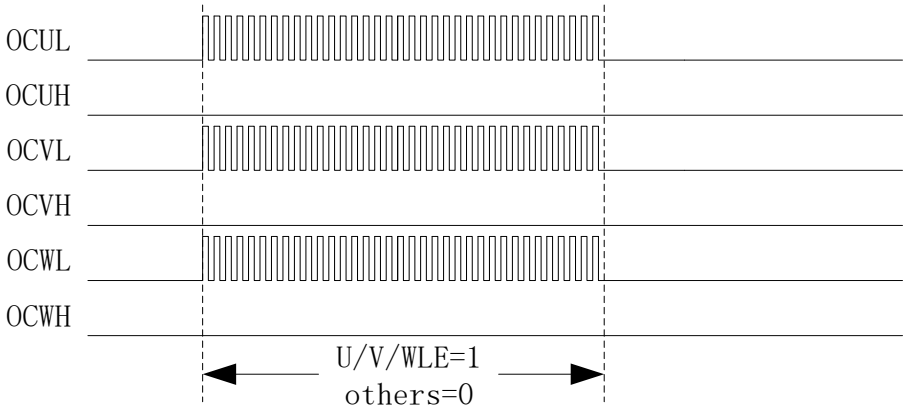


Figure 19-7 Waveform Diagram of Brake

19.1.2.4 MOE

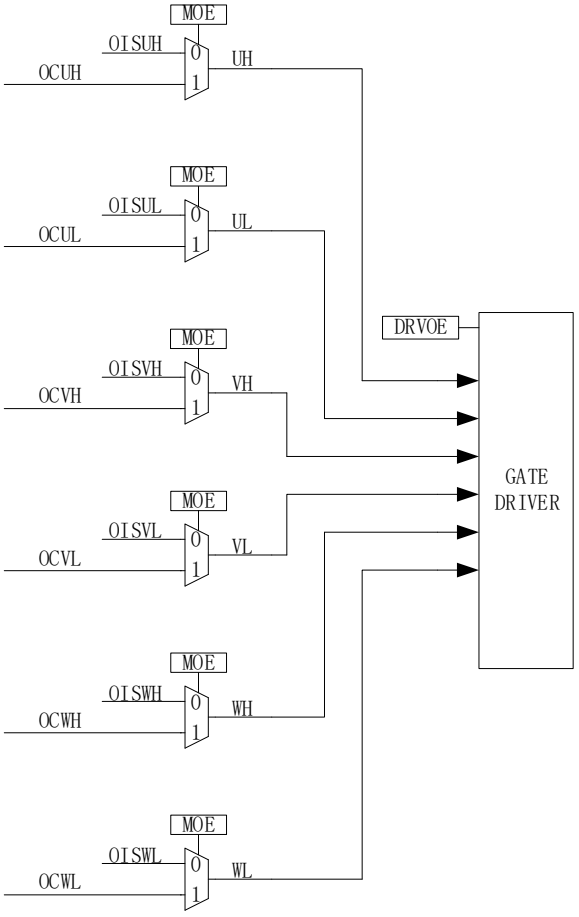


Figure 19-8 Block Diagram of Rear Stage of Output Control Module

Enable DRV_OUT[MOE] to output PWM waveform for driving the motor. Disable DRV_OUT[MOE] output the idle level set by software to stop motor driving.

19.1.2.5 Interrupt

19.1.2.5.1 Compare Match Interrupt

Set the generation condition of compare match interrupt by DRV_SR[DCIM], the compare value by DRV_COMR. When the count value is equal to DRV_COMR and the condition set by DRV_SR[DCIM] is met, a compare match interrupt is generated and the interrupt flag DRV_SR[DCIF] is set to 1 by hardware.

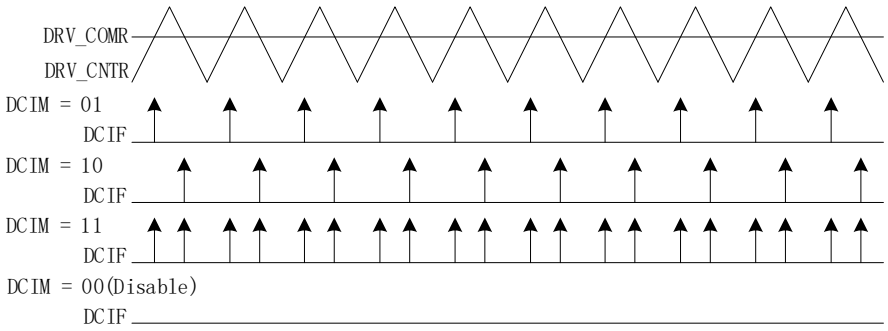


Figure 19-9 Compare Match Interrupt of Driver

19.1.2.5.2 FG Interrupt

Set DRV_SR[FGIE] to 1 to enable FG interrupt, the motor generates an interrupt for every electrical cycle.

19.2 Driver Register

19.2.1 DRV_CR (0x4062)

Bit	7	6	5	4	3	2	1	0
Name	DRVEN	DDIR	FOCEN	DRPE	OCS	MESEL	RSV	DRVOE
Type	R/W	R/W	R/W	R/W	R/W	R/W	—	R/W
Reset	0	0	0	0	0	0	—	0

Bit	Name	Description
[7]	DRVEN	Counter Enable 0: Disable 1: Enable
[6]	DDIR	Output direction (forward and reverse) Change the direction of motor rotation, both square wave control and FOC are valid. In sensorless FOC mode, change this bit can change the direction. In Hall effected FOC, this bit is required to work with the software to obtain the angle; Square wave control should modify the configuration of Timer1. 0: Forward 1: Reverse
[5]	FOCEN	FOC module Enable 0: Disable 1: Enable
[4]	DRPE	DRV_DR preload enable When preload is enabled, the value written by software to DRV_DR is updated after a counter underflow event. Disable this function, the value written by software to DRV_DR is updated immediately 0: Disable 1: Enable
[3]	OCS	Comparison source selection 0: DRV_DR 1: FOC module
[2]	MESEL	ME operating mode selection 0: Square wave control mode 1: FOC mode

[1]	RSV	Reserved
[0]	DRVOE	Driver enable 0: Disable 1: Enable

19.2.2 DRV_SR (0x4061)

Bit	7	6	5	4	3	2	1	0
Name	SYSTIF	SYSTIE	FGIF	DCIF	FGIE	DCIP	DCIM1	DCIM0
Type	R/W0	R/W	R/W0	R/W0	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	SYSTIF	Systick interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No Meaning
[6]	SYSTIE	Systick interrupt enable 0: Disable 1: Enable
[5]	FGIF	FG interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No Meaning
[4]	DCIF	Driver compare match interrupt flag When the driver counter value is equal to DRV_COMR, whether to generate an interrupt is determined according to the counting direction set by DRV_SR[DCIM] Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[3]	FGIE	FG interrupt enable After interrupt enable, one FG interrupt is generated in each electric cycle in FOC/square wave control mode 0: Disable 1: Enable
[2]	DCIP	Number of PWM cycles generating compare match interrupt 0: 1 interrupt in 1 PWM cycle 1: 1 interrupt in 2 PWM cycles
[1:0]	DCIM	Compare match interrupt mode selection When the count value is equal to DRV_COMR, whether to generate an interrupt request is determined according to the value of DRV_SR[DCIM] 00: No interrupt will be generated 01: Interrupt is generated when the counter counts up 10: Interrupt is generated when the counter counts down 11: Interrupt is generated when the counter counts up/down

19.2.3 DRV_OUT (0xF8)

Bit	7	6	5	4	3	2	1	0
Name	MOE	RSV	OISWL	OISWH	OISVL	OISVH	OISUL	OISUH
Type	R/W	—	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	—	0	0	0	0	0	0

Bit	Name	Description
[7]	MOE	Main output enable Used to select the source of 3-phase output signals of bridge high and low side. This bit can be set to 1 and cleared by software. When bus current protection is generated (refer to 29.1.1.1, the hardware automatically clears this bit to 0 and turns off the output. 0: Disable, the output is derived from the idle levels set by DRV_OUT[OISUH/OISVH/OISWH] and DRV_OUT[OISUL/OISVL/OISWL] 1: Enable, the output is derived from the comparison value of the counter
[6]	RSV	Reserved
[5]	OISWL	Output idle level of WL Refer to DRV_OUT[OISUH] description
[4]	OISWH	Output idle level of WH Refer to DRV_OUT[OISUH] description
[3]	OISVL	Output idle level of VL Refer to DRV_OUT[OISUH] description
[2]	OISVH	Output idle level of VH Refer to DRV_OUT[OISUH]description
[1]	OISUL	Output idle level of UL Refer to DRV_OUT[OISUH]description
[0]	OISUH	Output idle level of UH This bit sets the output idle level of UH. When DRV_OUT[MOE] = 0, the output idle level is used to turn off the MOS 0: Low level 1: High level

19.2.4 DRV_CMR (0x405C, 0x405D)

DRV_CMRH(0x405C)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV				WHP	WLP	VHP	VLP
Type	—	—	—	—	R/W	R/W	R/W	R/W
Reset	—	—	—	—	0	0	0	0
DRV_CMRL(0x405D)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	UHP	ULP	WHE	WLE	VHE	VLE	UHE	ULE
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:12]	RSV	Reserved
[11]	WHP	Output polarity of WH 0: High level active 1: Low level active
[10]	WLP	Output polarity of WL 0: High level active 1: Low level active
[9]	VHP	Output polarity of VH 0: High level active 1: Low level active
[8]	VLP	Output polarity of VL 0: High level active 1: Low level active
[7]	UHP	Otput polarity of UH 0: High level active 1: Low level active
[6]	ULP	Otput polarity of UL 0: High level active 1: Low level active
[5]	WHE	Output enable of WH 0: Disable 1: Enable
[4]	WLE	Output enable of WL 0: Disable 1: Enable
[3]	VHE	Output enable of VH 0: Disable 1: Enable
[2]	VLE	Output enable of VL 0: Disable 1: Enable
[1]	UHE	Output enable of UH 0: Disable 1: Enable
[0]	ULE	Output enable of UL 0: Disable 1: Enable

Notes:

- When DRV_CMR[W/V/ULE] and DRV_CMR[W/V/UHE] are 1 at the same time, 3-phase outputs

complementary PWM signals with deadtime insertion. Low side output is the reference polarity

- When used for square wave control, Timer1 will automatically control the DRV_CMR register

19.2.5 DRV_ARR (0x405E, 0x405F)

DRV_ARRH(0x405E)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV		DRV_ARR [13:8]					
Type	—	—	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	—	0	0	0	0	0	0
DRV_ARRL(0x405F)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	DRV_ARR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:14]	RSV	Reserved
[13:0]	DRV_ARR	Counter reload value, which determines the PWM frequency (central alignment mode) Driver counter counts from 0 to DRV_ARR/2 - 1 and generates an overflow event. Then, the counter counts down to 0 Calculation formula: $f_{carrier}$ = 48MHz/DRV_ARR Note: LSB is always 0, writing 1 to LSB is meaningless The value of DRV_ARR is calculated by clock of 48MHz, and the range is [0,16382]

19.2.6 DRV_COMR (0x405A, 0x405B)

DRV_COMRH(0x405A)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV				DRV_COMR[11:8]			
Type	—	—	—	—	R/W	R/W	R/W	R/W
Reset	—	—	—	—	0	0	0	0
DRV_COMRL(0x405B)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	DRV_COMR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:12]	RSV	Reserved
[11:0]	DRV_COMR	Counter compare match value The Driver compare match interrupt is generated when the count value is equal to DRV_COMR and the condition set by DRV_SR[DCIM] is met The clock frequency for the calculation of the DRV_COMR is 12MHz Duty cycle of matched point = DRV_COMR*4/DRV_ARR*100% Range: [0,4095]

19.2.7 DRV_DR (0x4058, 0x4059)

DRV_DRH(0x4058)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV		DRV_DR[13:8]					
Type	—	—	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	—	0	0	0	0	0	0
DRV_DRL(0x4059)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	DRV_DR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:14]	RSV	Reserved
[13:0]	DRV_DR	PWM duty cycle setting Duty cycle = DRV_DR/DRV_ARR*100% Note: When using this register as a comparison source, the PWM is referenced to the high side of bridge and a deadtime is inserted in the complementary output of the low side of bridge The clock frequency for the calculation of the DRV_DR is 48MHz Range: [0,16382]

19.2.8 DRV_DTR (0x4060)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	DRV_DTR							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	DRV_DTR	Deadtime setting Deadtime = (DRV_DTR + 1)*41.67ns E.g., when DRV_DTR = 11, Deadtime = 12*41.67ns = 500ns Note: DRV_DTR = 0, no deadtime insertion

19.2.9 DRV__CNTR (0x4066, 0x4067)

DRV__CNTRH(0x4066)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV				DRV__CNTR[11:8]			
Type	—	—	—	—	R/W	R/W	R/W	R/W
Reset	—	—	—	—	0	0	0	0
DRV__CNTRL(0x4067)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	DRV__CNTR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:12]	RSV	Reserved
[11:0]	DRV__CNTR	Counter value The clock frequency for the calculation of the DRV_ CNTR is 12MHz Duty cycle of driver = DRV__CNTR*4/DRV_ARR*100% Note: DRV__CNTR can only be written when DRV_CR[DRVEN] = 1 Range: [0,4095]

20 WDT

The watchdog is a timer that works in the internal slow clock and is used to monitor the main program operation and prevent the MCU running out. Watchdog principle: After starting the watchdog, the counter of watchdog starts counting. When the watchdog counter overflows, watchdog sends a signal to reset the MCU and the program restarts running from address 0. During the operation of program, software has to initialize watchdog at intervals to prevent the watchdog counter overflowing.

After enable, the watchdog starts counting from 0. When the counter reaches 0xFFFC, watchdog outputs a signal, width of 4 internal slow clock cycles, to reset the MCU and the program starts running from address 0. Software initialize watchdog at regular intervals during operation, and the watchdog counter value will return to WDT_ARR and restart counting.

20.1 WDT Notes

- When the MCU enters standby mode or sleep mode, the WDT will stop counting, but the count value will be retained
- WDT will be automatically disabled during emulation
- RST_SR[RSTWDT] will be set to 1 when the MCU is reset by WDT counter overflow

20.2 WDT Instructions

1. Set CCFG1[WDT_EN] = 1 to start the watchdog, which starts counting from 0;
2. Setting WDT_ARR (this operation can also be performed before starting the watchdog);
3. Set WDT_CR[WDTRF] = 1 in the running of program, the watchdog counter returns to WDT_ARR.

20.3 WDT Register

20.3.1 WDT_CR (0x4026)

Bit	7	6	5	4	3	2	1	0
Name	RSV							WDTRF
Type	—	—	—	—	—	—	—	R/W
Reset	—	—	—	—	—	—	—	0

Bit	Name	Description
[7:1]	RSV	Reserved
[0]	WDTRF	Watchdog initialization 0: No meaning 1: The watchdog counter returns to the WDT_ARR setting and starts counting again

20.3.2 WDT_ARR (0x4027)

Bit	7	6	5	4	3	2	1	0
Name	WDT_ARR							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	WDT_ARR	Watchdog Reload Counter Set the high 8 bits of the initialized value of the watchdog counter

20.3.3 CCFG1 (0x401E)

Bit	7	6	5	4	3	2	1	0
Name	RSV	LVWIE	WDT_EN	RSV				
Type	—	R/W	R/W	—	—	—	—	—
Reset	—	0	0	—	—	—	—	—

Bit	Name	Description
[7]	RSV	Reserved
[6]	LVWIE	LVW interrupt enable 0: Disable 1: Enable
[5]	WDT_EN	WDT Enable 0: Disable 1: Enable
[4:0]	RSV	Reserved

21 RTC and Clock Calibration

21.1 Functional Block Diagram of RTC

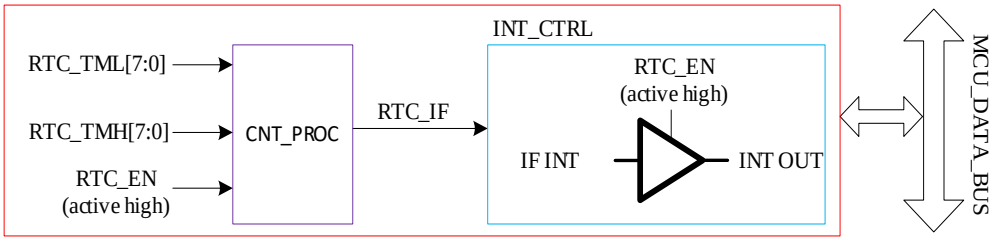


Figure 21-1 Functional Block Diagram of RTC

21.2 RTC Instructions

Write RTC_TM to set the reload value of RTC. Set RTC_STA[RTC_EN] to 1 to enable RTC counter.

21.3 RTC Registers

21.3.1 RTC_TM (0x402C, 0x402D)

RTC_TMH(0x402C)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RTC_TM[15:8]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1
RTC_TML(0x402D)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	RTC_TM[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	1	1	1	1	1	1	1	1

Bit	**Name**	**Description**
[15:0]	RTC_TM	RTC count register Read: Instantaneous value of counter Write: The RTC counter counts from 0 to the written value at 32768Hz and then overflows. RTC generates an interrupt request and clear the counter to 0 to restart counting

21.3.2 RTC_STA (0x402E)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	RTC_EN	RTC_IF	RSV	ISOSCEN	RSV			
Type	R/W	R/W0	—	R/W	—	—	—	—
Reset	0	0	—	0	—	—	—	—

Bit	**Name**	**Description**
[7]	RTC_EN	RTC Enable 0: Disable 1: Enable
[6]	RTC_IF	RTC interrupt flag This bit is set to 1 when counter value equals RTC_TM Read:

		0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[5]	RSV	Reserved
[4]	ISOSCEN	Internal slow clock enable 0: Disable 1: Enable
[3:0]	RSV	Reserved

21.4 Clock Calibration

21.4.1 Clock Calibration Introduction

Clock calibration is a function that uses the internal slow clock to calibrate the internal fast clock. Calibration principle: A 12-bit counter is used to count the time of 4 slow clock cycles by using the fast clock as the clock source.

Calibration process: Set CAL_CR0[CAL_STA] = 1 to start the calibration. Software read CAL_CR0[CAL_BUSY] to detect whether the calibration is finished. When the calibration is completed (CAL_CR0[CAL_BUSY] = 0), the value of CAL_CR0[CAL_ARR] is the value of counter, counting 4 slow clock cycles.

21.4.2 Clock Calibration Registers

21.4.2.1 CAL_CR0 (0x4044) CAL_CR1 (0x4045)

CAL_CR0(0x4044)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	CAL_STA/ CAL_BUSY	RSV			CAL_ARR[11:8]			
Type	R/W1	—	—	—	R/W	R/W	R/W	R/W
Reset	1	—	—	—	0	0	0	0
CAL_CR1(0x4045)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	CAL_ARR[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15]	CAL_STA /CAL_BUSY	Clock calibration enable Read: 0: Calibration completed 1: Calibration in progress Write: 0: No meaning 1: Start clock calibration
[14:12]	RSV	Reserved
[11:0]	CAL_ARR	Calibrate count values The value of fast clock to count 4 slow clock cycles

		Note: 0 of this value indicates that there is no corresponding slow clock input, and 0xFFF of this value indicates a count overflow (slow clock too slow or fast clock too fast)

22 IO

22.1 IO Introduction

The chip has 35 GPIO pins: P0.0 ~ P0.7, P1.0 ~ P1.7, P2.0 ~ P2.7, P3.0 ~ P3.7, P4.2 and P4.4 ~ P4.5. Each GPIO has related configuration registers.

22.2 IO Instructions

- P0.0 ~ P0.7, P1.0 ~ P1.7, P2.0 ~ P2.7, P3.0 ~ P3.7, P4.2 and P4.4 ~ P4.5 are mapped to registers P0, P1, P2, P3, P4 respectively
- P0_OE, P1_OE, P2_OE, P3_OE, and P4_OE are used to configure the digital inputs and outputs of P0.0 ~ P3.7, P4.2, and P4.4 ~ P4.5. When the port is configured as input, Schmitt trigger input is enabled.
- Configure corresponding bits of P0_PU, P1_PU, P2_PU, P3_PU, and P4_PU to 1, the pull-up resistors of P0.0 ~ P3.7, P4.2, and P4.4 ~ P4.5 can be enabled where the pull-up resistor of P0.0 ~ P0.2, P1.3 ~ P1.6, P2.1, P3.6 ~ P3.7 is about 5.6kΩ, and the other is about 33kΩ.
- The pull-down resistor of P1.1, P0.1 can be enabled. The pull-down resistor is about 10kΩ, configured by P3_AN[7:6]
- When IO output digital 0, the pull-up resistor will be automatically disabled
- P0.0 ~ P0.6 can be configured as INT0 input, P1.0 ~ P2.7 can be configured as INT1 input. Both INT0 and INT1 can be configured as rising edge triggered interrupt, falling edge triggered interrupt or level change triggered interrupt.
- Set the corresponding bits of P1_AN, P2_AN and P3_AN to 1, P1.3 ~ P1.7, P2.0 ~ P2.7 and P3.0 ~ P3.5 is configured as analog mode. When the port is configured as analog port, all digital functions are disabled and the port status, corresponding to registers P1, P2, and P3, are 0.
- Pull-up resistor of P1.6 ~ P1.7, P2.0 ~ P2.7, P3.0 ~ P3.5 will be automatically disabled when the port is configured as analog mode. The pull-up resister of P1.3 ~ P1.5 can be enabled or disabled according to the actual application requirements.

22.3 IO Registers

22.3.1 P0_OE (0xFC)

Bit	7	6	5	4	3	2	1	0
Name	P0_OE							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	P0_OE	P0.0 ~ P0.7 digital input/output selection 0: Input 1: Output

22.3.2 P1_OE (0xFD)

Bit	7	6	5	4	3	2	1	0
Name	P1_OE							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	P1_OE	P1.0 ~ P1.7 digital input/output selection 0: Input 1: Output

22.3.3 P2_OE (0xFE)

Bit	7	6	5	4	3	2	1	0
Name	P2_OE							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	P2_OE	P2.0 ~ P2.7 digital input/output selection 0: Input 1: Output

22.3.4 P3_OE (0xFF)

Bit	7	6	5	4	3	2	1	0
Name	P3_OE							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	P3_OE	P3.0 ~ P3.7 digital input/output selection 0: Input 1: Output

22.3.5 P4_OE (0xE9)

Bit	7	6	5	4	3	2	1	0
Name	RSV		P4_OE[5:4]		RSV	P4_OE[2]	RSV	

Type	—	—	R/W	R/W	—	R/W	—	—
Reset	—	—	0	0	—	0	—	—

Bit	Name	Description
[7:6]	RSV	Reserved
[5:4]	P4_OE[5:4]	P4.4 ~ P4.5 digital input/output selection 0: Input 1: Output
[3]	RSV	Reserved
[2]	P4_OE[2]	P4.2 digital input/output selection 0: Input 1: Output
[1:0]	RSV	Reserved

22.3.6 P1_AN (0x4050)

Bit	7	6	5	4	3	2	1	0
Name	P1_AN				HBMOD	RSV	ODE1	ODE0
Type	R/W	R/W	R/W	R/W	R/W	—	R/W	R/W
Reset	0	0	0	0	0	—	0	0

Bit	Name	Description
[7:4]	P1_AN	P1.4 ~ P1.7 analog mode enable 0: Disable 1: Enable
[3]	HBMOD	P1.3 mode configuration, this bit combined with P1_OE[3] determines the functional mode of P1.3 as shown in Table 22-1 Table 22-1 P1.3 Mode Configuration
[2]	RSV	Reserved
[1]	ODE1	P0.1 collector open-drain enable 0: Disable 1: Enable
[0]	ODE0	P0.0 collector open-drain enable 0: Disable 1: Enable

HBMOD	P1_OE[3]	P1.3 mode
0	0	Digital input
0	1	Digital output
1	0	Analog mode
1	1	Digital enhanced drive output mode. The maximum output current of high level output can be up to 20mA for Hall bias power supply. The drive mode of low level output is the same as the digital output mode.

22.3.7 P2_AN (0x4051)

Bit	7	6	5	4	3	2	1	0
Name	P2_AN							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	P2_AN	P2.0 ~ P2.7 analog mode enable 0: Disable 1: Enable

22.3.8 P3_AN (0x4052)

Bit	7	6	5	4	3	2	1	0
Name	P11_PL	P01_PL	P3_AN					
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	P11_PL	P1.1 pull-down resistor enable 0: Disable 1: Enable Note: Pull-up and pull-down resistors of P1.1 cannot be enabled at the same time
[6]	P01_PL	P0.1 pull-down resistor enable 0: Disable 1: Enable Note: Pull-up and pull-down resistors of P0.1 cannot be enabled at the same time
[5:0]	P3_AN	P3.0 ~ P3.5 analog mode enable 0: Disable 1: Enable

22.3.9 P0_PU (0x4053)

Bit	7	6	5	4	3	2	1	0
Name	P0_PU							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	P0_PU	P0.0 ~ P0.7 pull-up resistor enable 0: Disable 1: Enable

22.3.10 P1_PU (0x4054)

Bit	7	6	5	4	3	2	1	0
Name	P1_PU							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description

[7:0]	P1_PU	P1.0 ~ P1.7 pull-up resistor enable 0: Disable 1: Enable

22.3.11 P2_PU (0x4055)

Bit	7	6	5	4	3	2	1	0
Name	P2_PU							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	P2_PU	P2.0 ~ P2.7 pull-up resistor enable 0: Disable 1: Enable

22.3.12 P3_PU (0x4056)

Bit	7	6	5	4	3	2	1	0
Name	P3_PU							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	P3_PU	P3.0 ~ P3.7 pull-up resistor enable 0: Disable 1: Enable

22.3.13 P4_PU (0x4057)

Bit	7	6	5	4	3	2	1	0
Name	RSV		P4_PU[5]	P4_PU[4]	RSV	P4_PU[2]	RSV	
Type	—	—	R/W	R/W	—	R/W	—	—
Reset	—	—	0	0	—	0	—	—

Bit	Name	Description
[7:6]	RSV	Reserved
[5:4]	P4_PU[5:4]	P4.4 ~ P4.5 pull-up resistor enable 0: Disable 1: Enable
[3]	RSV	Reserved
[2]	P4_PU[2]	P4.2 pull-up resistor enable 0: Disable 1: Enable
[1:0]	RSV	Reserved

22.3.14 PH_SEL (0x404C)

Bit	7	6	5	4	3	2	1	0
Name	SPITMOD	UART1EN	UART2EN	T4SEL	T3SEL	T2SEL	T2SSEL	RSV
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	—
Reset	0	0	0	0	0	0	0	—

Bit	Name	Description
[7]	SPITMOD	MISO port status after the SPI slave device finishes sending 0: Remain output state 1: High impendence
[6]	UART1EN	Port multiplexed as RXD, TXD and UART1 enable 0: Disable 1: P0.5, P0.6 multiplexed as RXD, TXD and enable UART1
[5]	UART2EN	Port multiplexed to RXD2, TXD2 and UART2 enable 0: Disable 1: P3.6, P3.7 multiplexed as RXD2, TXD2 and enable UART2
[4]	T4SEL	Port multiplexed as Timer4 or Timer4S selection 0: Not multiplexed 1: P0.1 or P0.0 (PH_SEL1[T4CT] = 1) multiplexed as input and output of Timer4
[3]	T3SEL	Port multiplexed as Timer3 or Timer3S selection 0: Not multiplexed 1: P1.1 or P0.1 (PH_SEL1[T3CT] = 1) multiplexed as input and output of Timer3
[2]	T2SEL	Port multiplexed as Timer2 selection 0: Not multiplexed 1: P1.0 multiplexed as input and output of Timer2
[1]	T2SSEL	Port multiplexed as Timer2S selection 0: Not multiplexed 1: P0.7 multiplexed as input and output of Timer2
[0]	RSV	Reserved

22.3.15 PH_SEL1 (0x404D)

Bit	7	6	5	4	3	2	1	0
Name	RSV					SPICT	T4CT	T3CT
Type	—	—	—	—	—	R/W	R/W	R/W
Reset	—	—	—	—	—	0	0	0

Bit	Name	Description
[7:3]	RSV	Reserved
[2]	SPICT	SPI function transfer enable 0: No function transfer, P0.5 as SCLK, P0.6 as MOSI 1: Function transfer, P0.0 as MOSI, P0.1 as SCLK Note: FU6832N/S/F have no NSS pin. When using SPI, configure SPI_CR1[NSSMOD] = 00 and NSS signal will not connect to the pin
[1]	T4CT	Timer4 function transfer enable 0: No function transfer, P0.1 as input and output of Timer4 1: Function transfer, P0.0 as input and output of Timer4
[0]	T3CT	Timer3 function transfer enable 0: No function transfer, P1.1 as input and output of Timer3 1: Function transfer, P0.1 as input and output of Timer3

22.3.16 P0 (0x80)

Port output registers P0/1/2/3/4 support read/write access. The RMW command accesses the value of the register (see Table 22-2 for RMW command), and the other commands access the PORT.

Bit	7	6	5	4	3	2	1	0
Name	GP07	GP06	GP05	GP04	GP03	GP02	GP01	GP00
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	GP07	Port GP07
[6]	GP06	Port GP06
[5]	GP05	Port GP05
[4]	GP04	Port GP04
[3]	GP03	Port GP03
[2]	GP02	Port GP02
[1]	GP01	Port GP01
[0]	GP00	Port GP00

22.3.17 P1 (0x90)

Bit	7	6	5	4	3	2	1	0
Name	GP17	GP16	GP15	GP14	GP13	GP12	GP11	GP10
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	GP17	Port GP17
[6]	GP16	Port GP16
[5]	GP15	Port GP15
[4]	GP14	Port GP14
[3]	GP13	Port GP13
[2]	GP12	Port GP12
[1]	GP11	Port GP11
[0]	GP10	Port GP10

22.3.18 P2 (0xA0)

Bit	7	6	5	4	3	2	1	0
Name	GP27	GP26	GP25	GP24	GP23	GP22	GP21	GP20
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	GP27	Port GP27
[6]	GP26	Port GP26
[5]	GP25	Port GP25
[4]	GP24	Port GP24
[3]	GP23	Port GP23
[2]	GP22	Port GP22
[1]	GP21	Port GP21
[0]	GP20	Port GP20

22.3.19 P3 (0xB0)

Bit	7	6	5	4	3	2	1	0
Name	GP37	GP36	GP35	GP34	GP33	GP32	GP31	GP30
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	GP37	Port GP37
[6]	GP36	Port GP36
[5]	GP35	Port GP35
[4]	GP34	Port GP34
[3]	GP33	Port GP33
[2]	GP32	Port GP32
[1]	GP31	Port GP31
[0]	GP30	Port GP30

22.3.20 P4 (0xE8)

Bit	7	6	5	4	3	2	1	0
Name	RSV		GP45	GP44	RSV	GP42	RSV	
Type	—	—	R/W	R/W	—	R/W	—	—
Reset	—	—	0	0	—	0	—	—

Bit	Name	Description
[7:6]	RSV	Reserved
[5]	GP45	Port GP45
[4]	GP44	Port GP44
[3]	RSV	Reserved
[2]	GP42	Port GP42
[1:0]	RSV	Reserved

Table 22-2 RMW

ANL	Logic AND
ORL	Logic OR
XRL	Logic exclusive OR
JBC	Jump if bit is set and clear
CPL	Complement bit
INC,DEC	Increment, decrement byte
DJNZ	Decrement and jump if not zero
MOV Px,y, C	Move carry bit to bit y of port x.y
CLR Px,y	Clear bit y of port x.y
SETB Px,y	Set bit y of port x.y
ANL	Logic AND

23 ADC

23.1 ADC Introduction

ADC of the chip is a 12-bit successive approximation register ADC with 15 channels, where channel 0 ~ 13 are external pin channels and channel 14 is internal channel. VCC pin is divided by built-in resistor, and the result is sampled by channel 14. The ratio of voltage division is configured by ADC_CR[URATIO]. The sampling mode supports sequential sampling (i.e., from ADC channel 0 to ADC channel 14 in sequence) and triggered sampling (including FOC triggered sampling mode and Timer1 triggered sampling mode). The result of sequential sampling mode is stored in ADCx_DR (x = 0 ~ 14) in right alignment or second highest bit alignment format. The result of triggerred sampling mode will be sent to FOC module or Timer1 module, for sampling of motor drive, instead of ADCx_DR. The relevant registers of FOC module or Timer1 module store the result of triggered sampling mode fixed in second highest bit alignment format. Triggerred sampling is done automatically by hardware and sequential sampling is controlled by software. The priority of triggered sampling is higher than the sequential sampling. If both triggered sampling and sequential sampling are applied at the same time, triggered sampling is performed first, and ADC will automatically recover to sequential sampling mode after the triggered sampling is completed.

The clock source of ADC is 12MHz, and the sampling time is set by ADC_SCYC. Please refer to ADC Electrical Characteristics for the sampling time range and conversion time.

23.2 Block Diagram of ADC

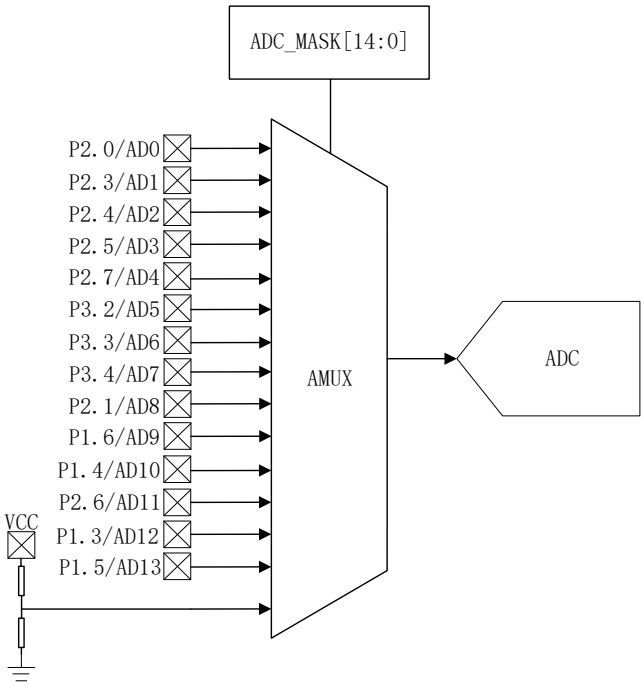


Figure 23-1 Block Diagram of ADC Multiplexer

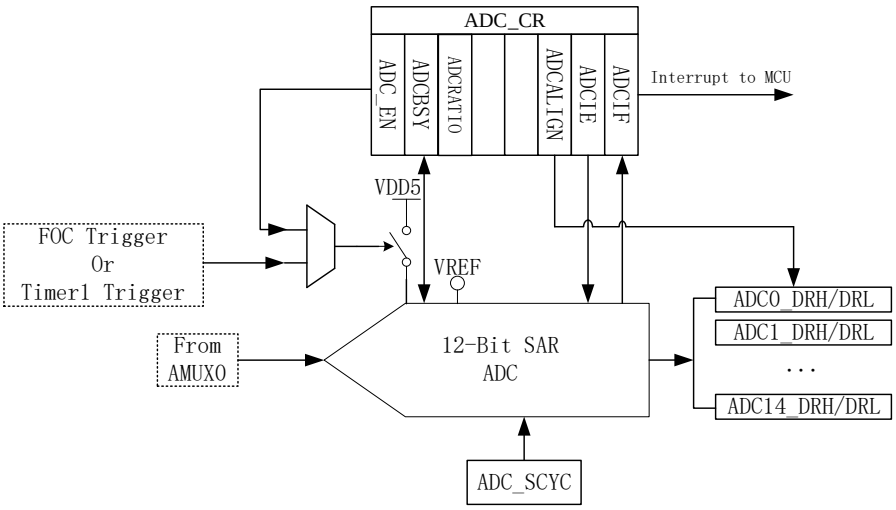


Figure 23-2 Functional Block Diagram of ADC

23.3 ADC Instructions

23.3.1 Sequential Sampling Mode

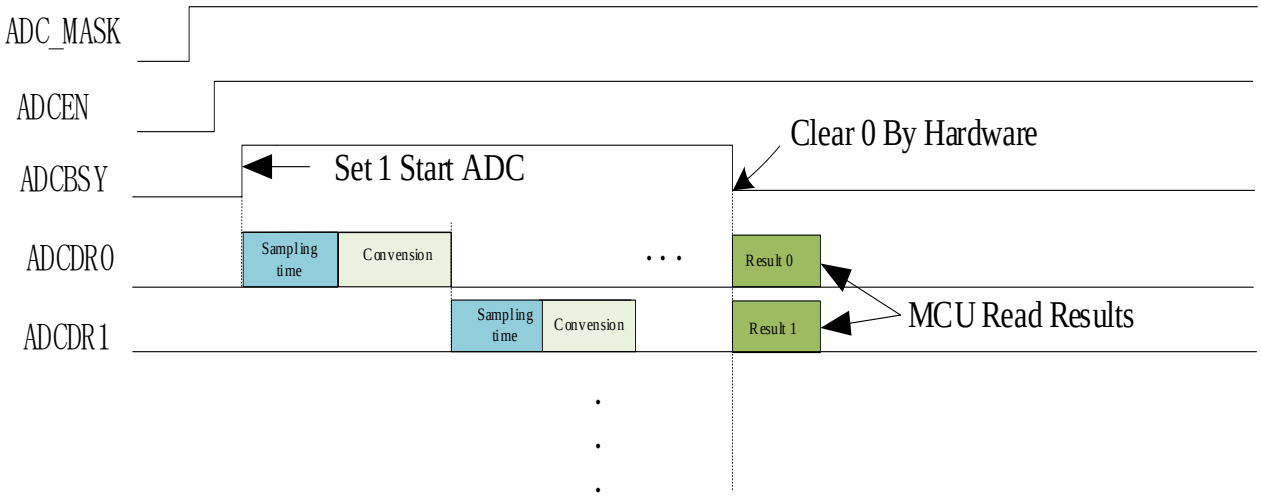


Figure 23-3 Timing Diagram of ADC Sequential Sampling Mode

Instructions:

1. Configure VREF_CR[VREFVSEL] to select appropriate ADC reference voltage (VREF);
2. Configure ADC_MASK to enable the corresponding channel required to sample;
3. Configure ADC_SCYC, with minimum value 3, to select the sampling period of each channel;
4. Set ADC_CR[ADCEN] to 1 to enable ADC;
5. Set ADC_CR[ADCBSY] to 1 to start ADC;

6. When ADC_CR[ADCBSY] = 0, software read ADC results by ADCx_DR.

Note: The ADC conversion order is from low to high according to the enabled channel (i.e., when channel 2/3/4 is enabled, the signal is sampled in order 2, 3, 4).

23.3.2 Triggered Sampling Mode

When FOC module is enabled, ADC channel 0/1/2/4/14 can be used for FOC triggered sampling. Set FOC_CR0[UCSEL] to select voltage sampling channel as ADC channel 2 or channel 14. In single-resistance current sampling mode, ADC channel 4 is used for itrip sampling. In dual- resistance current sampling mode, ADC channel 0 is used for ia sampling and channel 1 is used for ib sampling. In triple- resistance current sampling mode, ADC channel 0 is used for ia sampling, channel 1 is used for ib sampling and channel 4 is used for ic sampling.

When Timer1 is enabled, ADC channel 4 is used for bus current sampling. Configure TIM1_CR3[T1TIS] to select the input source of position detection as ADC. When CMP0_CR4[CMP0FS] = 0, ADC channel 10 is used for U phase voltage sampling, channel 9 is used for V phase voltage sampling and channel 8 is used for W phase voltage sampling. When CMP0_CR4[CMP0FS] = 1, ADC channel 10 is used for U phase voltage sampling, channel 12 is used for V phase voltage sampling and channel 13 is used for W phase voltage sampling.

23.3.3 Output Data Format

ADCx_DRH and ADCx_DRL are high 8 bits and low 8 bits of ADC sampling results. Data format can be selected as right alignment or second-highest bit alignment by the configuration of ADC_CR[ADCALIGN]. The relation between the input voltage and result data is shown as in Table 23-1. The bits, which are not used in ADCx_DRH and ADCx_DRL, are set to 0.

Table 23-1 The Relation between the Input Voltage and Result Data

Input Voltage	Right Alignment	Second-Highest Bit Alignment
0	0x0000	0x0000
VREF/2	0x0800	0x4000
VREF	0x0FFF	0x7FF8

23.4 ADC Register

23.4.1 ADC_CR (0x4039)

Bit	7	6	5	4	3	2	1	0
Name	ADCEN	ADCBSY	ADCRATIO	RSV		ADCALIGN	ADCIE	ADCIF
Type	R/W	R/W1	R/W	—	—	R/W	R/W	R/W0
Reset	0	0	0	—	—	0	0	0

Bit	Name	Description
[7]	ADCEN	ADC enable 0: Disable 1: Enable
[6]	ADCBSY	ADC start & ADC busy flag Read: 0: ADC idle 1: ADC busy Write: 0: No meaning 1: Start ADC conversion Note: When ADC_MASK = 0, writing 1 to this bit has no meaning
[5]	ADCRATIO	The interal voltage division ratio of ADC channel 14 0: 1/12 1: 1/6.5
[4:3]	RSV	Reserved
[2]	ADCALIGN	ADC data format selection 0: Right alignment, ADC result is ADCx_DR[11:0] 1: Second highest bit alignment, ADC result is ADCx_DR[14:3] Note: FOC triggered sampling mode results are fixed to second highest bit alignment
[1]	ADCIE	ADC interrupt enable (except triggered sampling mode) 0: Disable 1: Enable
[0]	ADCIF	ADC interrupt flag This bit is set to 1 by hardware when ADC conversion is completed Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning

23.4.2 ADC_MASK (0x4036, 0x4037)

ADC_MASKH(0x4036)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV	CH14EN	CH13EN	CH12EN	CH11EN	CH10EN	CH9EN	CH8EN
Type	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	0	0	0	0	0	0	0
ADC_MASKL(0x4037)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	CH7EN	CH6EN	CH5EN	CH4EN	CH3EN	CH2EN	CH1EN	CH0EN
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15]	RSV	Reserved
[14]	CH14EN	ADC channel 14 enable
[13]	CH13EN	ADC channel 13 enable
[12]	CH12EN	ADC channel 12 enable
[11]	CH11EN	ADC channel 11 enable
[10]	CH10EN	ADC channel 10 enable
[9]	CH9EN	ADC channel 9 enable
[8]	CH8EN	ADC channel 8 enable
[7]	CH7EN	ADC channel 7 enable
[6]	CH6EN	ADC channel 6 enable
[5]	CH5EN	ADC channel 5 enable
[4]	CH4EN	ADC channel 4 enable
[3]	CH3EN	ADC channel 3 enable
[2]	CH2EN	ADC channel 2 enable
[1]	CH1EN	ADC channel 1 enable
[0]	CH0EN	ADC channel 0 enable

Note: In triggered sampling mode, the configuration of ADC_MASK is not required

23.4.3 ADC_SCYC (0x4035, 0x4038)

DAC_CR(0x4035)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	DAC0_1EN	DACMOD	ADC_SCYC[11:8]				RSV	
Type	R/W	R/W	R/W	R/W	R/W	R/W	—	—
Reset	0	0	0	0	1	1	—	—
ADC_SCYC(0x4038)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC_SCYC[7:4]				ADC_SCYC[3:0]			
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15]	DAC0_1EN	Refer to 24.4.1 in Chapter DAC for details.
[14]	DACMOD	Refer to 24.4.1 in Chapter DAC for details.
[13:10]	ADC_SCYC [13:10]	ADC sampling period setting, for ADC channel 8/9/10/11/12/13 ADC_SCYC[11] = 0: The sampling period is ADC_SCYC[10:8] ADC clock cycles ADC_SCYC[11] = 1: The sampling period is (ADC_SCYC[10:8]*8 + 7) ADC clock cycles
[9:8]	RSV	Reserved
[7:4]	ADC_SCYC [7:4]	ADC sampling period setting, for ADC channel 5/6/7/14 ADC_SCYC[7] = 0: The sampling period is ADC_SCYC[6:4] ADC clock cycles ADC_SCYC[7] = 1: The sampling period is (ADC_SCYC[6:4]*8 + 7) ADC clock cycles
[3:0]	ADC_SCYC [3:0]	ADC sampling period setting, for ADC channel 0/1/2/3/4 ADC_SCYC [3] = 0: The sampling period is ADC_SCYC [2:0] ADC clock cycles ADC_SCYC [3] = 1: The sampling period is (ADC_SCYC[2:0]*8 + 7) ADC clock cycles

23.4.4 ADC0_DR (0x0300, 0x0301)

ADC0_DRH(0x0300)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	ADC0_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC0_DRL(0x0301)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC0_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	ADC0_DR	ADC channel 0 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.5 ADC1_DR (0x0302, 0x0303)

ADC1_DRH(0x0302)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	ADC1_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC1_DRL(0x0303)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC1_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	ADC1_DR	ADC channel 1 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.6 ADC2_DR (0x0304, 0x0305)

ADC2_DRH(0x0304)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	ADC2_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC2_DRL(0x0305)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC2_DR[7:0]							
Type	R	R	R	R	R	R	R	R

Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	ADC2_DR	ADC channel 2 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.7 ADC3_DR (0x0306, 0x0307)

ADC3_DRH(0x0306)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	ADC3_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC3_DRL(0x0307)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC3_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	ADC3_DR	ADC channel 3 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.8 ADC4_DR (0x0308, 0x0309)

ADC4_DRH(0x0308)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	ADC4_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC4_DRL(0x0309)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC4_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	ADC4_DR	ADC channel 4 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.9 ADC5_DR (0x030A, 0x030B)

ADC5_DRH(0x030A)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**

Name	ADC5_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC5_DRL(0x030B)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC5_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	ADC5_DR	ADC channel 5 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.10 ADC6_DR (0x030C, 0x030D)

ADC6_DRH(0x030C)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	ADC6_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC6_DRL(0x030D)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC6_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	ADC6_DR	ADC channel 6 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.11 ADC7_DR (0x030E, 0x030F)

ADC7_DRH(0x030E)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	ADC7_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC7_DRL(0x030F)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC7_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description

[15:0]	ADC7_DR	ADC channel 7 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

ADC7_DRH(0x030E)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	ADC7_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC7_DRL(0x030F)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC7_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	ADC7_DR	ADC channel 7 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.12 ADC8_DR (0x0310, 0x0311)

ADC8_DRH(0x0310)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	ADC8_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC8_DRL(0x0311)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC8_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	**Name**	**Description**
[15:0]	ADC8_DR	ADC channel 8 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.13 ADC9_DR (0x0312, 0x0313)

ADC9_DRH(0x0312)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	ADC9_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC9_DRL(0x0313)								

Bit	7	6	5	4	3	2	1	0
Name	ADC9_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	ADC9_DR	ADC channel 9 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.14 ADC10_DR (0x0314, 0x0315)

ADC10_DRH(0x0314)								
Bit	15	14	13	12	11	10	9	8
Name	ADC10_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC10_DRL(0x0315)								
Bit	7	6	5	4	3	2	1	0
Name	ADC10_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	ADC10_DR	ADC channel 10 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.15 ADC11_DR (0x0316, 0x0317)

ADC11_DRH(0x0316)								
Bit	15	14	13	12	11	10	9	8
Name	ADC11_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC11_DRL(0x0317)								
Bit	7	6	5	4	3	2	1	0
Name	ADC11_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	ADC11_DR	ADC channel 11 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.16 ADC12_DR (0x0318, 0x0319)

ADC12_DRH(0x0318)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	ADC12_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC12_DRL(0x0319)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC12_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	ADC12_DR	ADC channel 12 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.17 ADC13_DR (0x031A, 0x031B)

ADC13_DRH(0x031A)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	ADC13_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC13_DRL(0x031B)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC13_DR[7:0]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	ADC13_DR	ADC channel 13 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

23.4.18 ADC14_DR (0x031C, 0x031D)

ADC14_DRH(0x031C)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	ADC14_DR[15:8]							
Type	R	R	R	R	R	R	R	R
Reset	0	0	0	0	0	0	0	0
ADC14_DRL(0x031D)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	ADC14_DR[7:0]							
Type	R	R	R	R	R	R	R	R

Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:0]	ADC14_DR	ADC channel 14 conversion result after sequential sampling mode ADC conversion is completed Data alignment is selected according to ADC_CR[ADCALIGN] Note: Triggered sampling mode ADC results are not updated to this register

24 DAC

24.1 DAC Introduction

The chip integrates two DAC modules, where DAC0 is a 9-bit digital-to-analog converter, and DAC1 is a 6-bit digital-to-analog converter.

24.2 DAC0 Functional Block Diagram

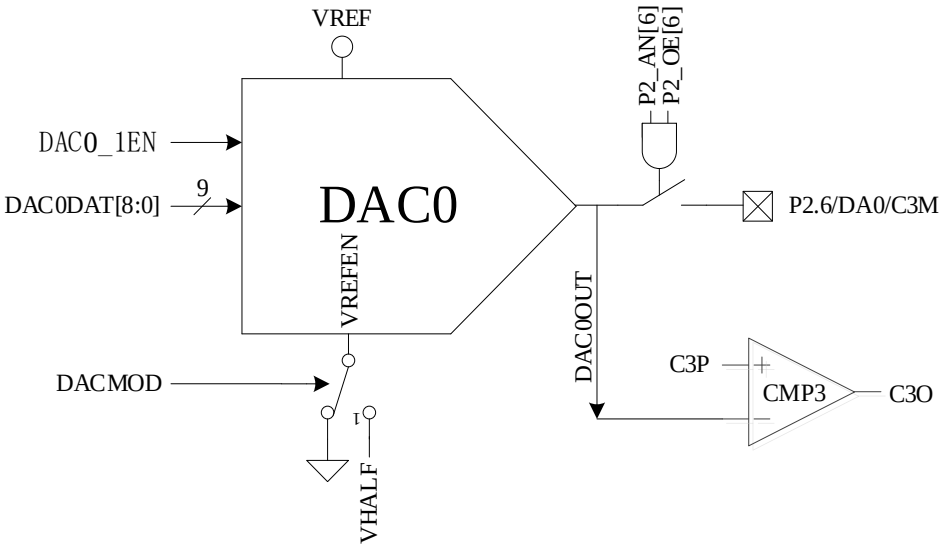


Figure 24-1 Functional Block Diagram of DAC0

As shown in Figure 24-1, DAC0 converts the 9-bit digital data to analog voltage. The voltage will be sent to the negative input of CMP3 for bus overcurrent protection, while P2.6 can be configured as the analog output.

Note: DAC0 output has few current drive capability and can only carry capacitive load. To carry resistive load, use operational amplifiers to follow the voltage output.

The DAC0 instructions are as follows:

1. Configure P2_AN[6] = 1, P2_OE[6] = 1, DAC0 output to P2.6;
2. Configure VREF_CR [VREFEN] to 1 to select VREF as the reference voltage for DAC0, and set DAC_CR[DAC0_1EN] to 1 to enable DAC0;
3. The output voltage range is selected by DAC_CR[DACMOD]. Configure DAC_CR[DACMOD] = 0 to select full-voltage output mode, and the output voltage range is 0 ~ VREF. Configure DAC_CR[DACMOD] = 1 to select half-voltage output mode, and the output voltage range is VHALF ~ VREF. The output voltage of DAC0 in different modes are shown in Table 24-1.

Table 24-1 Voltage Output of DAC0 in Different Configurations

DAC0DAT[8:0]	DAC Output Voltage (DAC_CR[DACMOD] = 0)	DAC Output Voltage (DAC_CR[DACMOD] = 1)
0x000	0	VHALF
0x100	VREF/2	(VREF - VHALF)/2 + VHALF
0x1FF	VREF*511/512	(VREF -V HALF)*511/512 + VHALF

24.3 DAC1 Functional Block Diagram

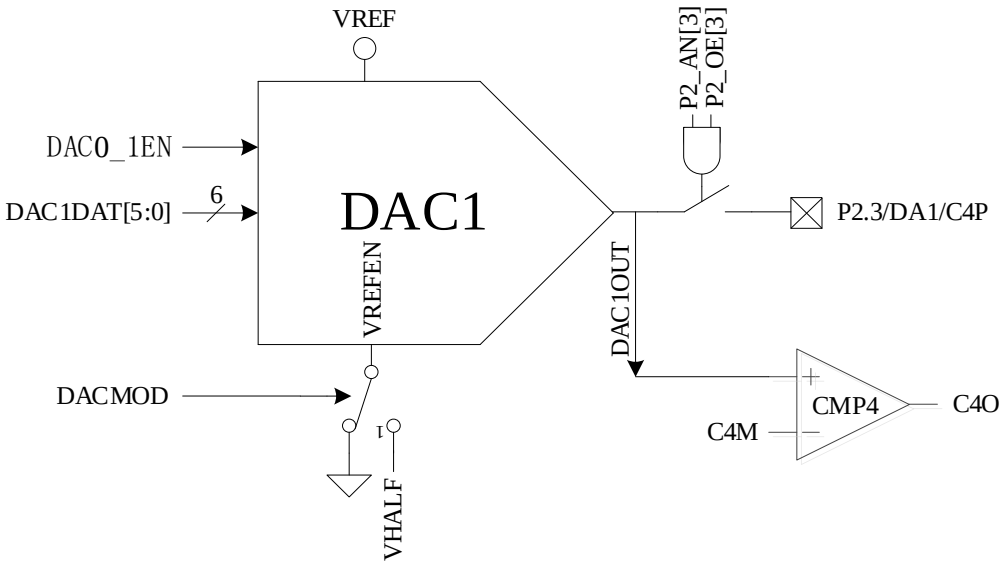


Figure 24-2 Functional Block Diagram of DAC1

As shown in Figure 24-2, DAC1 converts the 6-bit digital data to analog voltage and sends the voltage to the positive input of CMP4 for the cycle-by-cycle current limiting function, while P2.3 can be configured as the analog output.

Note: DAC1 output has few current drive capability and can only carry capacitive load. To carry resistive load, use operational amplifiers to follow the voltage output.

The DAC1 instructions are as follows:

1. Configure P2_AN[3] = 1 and P2_OE[3] = 1, DAC1 output to P2.3;
2. Configure VREF_CR[VREFEN] to 1 to select VREF as the reference voltage for DAC0, and set DAC_CR[DAC0_1EN] to 1 to enable DAC1;
3. The output voltage range is set by DAC_CR[DACMOD]. Configure DAC_CR[DACMOD] = 0 to select full-voltage output mode, and the output voltage range is 0 ~ VREF. Configure DAC_CR[DACMOD] = 1 to select half-voltage output mode, and the output voltage range is VHALF ~ VREF. The output voltage of DAC1 under different configurations is shown in Table 24-2.

Table 24-2 Voltage Output of DAC1 in Different Configurations

DAC1DAT[5:0]	DAC Output Voltage (DAC_CR[DACMOD] = 0)	DAC Output Voltage (DAC_CR[DACMOD] = 1)
0x00	0	VHALF
0x20	VREF/2	(VREF-VHALF)/2 + VHALF
0x3F	VREF*63/64	(VREF - VHALF)*63/64 + VHALF

24.4 DAC Registers

24.4.1 DAC_CR (0x4035)

Bit	7	6	5	4	3	2	1	0
Name	DAC0_1EN	DACMOD	ADC_SCYCH[11:8]				RSV	
Type	R/W	R/W	R/W	R/W	R/W	R/W	—	—
Reset	0	0	0	0	1	1	—	—

Bit	Name	Description
[7]	DAC0_1EN	DAC0 & 1 enable 0: Disable 1: Enable
[6]	DACMOD	DAC mode selection 0: Full-voltage output mode 1: Half-voltage output mode
[5:2]	ADC_SCYCH[11:8]	Refer to 23.4.3 instructions
[1:0]	RSV	Reserved

24.4.2 DAC0_DR (0x404B)

Bit	7	6	5	4	3	2	1	0
Name	DAC0DAT[8:1]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	DAC0DAT[8:1]	High 8 bits data of DAC0 controller

24.4.3 DAC1_DR (0x404A)

Bit	7	6	5	4	3	2	1	0
Name	DAC0_DR_0	RSV	DAC1DAT					
Type	R/W	—	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	—	0	0	0	0	0	0

Bit	Name	Description
[7]	DAC0_DR_0	LSB of DAC0 controller
[6]	RSV	Reserved
[5:0]	DAC1DAT	6-bit data of DAC1 controller

25 DMA

25.1 DMA Instructions

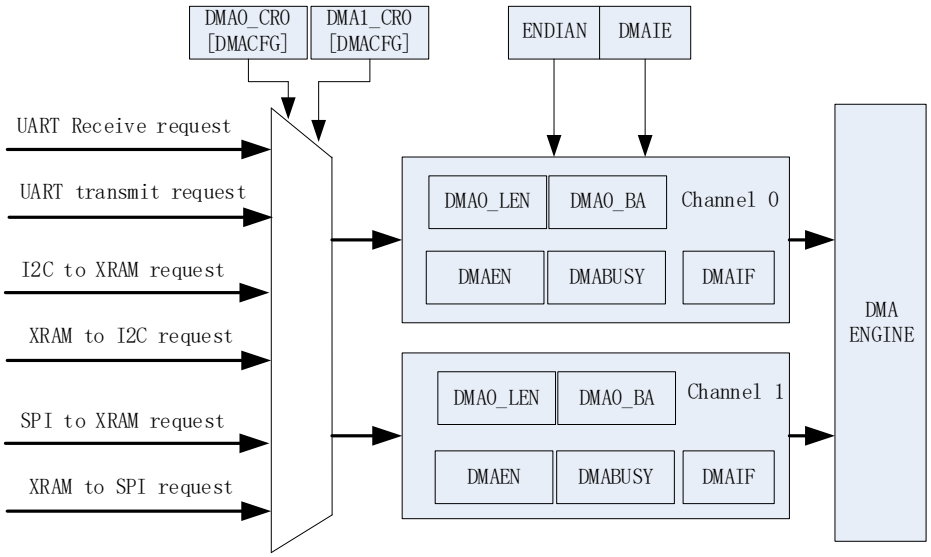


Figure 25-1 Functional Block Diagram of DMA

The DMA module is a dual-channel DMA controller, which performs direct data transfer between peripherals (SPI, UART, I2C) and XRAM (IRAM data invalid). DMA accessing to XRAM does not interfere with the normal CPU read/write operation to XRAM. The length of the transferred data and the start address of XRAM access is configurable. Data transfer mode is configurable and interrupt can be enabled.

DMA instructions are as follows:

1. Configure the peripheral and enable the peripheral. Set input and output channels taken over by DMA by DMAx_CR0[DMACFG];
2. Configure DMA interrupt enable, transfer order, transfer length and XRAM start address. Write DMAx_CR0[DMAEN] and DMAx_CR0[DMABSY] to 1 to start DMA;
3. After data transfer, the interrupt flag bit DMAx_CR0[DMAIF] is set to 1 by hardware and it should be cleared by software;
4. Set DMAx_CR0[DMABSY] to 1 to start DMA again.

25.2 DMA Registers

25.2.1 DMA0_CR0 (0x403A)

Bit	7	6	5	4	3	2	1	0
Name	DMAEN	DMABSY	DMACFG			DMAIE	ENDIAN	DMAIF
Type	R/W	R/W1	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	DMAEN	DMA channel 0 enable 0: Disable 1: Enable
[6]	DMABSY	DMA channel 0 busy state/start Read: 0: Channel 0 is idle 1: Channel 0 is transmitting Write: 0: No meaning 1: Start DMA channel 0
[5:3]	DMACFG [2:0]	DMA channel 0 peripheral and transfer direction selection 000: from UART1 to XRAM 001: From XRAM to UART1 010: From I2C to XRAM 011: From XRAM to I2C 100: From SPI to XRAM 101: From XRAM to SPI 110: From UART2 to XRAM 111: From XRAM to UART2 Note: Cannot be configured when DMA channel 0 is busy
[2]	DMAIE	DMA interrupt enable 0: Disable 1: Enable
[1]	ENDIAN	DMA data transfer sequence 0: High byte is received or transmitted first 1: Low byte is received or transmitted first Notes: The setting of this bit is for the 16-bit data mode and should be configured to 0 for the 8-bit data mode. Cannot be configured when channel 0 or 1 is busy
[0]	DMAIF	DMA channel 0 transfer interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: generate interrupt

25.2.2 DMA1_CR0 (0x403B)

Bit	7	6	5	4	3	2	1	0
Name	DMAEN	DMABSY	DMACFG			DBGSW	DBGEN	DMAIF
Type	R/W	R/W1	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	DMAEN	DMA channel 1 enable 0: Disable 1: Enable
[6]	DMABSY	DMA channel 1 busy state/start Read: 0: Channel 1 is idle 1: Channel 1 is transmitting Write: 0: No meaning 1: Start DMA channel 1
[5:3]	DMACFG [2:0]	DMA channel 1 peripheral and transfer direction selection 000: from UART1 to XRAM 001: From XRAM to UART1 010: From I2C to XRAM 011: From XRAM to I2C 100: From SPI to XRAM 101: From XRAM to SPI 110: From UART2 to XRAM 111: From XRAM to UART2 Note: Cannot be configured when DMA channel 1 is busy
[2]	DBGSW	Debug area selection 0: Debug access to XSFR (exported address space: 0x4020 ~ 0x40FF) 1: Debug access to XRAM (exported address space: 0x0000 ~ 0x0317)
[1]	DBGEN	Debug mode enable Configure DMA1_CR0[DMACFG] to 101 and DMA1_CR0[DBGEN] = 1, DMA works in Debug mode. Enable SPI, the DMA will automatically send the relevant data in the area specified by DMA_CR0[DBGSW] through MOSI, and DMA1_BA/ DMA1_LEN is used to specify the start address and range of the relevant data. 0: Disable 1: Enable Note: In Debug mode, DMA channel 1 interrupt is turned off
[0]	DMAIF	DMA channel 0 transfer interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: Generate interrupt

25.2.3 DMA0_LEN (0x403C)

Bit	7	6	5	4	3	2	1	0
Name	RSV		DMA0_LEN					
Type	—	—	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	—	0	0	0	0	0	0

Bit	Name	Description
[7:6]	RSV	Reserved
[5:0]	DMA0_LEN	DMA channel 0 transfer length setting Read: The number of the bytes currently transferred by DMA channel 0 (0 means the first byte) Write: DMA channel 0 transfer XRAM data length Note: Cannot be configured when DMA channel 0 is busy. When DMA0_CR0[ENDIAN] = 1 (low byte is received or sent first), it is recommended that DMA0_LEN be set to an odd number

25.2.4 DMA0_BA (0x403E, 0x403F)

DMA0_BAH(0x403E)								
Bit	15	14	13	12	11	10	9	8
Name	RSV					DMA0_BA[10:8]		
Type	—	—	—	—	—	R/W	R/W	R/W
Reset	—	—	—	—	—	0	0	0
DMA0_BAL(0x403F)								
Bit	7	6	5	4	3	2	1	0
Name	DMA0_BA[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:11]	RSV	Reserved
[10:0]	DMA0_BA	DMA channel 0 transfer data start address setting DMA channel 0 XRAM start address Cannot be configured when DMA channel 0 is busy Note: The XRAM address space area for DMA channel 0 is: DMA0_BA[10:0] ~ (DMA0_BA[10:0] + DMA0_LEN[5:0])

25.2.5 DMA1_LEN (0x403D)

Bit	7	6	5	4	3	2	1	0
Name	RSV		DMA1_LEN					
Type	—	—	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	—	0	0	0	0	0	0

Bit	Name	Description
[7:6]	RSV	Resverd
[5:0]	DMA1_LEN	DMA channel 1 transfer data length setting Read: The number of the bytes currently transferred by DMA channel 1 (0 means the first byte) Write: DMA channel 1 XRAM transfer data length Note: Cannot be configured when DMA channel 1 is busy. When DMA0_CR0[ENDIAN] = 1 (low byte is received or sent first), it is recommended that DMA1_LEN be set to an odd number

25.2.6 DMA1_BA (0x4040, 0x4041)

DMA1_BAH(0x4040)								
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Name	RSV					DMA1_BA[10:8]		
Type	—	—	—	—	—	R/W	R/W	R/W
Reset	—	—	—	—	—	0	0	0
DMA1_BAL(0x4041)								
Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Name	DMA1_BA[7:0]							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[15:11]	RSV	Resvered
[10:0]	DMA1_BA	DMA channel 1 transfer data start address setting DMA channel 1 XRAM start address Cannot be configured when DMA channel 0 is busy Notes:The XRAM address space area for DMA channel 1 transfer is: DMA0_BA[10:0] ~ (DMA0_BA[10:0] + DMA0_LEN[5:0])

Note: When the DMA channel peripheral is selected to I2C (including from I2C to XRAM and from XRAM to I2C), the START + address interrupt for I2C communication still requires to be cleared by software. In I2C slave mode, if STOP is received, configure I2C_SR[I2CSTP] = 0 to clear the I2C interrupt and restart the DMA transfer.

26 VREF

26.1 VREF Instructions

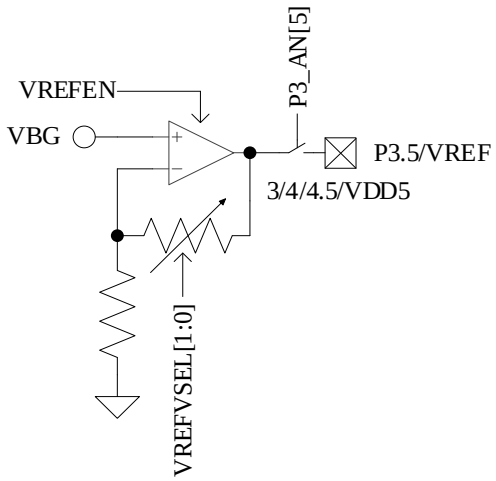


Figure 26-1 Diagram of VREF Module Input and Output

The input and output ports of the VREF module are shown in Figure 26-1. VREF is the voltage reference generation block that provides the internal voltage reference to ADC and DAC module. VBG is the voltage provided by the chip internally.

Configure VREF_CR[VREFEN] = 1 to enable VREF. Configure VREF_CR[VREFVSEL] to select the output voltage value. Configure P3_AN[5] = 1 to output VREF voltage to P3.5.

26.2 VREF Register

26.2.1 VREF_CR (0x404F)

Bit	7	6	5	4	3	2	1	0
Name	VREFVSEL		RSV	VREFEN	RSV			VHALFEN
Type	R/W	R/W	—	R/W	—	—	—	R/W
Reset	0	0	—	0	—	—	—	0

Bit	Name	Description
[7:6]	VREFVSEL	VREF module output voltage selection 00: 4.5V 01: VDD5 10: 3V 11: 4V
[5]	RSV	Reserved
[4]	VREFEN	VREF module enable 0: Disable. Set P3_AN[5] = 1, external voltage reference input from P3.5 1: Enable. Set P3_AN[5] = 1 to send the internal VREF reference to P3.5. Shunt a 0.1μF ~ 1μF capacitance to improve the stability of VREF
[3:1]	RSV	Reserved
[0]	VHALFEN	VHALF enable 0: Disable 1: Enable

27 VHALF

27.1 VHALF Instructions

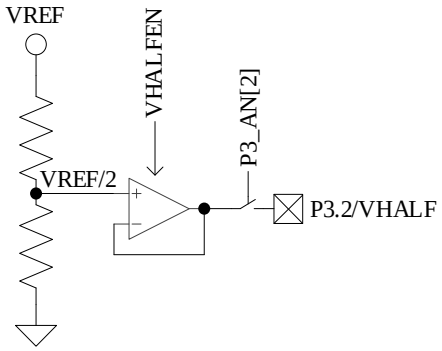


Figure 27-1 Diagram of VHALF Module Input and Output

The VHALF module input and output ports are shown in Figure 27-1. This module generates the voltage reference VREF/2.

Configure VREF_CR[VHALFEN] = 1 to enable VHALF and output to P3.2. Shunt a 1μF capacitance to ground.

27.2 VHALF Register

VHALF register refer to 26.2.1.

28 Operational Amplifiers

28.1 Operational Amplifier Introduction

The chip integrates three high-speed independent operational amplifiers, AMP0, AMP1, and AMP2. Each operational amplifier has an independent enable terminal. AMP0 can be configured as a gain programmable amplifier.

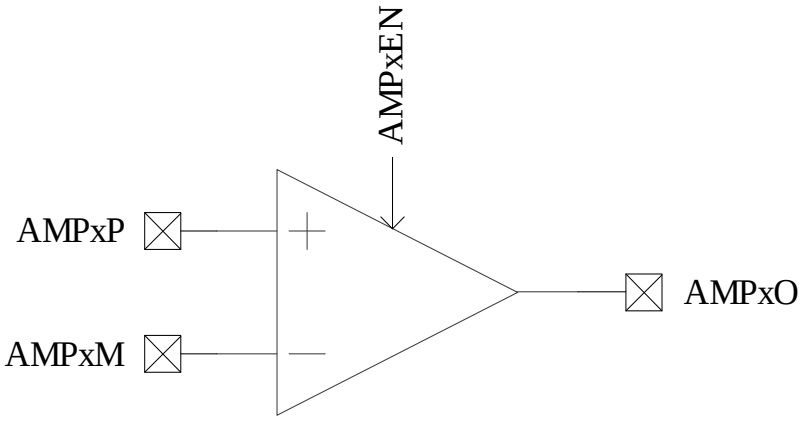


Figure 28-1 Schematic Diagram of Operational Amplifier module

28.2 Operational Amplifier Instructions

28.2.1 Bus Current Sampling Operational Amplifier (AMP0)

AMP0 can be configured as normal mode or PGA differential input mode

28.2.1.1 AMP0 Normal Mode

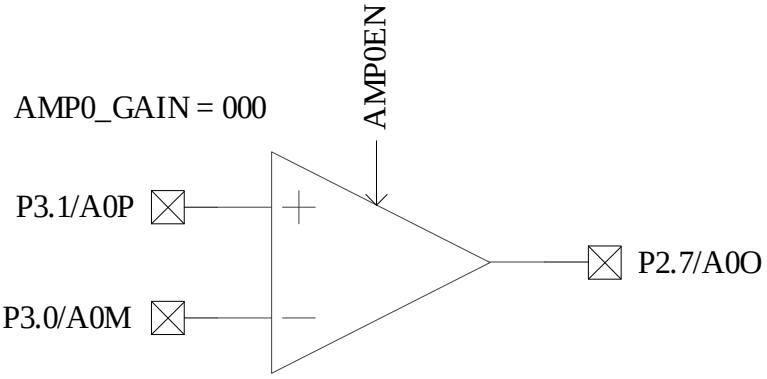


Figure 28-2 Diagram of Bus Current Operational Amplifier AMP0

The ports corresponding to the inputs and outputs of the bus current operational amplifier are shown in Figure 28-2. Configure AMP_CR[AMP0EN] = 1 to enable AMP0, and the hardware automatically configures P2.7, P3.0, and P3.1 to analog signal mode.

28.2.1.2 AMP0 PGA Differential Input Mode

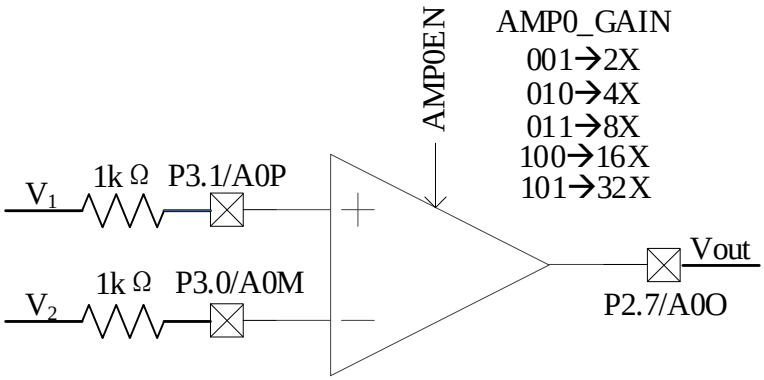


Figure 28-3 Diagram of AMP0 PGA Differential Input Mode

As shown in Figure 28-3, the positive and negative inputs of AMP0 are connected with a 1kΩ resistor in the external circuit respectively.

When using AMP0 PGA differential input mode, the amplification gain is set by AMP0_GAIN and set AMP_CR[AMP0EN] = 1 to enable AMP0. The relationship between the output and the input of operational amplifier is: Vout = VHALF + (V1 - V2)*AMP0_GAIN.

28.2.2 Phase Current Operational Amplifier (AMP1/AMP2)

28.2.2.1 AMP1

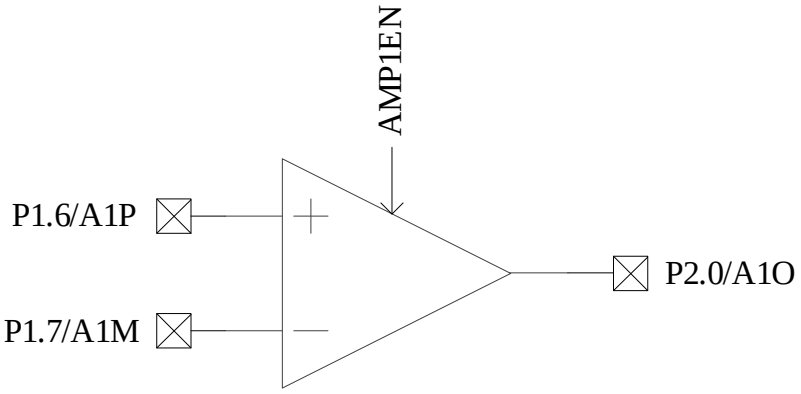


Figure 28-4 Diagram of AMP1 Input and Output

The ports corresponding to the phase current operational amplifier inputs and outputs are shown in Figure 28-4. Configure AMP_CR[AMP1EN] = 1 to enable the phase current operational amplifier AMP1, and the hardware automatically configures P1.6, P1.7, and P2.0 to analog signal mode.

28.2.2.2 AMP2

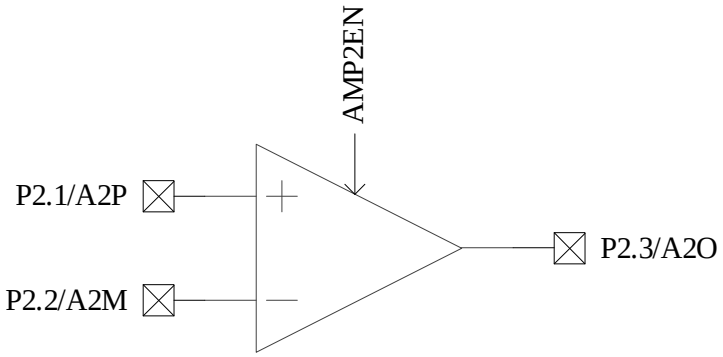


Figure 28-5 Diagram of AMP2 Input and Output

The ports corresponding to the phase current operational amplifier inputs and outputs are shown in Figure 28-5. Configure CMP_CR[AMP2EN] = 1 to enable the phase current operational amplifier AMP2, and the hardware automatically configures P2.1, P2.2, and P2.3 to analog signal mode.

28.3 Operational Amplifier Register

28.3.1 AMP_CR0 (x404E)

Bit	7	6	5	4	3	2	1	0
Name	RSV					AMP2EN	AMP1EN	AMP0EN
Type	—	—	—	—	—	R/W	R/W	R/W
Reset	—	—	—	—	—	0	0	0

Bit	Name	Description
[7:3]	RSV	Reserved
[2]	AMP2EN	AMP2 enable 0: Disable 1: Enable
[1]	AMP1EN	AMP1 enable 0: Disable 1: Enable
[0]	AMP0EN	AMP0 enable 0: Disable 1: Enable

28.3.2 AMP0_GAIN (0x4034)

Bit	7	6	5	4	3	2	1	0
Name	RSV					AMP0_GAIN		
Type	—	—	—	—	—	R/W	R/W	R/W
Reset	—	—	—	—	—	0	0	0

Bit	Name	Description
[7:3]	RSV	Reserved
[2:0]	AMP0_GAIN	Operational amplifier gain setting 000: The gain is configured by the external circuit 001: 2 010: 4 011: 8 100: 16 101: 32 110: Reserved 111: Reserved Note: The built-in amplification is isotropic amplification. When the input voltage deviation is 0, the output voltage is VHALF. For other application, configure AMP0_GAIN = 000 to use external circuit to configure gain

29 Comparator

29.1 Comparator Instructions

29.1.1 CMP3

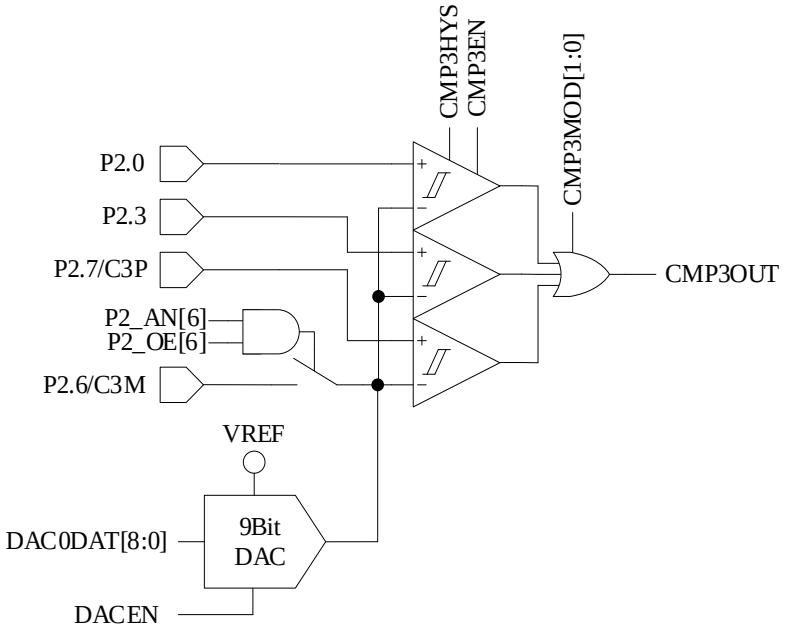


Figure 29-1 Diagram of CMP3 Input and Output Ports

The input and output ports of the comparator are shown in Figure 29-1.

The configuration of CMP3 is as follows:

1. Configure P2_AN[6] and P2_OE[6] to 1. Enable the reference voltage of the negative input and CMP3. The reference voltage source can be the on-chip DAC0 output voltage or the external circuit input voltage. When DAC0 output is selected, shunt an external capacitance between P2.6 and ground (recommended capacitance 100pF. output voltage will be stable after a period of DAC0 output);
2. Configure CMP_CR1[CMP3MOD] to select single comparator input, dual comparator input, and triple comparator input modes;
   - Configure CMP3MOD = 00, CMP3 works in single comparator input mode. The connection of input and output ports are as shown in Figure 29-2
   - Configure CMP3MOD = 01, CMP3 works in dual comparator input mode. The connection of input and output ports are as shown in Figure 29-3
   - Configure CMP3MOD = 1X, CMP3 works in triple comparator input mode. The connection of input and output ports are as shown in Figure 29-4
3. Configure CMP_CR1[CMP3HYS] to enable or disable hysteresis;
4. Set CMP_CR1[CMP3EN] = 1 to enable CMP3.

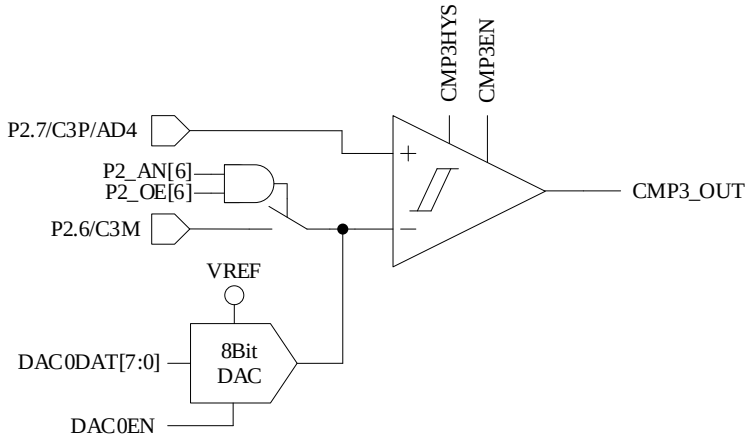


Figure 29-2 Diagram of Single Comparator Input Mode

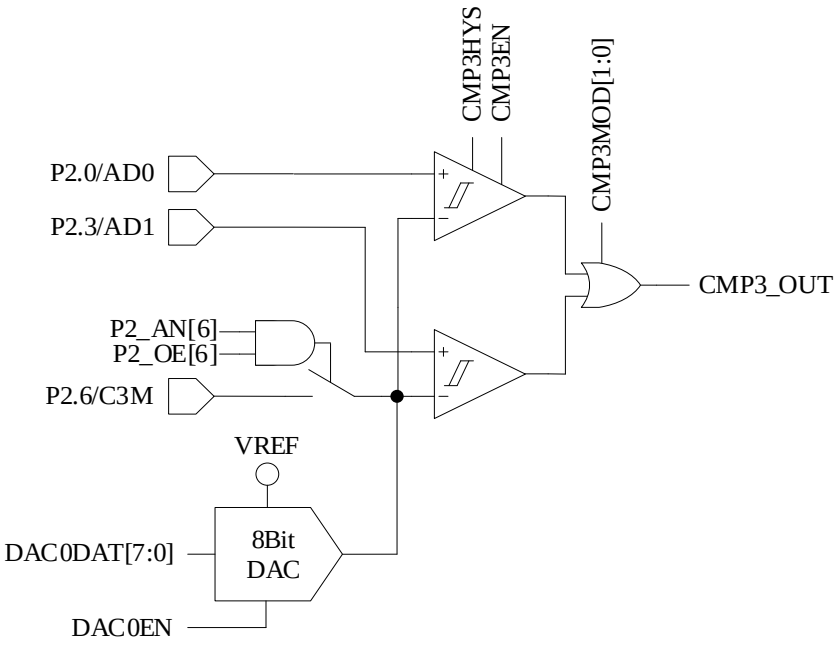


Figure 29-3 Diagram of Dual Comparator Input Mode

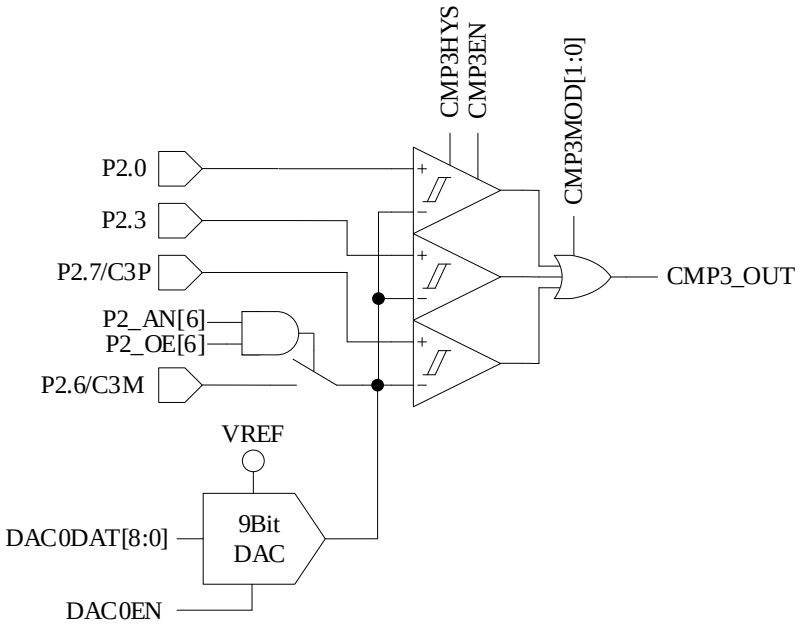


Figure 29-4 Diagram of Triple Comparator Input Mode

29.1.1.1 Overcurrent Protection

When the overcurrent protection signal is generated, hardware automatically clears DRV_OUT[MOE] and outputs the idle voltage to stop the motor drive and protect the chip and motor. Configure EVT_FILT[MOEMD] = 01 to enable the overcurrent protection function, which automatically turns off the output and generates an overcurrent protection interrupt request when the current exceeds the threshold. Configure EVT_FILT[MOEMD] = 00, output will be not automatically turned off when the current exceeds the threshold However, hardware generate an overcurrent protection interrupt request.

The overcurrent protection can be chosen to be generated by CMP3 or INT0 by configuring EVT_FILT[EFSRC]. If EVT_FILT[EFSRC] = 1, the external interrupt INT0 generates the overcurrent protection signal, and the protection interrupt is the external interrupt INT0. If EVT_FILT[EFSRC] = 0, the overcurrent protection signal is generated by CMP3, and the protection interrupt is the CMP3 interrupt. When in triple-resistance current sampling mode, configure CMP_CR1[CMP3MOD] working in triple-comparator input mode. When any phase current is over the threshold, CMP3 generates an overcurrent protection signal. For other sampling mode, configure CMP_CR1[CMP3MOD] working in single-comparator input mode. When bus current is over the threshold, CMP3 generates an overcurrent protection signal.

The input signal for overcurrent protection can be configured by EVT_FILT[EFDIV] to enable the filtering function. The filtering width is selectable as 4/8/16 clock cycles by configuring EVT_FILT[EFDIV] = 01/10/11. When the filtering function is enabled, the filtered signal will be delayed by 4/8/16 clock cycles compared to the signal before filtering.

29.1.1.2 Cycle-by-Cycle Current Limiting

The Cycle-by-Cycle current limiting function is applied to square wave control based BLDC motor drive. When the overcurrent protection event occurs, DRV_OUT[MOE] is cleared for a period of time and then set to 1 by hardware to automatically resume the motor drive. Configure CMP_CR0[CMP3IM] = 11, DRV_OUT[MOE] is cleared to 0 on the rising edge of CMP3OUT and generate the protection event. Configure EVT_FILT[MOEMD] = 10 to automatically turn off the output after generating a protection event. Hardware will automatically enable MOE to resume the drive after 10μs or at the Driver counter overflow and underflow event. Configure EVT_FILT[MOEMD] = 11 to turn off the output after generating a protection event. Hardware will automatically enable DRV_OUT[MOE] to resume the drive after 5μs or at the Driver counter overflow and underflow event.

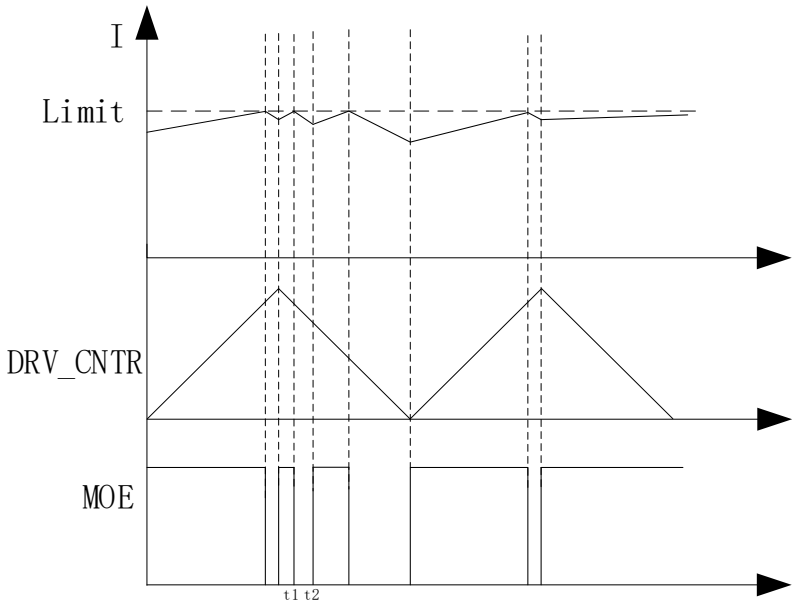


Figure 29-5 EVT_FILT[MOEMD] = 10, Cycle-by-Cycle Current Limiting Waveform (t2 - t1 = 10μs)

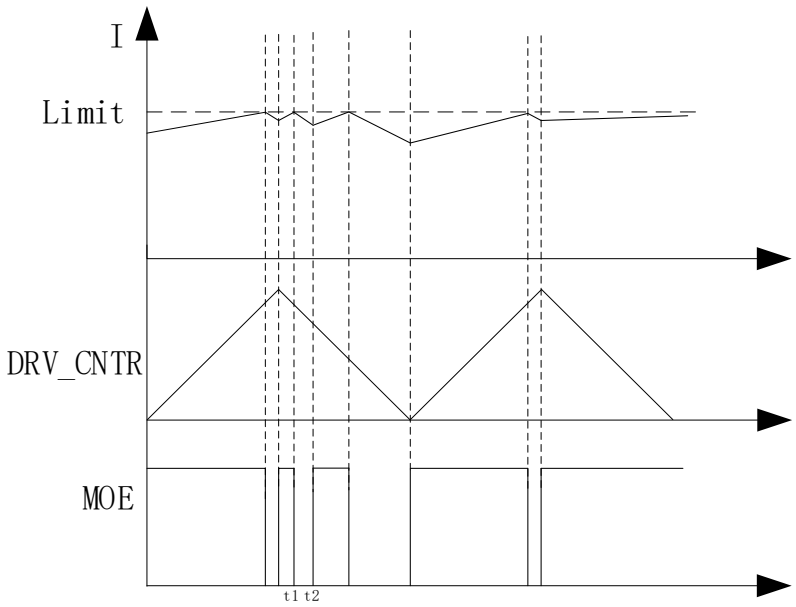


Figure 29-6 EVT_FILT[MOEMD] = 11, Cycle-by-Cycle Current Limiting Waveform (t2 - t1 = 5μs)

29.1.2 CMP4

CMP4 is a hysteresis comparator as shown in Figure 29-7. CMP4OUT can be read by software or connected to INT0 to detect whether it is changed. When CMP3 is used for Cycle-by-Cycle current limiting protection, CMP4 is used for bus current protection. Output is required to be turned off by software after CMP4 bus current overcurrent protection generated.

The configuration of CMP4 is as follows:

1. Configure P2_AN[3] = 1 and P2_OE[3] = 1. Enable the the positive input reference voltage of CMP4. The reference voltage can be the internal DAC1 output voltage of the chip or the external circuit input voltage. When DAC1 output is selected, shunt an external capacitance between P2.3 and ground (recommended 100pF. Voltage will be stable after DAC1 outputs a period of time);
2. Configure P2_AN[7] = 1 to enable the analog signal mode of P2.7;
3. Configure CMP_CR2[CMP4EN] = 1 to enable CMP4;
4. Clear INT0 flag bit and enable INT0;
5. Set LVSR[EXT0CFG] = 111 to select CMP4 as the source of INT 0;
6. Configure TCON[IT0] = 01 to select falling edge triggered INT0.

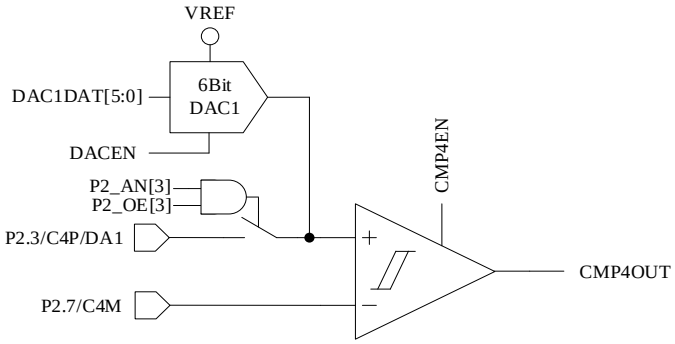


Figure 29-7 Diagram of CMP4 Module

29.1.3 Comparator Group CMPG

The comparator group CMPG is a collection of CMP0, CMP1 and CMP2, with multiple comparison modes for different application.

When setting CMP_CR2[CMP0MOD] = 00, the triple comparator mode without built-in resistors is selected for BEMF detection with the external virtual neutral point resistors. The input and output ports are shown in Figure 29-8. The negative inputs of the three comparators are connected together to P1.5, and the positive inputs are connected to P1.4, P1.6 and P2.1, and the outputs are CMP0OUT, CMP1OUT, and CMP2OUT respectively. The number of comparators working in this mode is determined by CMP_CR2[CMP0SEL]. When setting CMP_CR2[CMP0SEL] = 00, CMP0, CMP1 and CMP2 are all working simultaneously, which is the recommended setting. When setting CMP_CR2[CMP0SEL] = 01, only CMP0 works and CMP1, CMP2 are idle. When setting CMP_CR2[CMP0SEL] = 10, only CMP1 works and CMP0,CPM2 are idle. When setting CMP_CR2[CMP0SEL] = 11, only CMP2 works and the CMP0, CMP1 are idle.

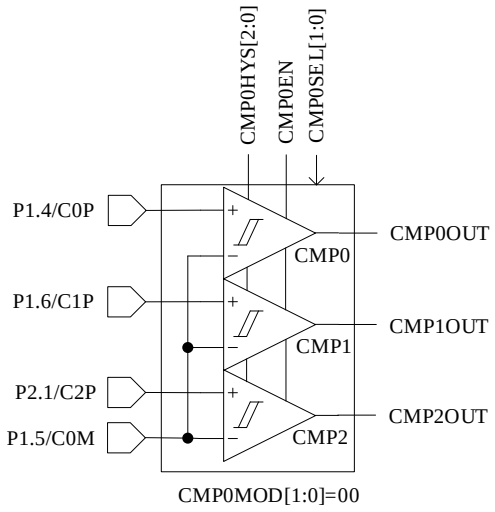


Figure 29-8 Diagram of the Triple Comparator Mode without Built-in Resistors

When setting CMP_CR2[CMP0MOD] = 01, the triple comparator mode with built-in resistors is selected for

BEMF detection without the external virtual neutral point resistors. The input port can be selected by setting the function transfer bit CMP_CR4[CMP0FS]. The number of comparators operating in this mode is determined by CMP_CR2[CMP0SEL]. When setting CMP_CR2[CMP0SEL] = 00, CMP0, CMP1 and CMP2 comparators are all working simultaneously, which is the recommended setting. When setting CMP_CR2[CMP0SEL] = 01, only CMP0 works and CMP1, CMP2 are idle. When setting CMP_CR2[CMP0SEL] = 10, only CMP1 works and CMP0, CMP2 are idle. When setting CMP_CR2[CMP0SEL] = 11, only CMP2 works and CMP0, CMP1 are idle. When setting CMP_CR4[CMP0FS] = 0, The input and output ports are shown in Figure 29-9. The negative inputs of the three comparators are connected together to the neutral point of the built-in resistors. The positive inputs are connected to P1.4, P1.6 and P2.1, and the outputs are CMP0OUT, CMP1OUT, and CMP2OUT, respectively.

When setting CMP_CR4[CMP0FS] = 1, The input and output ports are shown in Figure 29-10. The negative inputs of the three comparators are connected together to the neutral point of the built-in resistors, the positive inputs are connected to P1.4, P1.3 and P1.5, and the outputs are CMP0OUT, CMP1OUT, and CMP2OUT, respectively.

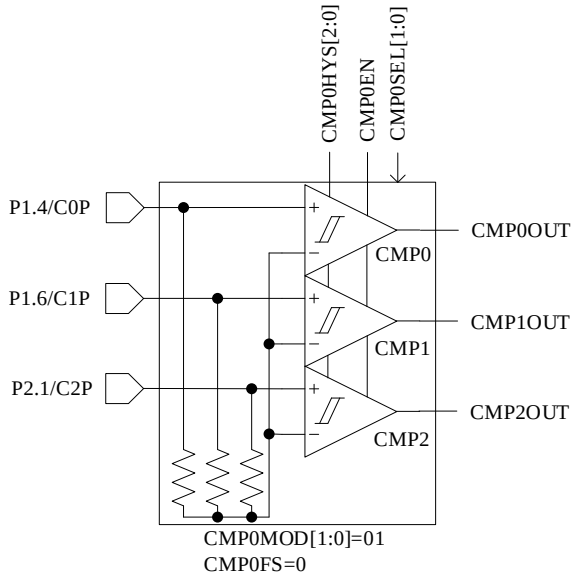


Figure 29-9 Diagram of Triple Comparator Mode with Built-in Resistor and without Function Transfer

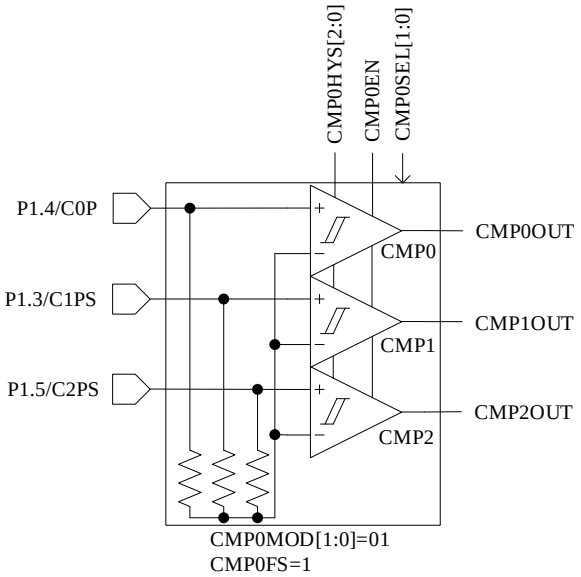


Figure 29-10 Diagram of Triple Comparator Mode with Built-in Resistor and Port Function Transfer

When setting CMP_CR2[CMP0MOD] = 10, the differential three comparator mode is selected for the differential Hall sensor to detect the motor rotor position. The input and output ports are shown in Figure 29-11. The negative input of the three comparators are respectively connected to P1.5, P1.7 and P2.2, the positive input are respectively connected to P1.4, P1.6 and P2.1, and the outputs are CMP0OUT, CMP1OUT and CMP2OUT respectively. In this mode, the number of comparators is determined by CMP_CR2[CMP0SEL]. When setting CMP_CR2[CMP0SEL] = 00, CMP0, CMP1 and CMP2 all works simultaneously, which is the recommended setting. When setting CMP_CR2[CMP0SEL] = 01, only CMP0 works and CMP1, CMP2 are idle. When setting CMP_CR2[CMP0SEL] = 10, only CMP1 works and CMP0, CMP2 are idle. When setting CMP_CR2[CMP0SEL] = 11, only CMP2 works and CMP0, CMP2 are idle.

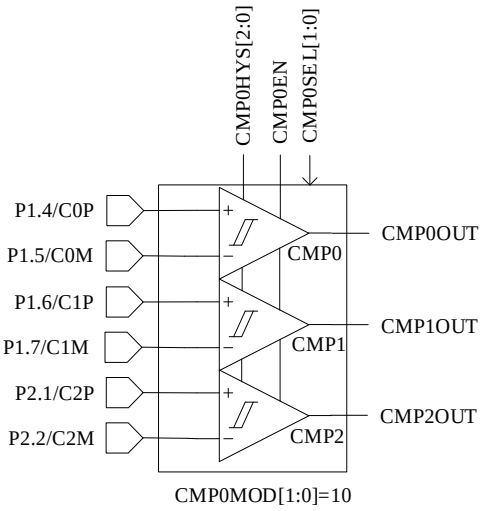


Figure 29-11 Diagram of Differential Triple Comparator Mode

When setting CMP_CR2[CMP0MOD] = 11, the dual comparator mode is selected for motor speed detection. The input and output ports are shown in Figure 29-12. The negative input of the comparators are connected together to P1.5, the positive input are connected to P1.4 and P1.3, and the outputs are CMP0OUT and CMP1OUT respectively. In this mode, the number of comparators is determined by CMP_CR2[CMP0SEL]. When setting CMP_CR2[CMP0SEL] = 00, CMP0 and CMP1 all works simultaneously, which is the recommended setting. When setting CMP_CR2[CMP0SEL] = 01, only CMP0 works and CMP1 is idle. When setting CMP_CR2[CMP0SEL] = 10, only CMP1 works and CMP0 is idle.

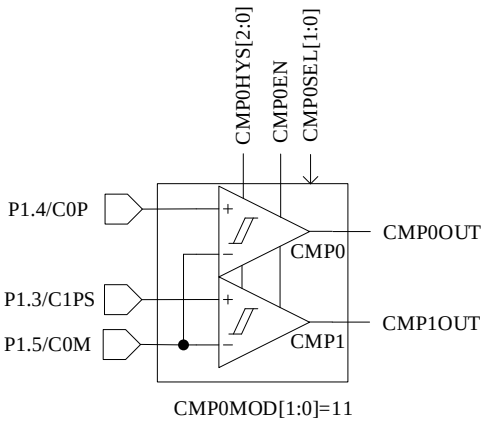


Figure 29-12 Diagram of Dual Comparator Mode

The output signal of CMP0/CMP1/CMP2 is sent to Timer1 after filtering and sampling module.

29.1.4 Comparator Sampling

The comparator sampling function is mainly used for the square wave control and RSD, which can eliminate the nonlinarity of inverter. Please refer to CMP/GPIO Position Detected Event for square wave control in 15.1.2.3 and

RSD Comparator Sampling for RSD in 16.1.7.1.

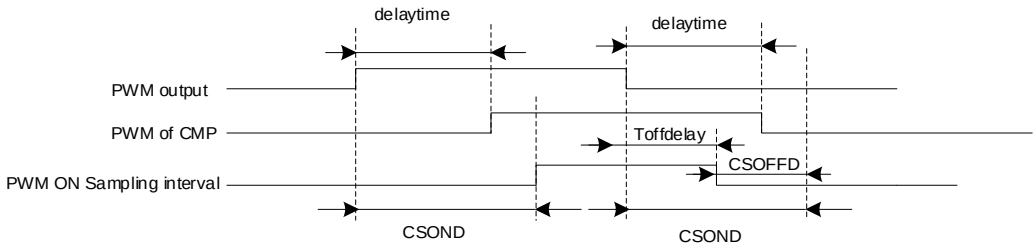


Figure 29-13 PWM ON Sampling Mode

There is a delay from PWM output to the output of the comparator, which is mainly affected by the following factors: the size of the drive resistor, the switching speed of the power device, and the input delay and hysteresis settings of the comparator. As shown in Figure 29-13, the delay time is from the chip output to the comparator output. When performing high-level sampling, the sampling interval should be enveloped by the actual high-level output of the comparator. First, set the sampling ON-delayed time CMP_SAMR[CSOND] to overcome the output delay and the oscillation interval of the power device. The end time of the sampling interval is CMP_SAMR[CSOND] after the chip outputs the falling edge of PWM. At this time, the actual sampling window has exceeded the comparator high level interval. Set the sampling OFF-advanced time CMP_SAMR[CSOFFD] to stop sampling Toffdelay after the PWM output falling edge, where Toffdelay = CMP_SAMR[CSOND] – CMP_SAMR[CSOFFD]. By configuring CMP_SAMR[CSOND] and CMP_SAMR[CSOFDD], the sampling interval can be located in the high-level interval of the actual output of the comparator.

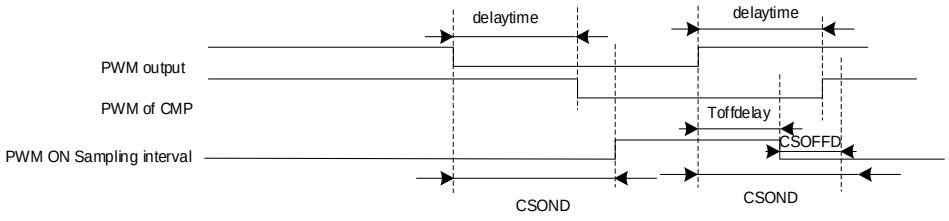


Figure 29-14 PWM OFF Sampling Mode

Similarly, when performing low-level sampling, set the sampling ON-delayed time CMP_SAMR[CSOND] and the sampling OFF-advanced time CMP_SAMR[CSOFFD] reasonably to ensure that the actual sampling interval is located in the actually low-voltage output interval by the comparator.

Method for measuring the delay of PWM output to comparator: Set CMP_CR3[SAMSEL] = 00 to disable the comparator sampling delay function. Set CMP_CR3[CMPSEL] to select the corresponding comparator output to

test port P0.7 (FU6832L/N) or P1.1 (FU6832F/S). Enable the PWM output and comparator, manually rotate the motor to change the comparator value, and measure the delay between the PWM output and the comparator output.

29.1.5 Comparator Output

Configure CMP_CR3[CMPSEL] to select the output of one of the comparator results to P0.7 (FU6832L/N) or P1.1 (FU6832F/S).

29.2 Comparator Register

29.2.1 CMP_CR0 (0xD5)

Bit	7	6	5	4	3	2	1	0
Name	CMP3IM		CMP2IM		CMP1IM		CMP0IM	
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:6]	CMP3IM	CMP3 interrupt mode 00: No interrupt generates 01: Interrupt on rising edge 10: Interrupt on falling edge 11: When a rising edge is detected, clear DRV_OUT[MOE] to 0, and the interrupt event flag bit CMP_SR[CMP3IF] is set to 1. However, the interrupt will not be enabled (Note: for Cycle-by-Cycle current limiting function, EVT_FILT[MOEMD] = 10/11 should be configured)
[5:4]	CMP2IM	CMP2 interrupt mode Refer to CMP_CR0[CMP0IM] description
[3:2]	CMP1IM	CMP1 interrupt mode Refer to CMP_CR0[CMP0IM] description
[1:0]	CMP0IM	CMP0 interrupt mode 00: No interrupt generates 01: Interrupt on rising edge 10: Interrupt on falling edge 11: Interrupt on both rising/falling edges

29.2.2 CMP_CR1 (0xD6)

Bit	7	6	5	4	3	2	1	0
Name	HALLSEL	CMP3MOD		CMP3EN	CMP3HYS	CMP0HYS		
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	HALLSEL	Hall signal function transfer 0: do not have function transfer, P0.2, P3.7, P3.6 as input for Hall signal 1: have function transfer, P1.4, P1.6, P2.1 as input for Hall signal
[6:5]	CMP3MOD	CMP3 mode selection Negative input connect to P2.6 or DAC0 output 00: Single comparator mode, P2.7 is connected to the positive input, refer to Figure 29-2 01: Dual comparator mode, P2.0 and P2.3 are connected to positive inputs, refer to Figure 29-3 1X: Triple Comparator mode, P2.0, P2.3 and P2.7 are connected to the positive inputs, refer to Figure 29-4
[4]	CMP3EN	CMP3 enable 0: Disable 1: Enable
[3]	CMP3HYS	CMP3 hysteresis selection 0: no hysteresis 1: hysteresis voltage is 15mV
[2:0]	CMP0HYS	CMP0/1/2 hysteresis voltage selection: 000: No hysteresis 001: ±2.5mV 010: –5mV

		100: +5mV 011: ±5mV 101: –10mV 110: +10mV 111: ±10mV

29.2.3 CMP_CR2 (0xDA)

Bit	7	6	5	4	3	2	1	0
Name	CMP4EN	CMP0MOD		CMP0SEL		RSV		CMP0EN
Type	R/W	R/W	R/W	R/W	R/W	—	—	R/W
Reset	0	0	0	0	0	—	—	0

Bit	Name	Description
[7]	CMP4EN	CMP4 enable 0: Disable 1: Enable
[6:5]	CMP0MOD	CMPG mode selection 00: No built-in resistors triple comparator mode, refer to Figure 29-8 01: Built-in resistors triple comparator mode. Function transfer can be configured by CMP_CR4[CMP0FS], refer to Figure 29-9 and Figure 29-10 10: Differential triple comparator mode, refer to Figure 29-11 11: Dual comparator mode. CMP0/CMP1 works and CMP2 is idle, refer to Figure 29-12
[4:3]	CMP0SEL	Port combination selection of CMPG. This bit is used with CMP_CR2[CMP0MOD]. The default value is 00. Under the square wave control application, the CMP_CR2[CMP0SEL] is automatically controlled by TIM1_DBRx[T1CPE], which controls the enable of each comparator.

Table 29-1 Function Description of CMPG Port and CMP_CR2[CMP0MOD] Combination

CMP_CR2 [CMP0MOD]	CMP_CR2 [CMP0SEL]	Description
00	00	CMP0/1/2 work simultaneously, refer to Figure 29-8. The negative input of 3 comparators are connected to C0M. The hardware compares the positive inputs C0P, C1P and C2P with C0M, and the output results are CMP0OUT, CMP1OUT and CMP2OUT respectively.
	01	CMP0 works and CMP1/2 are idle. The positive input is connected to C0P, the negative input is connected to C0M, and the output is CMP0OUT.
	10	CMP1 work and CMP0/2 are idle. The positive input is connected to C1P, the negative input is connected to C0M, and the output is CMP0OUT.
	11	CMP2 work and CMP0/1 are idle. The positive input is connected to C2P, the negative input is connected to C0M, and the output is CMP2OUT.
01	00	CMP0/1/2 work simultaneously, refer to Figure 29-9 and Figure 29-10. The negative inputs of the 3 comparators connected to the neutral point of built-in resistors. When CMP0FS = 0, the hardware automatically compares the positive inputs C0P, C1P, and C2P with C0M. When CMP0FS = 1, the hardware automatically compares the positive inputs C0P, C1PS and

					C2PS with C0M. The output results are CMP0OUT, CMP1OUT, and CMP2OUT respectively.	
				01	CMP0 works and CMP1/2 are idle. The positive input is connected to C0P, the negative input is connected to C0M, and the output is CMP0OUT.	
				10	CMP1 works and CMP0/2 are idle. When CMP0FS = 0, the positive input is connected to C1P. When CMP0FS = 1, the positive input is connected to C1PS. The negative input is connected to the neutral point of the built-in BEMF resistors, and the output is CMP1OUT.	
				11	CMP2 works and CMP0/1 are idle. When CMP0FS = 0, the positive input is connected to C2P. When CMP0FS = 1, the positive input is connected to C2PS. The negative input is connected to the neutral point of the built-in BEMF resistors, and the output is CMP2OUT.	
			10	00	CMP0/1/2 work simultaneously. Refer to Figure 29-11, the positive inputs of the three comparators are connected to C0P, C1P, and C2P, and the corresponding negative inputs are connected to C0M, C1M, and C2M, and the output results are CMP0OUT, CMP1OUT, and CMP2OUT, respectively.	
				01	CMP0 works and CMP1/2 are idle. Positive input is connected to C0P, negative input is connected to C0M, output is CMP0OUT.	
				10	CMP1 works and CMP0/2 are idle. Positive input is connected to C1P, negative input is connected to C1M, output is CMP01UT.	
				11	CMP2 work and CMP1/0 are idle. Positive input is connected to C2P, negative input is connected to C2M, output is CMP2OUT.	
			11	00	CMP0/1 work simultaneously, refer to Figure 29-12. 2 comparators are connected to C0P and C1PS for the positive input and C0M for the negative input, and the output results are CMP0OUT and CMP1OUT respectively.	
				01	CMP0 works and CMP1 is idle. Positive input is connected to C0P, negative input is connected to C0M, and output is CMP0OUT.	
				10	CMP1 works and CMP0 is idle. Positive input is connected to C1PS, negative input is connected to C0M, and output is CMP1OUT.	
				11	Reserved	
[2:1]	RSV	Reserved				
[0]	CMP0EN	CMP0 enable 0: Disable 1: Enable				

29.2.4 CMP_CR3 (0xDC)

Bit	7	6	5	4	3	2	1	0
Name	RSV	DBGSEL		SAMSEL		CMPSEL		
Type	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	0	0	0	0	0	0	0

Bit	Name	Description
[7]	RSV	Reserved
[6:5]	DBGSEL	Debug signal selection, output to P0.1 00: Disable debug signal output 01: End of diode freewheeling and detection of ZCP 10: ADC trigger signal 11: Comparator sampling interval
[4:3]	SAMSEL	CMP0, CMP1, CMP2 and ADC at PWM ON/OFF sampling delay enable 00: Sampling at both ON and OFF, no sampling delay 01: Sampling at OFF, sampling delay according to CMP_SAMR 10: Sampling at ON, sampling delay according to CMP_SAMR 11: Sampling at both ON and OFF, sampling delay according to CMP_SAMR
[2:0]	CMPSEL	Comparator output selection Select one of the comparator output signals to P0.7 (FU6832L/N) or P1.1 (FU6832F/S) 000: Disable 001: CMP0 010: CMP1 011: CMP2 100: CMP3 101: CMP4 111: omega start flag (estimator estimated angle output flag bit, refer Estimator Angle)

29.2.5 CMP_CR4 (0xE1)

Bit	7	6	5	4	3	2	1	0
Name	CMP4OUT	RSV				FAEN	CMP0FS	RSV
Type	R	—	—	—	—	R/W	R/W	—
Reset	1	—	—	—	—	0	0	—

Bit	Name	Description
[7]	CMP4OUT	CMP4 comparison results
[6:3]	RSV	Reserved
[2]	FAEN	Filter sampling coefficient expansion enable When enabled, the clock source of TIM1_CR3[T1INM] and CMP_SAMR is increased by 4 times 0: Disable 1: Enable
[1]	CMP0FS	CMP1, CMP2 function transfer enable 0: do not have function transfer, refer to Figure 29-9 1: have function transfer; this bit is valid when CMP_CR2[CMP0_MOD] = 01, otherwise it is meaningless, refer to Figure 29-10
[0]	RSV	Reserved

29.2.6 CMP_SAMR (0x40AD)

Bit	7	6	5	4	3	2	1	0
Name	CSOND				CSOFFD			
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	1

Bit	Name	Description
[7:4]	CSOND	CMP0/CMP1/CMP2 ON-delayed sampling time When the PWM commucate from OFF to ON or from ON to OFF, the ON and OFF of power device will affect the signal input of comparison. Set CMP_SAMR[CSOND] to avoid this interference. The delay time could be multiplied by 4 times according to the CMP_CR4[FAEN] setting. CMP_CR4[FAEN] = 0: ON-delayed sampling time = 8*CMP_SAMR[CSOND]*T CMP_CR4[FAEN] = 1: ON-delayed sampling time = 32*CMP_SAMR[CSOND]*T Note: ■ CMP_SAMR[CSOND] must be larger than or equal to CMP_SAMR[CSOFFD] ■ For BLDC control, please refer to Sampling ■ For RSD, please refer to RSD Comparator Sampling
[3:0]	CSOFFD	CMP0, CMP1, CMP2 OFF-advanced sampling time End sampling CMP_SAMR[CSOND] – CMP_SAMR[CSOFFD] after the back edge of PWM output to ensure sampling interval enveloped by the PWM interval. OFF-advanced sampling time could be multiplied by 4 times according to the CMP_CR4[FAEN] setting. CMP_CR4[FAEN] = 0: Off-advanced sampling time = 8*CMP_SAMR[CSOFFD]*T CMP_CR4[FAEN[= 1: Off-advanced sampling time = 32*CMP_SAMR[CSOFFD]*T Note: ■ CMP_SAMR[CSOND] must be larger than or equal to CMP_SAMR[CSOFFD] ■ For BLDC control, please refer to Sampling ■ For RSD, please refer to RSD Comparator Sampling

29.2.7 CMP_SR (0xD7)

Bit	7	6	5	4	3	2	1	0
Name	CMP3IF	CMP2IF	CMP1IF	CMP0IF	CMP3OUT	CMP2OUT	CMP1OUT	CMP0OUT
Type	R/W0	R/W0	R/W0	R/W0	R	R	R	R
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7]	CMP3IF	CMP3 interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[6]	CMP2IF	

		CMP2 interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[5]	CMP1IF	CMP1 interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[4]	CMP0IF	CMP0 interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning
[3]	CMP3OUT	CMP3 comparison result
[2]	CMP2OUT	CMP2 comparison result
[1]	CMP1OUT	CMP1 comparison result
[0]	CMP0OUT	CMP0 comparison result

29.2.8 EVT_FILT (0xD9)

Bit	7	6	5	4	3	2	1	0
Name	RSV			MOEMD		EFSRC	EFDIV	
Type	—	—	—	R/W	R/W	R/W	R/W	R/W
Reset	—	—	—	0	0	0	0	0

Bit	Name	Description
[7:5]	RSV	Reserved
[4:3]	MOEMD	Hardware clear and enable MOE selection The bus current protection event will cause MOE cleared and enabled by hardware 00: MOE is not automatically cleared 01: MOE is automatically cleared 10: MOE is automatically cleared and hardware enables MOE automatically after 10μs or at an overflow or underflow event of the Driver counter (for square wave control) 11: MOE is automatically cleared and hardware enables MOE automatically after 5μs or at an overflow or underflow event of the Driver counter (for square wave control)
[2]	EFSRC	Input source of current protection events 0: CMP3 interrupt 1: external interrupt INT0
[1:0]	EFDIV	Current protection filtering width: 00: No filtering 01: 4 system clock cycles 10: 8 system clock cycles 11: 16 system clock cycles

29.2.9 TSD_CR (0x402F)

Bit	7	6	5	4	3	2	1	0

Name	TSDEN	RSV			TSDADJ			
Type	R/W	—	—	—	R/W	R/W	R/W	R/W
Reset	0	—	—	—	1	1	1	1

Bit	Name	Description
[7]	TSDEN	Temperature detection function enable 0: Disable 1: Enable
[6:4]	RSV	Reserved
[3:0]	TSDADJ	Overtemperature value (chip junction temperature) 0000: 71°C 0001: 75°C 0010: 80°C 0011: 84°C 0100: 89°C 0101: 94°C 0110: 99°C 0111: 105°C 1000: 111°C 1001: 116°C 1010: 123°C 1011: 131°C 1100: 136°C 1101: 142°C 1110: 150°C 1111: Reserved

30 Power Supply

30.1 LDO

The MCU contains two internal LDO output modules: VDD5 and VDD18.

30.1.1 LDO Instructions

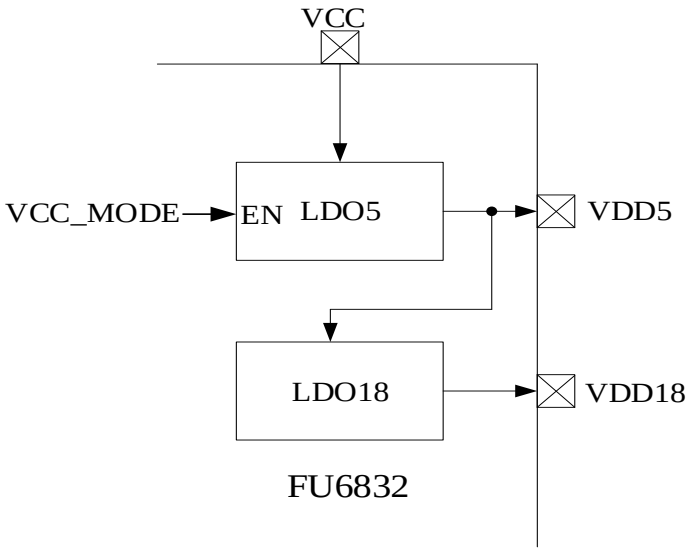


Figure 30-1 Functional Block Diagram of Power Supply

The corresponding input and output ports of the LDO module are shown in Figure 30-1. The LDO convert the input supply voltage to 5V (VDD5) and 1.8V (VDD18) as the power supply for the chip internal analog and digital modules, respectively. Configuring VCC_MODE to select VDD5 to be generated by internal LDO5 or externally supply. As shown in Figure 30-2, in the debug tool, if Vcc Mode is not selected, VCC_MODE = 0 and VDD5 voltage is generated by internal LDO. If Vcc Mode is selected, VCC_MODE = 1 and connect external 5V power supply to VDD5.

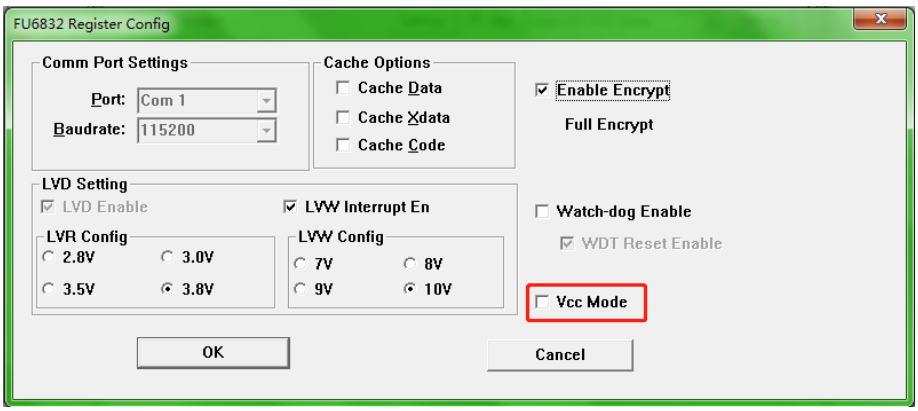


Figure 30-2 VCC_MODE Configuration

30.2 Low Voltage Detection

30.2.1 Low Voltage Detection Introduction

The low voltage detection consists of two parts: low voltage warning and low voltage reset.

30.2.2 Low Voltage Detection Instructions

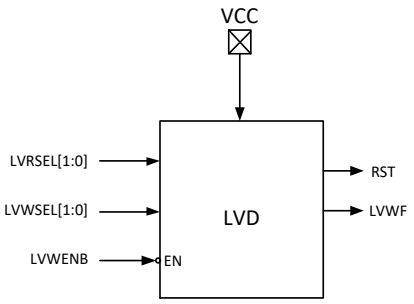


Figure 30-3 Diagram of Low Voltage Detection Module

The instructions for the low voltage detection module are as follows:

- Low voltage warning and low voltage reset are always enabled by default
- The low voltage warning voltage can be set to 7V/8V/9V/10V and the interrupt can be enabled. When the VCC voltage is lower than the warning voltage setting, the interrupt will be triggered after the interrupt is enabled.
- The low voltage reset voltage can be set to 2.8V/3.0V/3.5V/3.8V. When the VCC voltage is lower than the reset voltage setting, the chip will be reset

The low voltage warning voltage, interrupt configuration and low voltage reset voltage are configured through the configuration options in the debug tool, as shown in Figure 30-4.

LVR Config sets the low voltage reset voltage, LVW Interrupt sets the low voltage interrupt enable, and LVW Config sets the low voltage warning voltage.

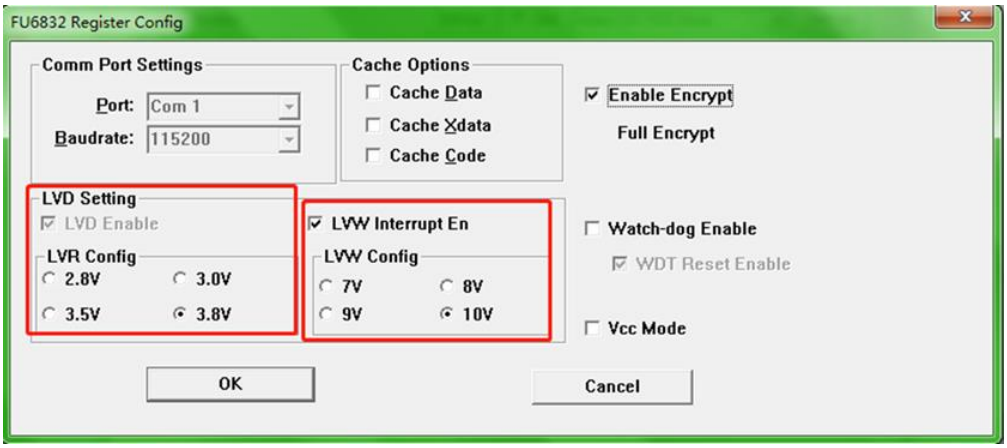


Figure 30-4 Configuration of Warning Voltage, Interrupt and Reset Voltage

30.2.3 Low Voltage Detection Register

30.2.3.1 LVSR (0xDB)

Bit	7	6	5	4	3	2	1	0
Name	RSV		EXT0CFG			TSDF	LVWF	LVWIF
Type	—	—	R/W	R/W	R/W	R	R	R/W0
Reset	—	—	0	0	0	0	0	0

Bit	Name	Description
[7:6]	RSV	Reserved
[5:3]	EXT0CFG	INT0 interface selection 000: P0.0 001: P0.1 010: P0.2 011: P0.3 100: P0.4 101: P0.5 110: P0.6 111: CMP4 output
[2]	TSDF	Overtemperature state bit 0: The current temperature does not exceed the threshold 1: The current temperature exceeds the threshold Note: This flag bit is often used with the TSD interrupt flag TCON[5]
[1]	LVWF	VCC low voltage flag The VCC low voltage flag reflects whether the current state is low voltage or not 0: No low voltage warning 1: Low voltage warning
[0]	LVWIF	VCC low voltage interrupt flag Read: 0: No interrupt pending 1: Interrupt pending Write: 0: Clear this bit to 0 1: No meaning Note: When the low voltage detect interrupt is not enabled, this bit will not be set to 1

31 Flash

31.1 Flash Introduction

The MCU provides 16k bytes on-chip Flash space. It supports full chip erase/write and sector erase/write.

Features:

- Each sector is 128 bytes, for a total of 128 sectors
- The last sector (address range: 0x3F80 to 0x3FFF) will not be erased at any moment
- Sector erase and chip erase takes about 120ms ~ 150ms time

31.2 Flash Instructions

- To ensure the safety of Flash operation, all interrupt events must be disabled before operation to avoid misoperation of Flash by MOVX instruction in interrupt processing
- Flash should be unlocked before operation. Write 0x5a, 0x1F to Flash unlock register FLA_KEY in sequence and then start the software programming Flash. If the sequence is not correct or write other values to FLA_KEY, Flash space will be freezed until the next reset. After unlocking, any write to FLA_CR will cause FLA_KEY to be locked again.
- If Flash is changed, the CRC result will be changed as well

31.3 Flash Register

31.3.1 FLA_CR (0x85)

Bit	7	6	5	4	3	2	1	0
Name	RSV			FLAERR	RSV	FLAPRE	FLAERS	FLAEN
Type	—	—	—	R	—	R/W	R/W	R/W
Reset	—	—	—	0	—	0	0	0

Bit	Name	Description
[7:5]	RSV	Reserved
[4]	FLAERR	Programming error flag 0: Programming or pre-programming operation succedded when writing to Flash 1: Programming or pre-programming operation failed when writing to Flash
[3]	RSV	Reserved
[2]	FLAPRE	Sector pre-programming enable (The sector must be pre-programmed before erasing) 0: Disable 1: Enable Note: FLA_CR[FLAPRE] works only when FLA_CR[FLAEN] is 1
[1]	FLAERS	Erase enable 0: Disable 1: Enable Note: FLA_CR[FLAERS] works only when FLA_CR[FLAEN] is 1
[0]	FLAEN	Flash operation enable 0: Disable 1: Enable

31.3.2 FLA_KEY (0x84)

Bit	7	6	5	4	3	2	1	0
Name	FLA_KEY							
Type	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	FLA_KEY	Write 0x5A, 0x1F to FLA_KEY in sequence to unlock Flash operation; Write any value to FLA_CR to lock Flash operation

Bit	7	6	5	4	3	2	1	0
Name	RSV						FLAKSTA	
Type	—	—	—	—	—	—	R	R
Reset	—	—	—	—	—	—	0	0

Bit	Name	Description
[7:2]	RSV	Reserved
[1:0]	FLAKSTA	Read: reflect the Flash unlock status 00: Locked 01: 0x5A has been written, waiting for 0x1F to be written 10: Freezed 11: Unlocked

32 CRC

32.1 Functional Block Diagram of CRC

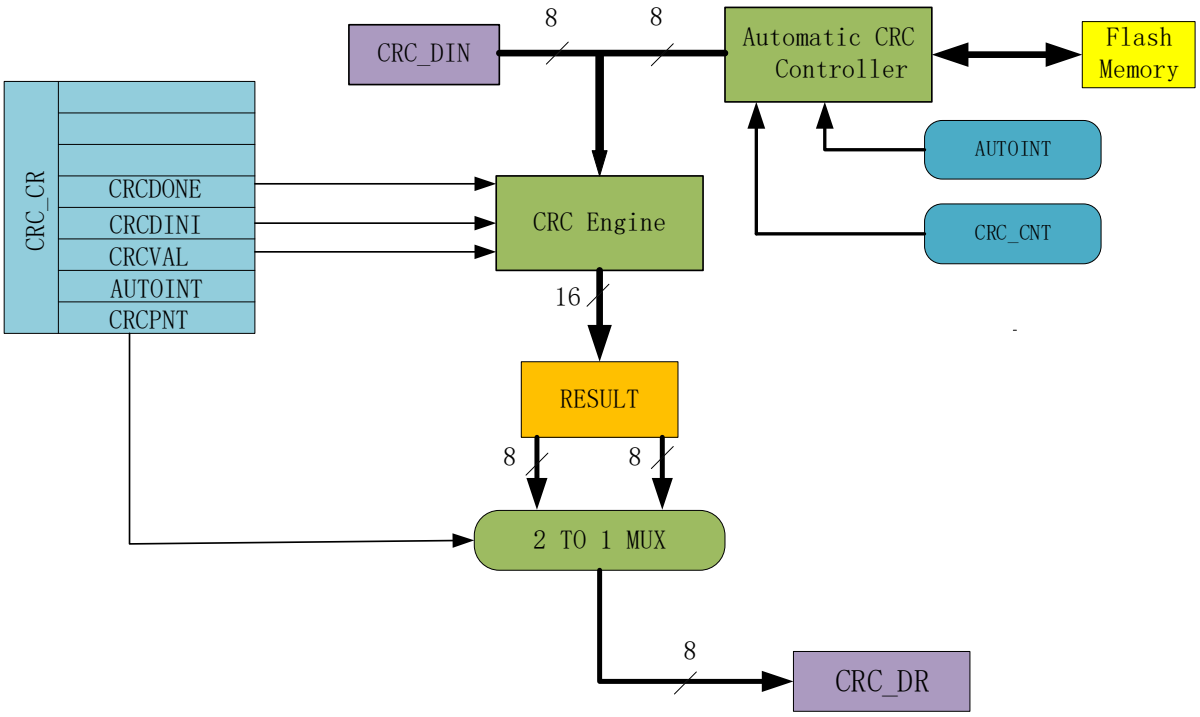


Figure 32-1 Functional Block Diagram of CRC

CRC module gets the result of CRC calculation for any 8-bit data based on a fixed polynomial. As shown in Figure 32-1, CRC receives the 8-bit data from CRC_DIN and sends the 16-bit result to the internal register after the calculation is completed. The result can be indirectly accessed through CRC_CR[CRCPNT] and CRC_DR.

Table 32-1 CRC Criteria and Polynomials

No.	CRC Criteria	Polynomials	Hexadecimal Representation
1	CRC12	x^12+x^11+x^3+x^2+x+1	0x80F
2	CRC16	x^16+x^15+x^2+1	0x8005
3	CRC16-CCITT	x^16+x^12+x^5+1	0x1021
4	CRC32	x^32+x^26+x^23+x^22+x^16+x^12 +x^11+x^10+x^8+x^9+x^5+x^4+x+1	0x04C11DB7

32.2 CRC16 Polynomial

Polynomial based on the CRC16-CCITT standard: x^16+x^12+x^5+1.

32.3 CRC16 Basic Logic Diagram

The circuit schematic digram of the serial CRC16 is shown in Figure 32-2. The chip implements using parallel algorithm. For each input byte, the MCU can calculate the result with 1 system clock cycle.

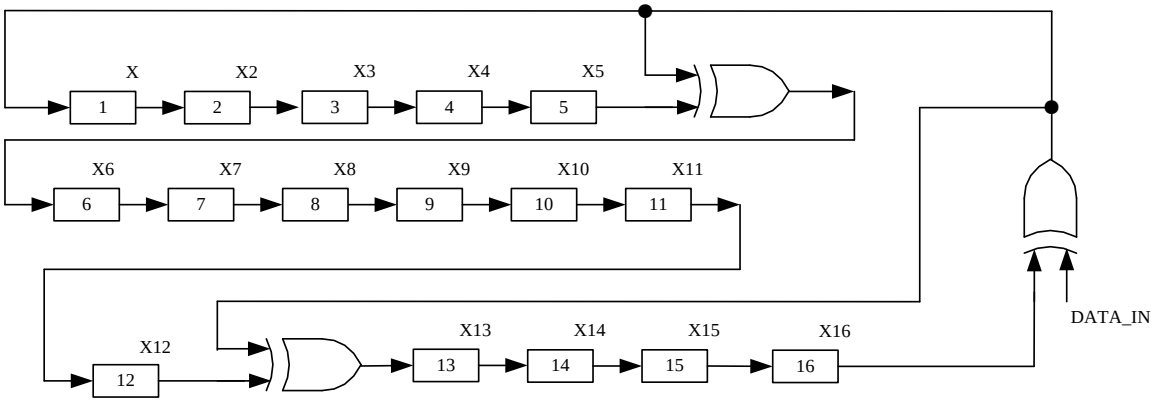


Figure 32-2 Circuit Schematic Diagram of CRC16

32.4 CRC Instructions

32.4.1 Calculation CRC of a Single Byte

The process of calculating the CRC value of a single byte is as follows:

1. Initialize CRC_DR, with two options: Configure CRC_CR[CRCVAL] and set CRC_CR[CRCDINI] to 1, with an initial value of 0x0000 or 0xFFFF. Initialize CRC by configuring CRC_CR[CRCPNT] and CRC_DR, which can configure any initial value;
2. Write data to CRC_DIN, and the CRC calculation is completed in the next clock cycle;
3. Read CRC result: Configure CRC_CR[CRCPNT] = 1, and the software reads CRC_DR to gets the high byte data. Configure CRC_CR[CRCPNT] = 0, and read CRC_DR to get the low byte data.

32.4.2 Calculation CRC of ROM Sector

To calculate the CRC value of a continuous area of data in the ROM:

1. Initialize CRC_DR, in the same way of single-byte CRC;
2. Configure CRC_BEG to set the start sector of the ROM to be calculated;
3. Configure CRC_CNT to set the increment from the start sector to the end sector;
4. Write 1 to CRC_CR[AUTOINT] and keep the other bits unchanged. Hardware start the automatic calculation process;
5. Read the CRC result.

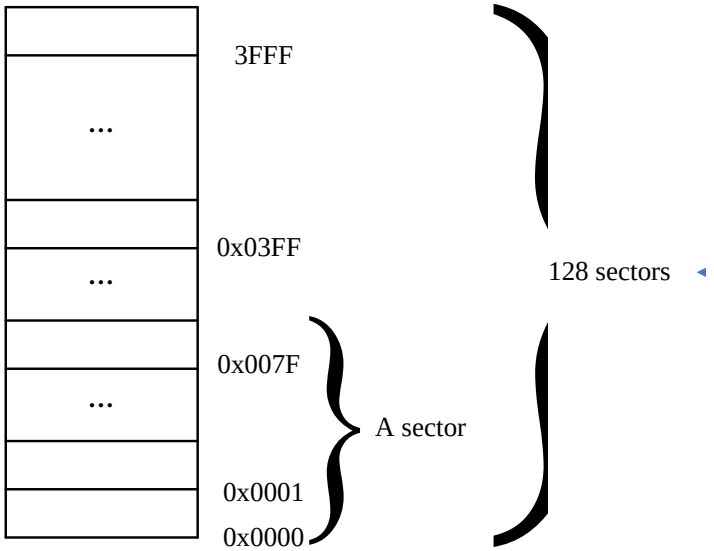


Figure 32-3 Diagram of ROM Sector

As shown in Figure 32-3, ROM has 16k bytes and is divided into 128 sectors, numbered from sector0 to sector127. Each sector contains 128 bytes. For CRC calculation of sector, the value CRCBEG of the start sector can be any value between 0x00 ~ 0x7F, including 0x00 and 0x7F. The value CRCCNT of the total number of sectors to be calculated can be 0x00 ~ 0x7F, including 0x00 and 0x7F.

As CRC_BEG increases, CRC_CNT should decrease accordingly. Eg: If the value of CRC_BEG is 0x7F, the value of CRC_CNT can only be 0x00, i.e. the CRC value of the data in the last sector is calculated. In this case, if the value of CRC_CNT is set to a large value, the CRC controller will automatically limit the number of sectors to be calculated. Finally, CRC module will only calculate the CRC value of the last sector.

32.5 CRC Register

32.5.1 CRC_CR (0x4022)

Bit	7	6	5	4	3	2	1	0
Name	RSV			CRCDONE	CRCDINI	CRCVAL	AUTOINT	CRCPNT
Type	—	—	—	R	W1	R/W	W1	R/W
Reset	—	—	—	1	0	0	0	0

Bit	Name	Description
[7:5]	RSV	Reserved
[4]	CRCDONE	Automatic CRC calculation completed flag During the automatic CRC calculation process, the hardware automatically writes 0 to this bit and the software program stops. In other cases, the hardware automatically sets this bit to 1, so it always returns 1 when reading this bit.
[3]	CRCDINI	CRC result initialization enable 0: No meaning 1: Trigger CRC result initialization
[2]	CRCVAL	CRC result initialization selection 0: CRC result is initialized to 0x0000 1: CRC result is initialized to 0xFFFF
[1]	AUTOINT	Sector CRC auto-calculation start 0: No meaning 1: Start sector CRC auto-calculation Please refer to Calculation CRC of ROM Sector
[0]	CRCPNT	CRC result pointer 0: Read CRC_DR accesses the low 8 bits of the 16-bit CRC result 1: Read CRC_DR accesses the high 8 bits of the 16-bit CRC result

Note: To calculate single byte CRC checksum, configure CRC_CR[AUTOINT] = 0.

32.5.2 CRC_DIN (0x4021)

Bit	7	6	5	4	3	2	1	0
Name	CRC_DIN							
Type	W	W	W	W	W	W	W	W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	CRC_DIN	CRC input data When writing data to this register, the CRC module automatically calculates a new CRC result according to the input data based on the last CRC result and overwrites the original CRC result. Note: This register is a virtual register, and the written data is not saved. Reading this address will return 0x00

32.5.3 CRC_DR (0x4023)

Bit	7	6	5	4	3	2	1	0
Name	CRC_DR							
Type	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	0	0	0	0	0	0	0	0

Bit	Name	Description
[7:0]	CRC_DR	CRC result output When read or write this register, the configuration of CRC_CR[CRCPNT] determines whether to access the high or low 8 bits of the CRC result.

32.5.4 CRC_BEG (0x4024)

Bit	7	6	5	4	3	2	1	0
Name	RSV	CRC_BEG						
Type	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	0	0	0	0	0	0	0

Bit	Name	Description
[7]	RSV	Reserved
[6:0]	CRC_BEG	Automatic calculation start sector of ROM for CRC Eg: If the value of CRC_BEG is 1, then the start address is 1*128 = 128, i.e., starts from the first byte of the second sector

32.5.5 CRC_CNT (0x4025)

Bit	7	6	5	4	3	2	1	0
Name	RSV	CRC_CNT						
Type	—	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Reset	—	0	0	0	0	0	0	0

Bit	Name	Description
[7]	RSV	Reserved
[6:0]	CRC_CNT	Sector area increment setting for auomatic CRC clculation The end sector for the automatic CRC calculation is determined by this bits

33 Sleep Mode

33.1 Sleep Mode Introduction

The chip provides three operating modes: normal mode, standby mode and sleep mode. The different operating modes are selected by setting the value of PCON[IDLE] and PCON[STOP].

The state of the module in various power consumption modes is summarized in Table 33-1.

Table 33-1 Power Mode

Power Mode	Description	Wake-up Source	Power Consumption Performance
Normal	All modules work at full speed except for peripherals that are disabled	NA	Higher power consumption, Best performance
Standby	The CPU clock is gated and the other function modules are turned off or working determined by their control bits. Watchdog Timer is gated	Any interrupt, External Reset/Debug reset	Low power consumption, Flexible performance
Sleep	Flash Deep Sleep. The analog fast clock circuit is turned off and software should ensure that the ADC, FOC, and driver module are disabled before the chip working in this mode. Watchdog Timer is disabled	External Interrupt, External Reset/Debug reset	Super low power consumption, Flexible performance

Note: It is recommended to insert 3 empty instructions after entering sleep mode.

```
PCON = 0x02;
_nop_();
_nop_();
_nop_();
```

33.2 Sleep Mode Register

33.2.1 PCON (0x87)

Bit	7	6	5	4	3	2	1	0
Name	RSV		GF3	GF2	GF1	RSV	STOP	IDLE
Type	—	—	R/W	R/W	R/W	—	R/W	R/W
Reset	—	—	0	0	0	—	0	0

Bit	Name	Description
[7:6]	RSV	Reserved
[5]	GF3	General flag bit 3
[4]	GF2	General flag bit 2
[3]	GF1	General flag bit 1
[2]	RSV	Reserved
[1]	STOP	Write 1 to make the chip into sleep mode, and this bit will be cleared by hardware after waked up
[0]	IDLE	Write 1 to make the chip into standby mode, and this bit will be cleared by hardware after waked up

Power consumption mode PCON[STOP:IDLE]:

00: Normal

01: Standby

1X: Sleep

34 Code Protection

34.1 Code Protection Introduction

The chip supports full Flash space encryption to protect the customer software property, avoiding illegal operations. When Flash is encrypted, the data inside cannot be read, and only the hardware CRC check can be used to confirm the content of software.

34.2 Code Protection Instructions

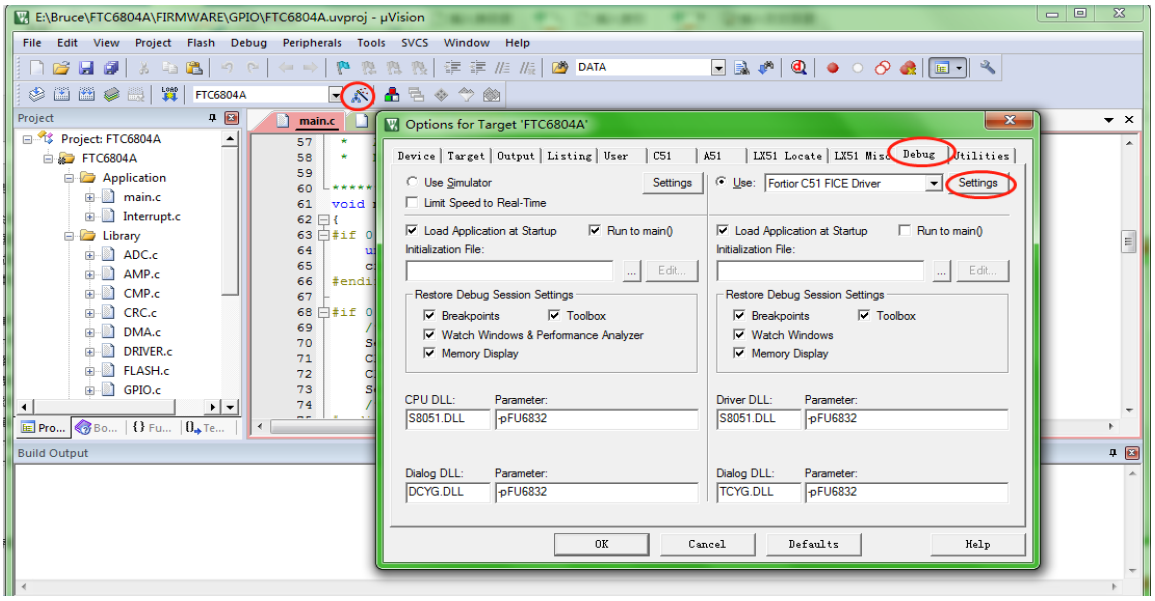


Figure 34-1 Code Protection Configuration

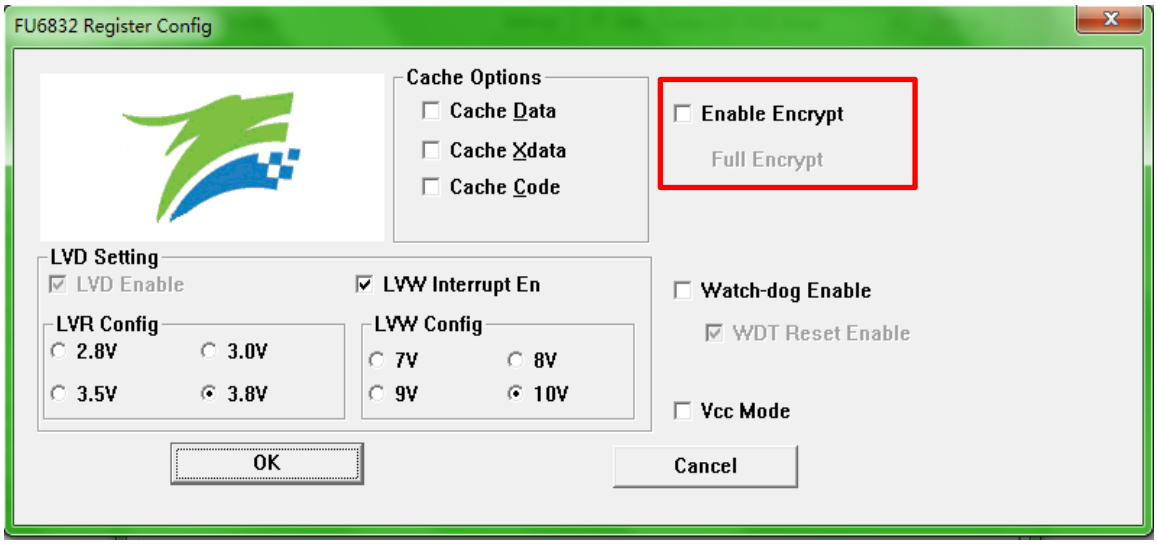


Figure 34-2 Full Code Protection Mode

Encryption steps are as follows:

1. Open the 8051 IDE, enter the Target Options and select the Debug tab before compiling. As shown in Figure 34-1, click Settings to enter the next step of setting;
2. As shown in Figure 34-2, click OK. Then compile the project and download it. Get the BIN file and burn it to Flash.

35 Modify Records

Date	Revision	Changes
2019/10/31	V1.0	Version for initial publishing This version corresponds to internal version V0.42
2020/03/25	V1.1	Add FU6832N/FU6832S
2020/07/22	V1.2	Update according to comments of application department Add document revision history
2020/10/19	V1.3	Modify registers and electrical characteristics parameters
2020/12/18	V1.4	Add FU6832F
2021/01/06	V1.5	Correct sequence number error in pin list tables
2022/04/18	V1.6	Document format standardizing
2022/05/27	V1.7	Document format standardizing
2022/06/08	V1.8	Update description of LIN_SR[ABORT]

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